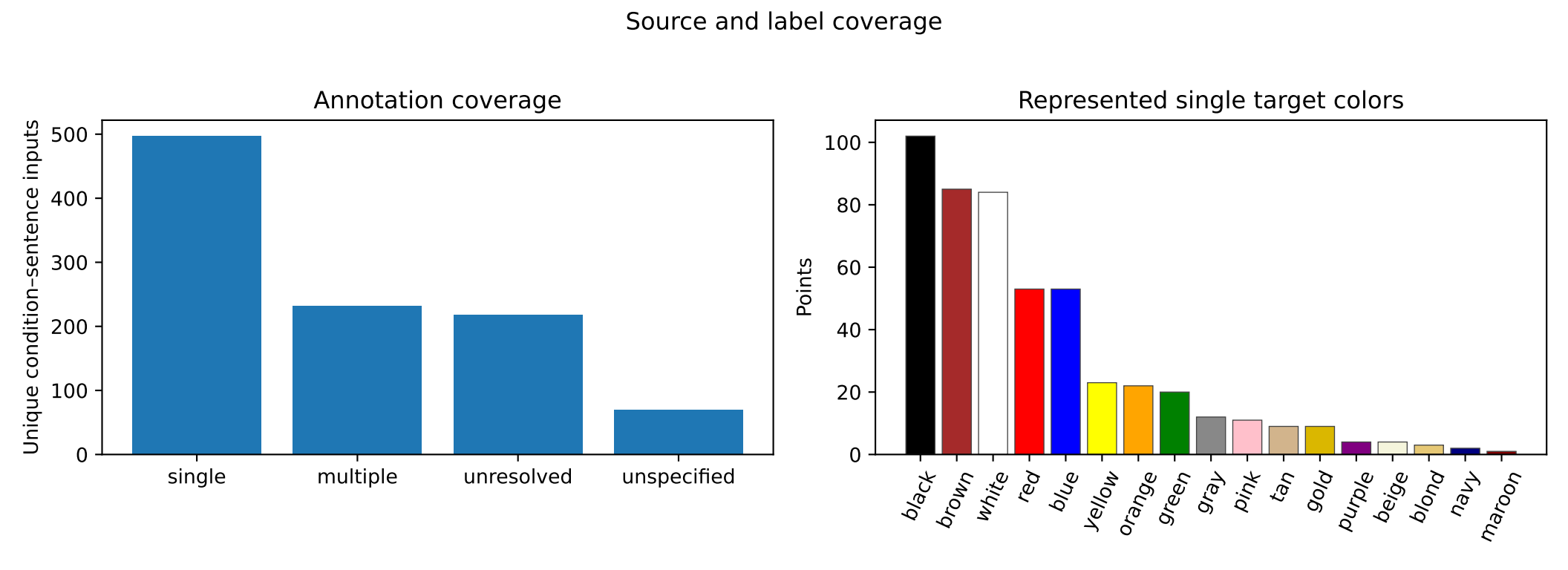
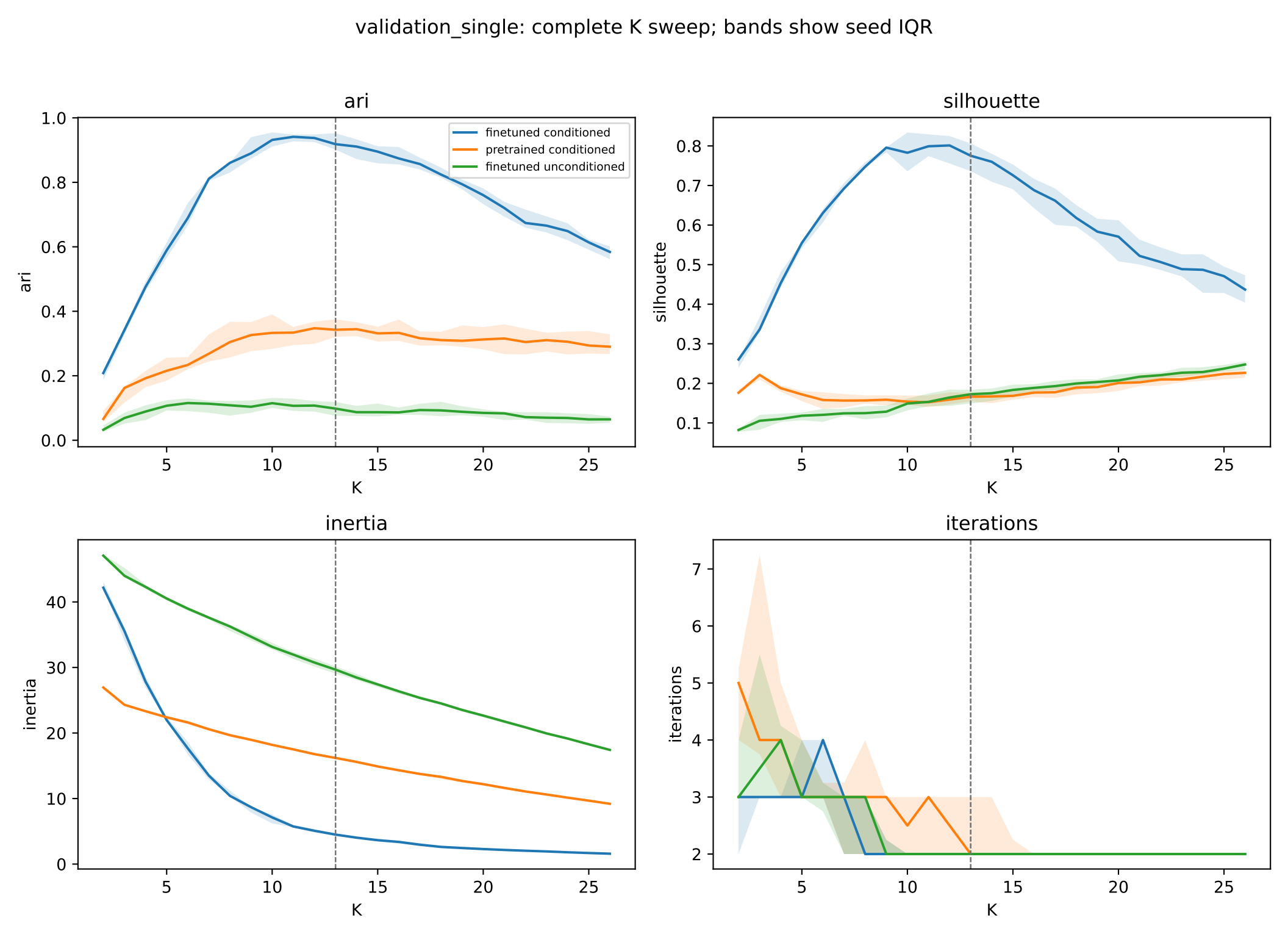
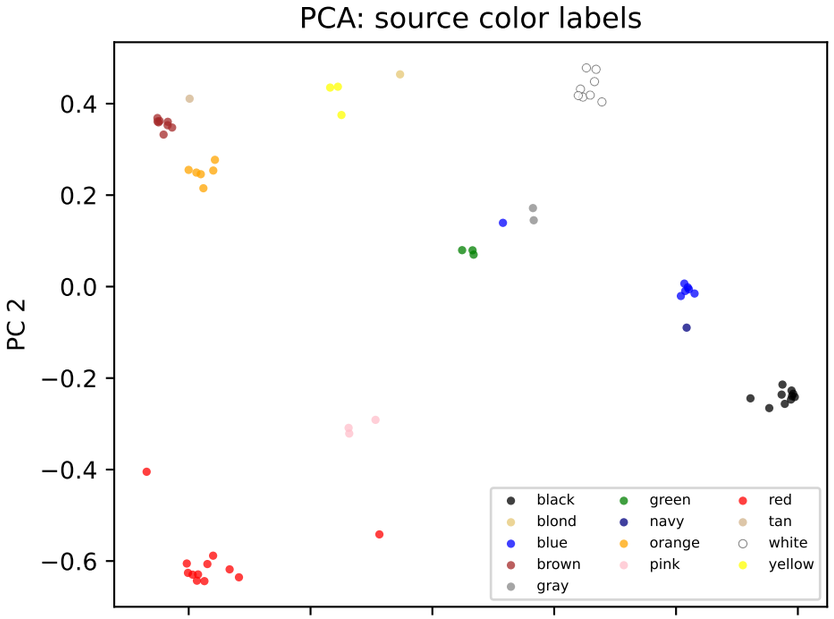
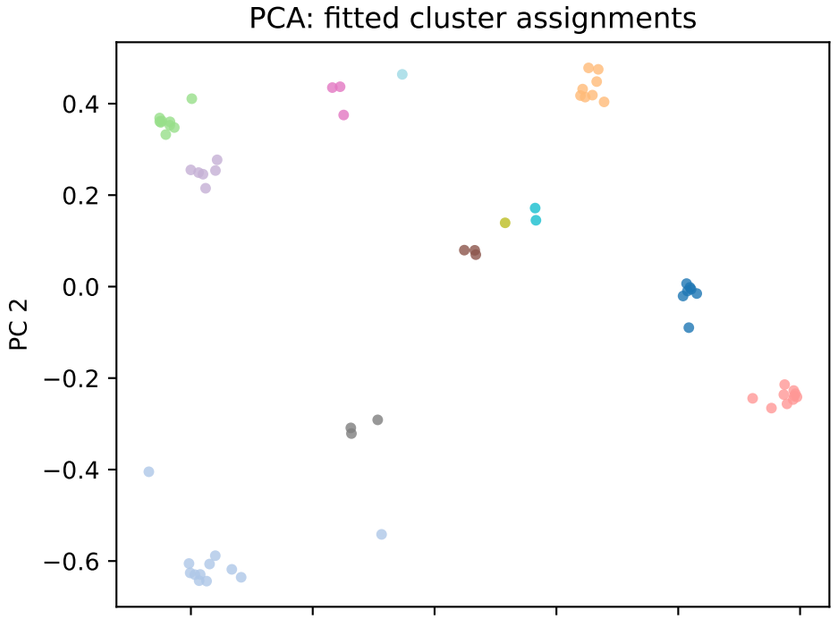
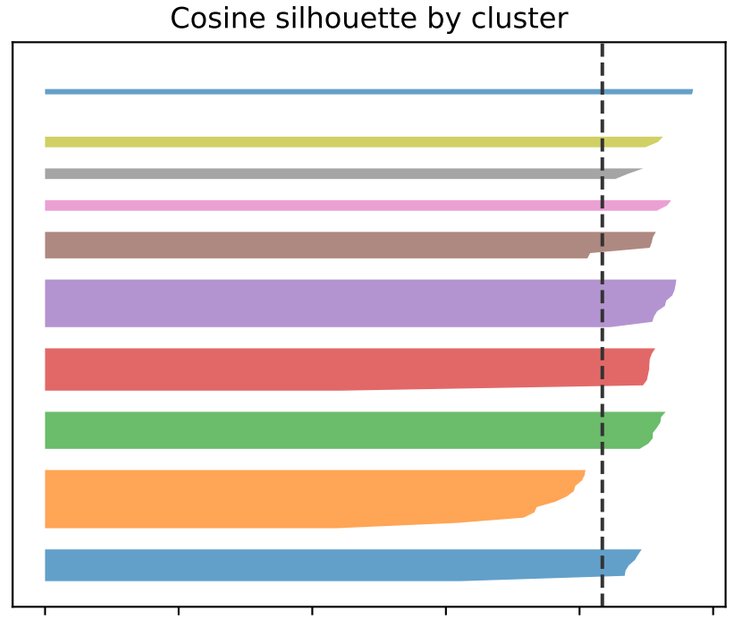

Source and label coverage
Annotation coverage
Unique condition–sentence inputs
0
100
200
300
400
500
single
multiple
unresolved
unspecified
Represented single target colors
Points
0
20
40
60
80
100
black
brown
white
red
blue
yellow
orange
green
gray
pink
tan
gold
purple
beige
blond
navy
maroon

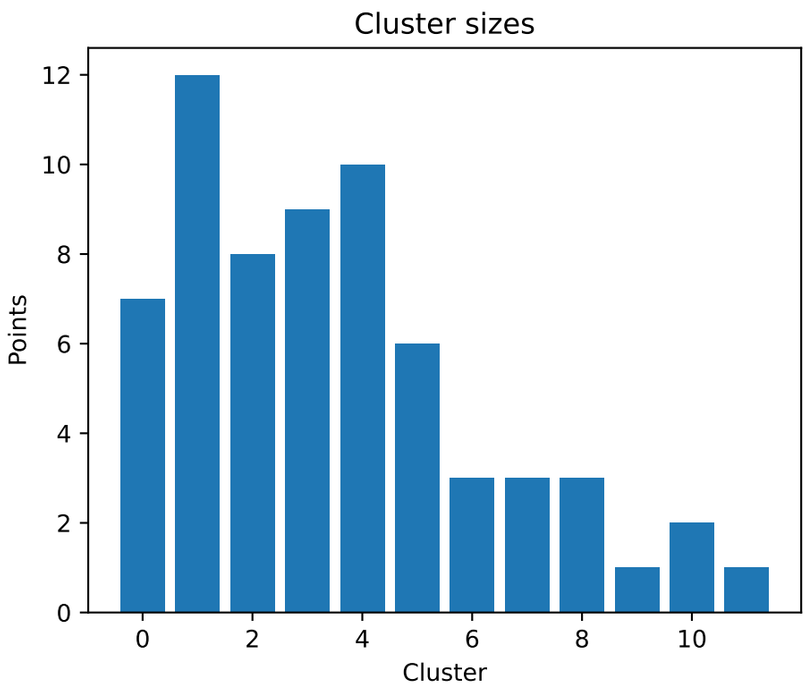

validation_single: complete K sweep; bands show seed IQR
ari
silhouette
inertia
iterations
finetuned conditioned
pretrained conditioned
finetuned unconditioned
K

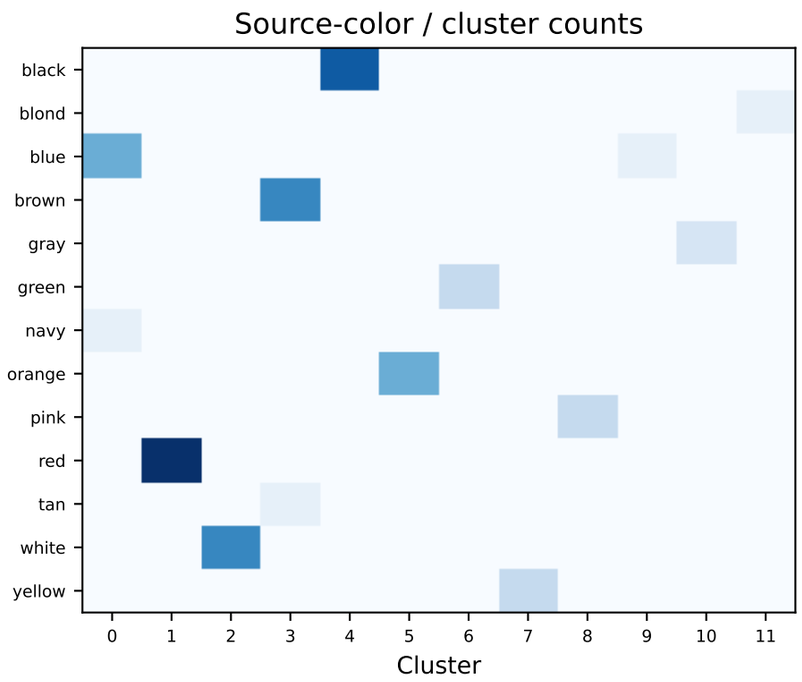
validation_single · finetuned_conditioned · K=12, seed=2; PCA explains 36.8%

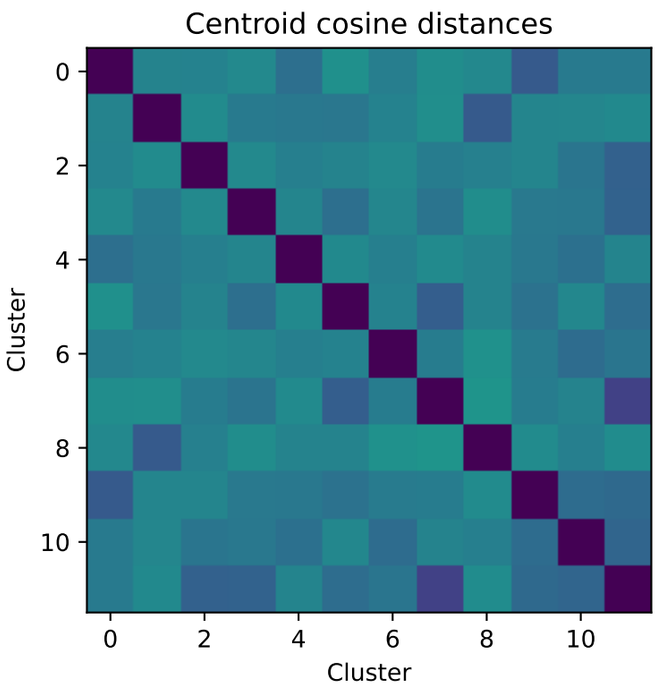




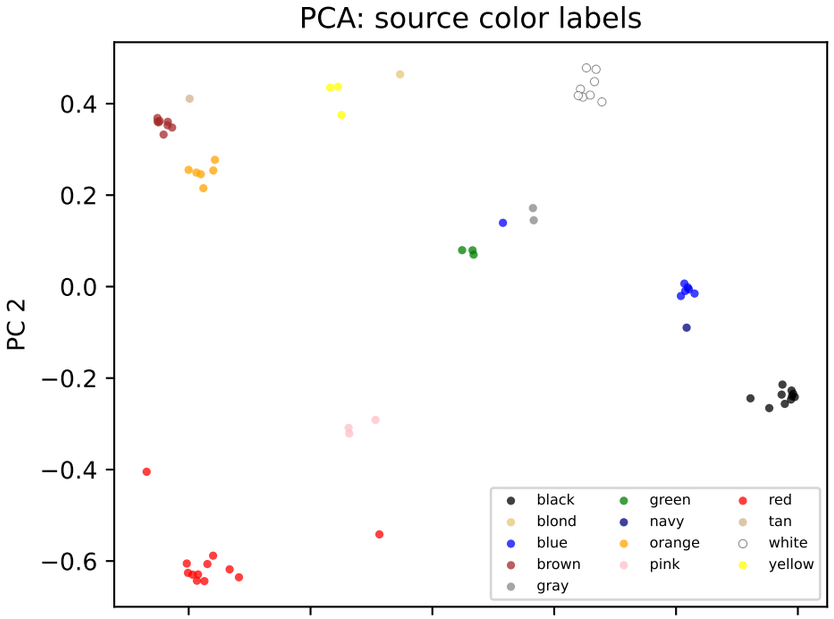


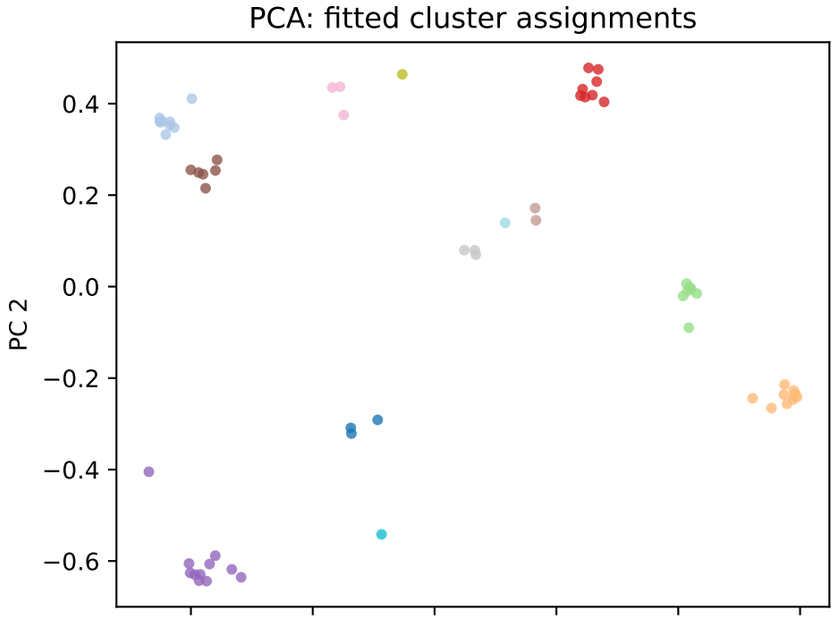


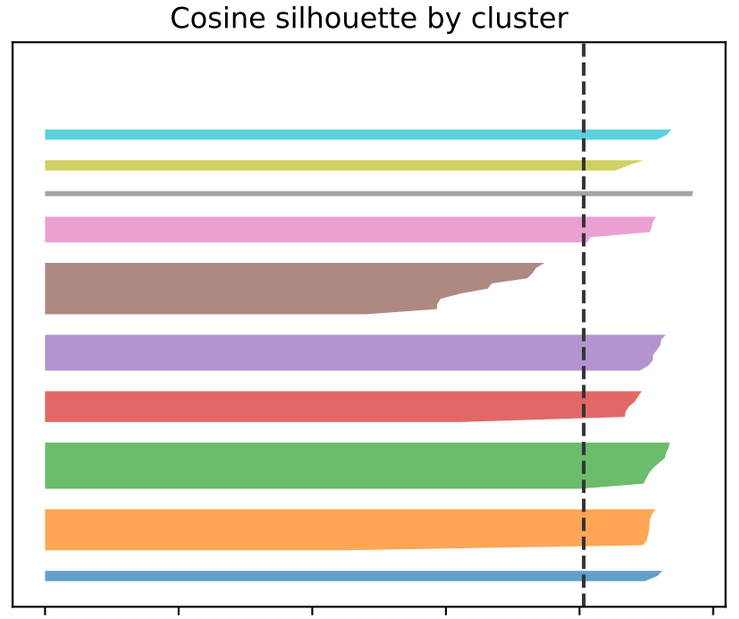


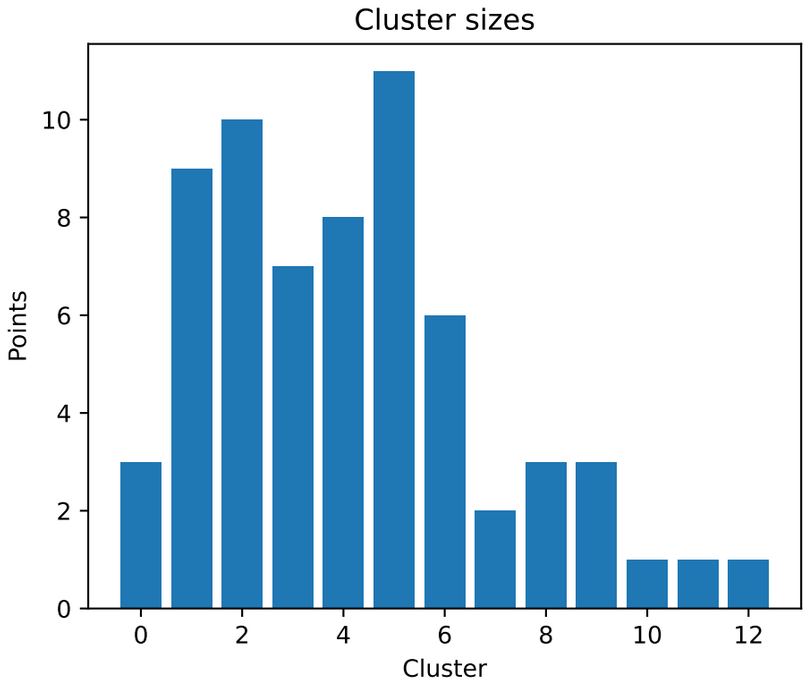


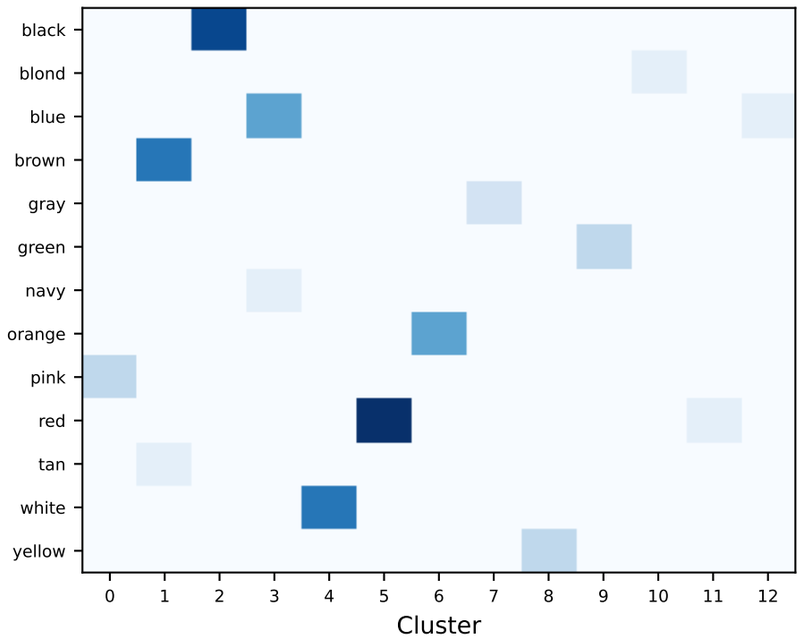
validation_single · finetuned_conditioned · K=13, seed=5; PCA explains 36.8%

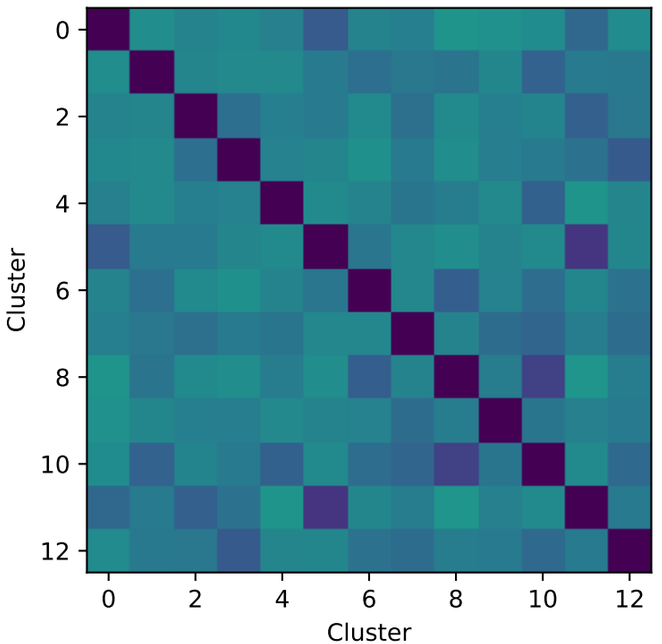


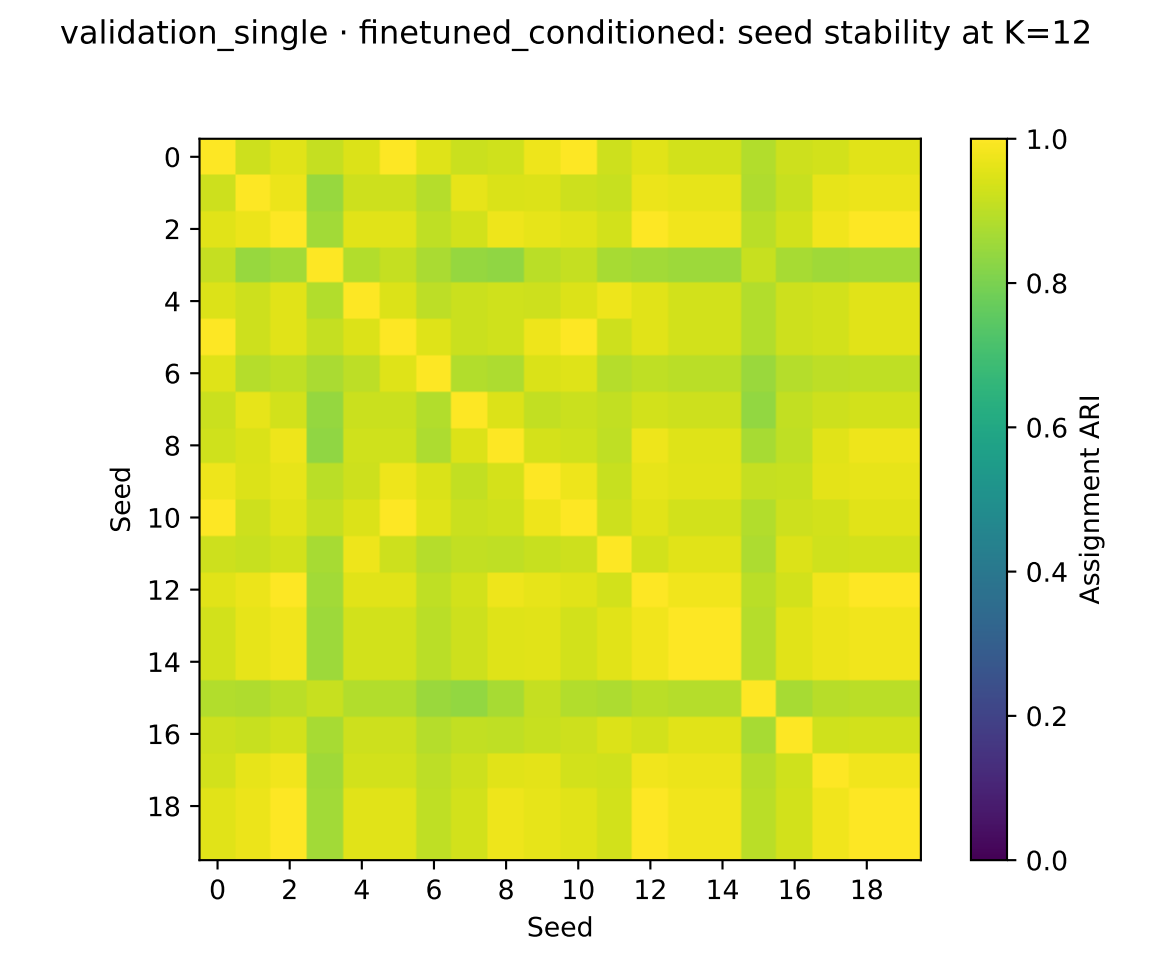




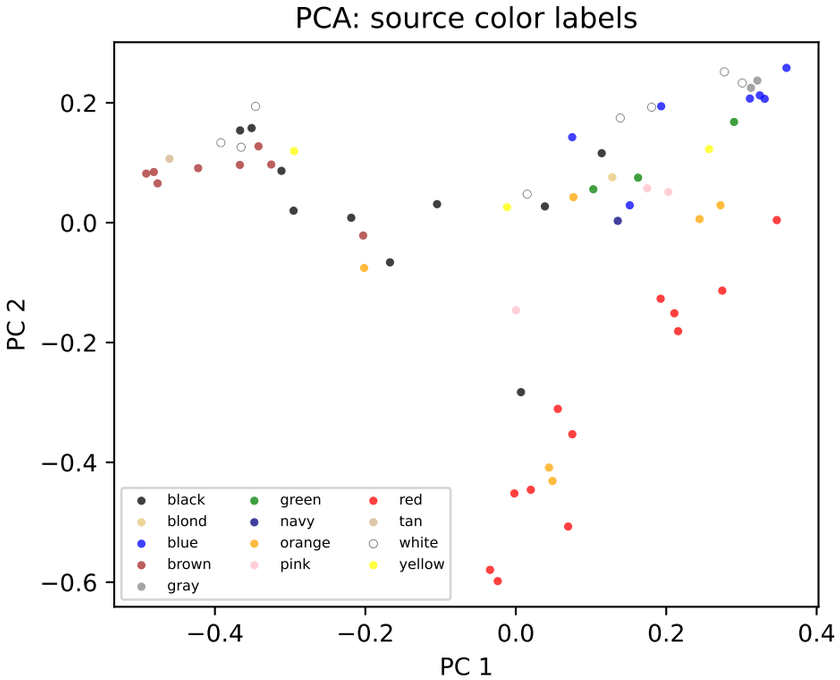


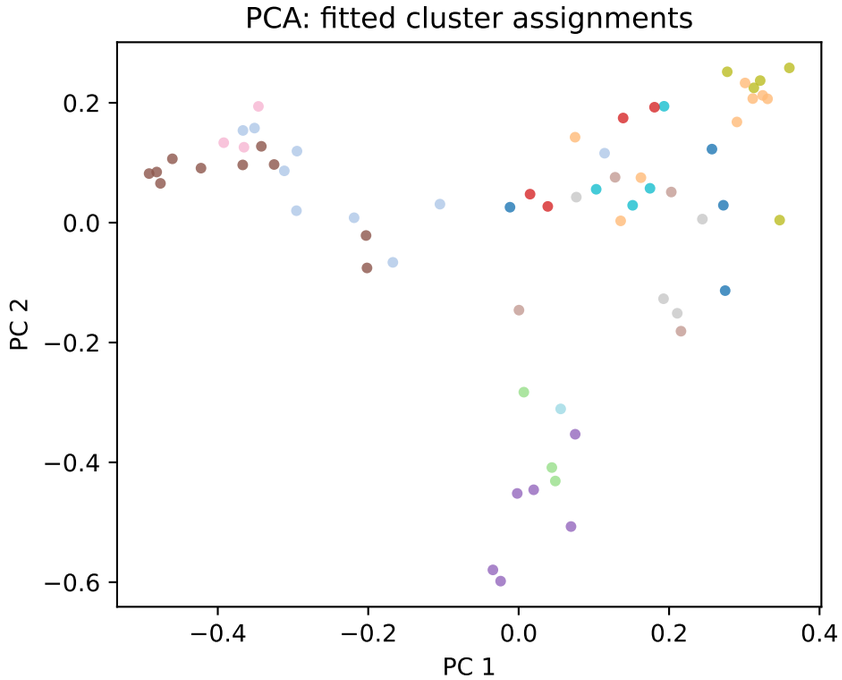


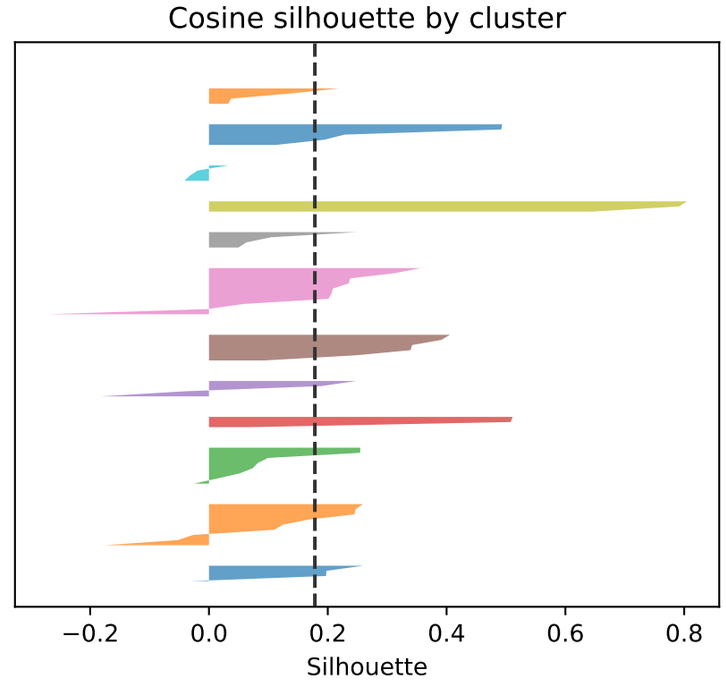



validation_single · finetuned_conditioned: seed stability at K=12
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

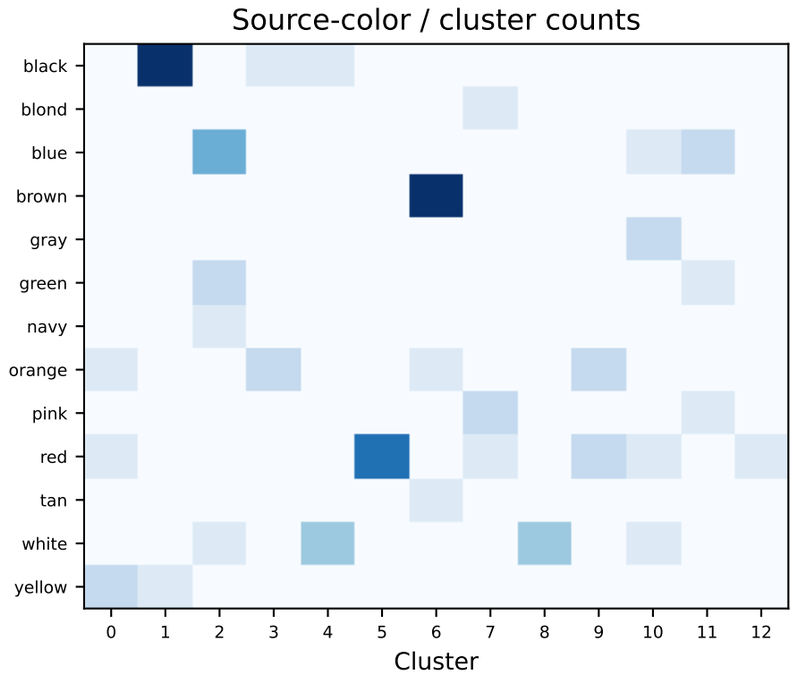
validation_single · pretrained_conditioned · K=13, seed=14; PCA explains 23.9%

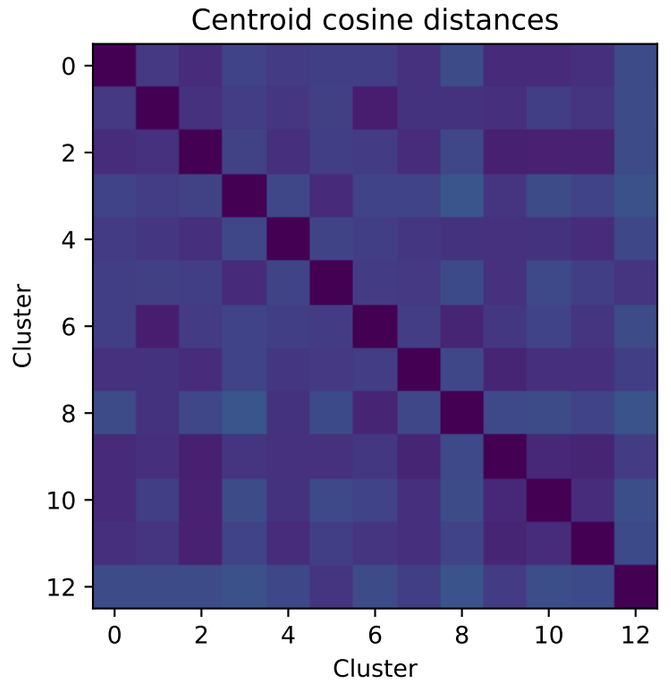






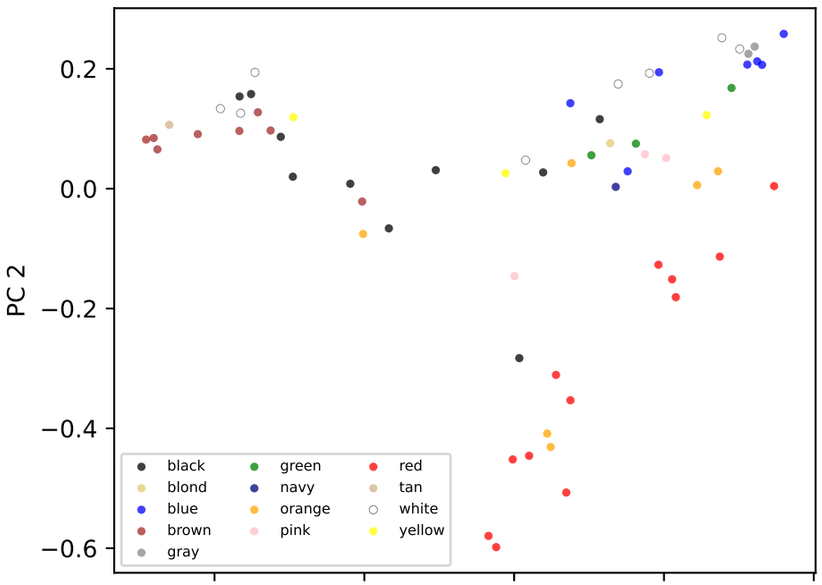




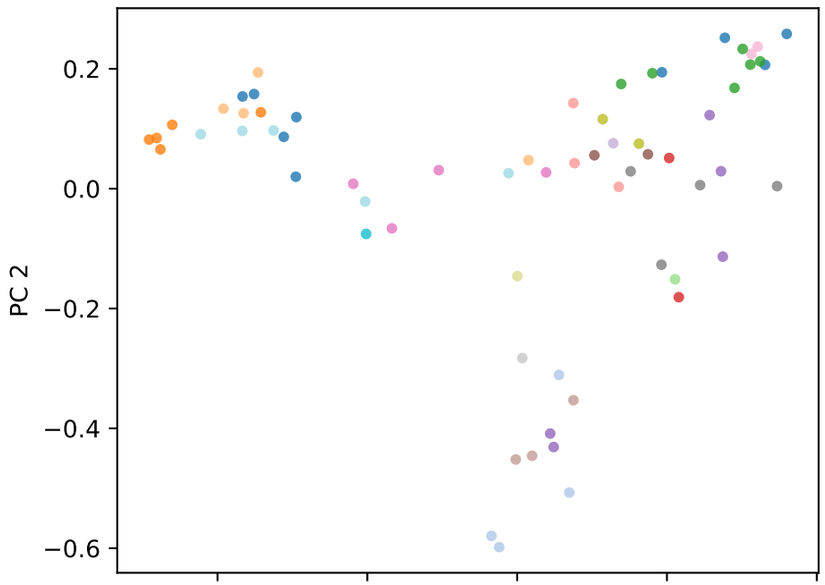


validation_single · pretrained_conditioned · K=26, seed=3; PCA explains 23.9%

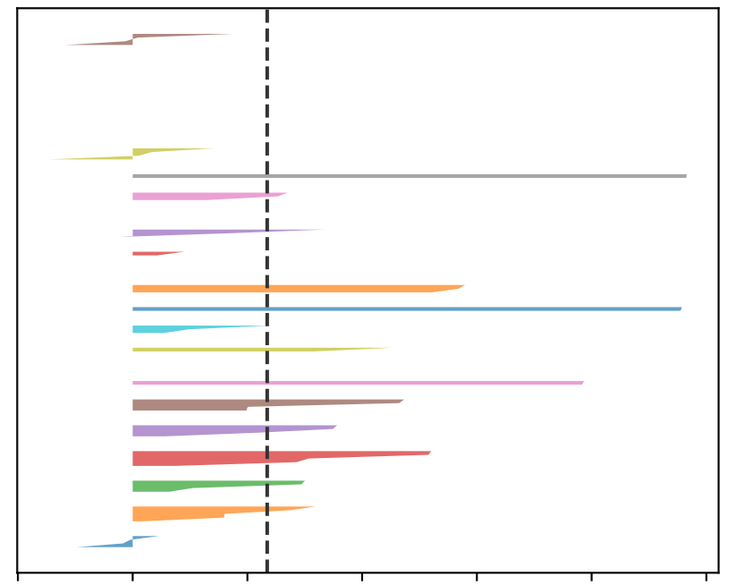
PCA: source color labels



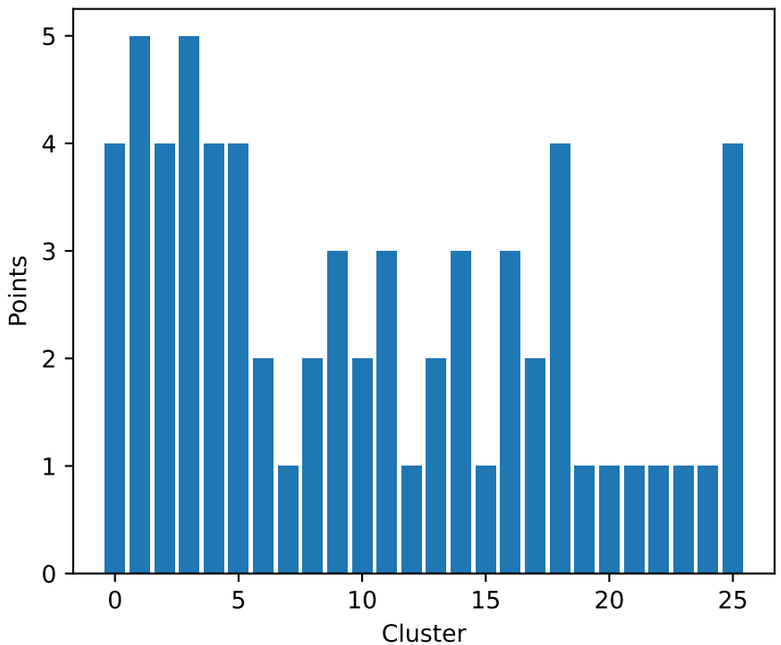
PCA: fitted cluster assignments



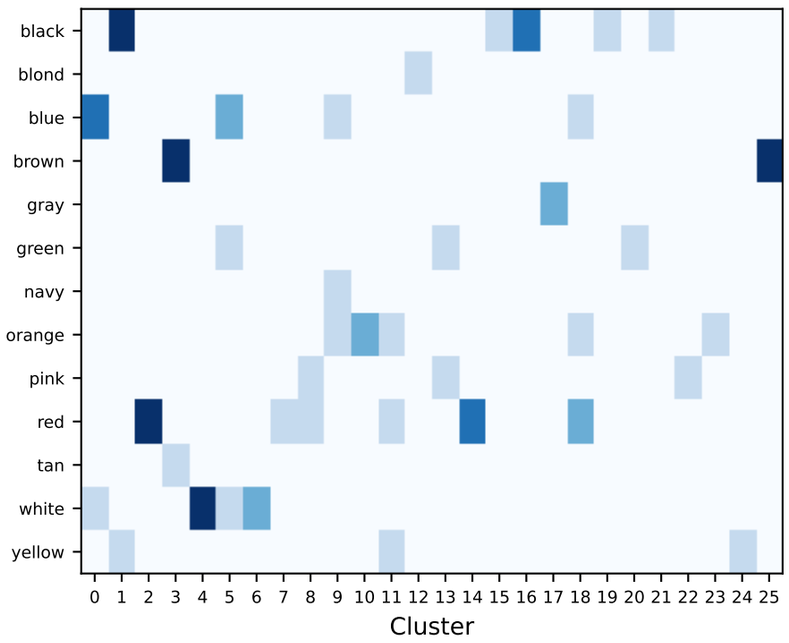
Cosine silhouette by cluster



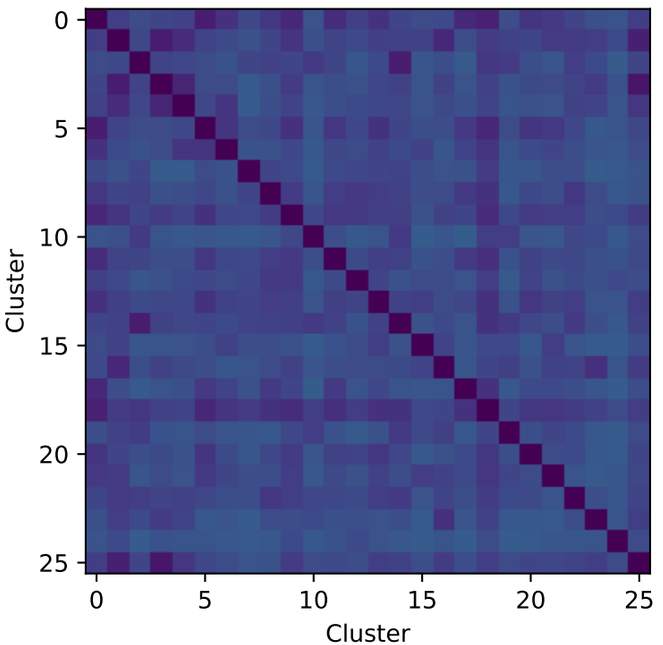
Cluster sizes



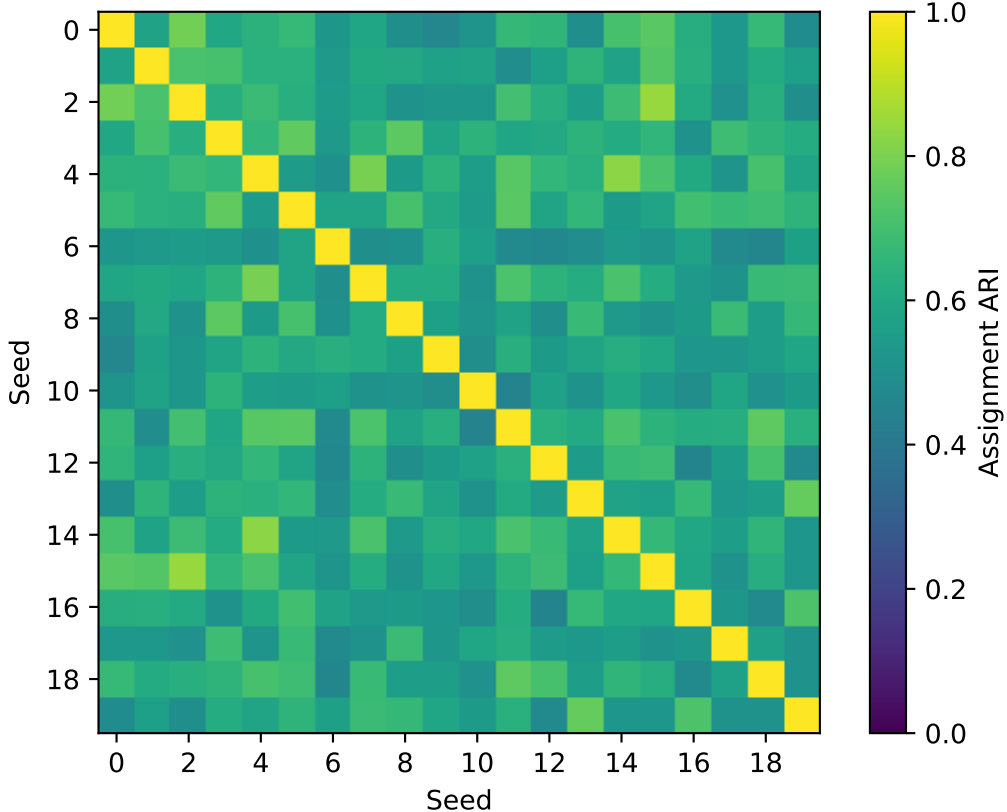
Source-color / cluster counts



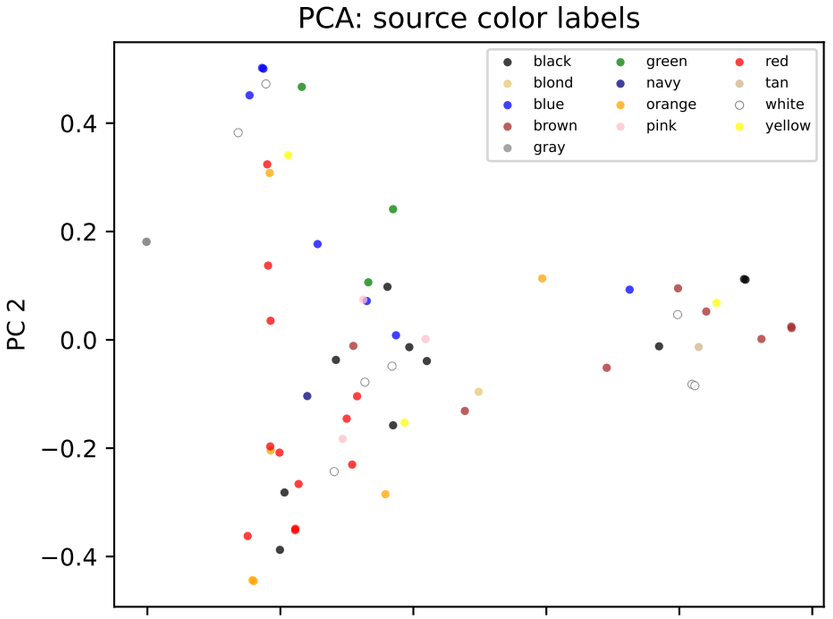
Centroid cosine distances

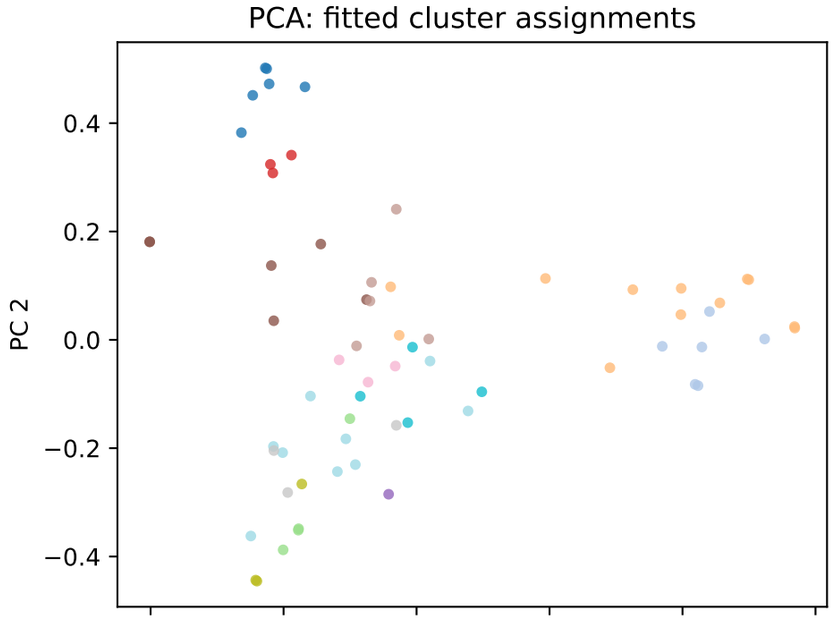


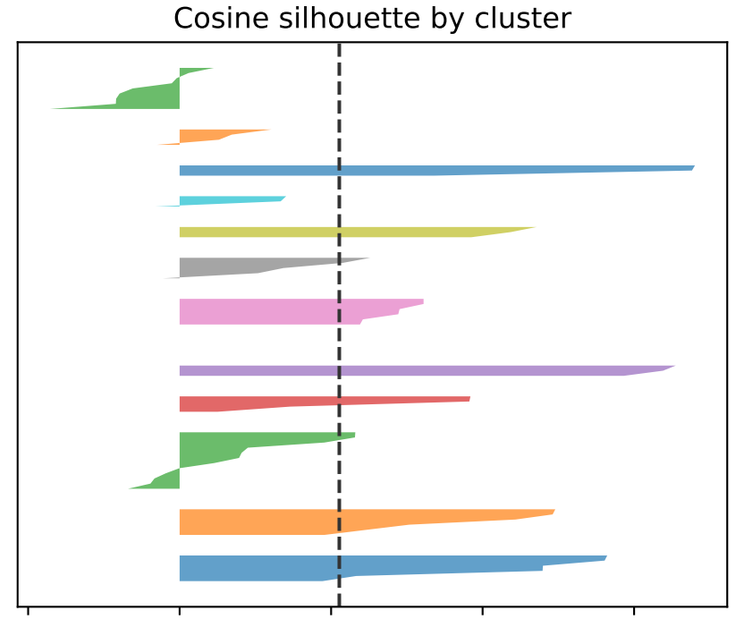
validation_single · pretrained_conditioned: seed stability at K=26



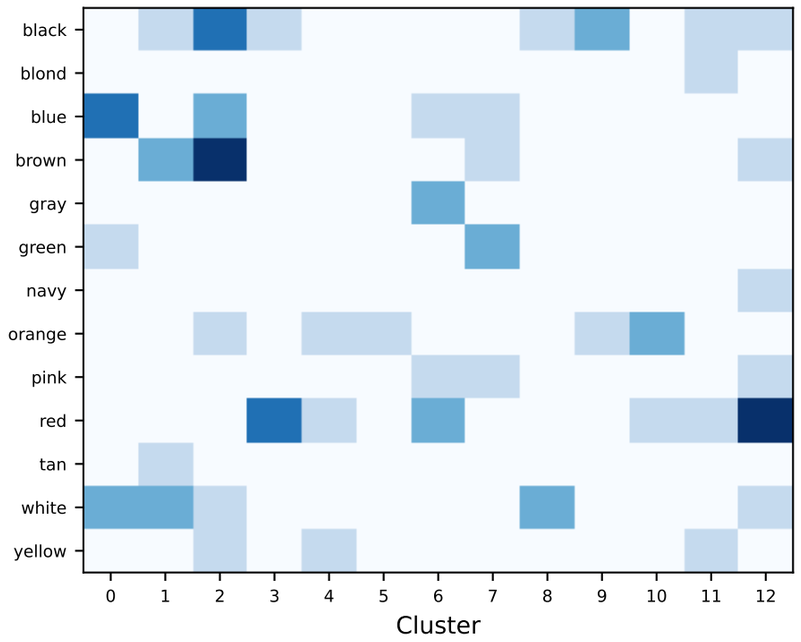
validation_single · finetuned_unconditioned · K=13, seed=2; PCA explains 16.0%

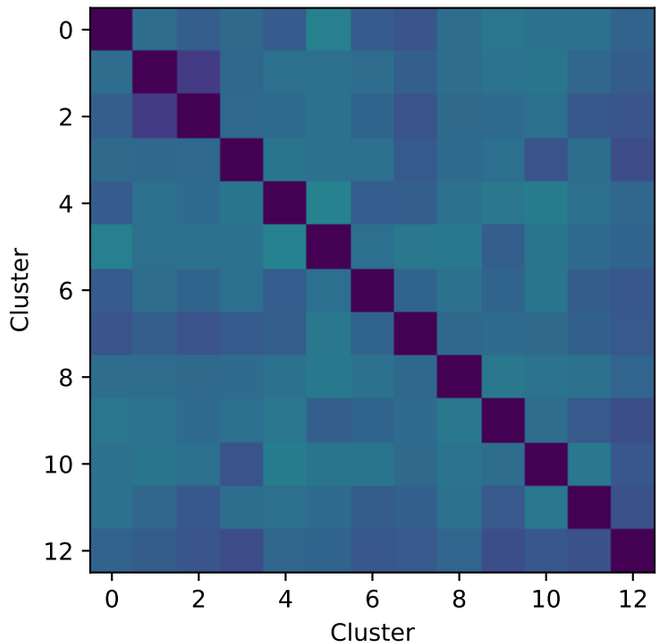






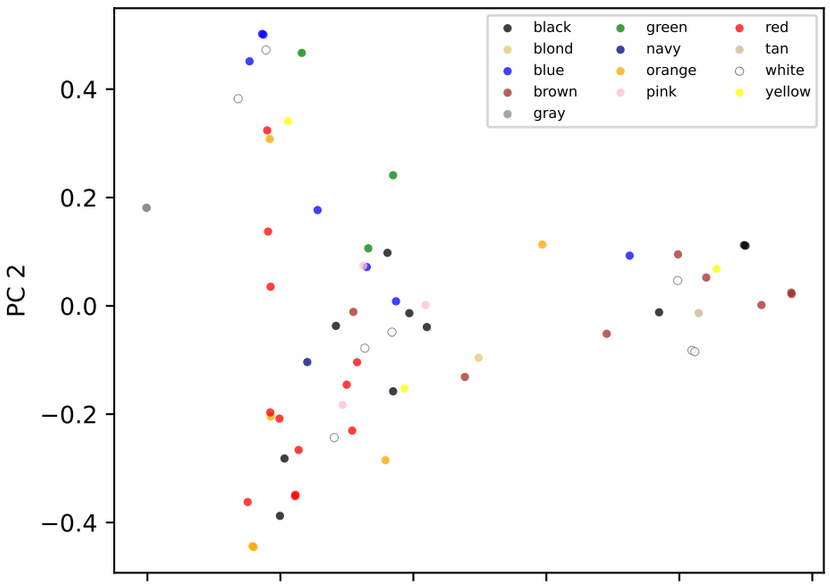




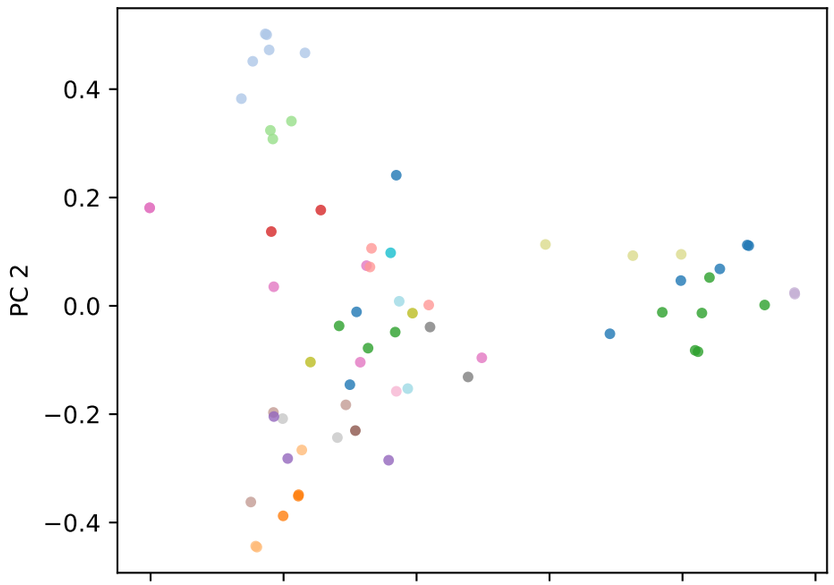


validation_single · finetuned_unconditioned · K=26, seed=18; PCA explains 16.0%

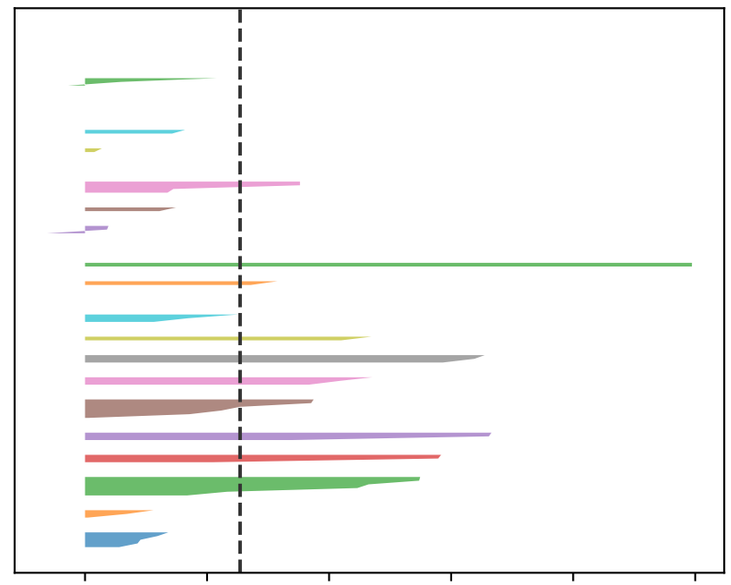
PCA: source color labels



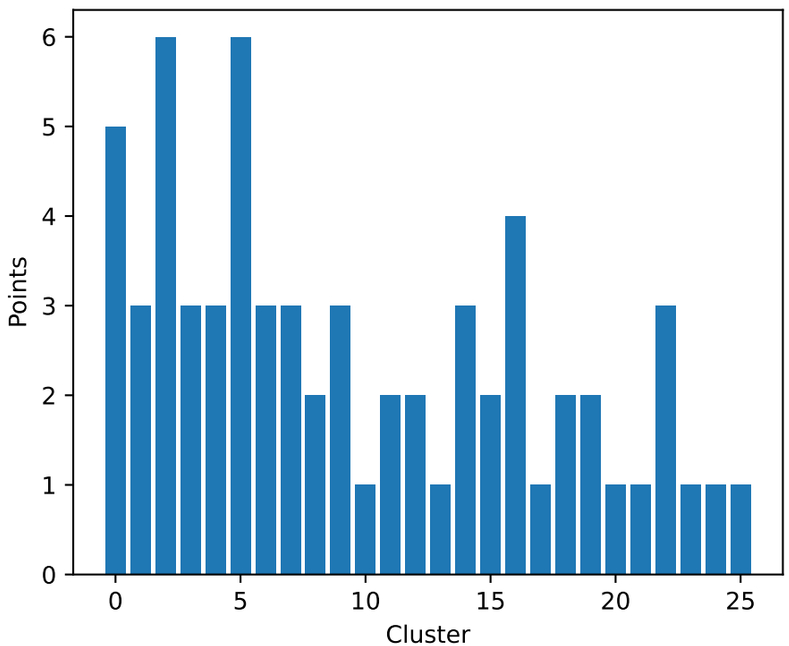
PCA: fitted cluster assignments



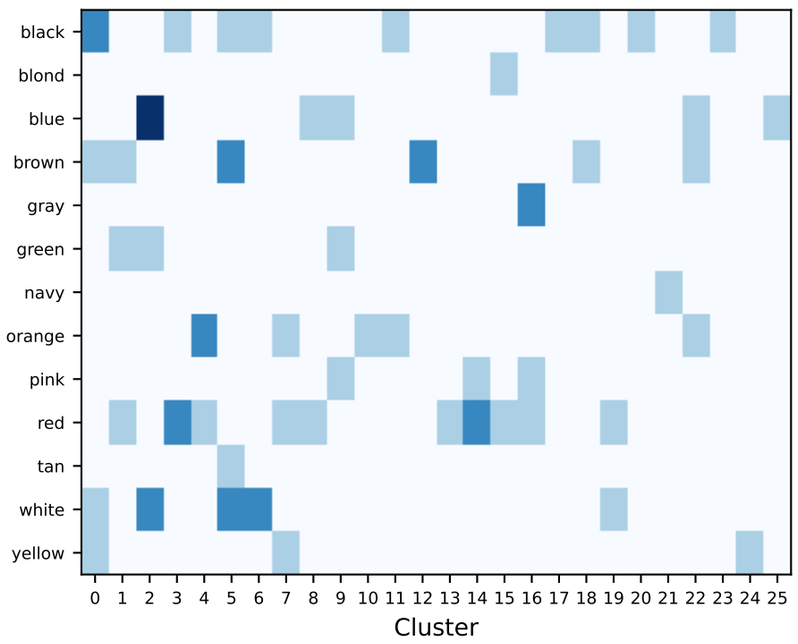
Cosine silhouette by cluster



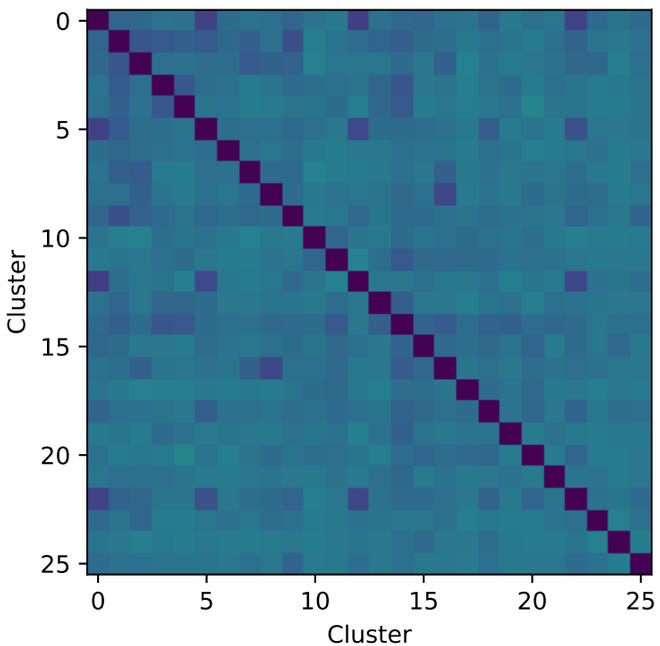
Cluster sizes

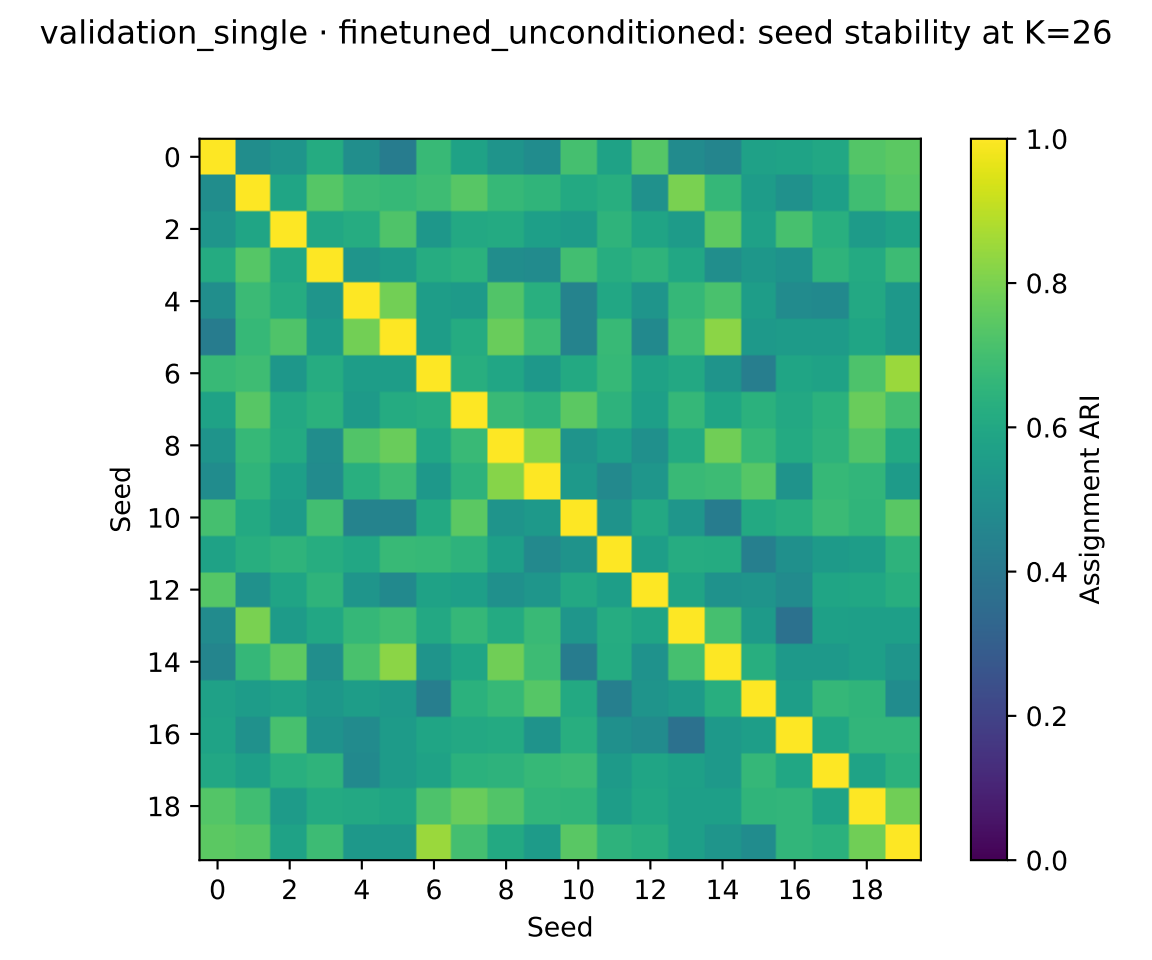


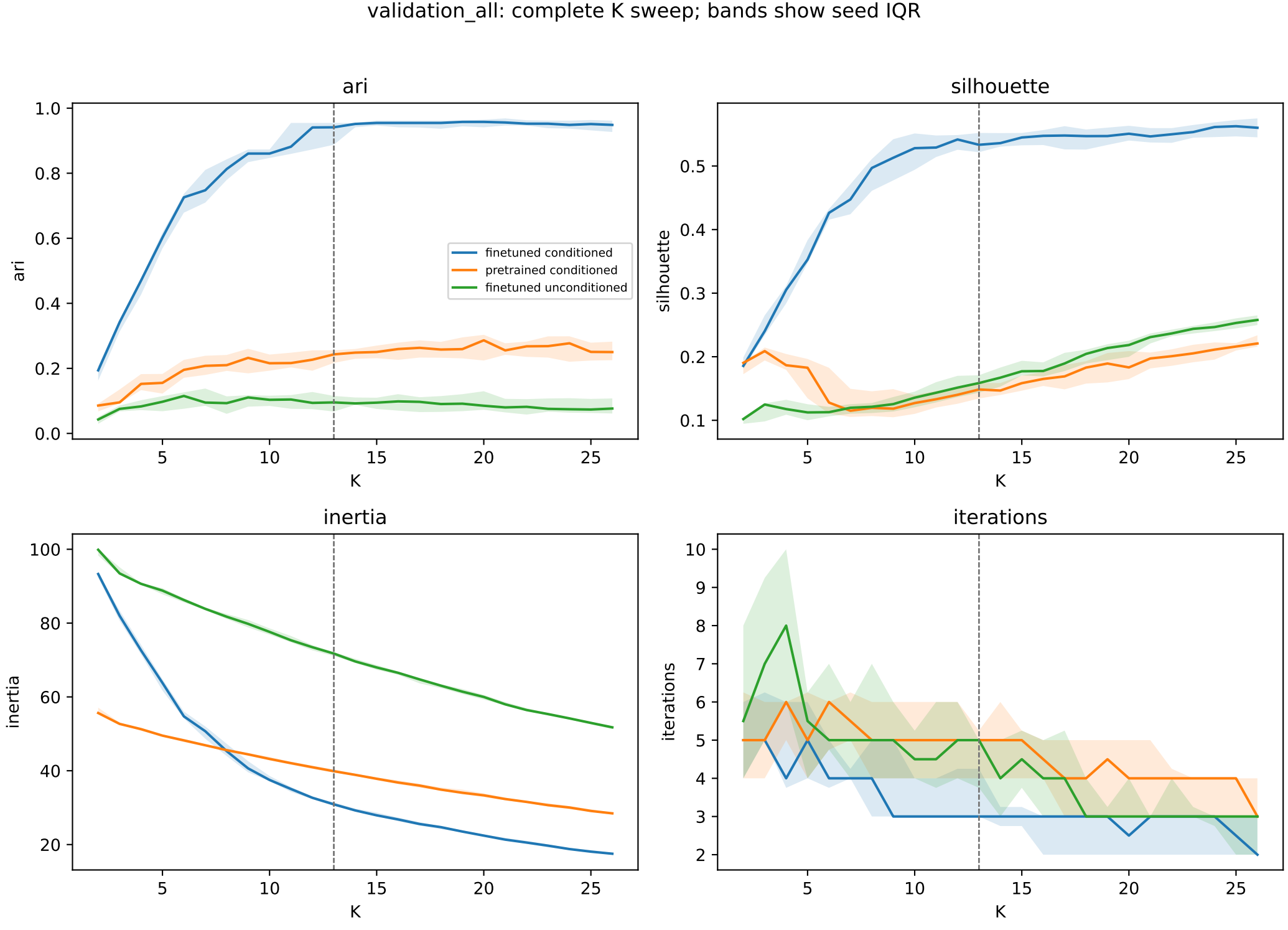
Source-color / cluster counts



Centroid cosine distances

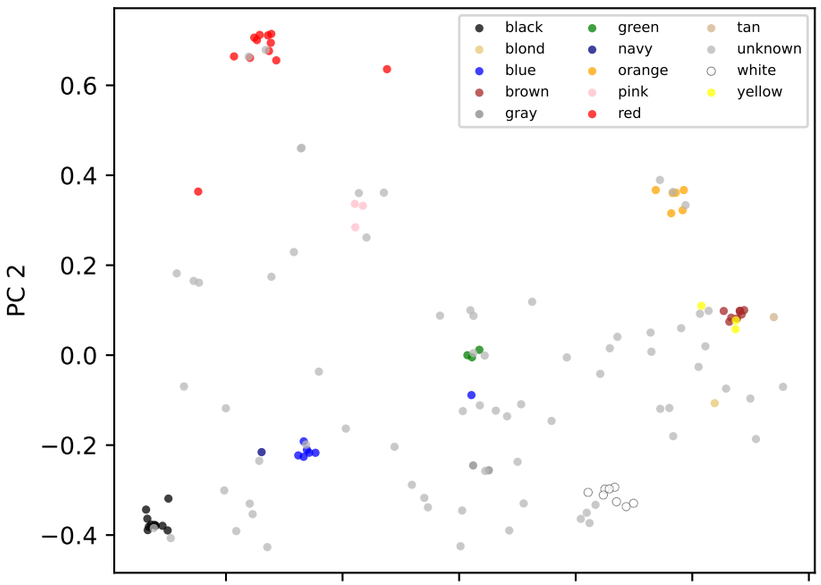



validation_single · finetuned_unconditioned: seed stability at K=26
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

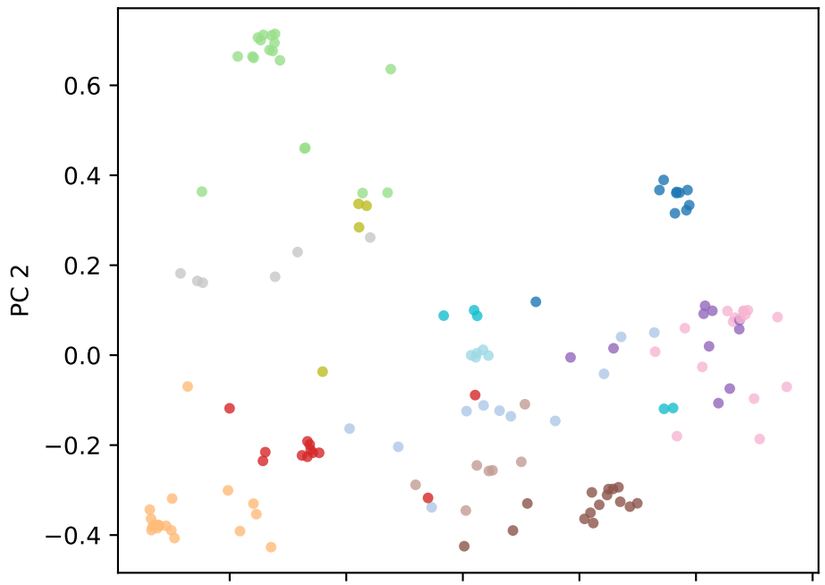

validation_all: complete K sweep; bands show seed IQR
ari
silhouette
inertia
iterations
finetuned conditioned
pretrained conditioned
finetuned unconditioned
K

validation_all · finetuned_conditioned · K=13, seed=7; PCA explains 28.4%

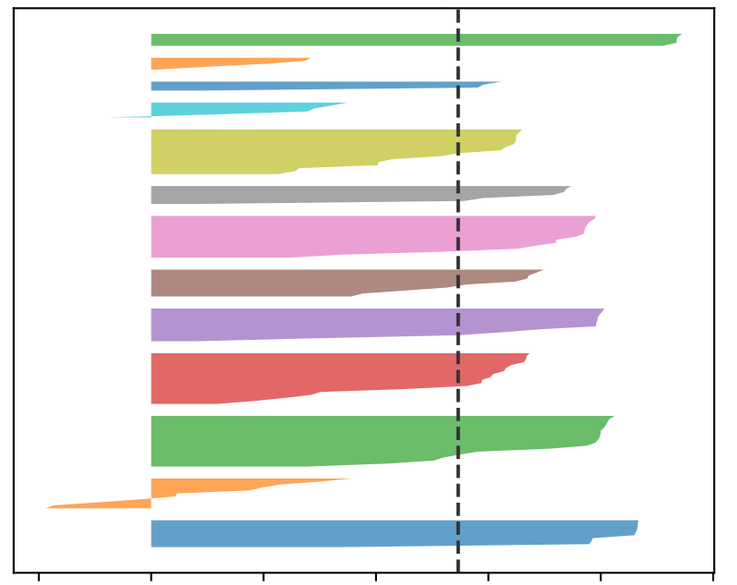
PCA: source color labels



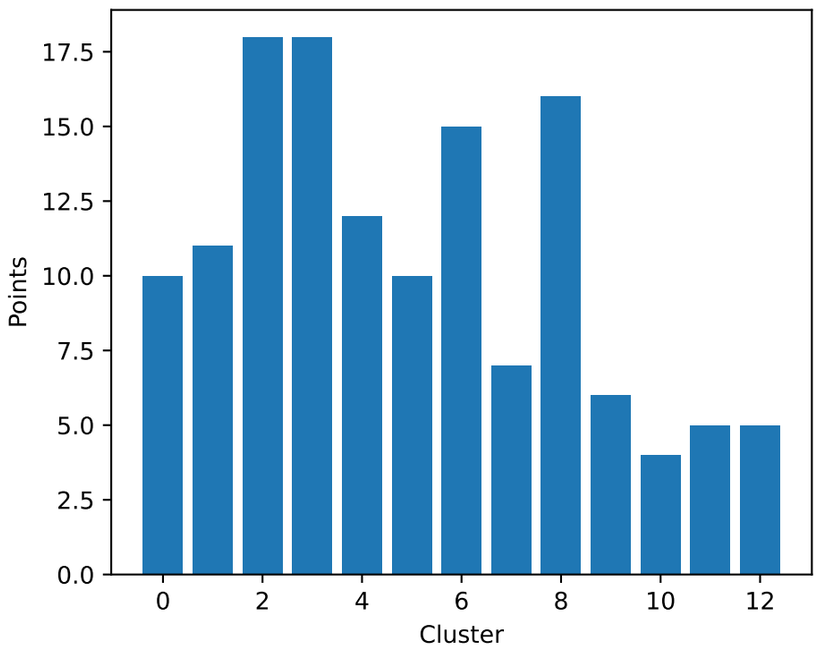
PCA: fitted cluster assignments



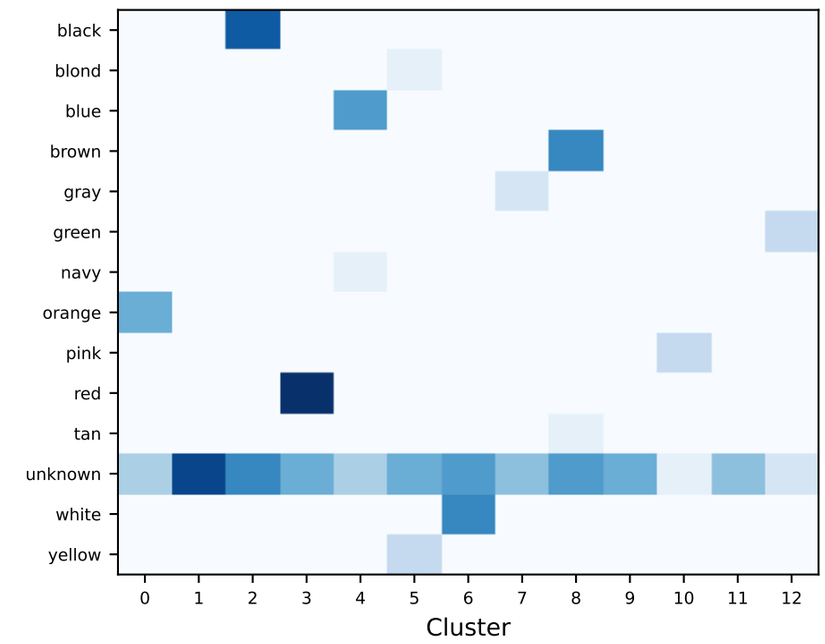
Cosine silhouette by cluster



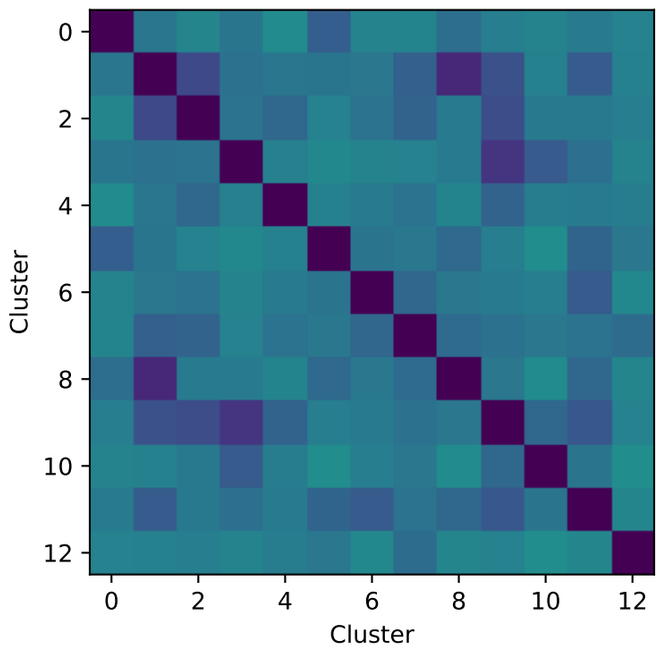
Cluster sizes



Source-color / cluster counts

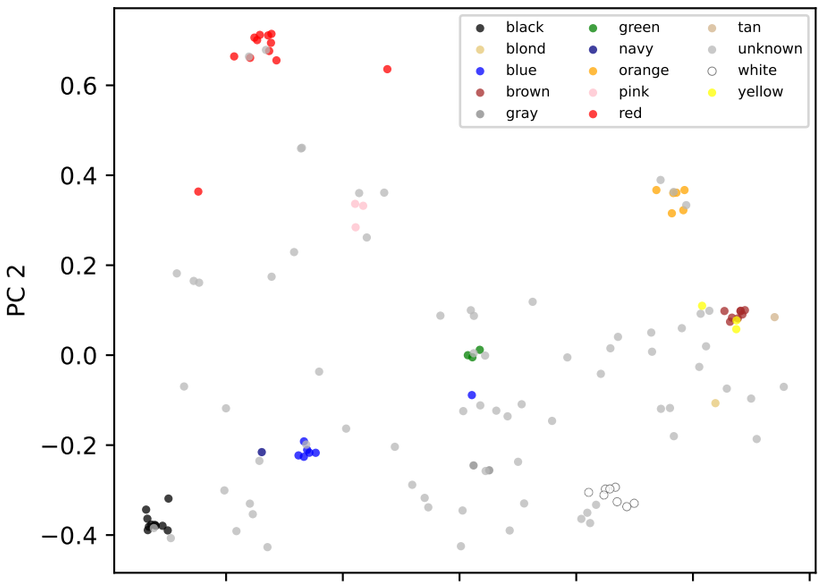


Centroid cosine distances

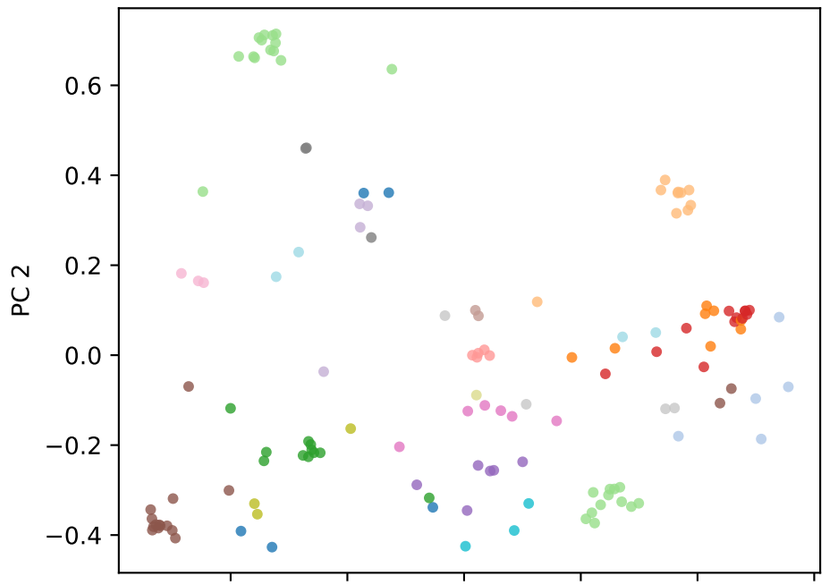


validation_all · finetuned_conditioned · K=25, seed=1; PCA explains 28.4%

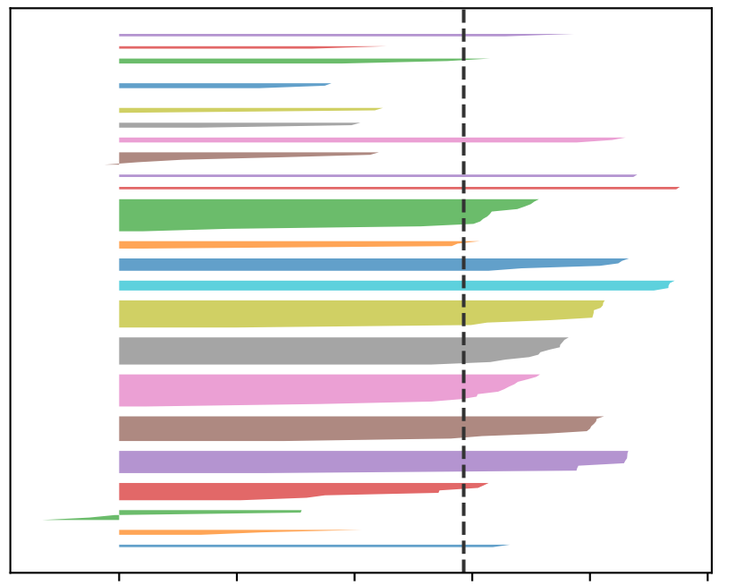
PCA: source color labels



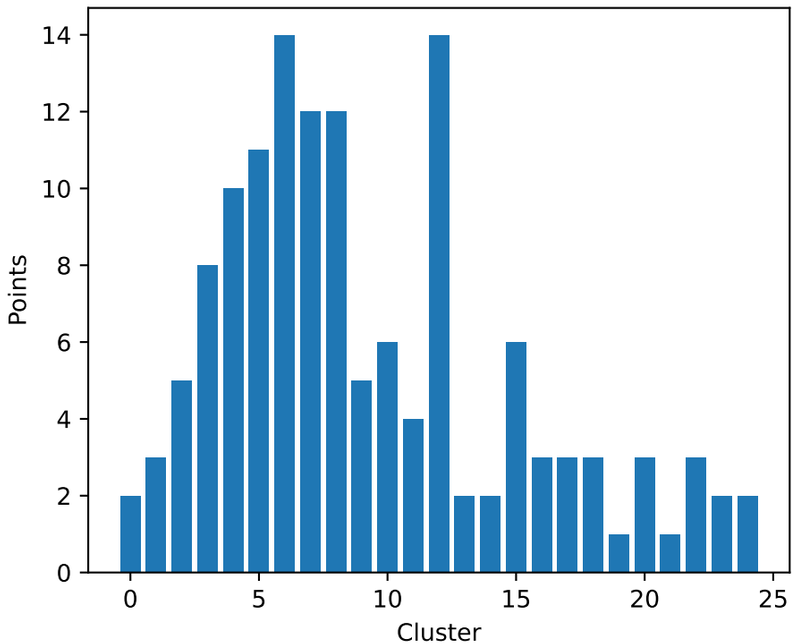
PCA: fitted cluster assignments



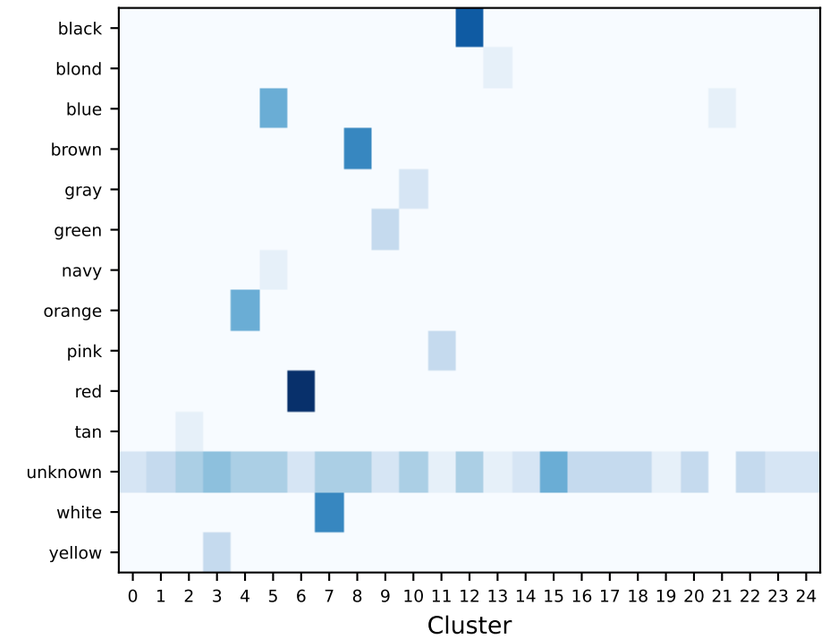
Cosine silhouette by cluster



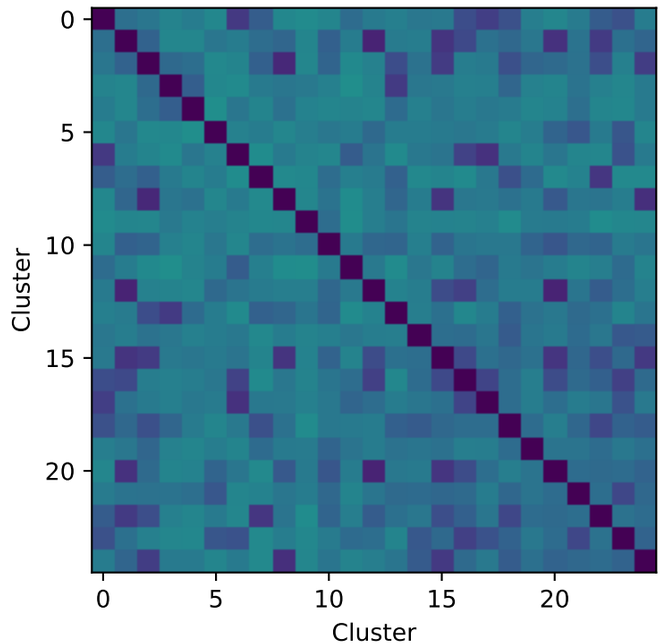
Cluster sizes

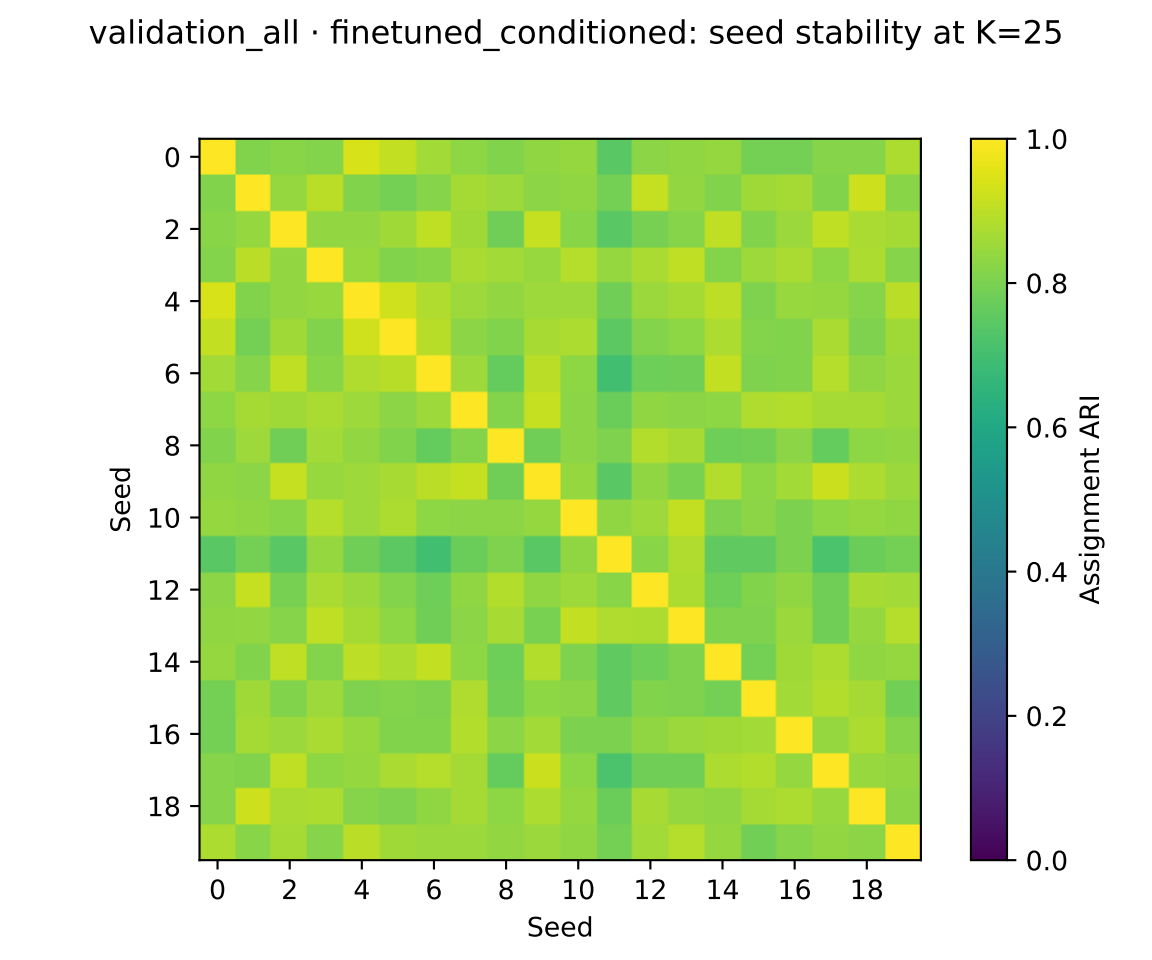


Source-color / cluster counts

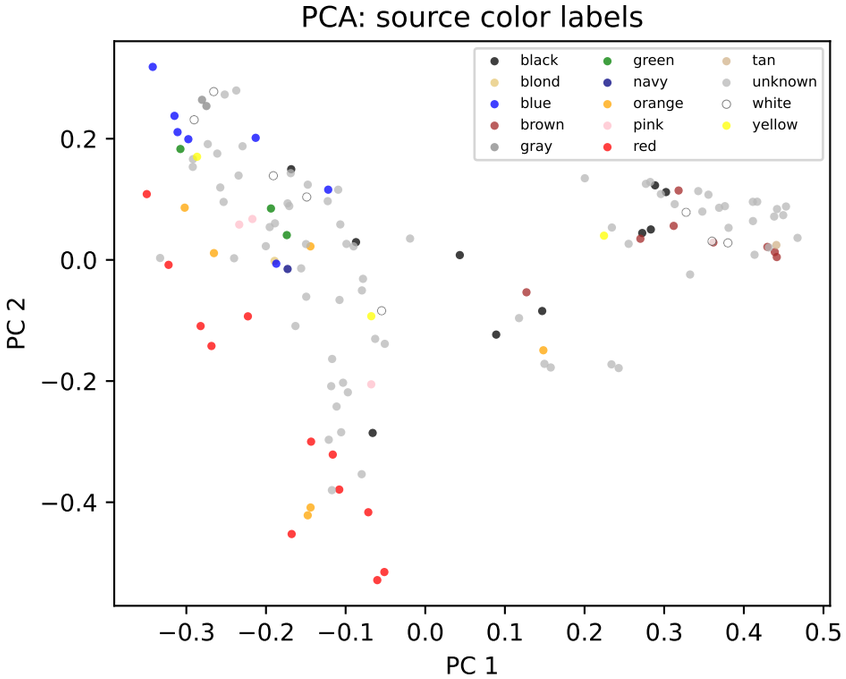


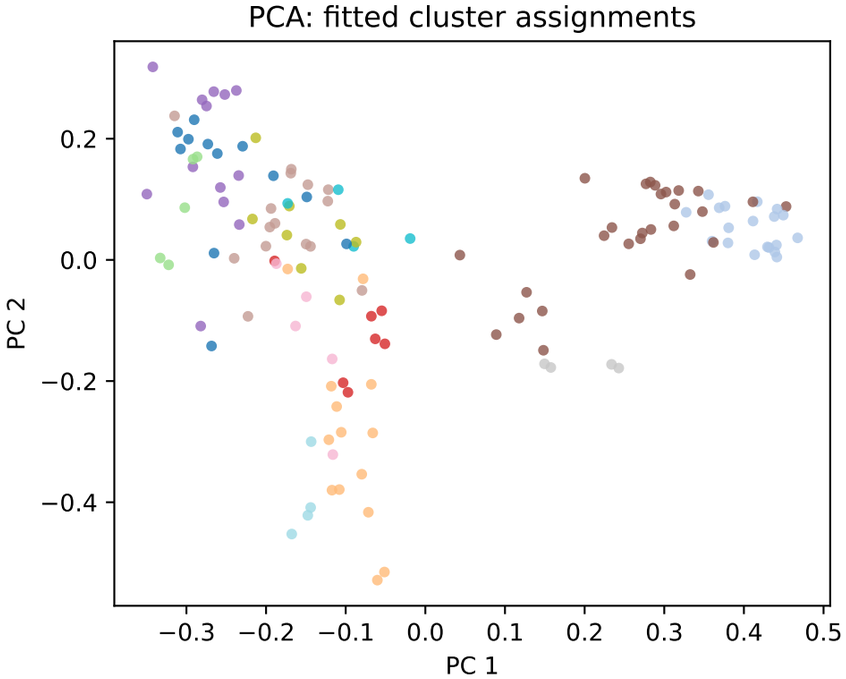
Centroid cosine distances

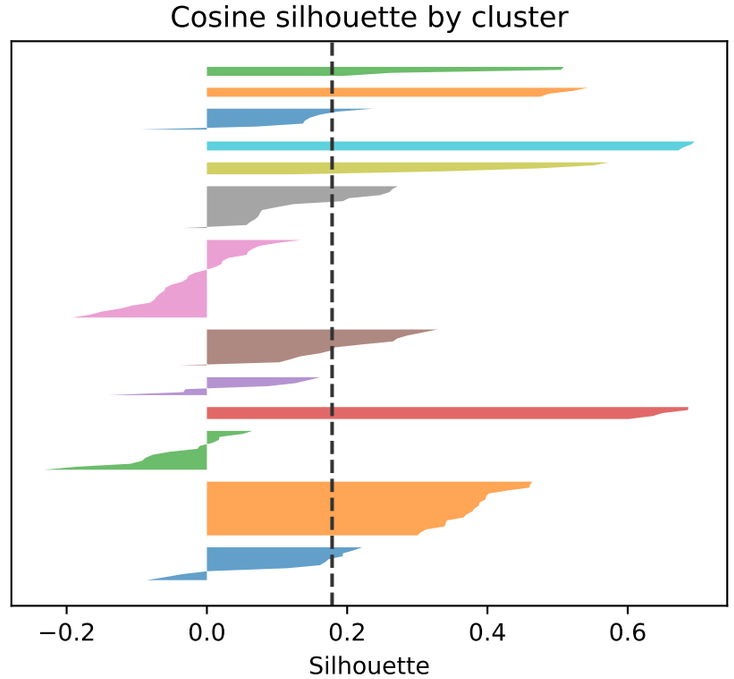



validation_all · finetuned_conditioned: seed stability at K=25
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

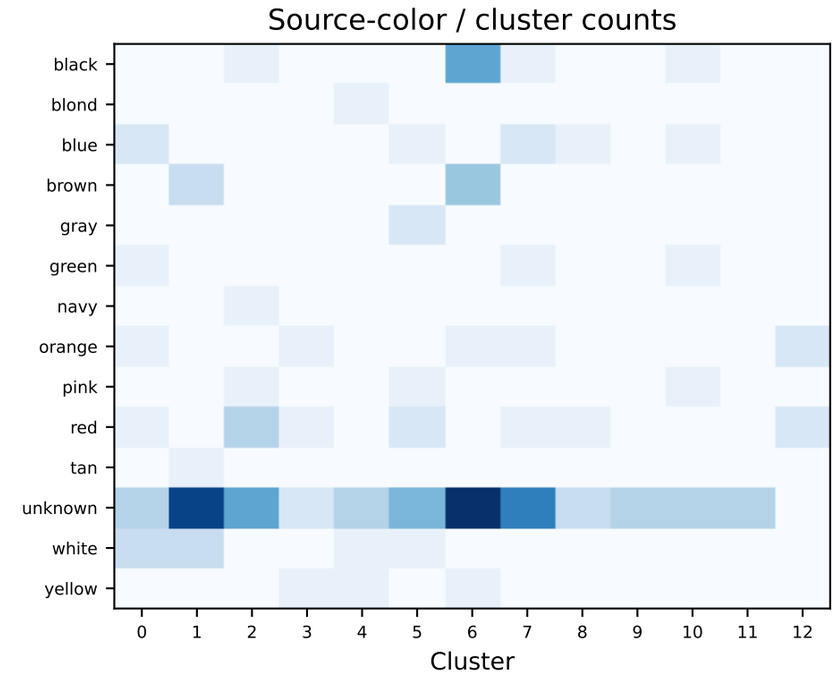
validation_all · pretrained_conditioned · K=13, seed=0; PCA explains 20.4%

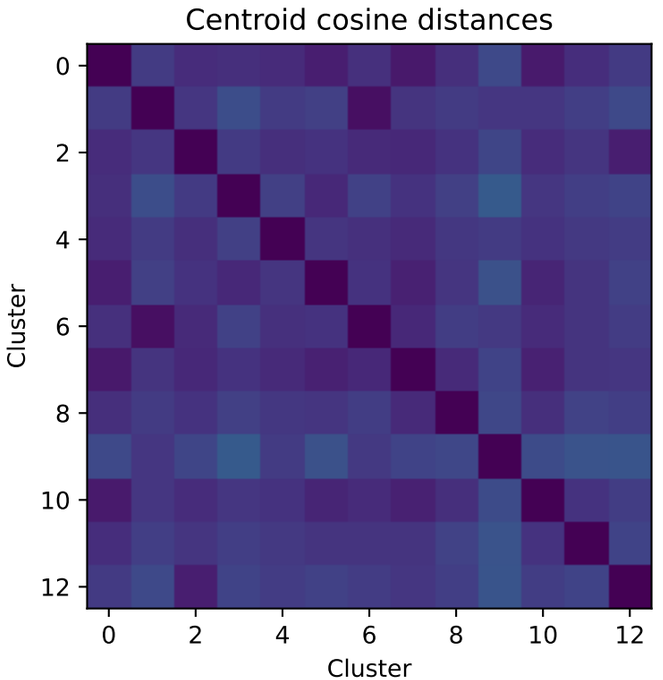






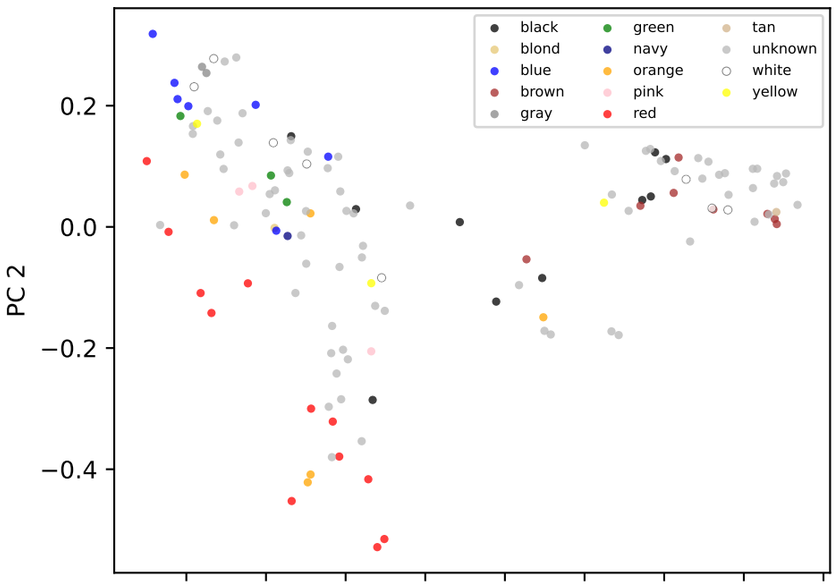




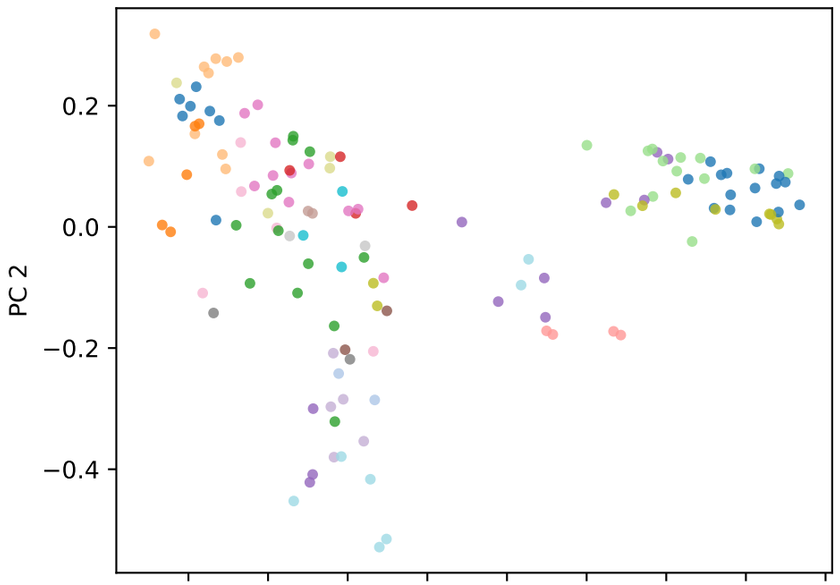


validation_all · pretrained_conditioned · K=26, seed=0; PCA explains 20.4%

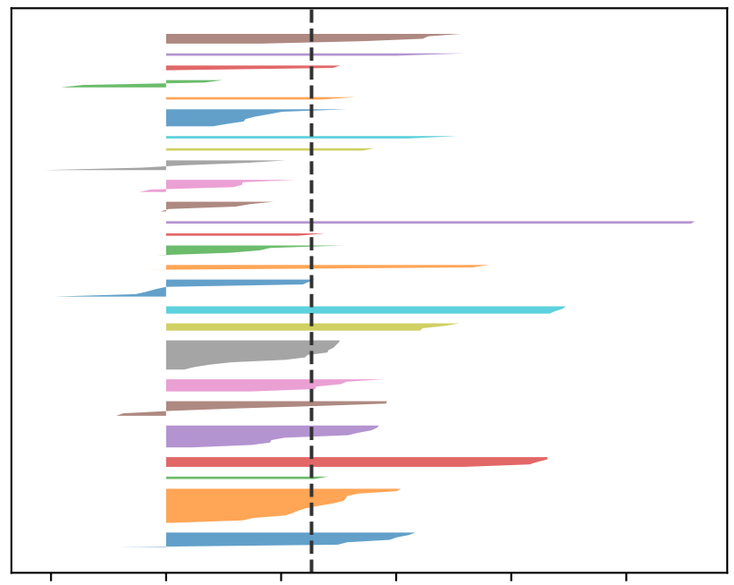
PCA: source color labels



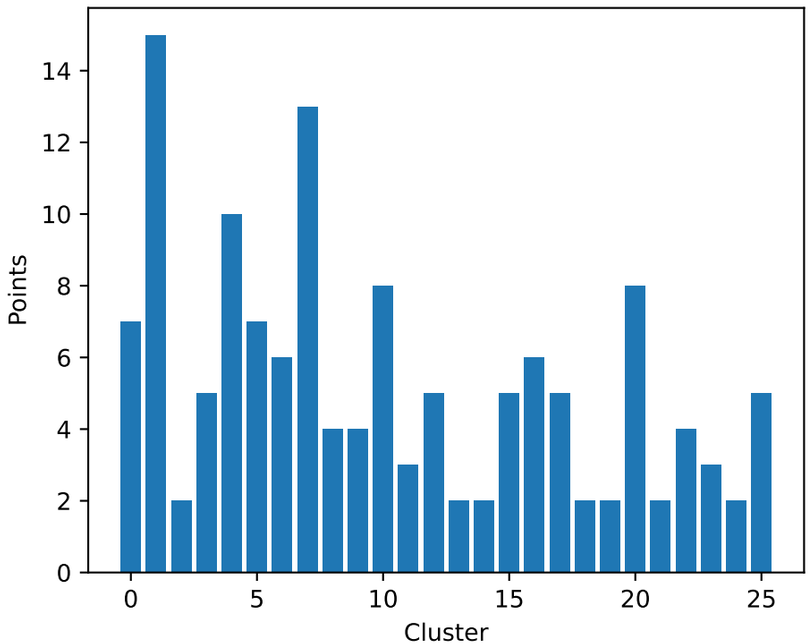
PCA: fitted cluster assignments



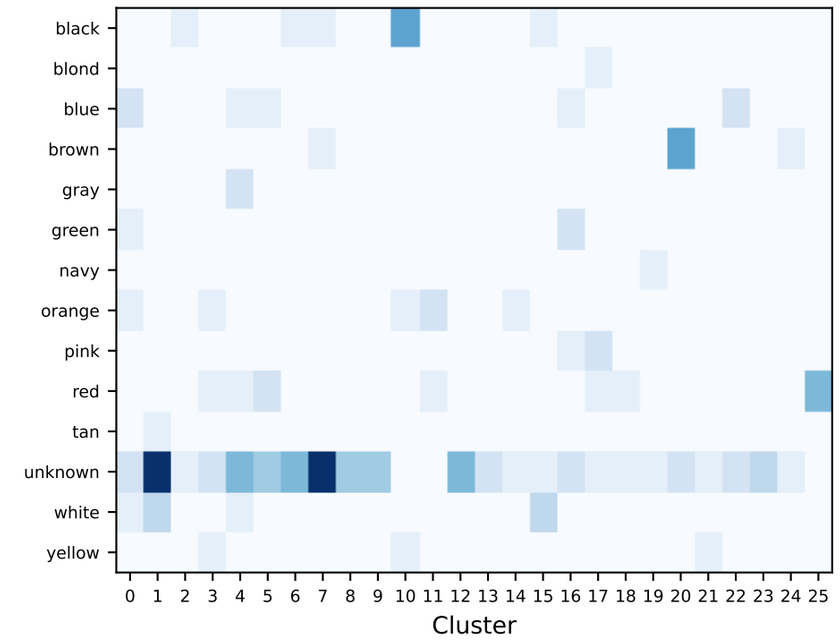
Cosine silhouette by cluster



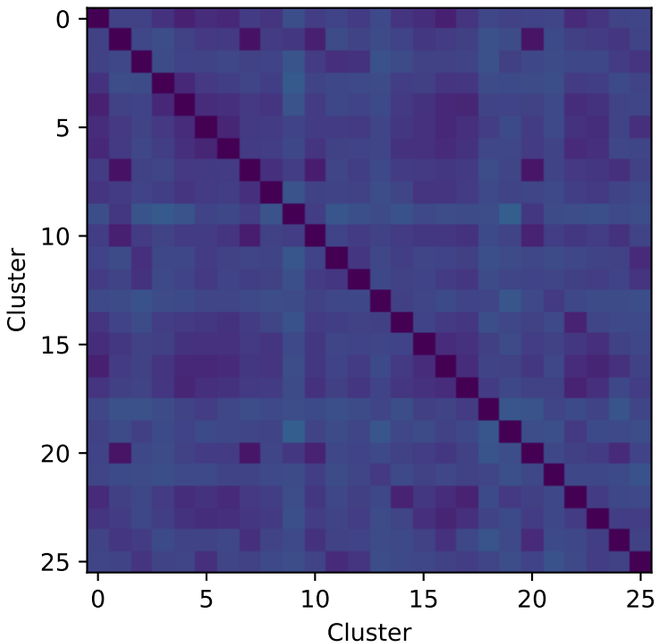
Cluster sizes

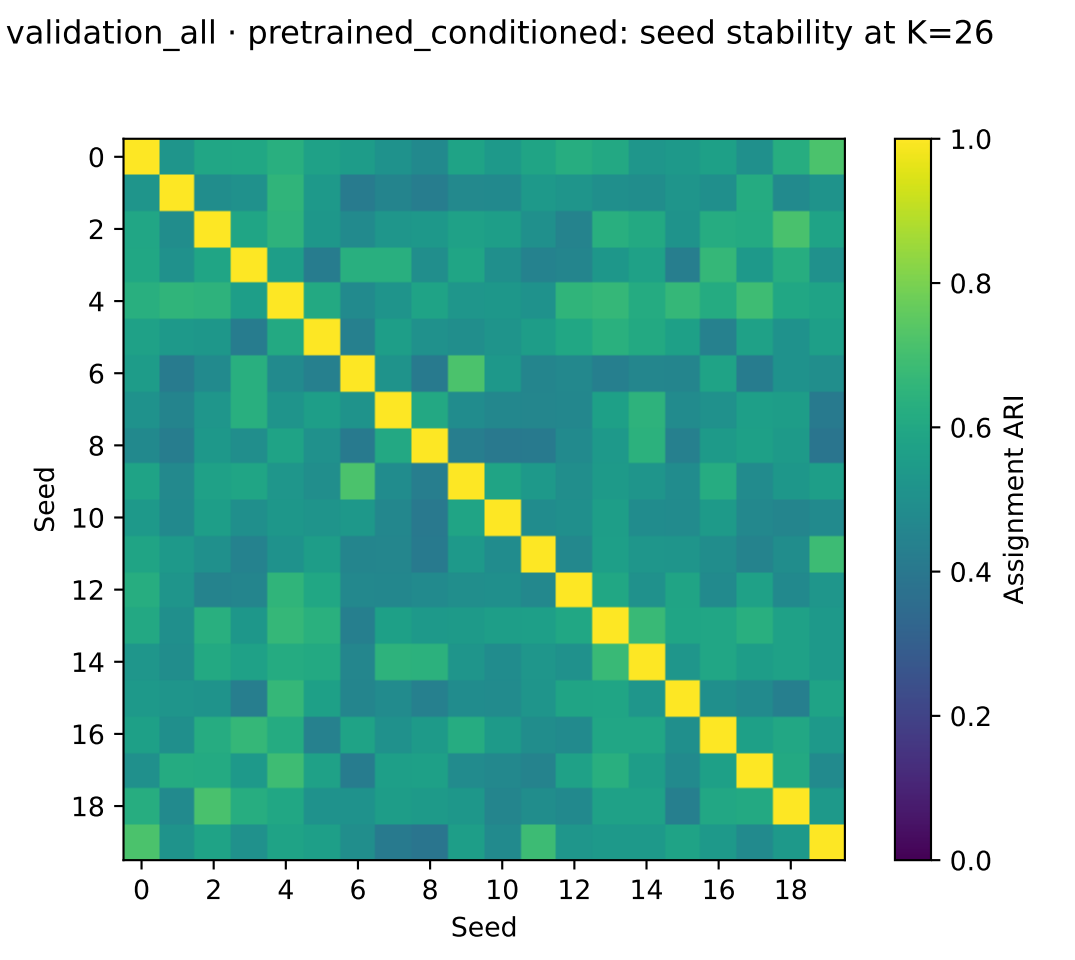


Source-color / cluster counts



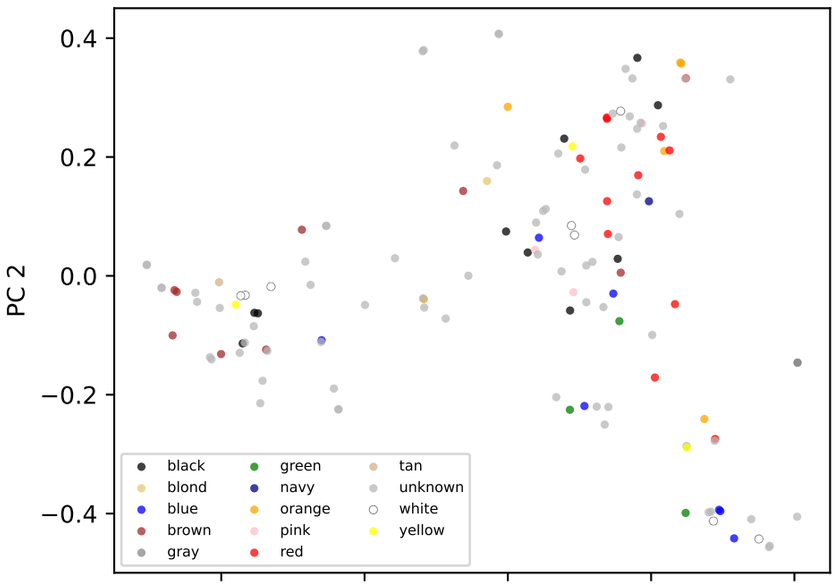
Centroid cosine distances



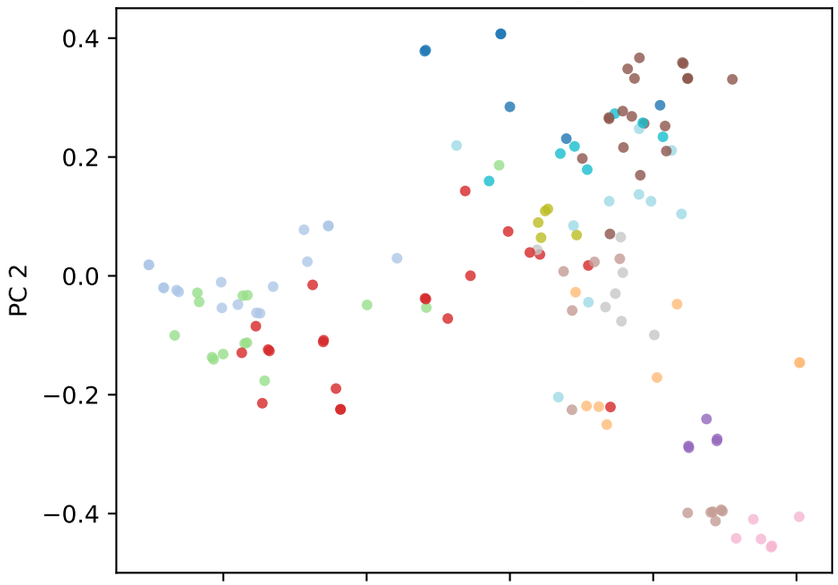

validation_all · pretrained_conditioned: seed stability at K=26
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

validation_all · finetuned_unconditioned · K=13, seed=7; PCA explains 14.5%

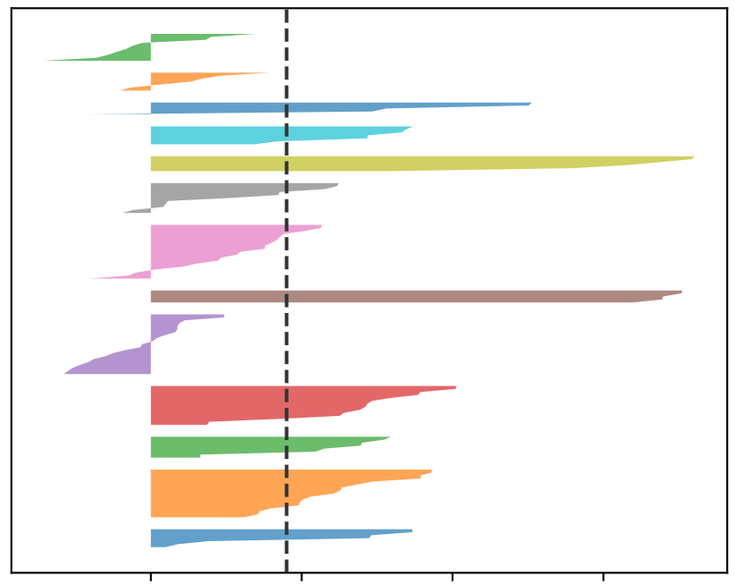
PCA: source color labels



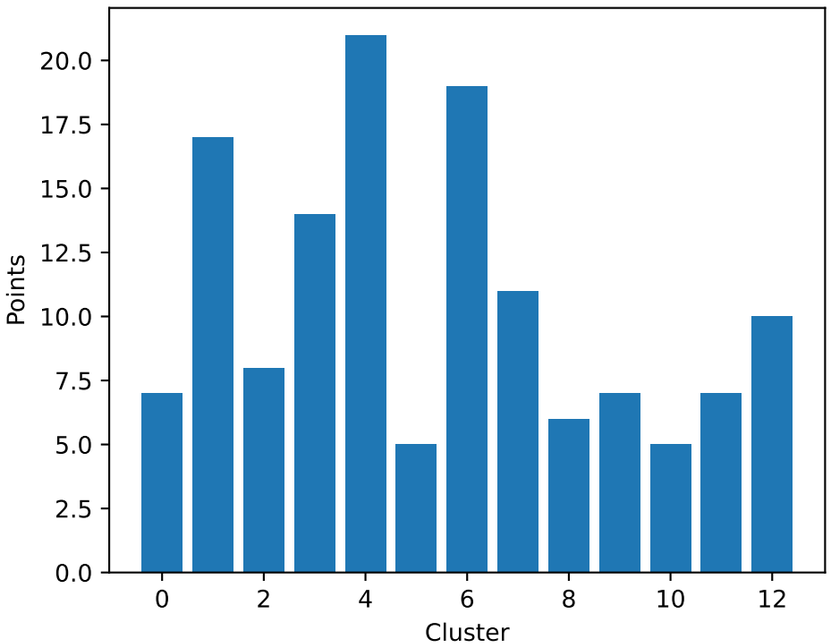
PCA: fitted cluster assignments



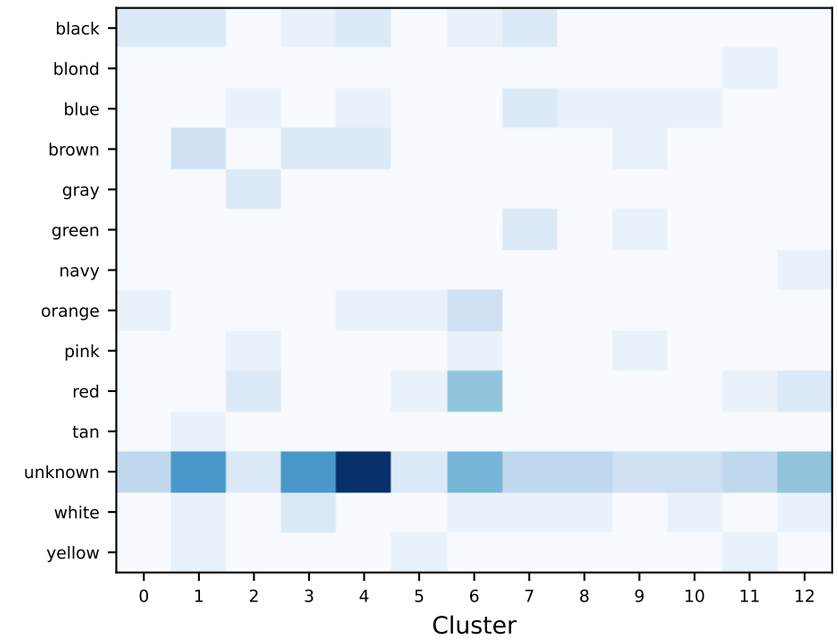
Cosine silhouette by cluster



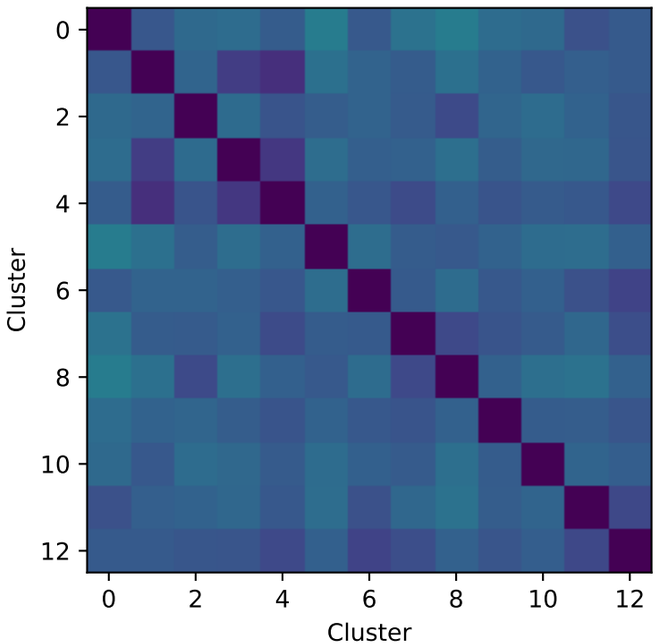
Cluster sizes



Source-color / cluster counts

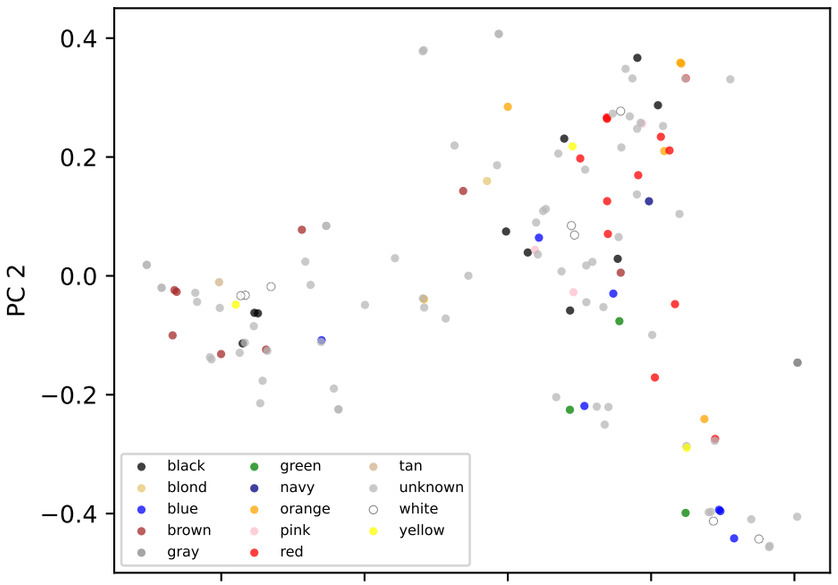


Centroid cosine distances

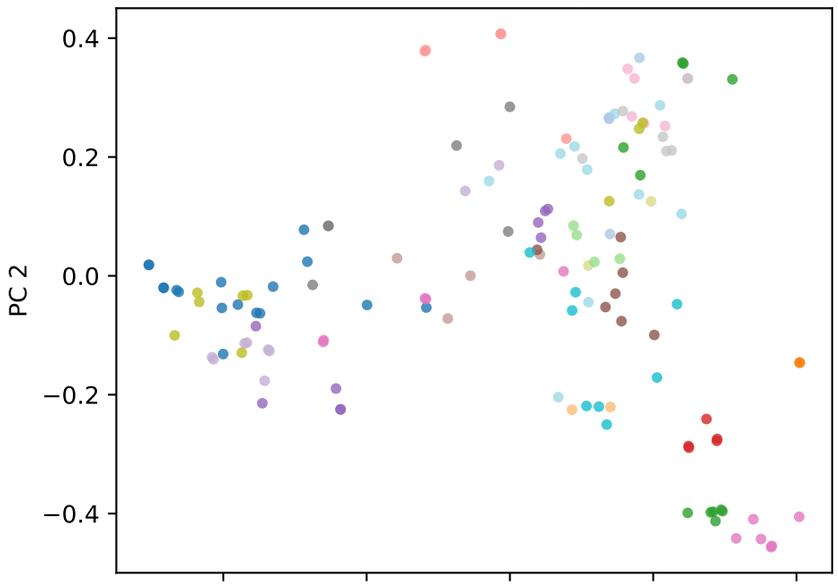


validation_all · finetuned_unconditioned · K=26, seed=18; PCA explains 14.5%

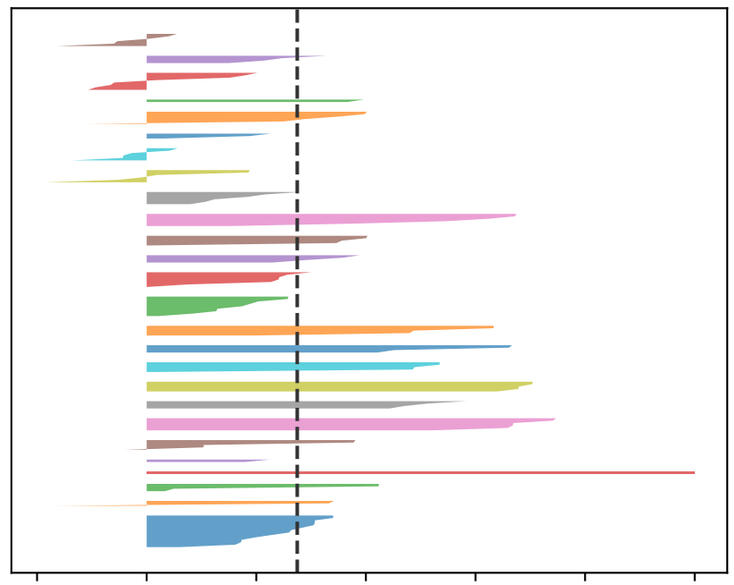
PCA: source color labels



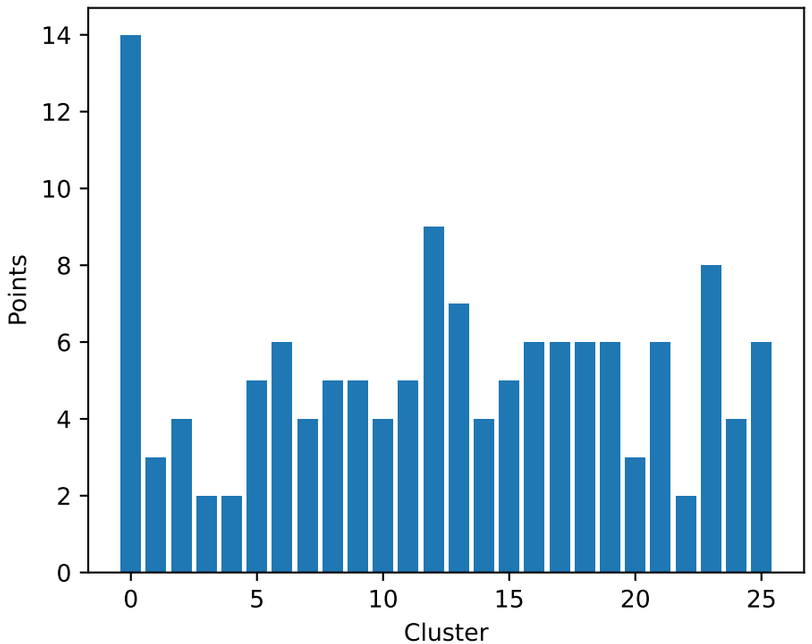
PCA: fitted cluster assignments



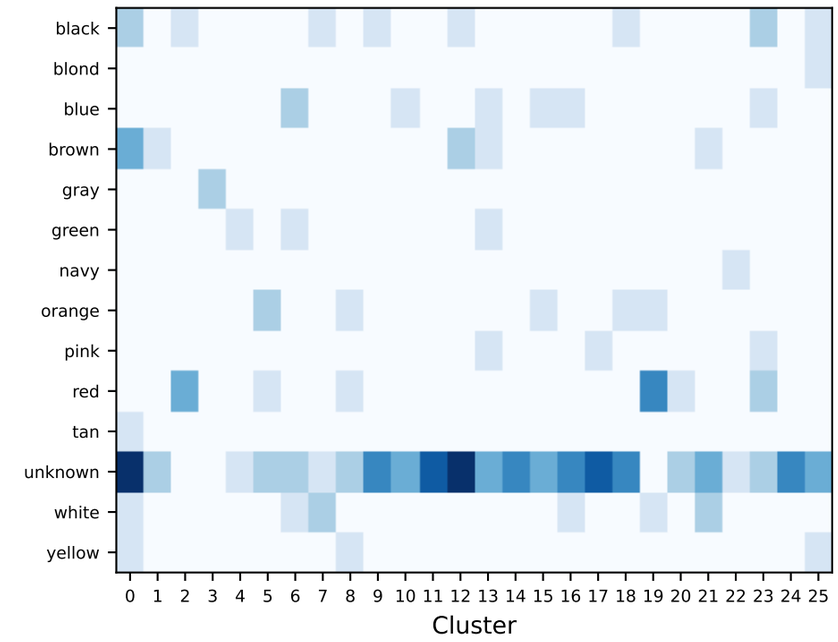
Cosine silhouette by cluster



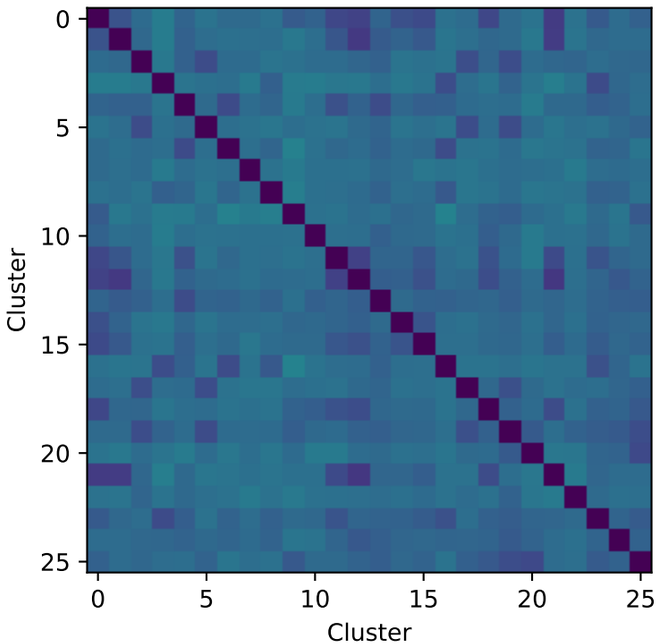
Cluster sizes

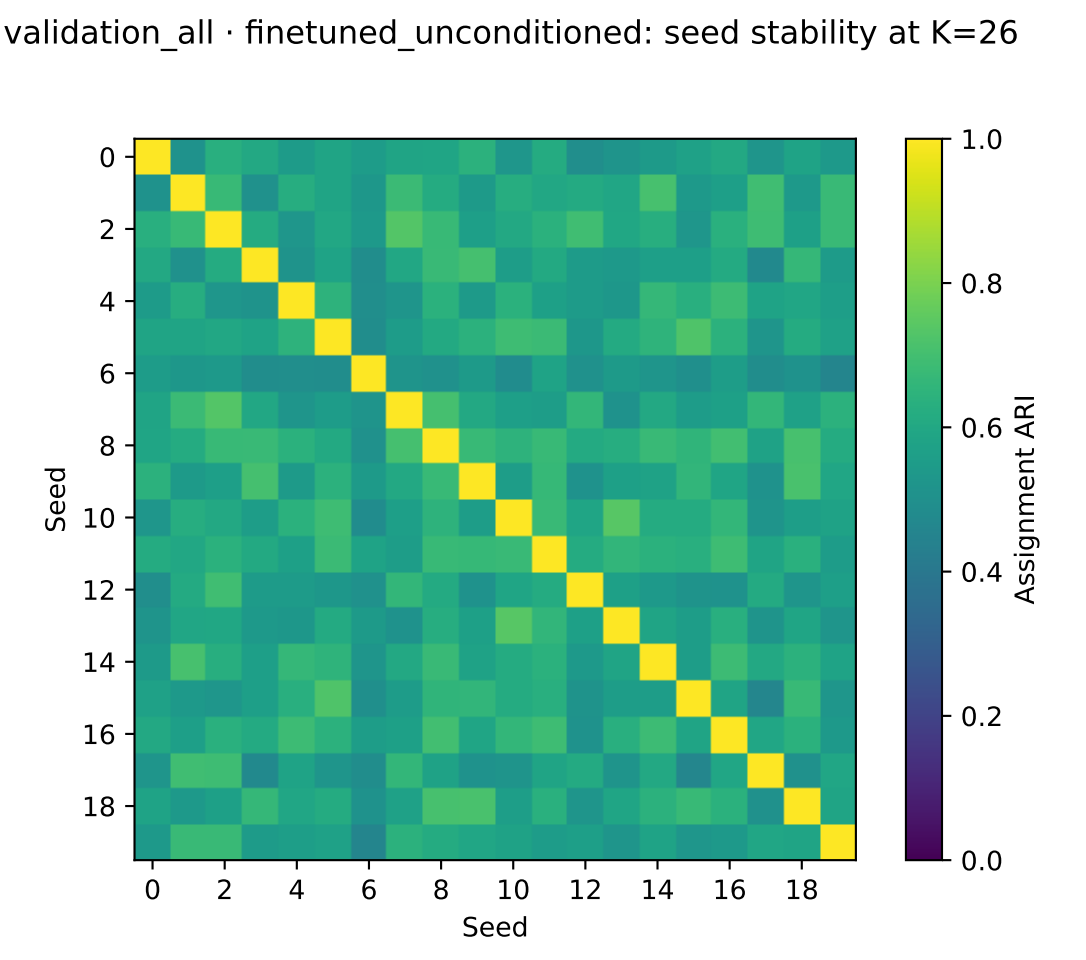


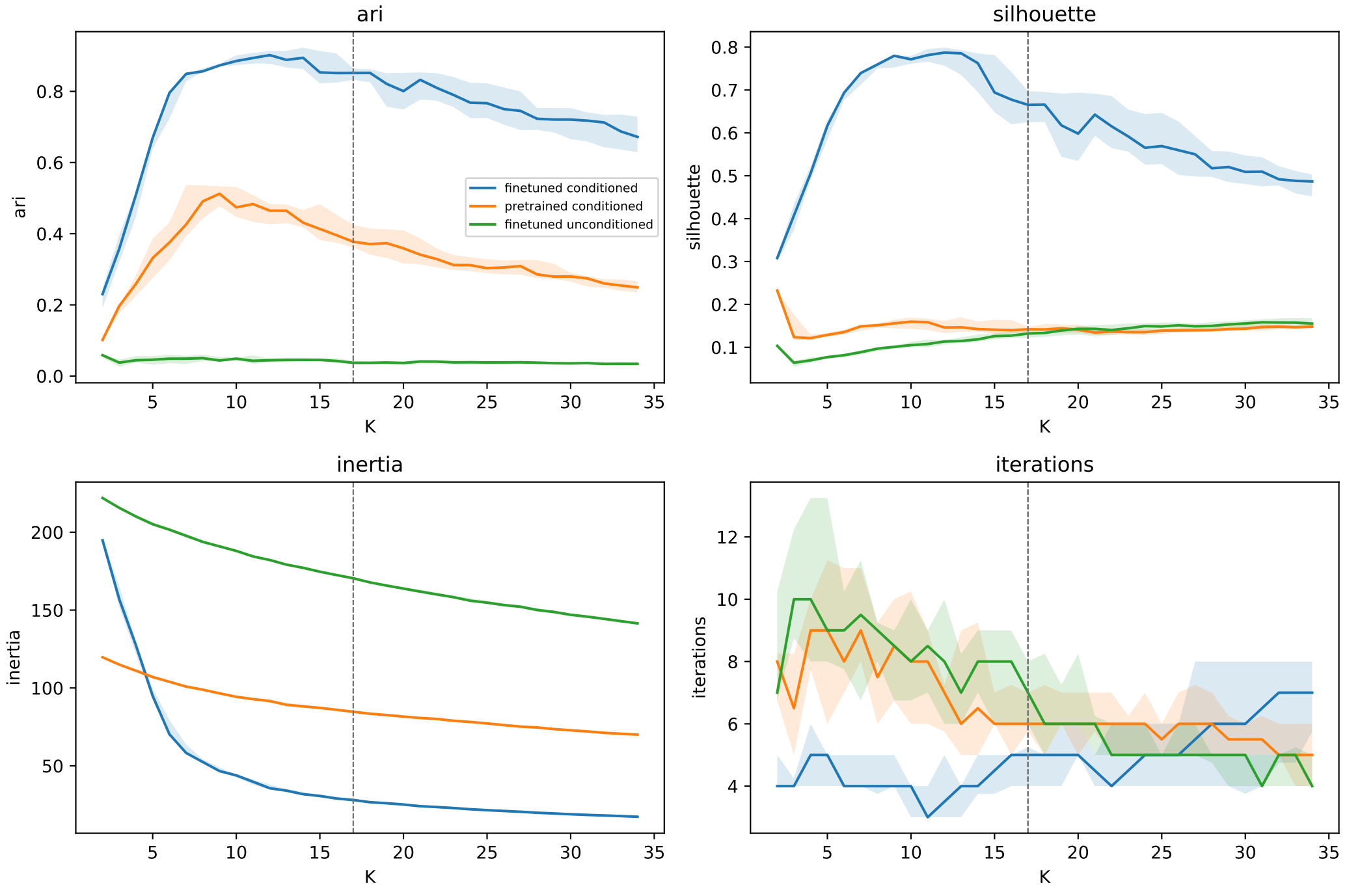
Source-color / cluster counts



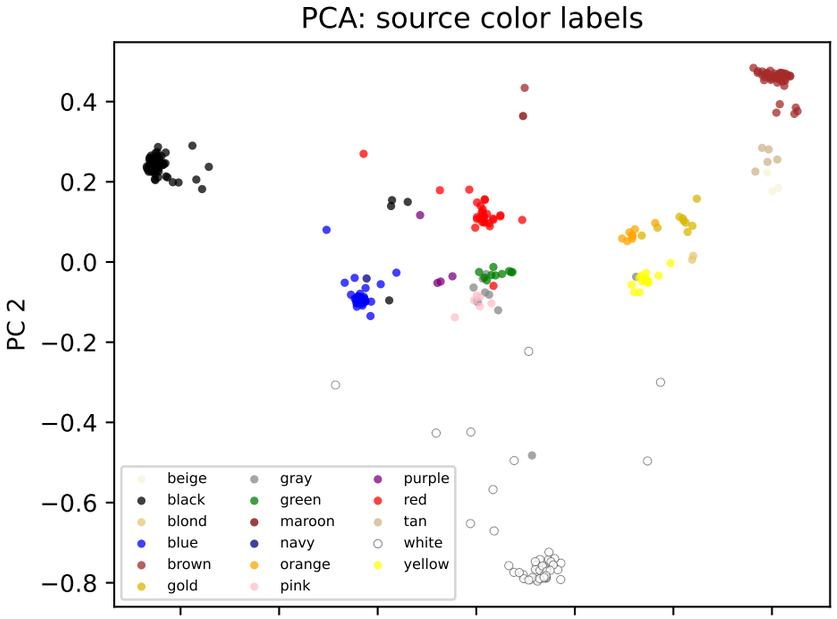
Centroid cosine distances

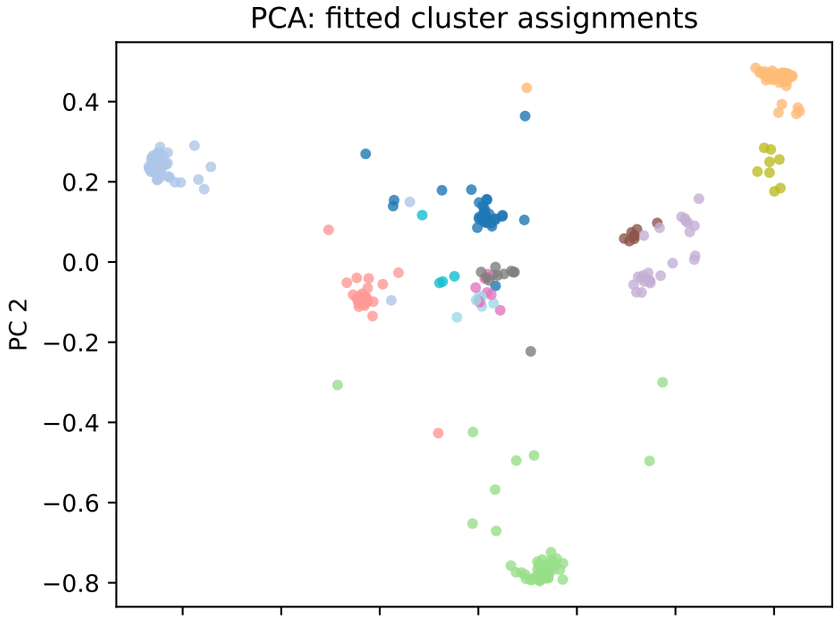


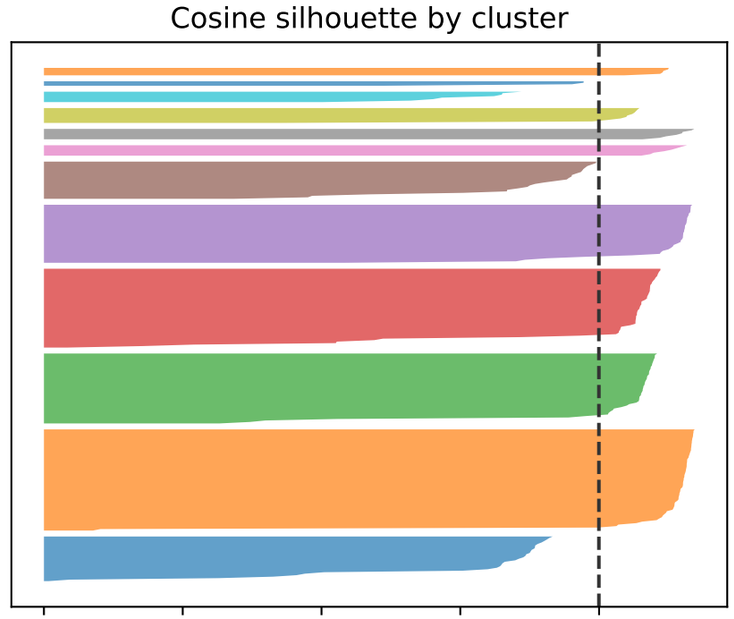

validation_all · finetuned_unconditioned: seed stability at K=26
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0


train_single: complete K sweep; bands show seed IQR
ari
silhouette
inertia
iterations
finetuned conditioned
pretrained conditioned
finetuned unconditioned
K

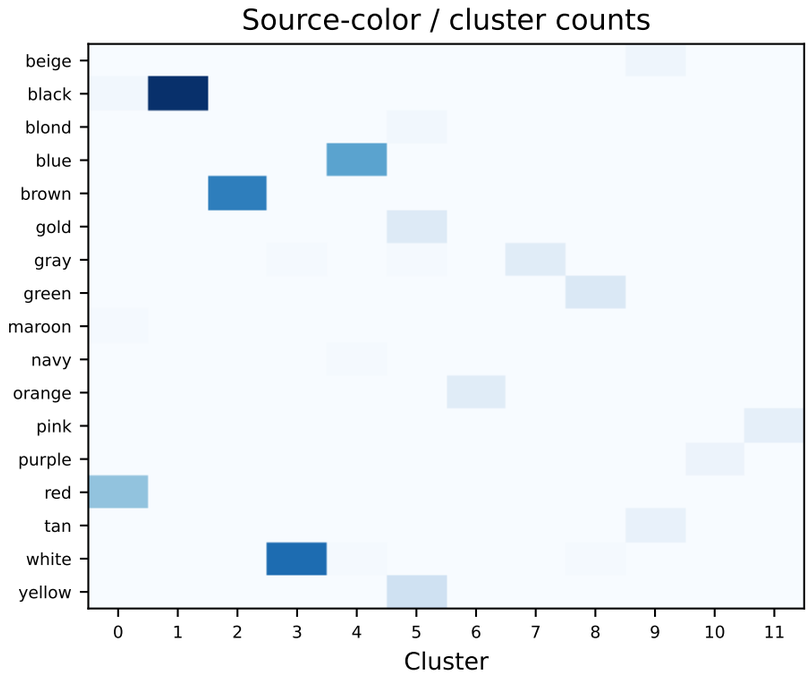
train_single · finetuned_conditioned · K=12, seed=13; PCA explains 42.5%

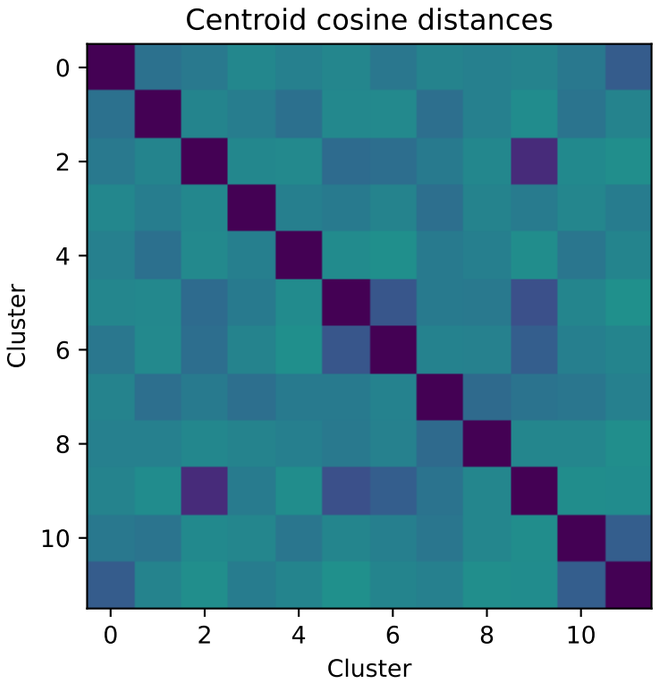




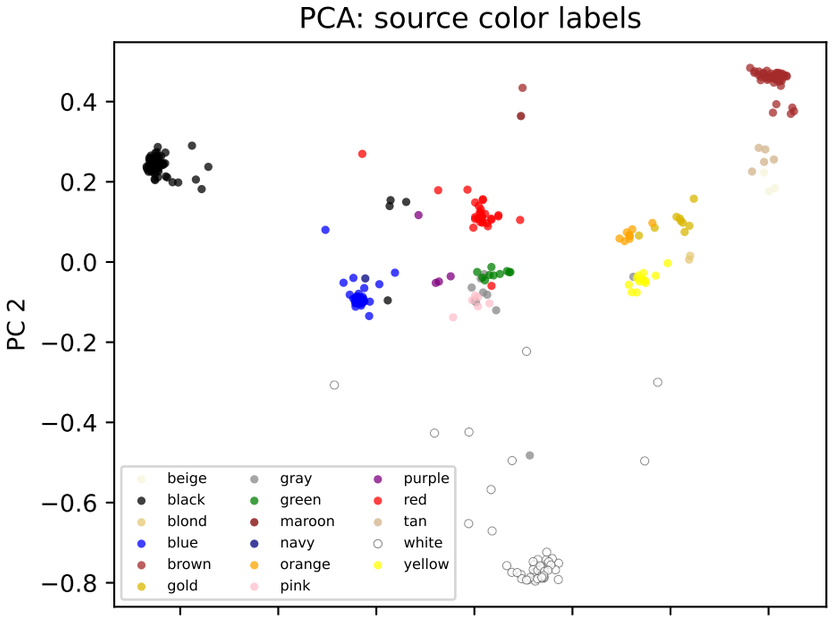


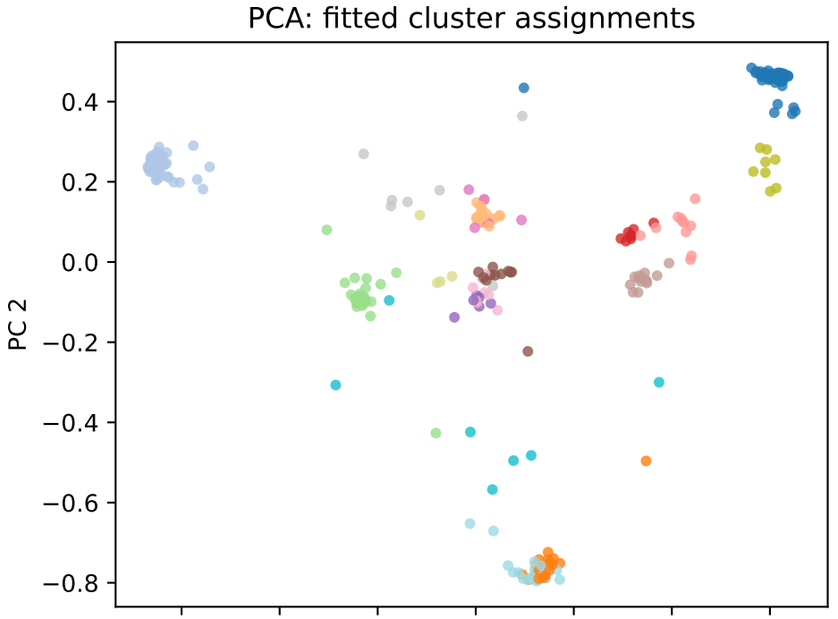


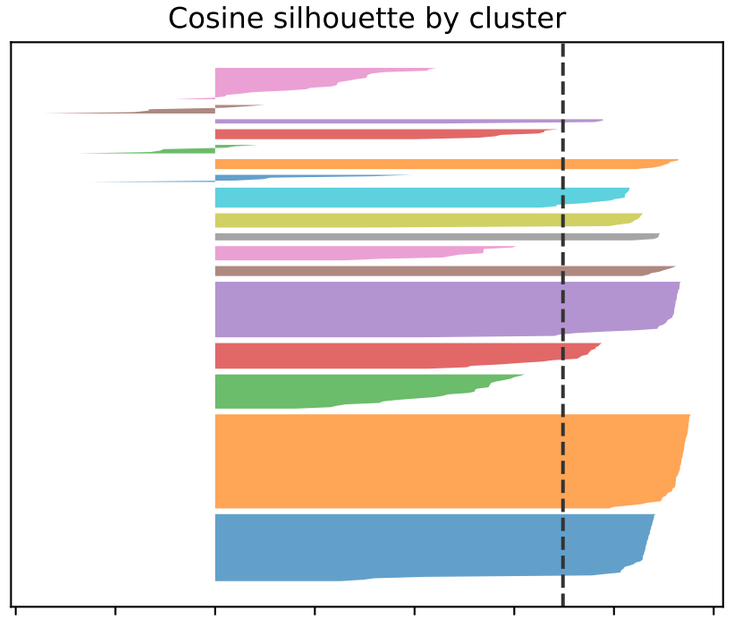


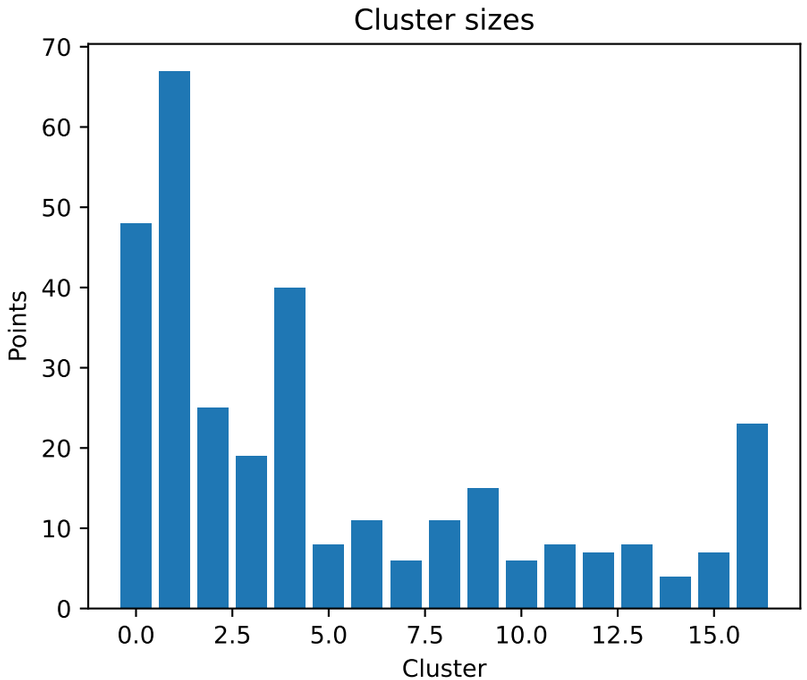


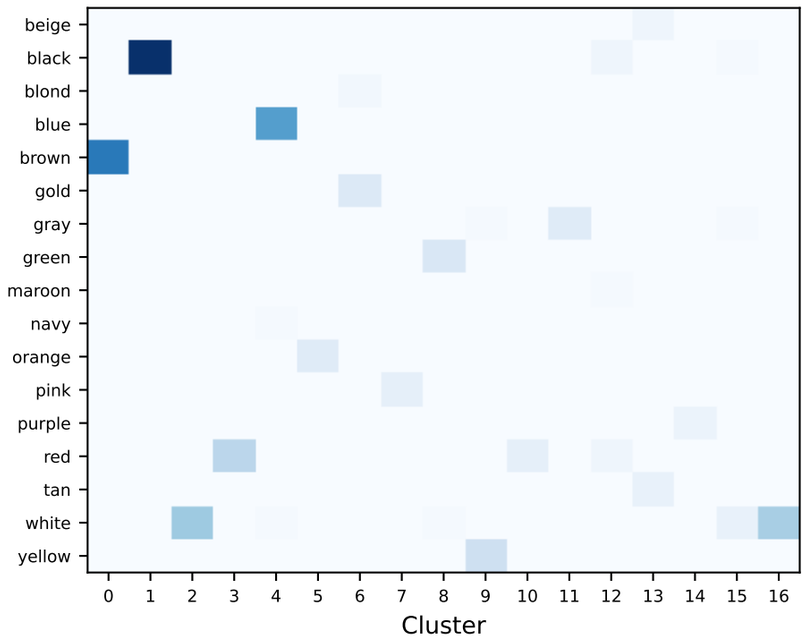
train_single · finetuned_conditioned · K=17, seed=19; PCA explains 42.5%

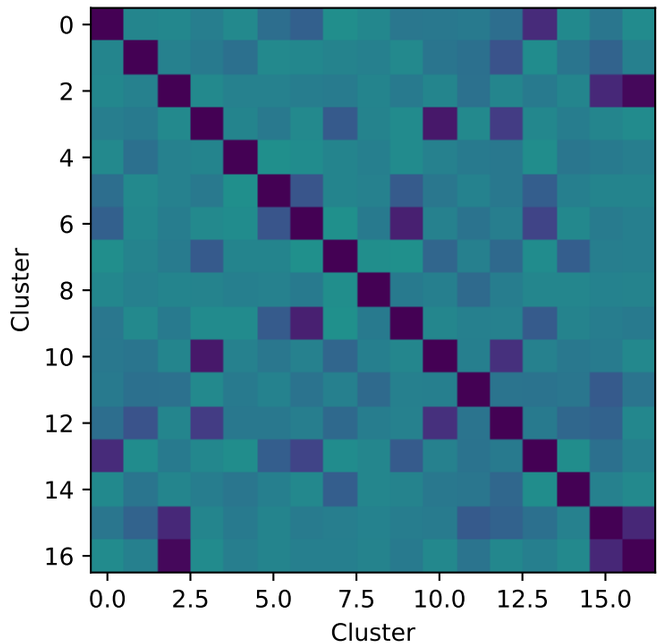


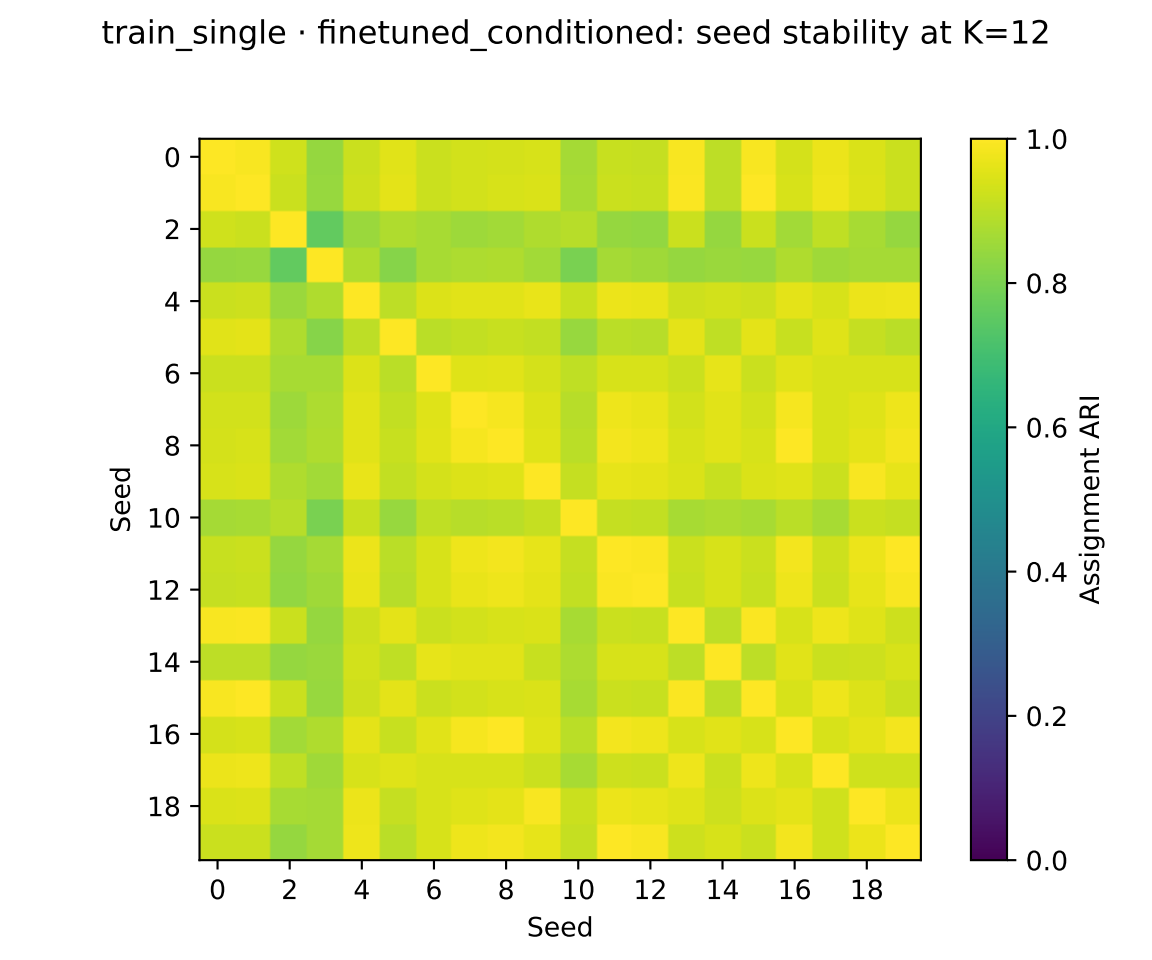






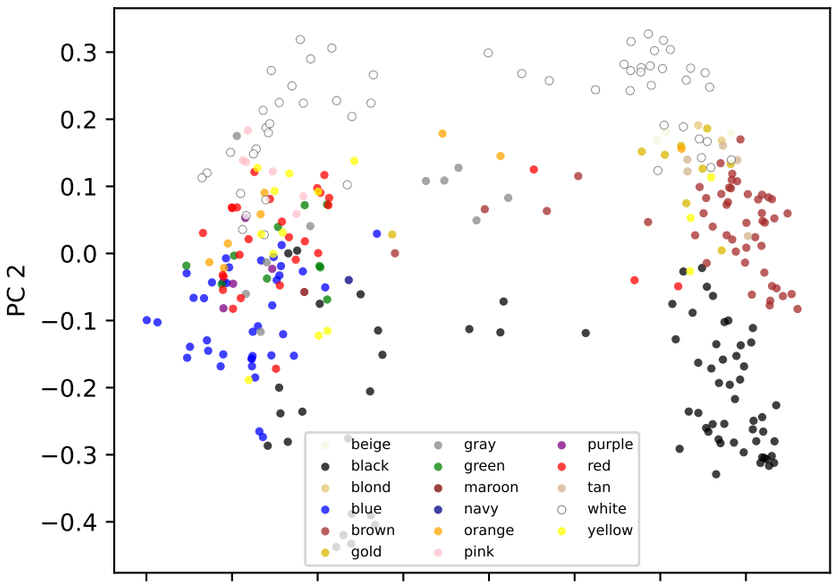




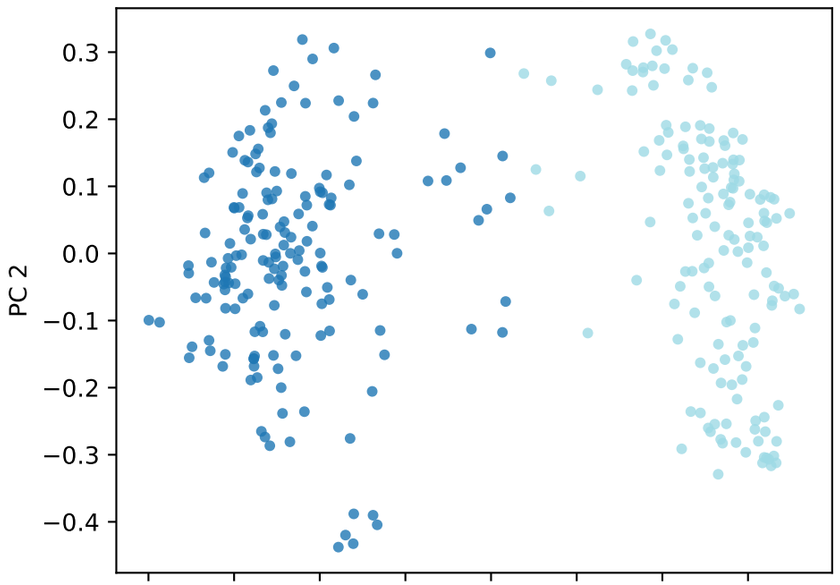

train_single · finetuned_conditioned: seed stability at K=12
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

train_single · pretrained_conditioned · K=2, seed=6; PCA explains 21.0%

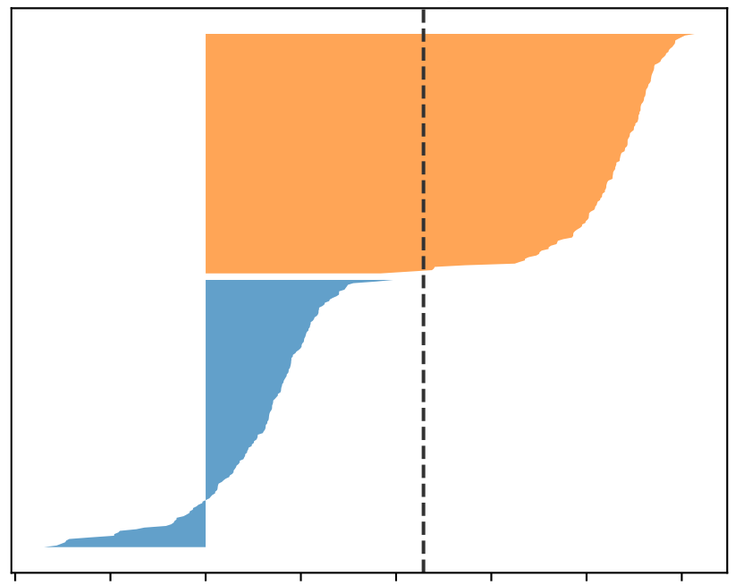
PCA: source color labels



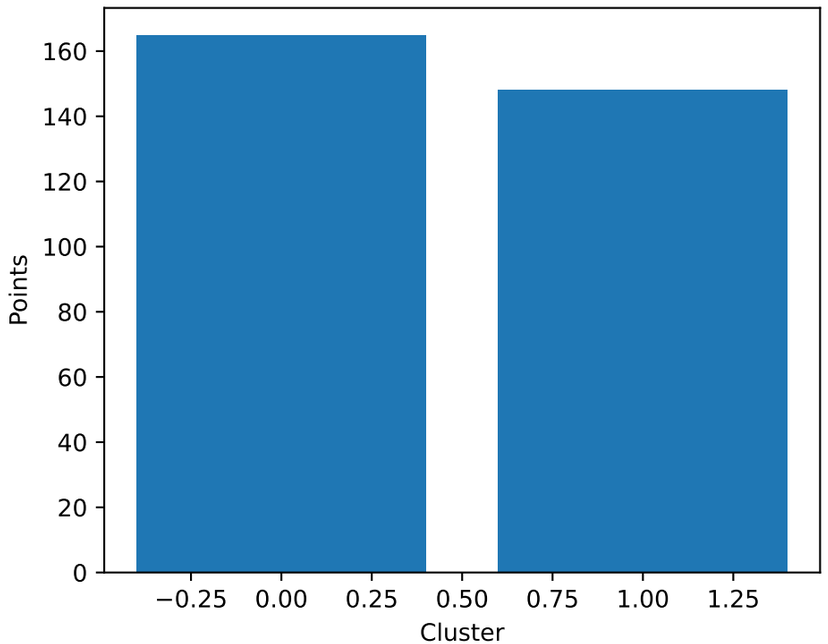
PCA: fitted cluster assignments



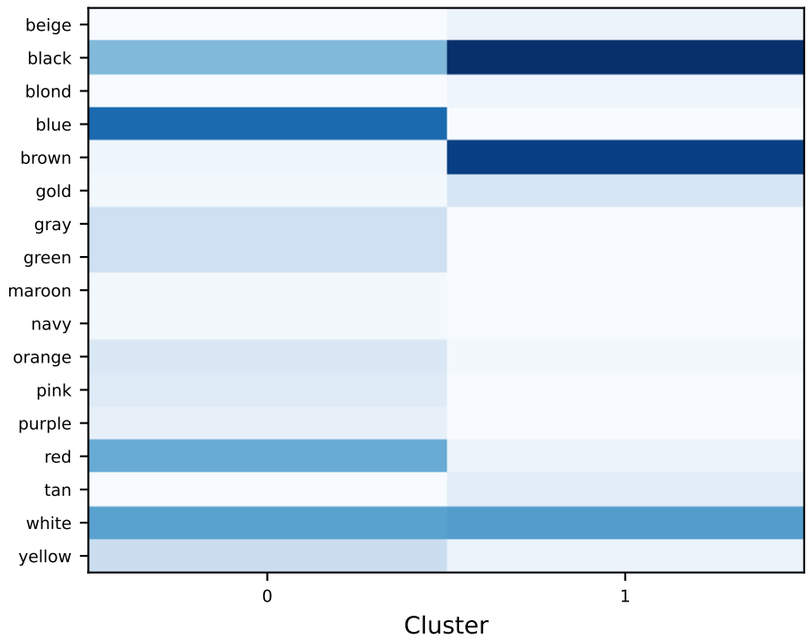
Cosine silhouette by cluster



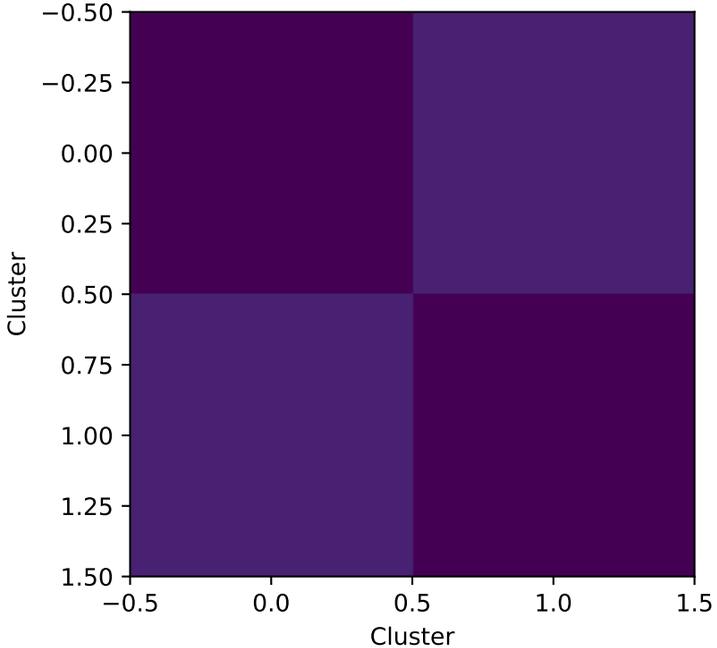
Cluster sizes



Source-color / cluster counts

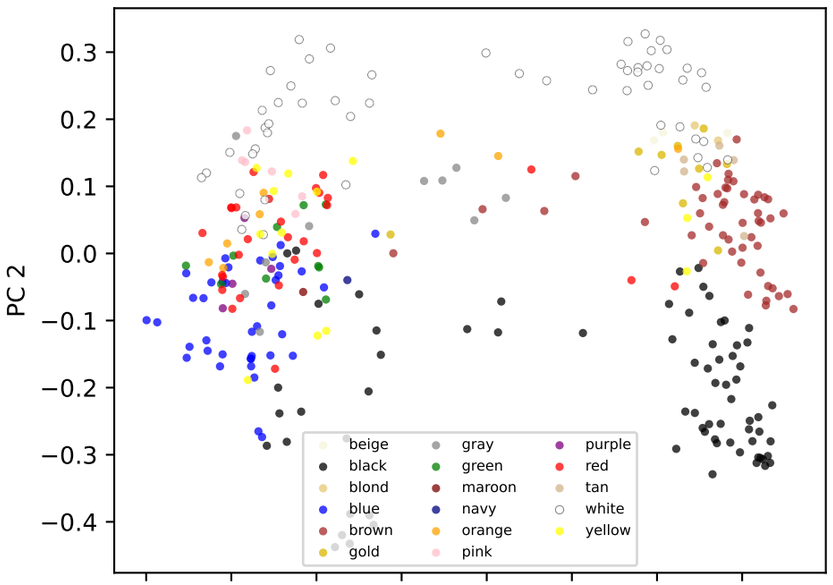


Centroid cosine distances

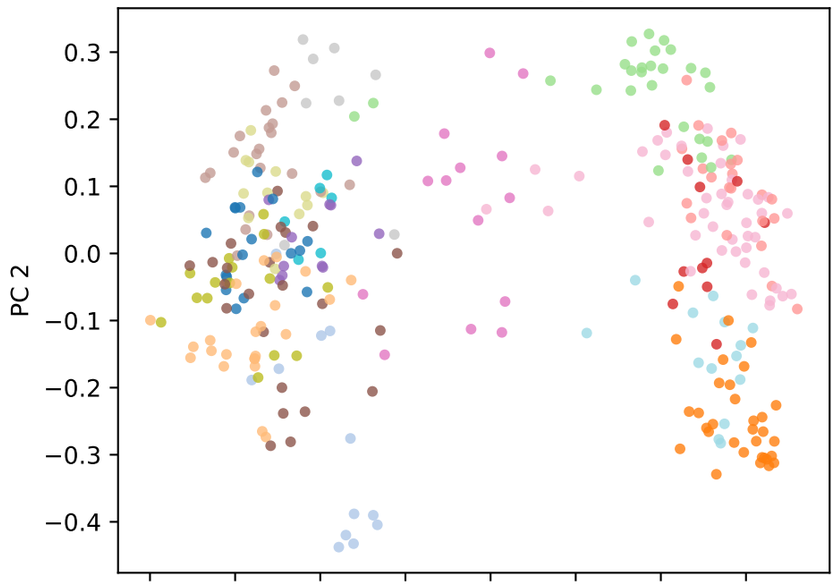


train_single · pretrained_conditioned · K=17, seed=0; PCA explains 21.0%

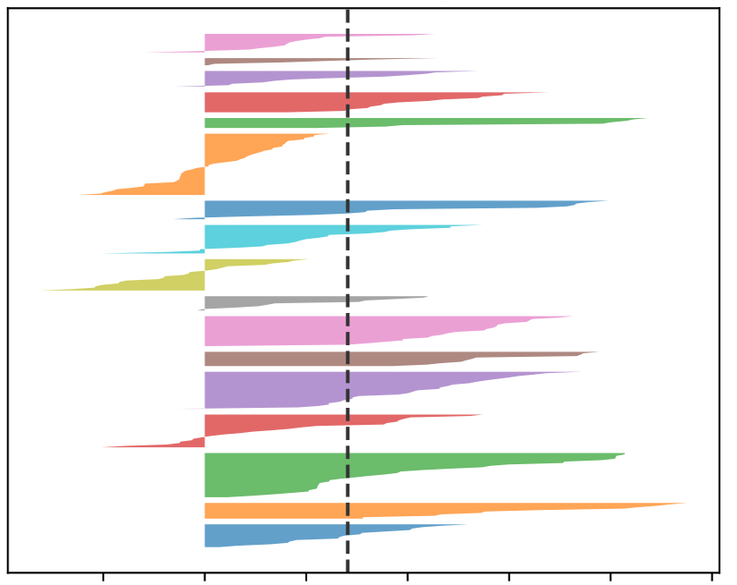
PCA: source color labels



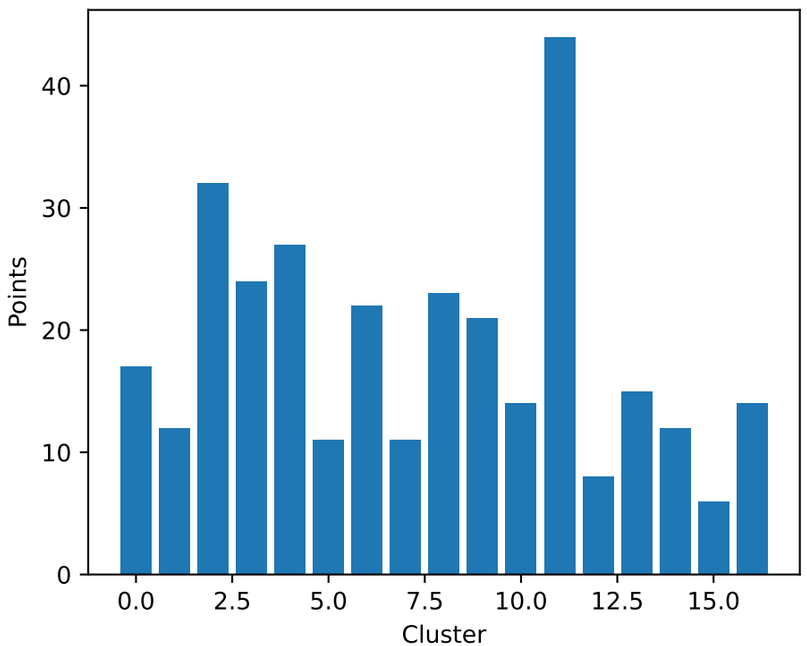
PCA: fitted cluster assignments



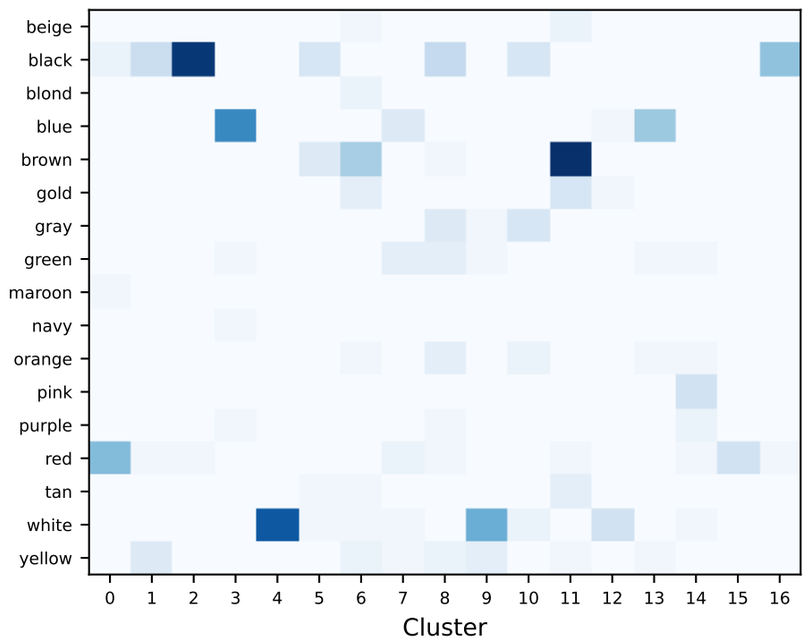
Cosine silhouette by cluster



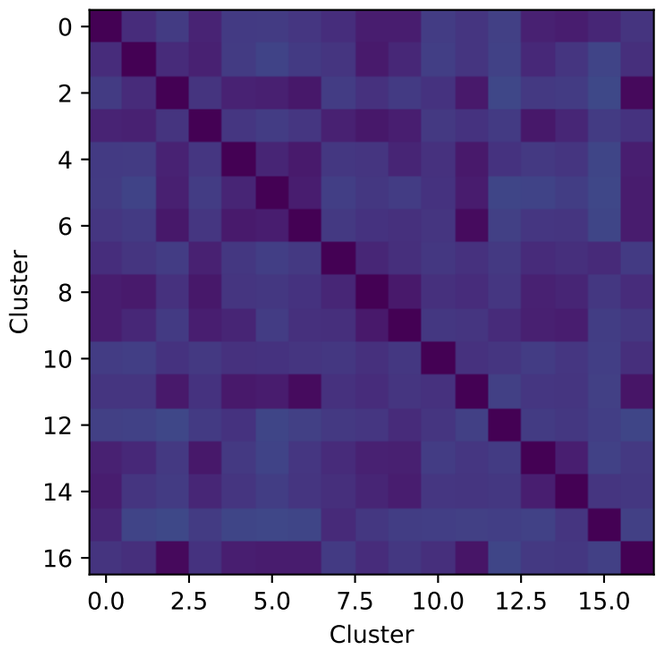
Cluster sizes

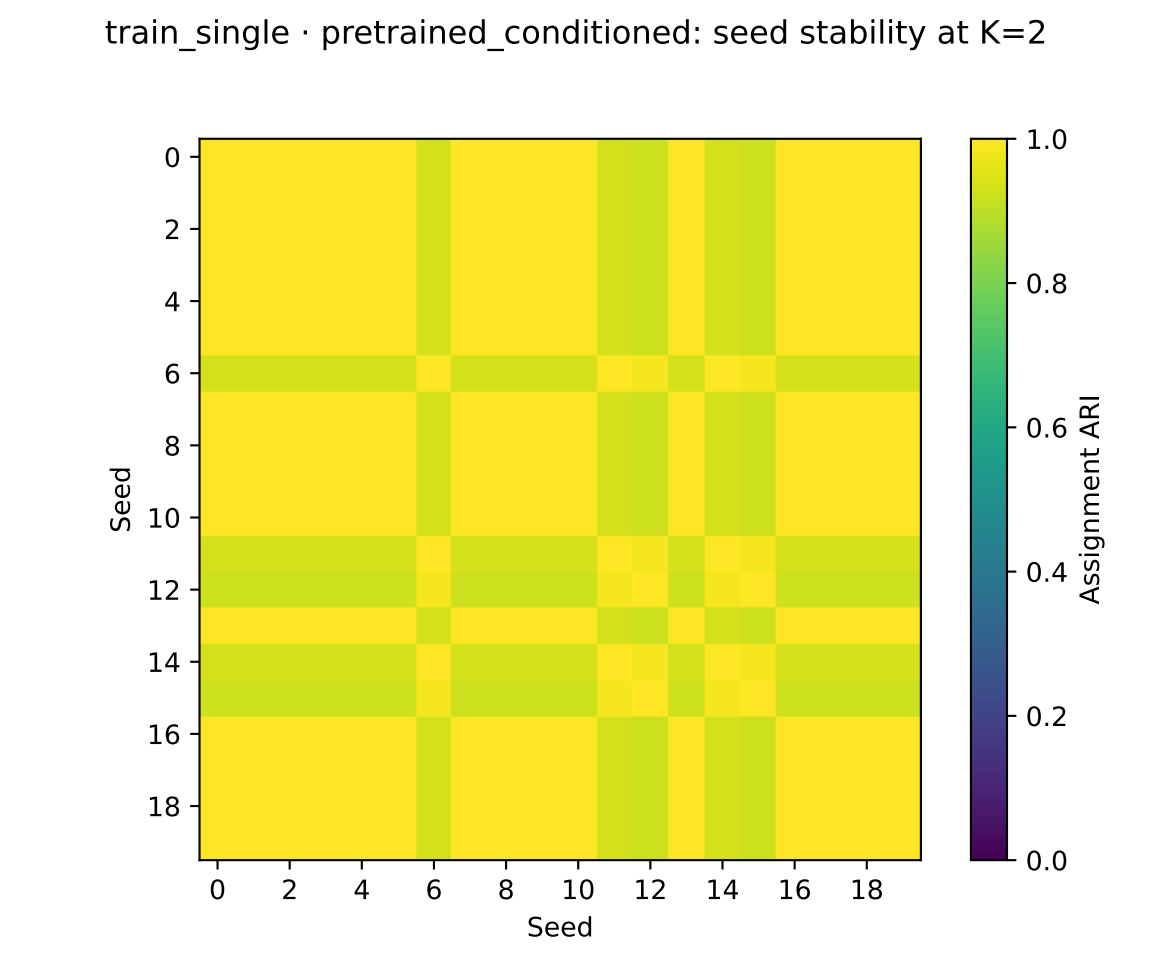


Source-color / cluster counts



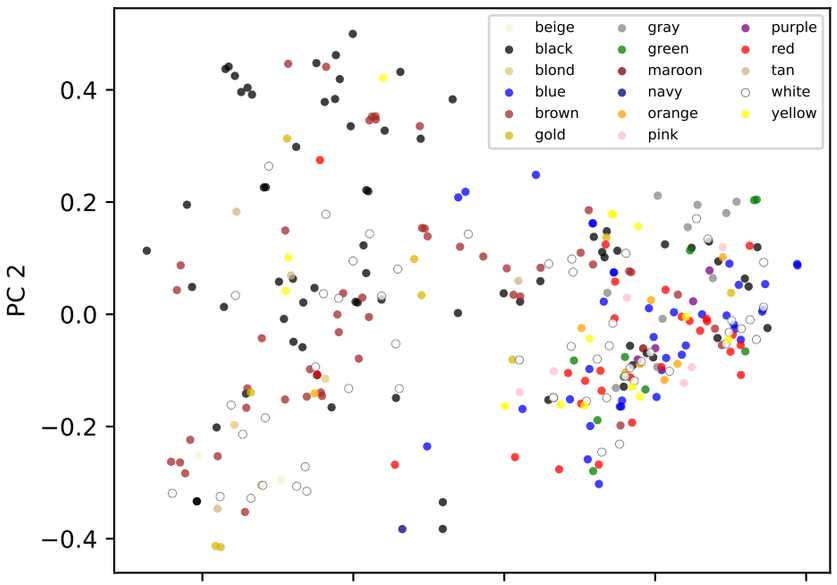
Centroid cosine distances



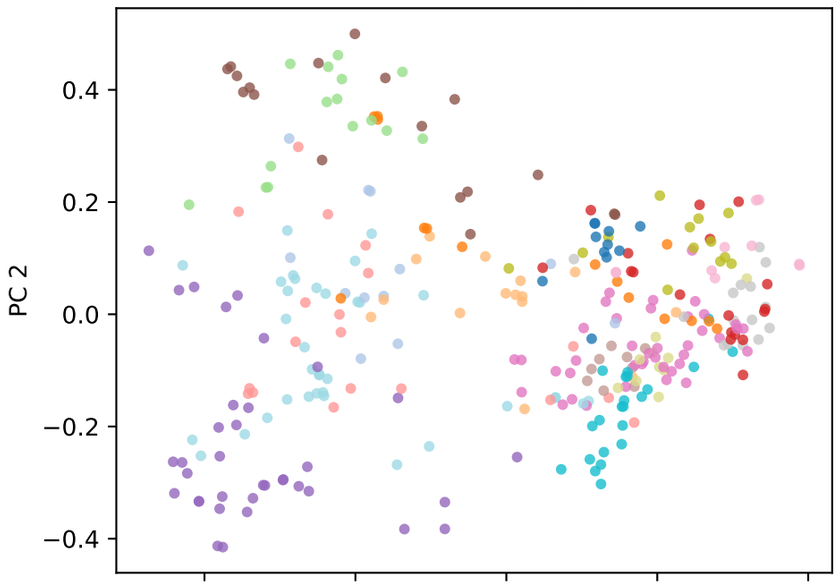

train_single · pretrained_conditioned: seed stability at K=2
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

train_single · finetuned_unconditioned · K=17, seed=5; PCA explains 12.1%

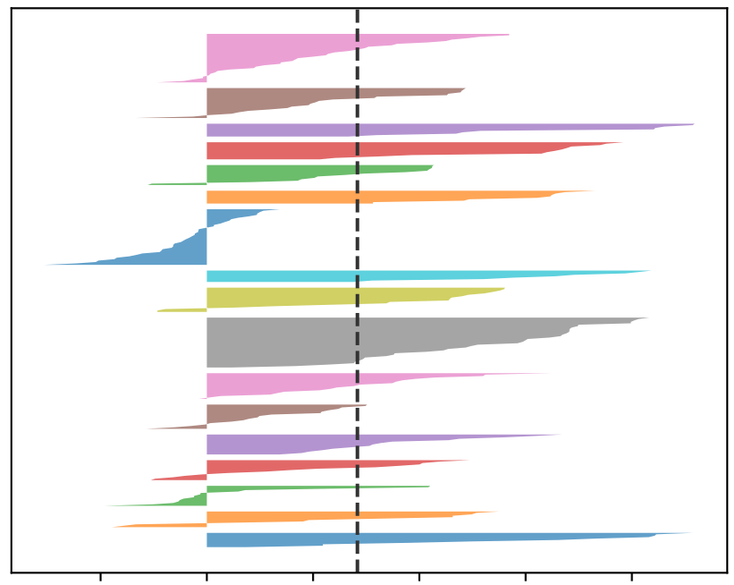
PCA: source color labels



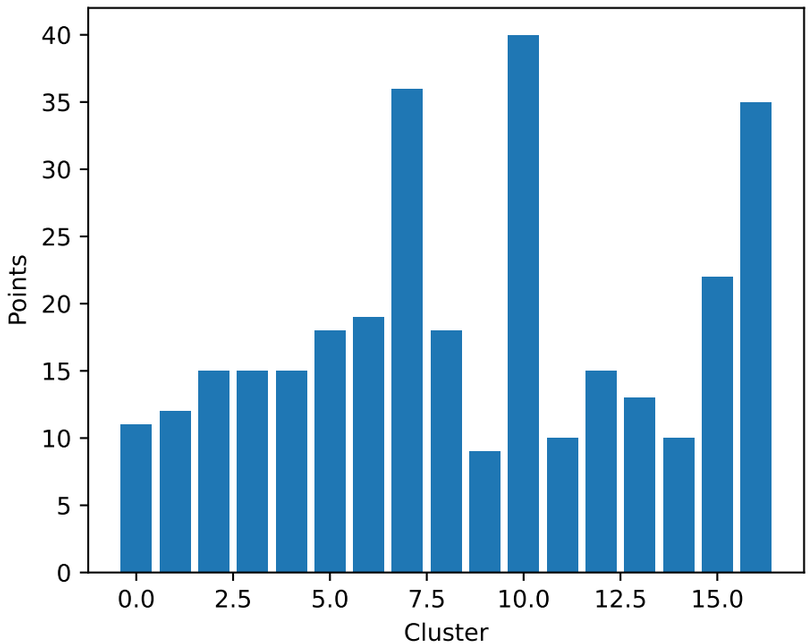
PCA: fitted cluster assignments



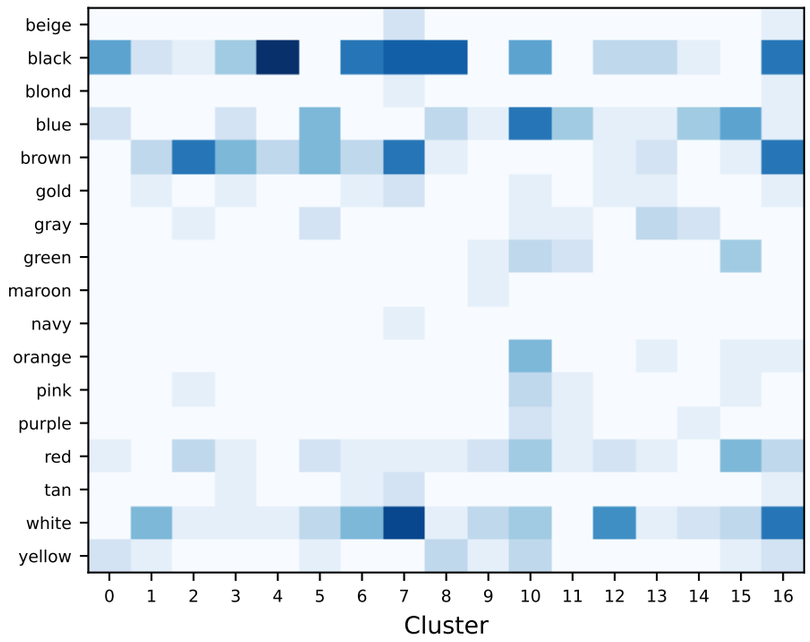
Cosine silhouette by cluster



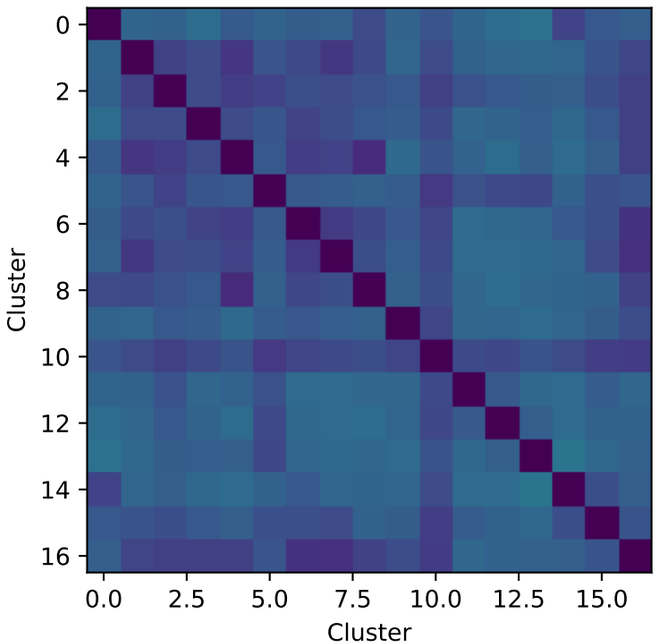
Cluster sizes



Source-color / cluster counts

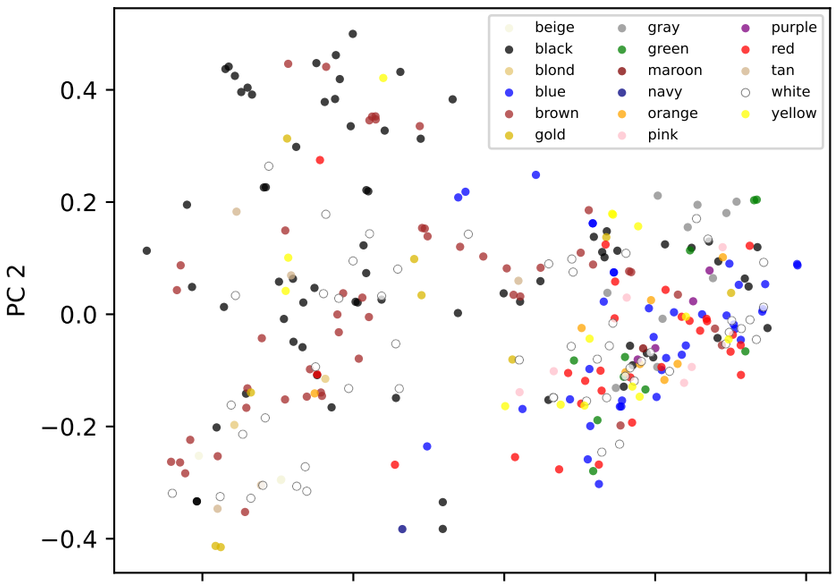


Centroid cosine distances

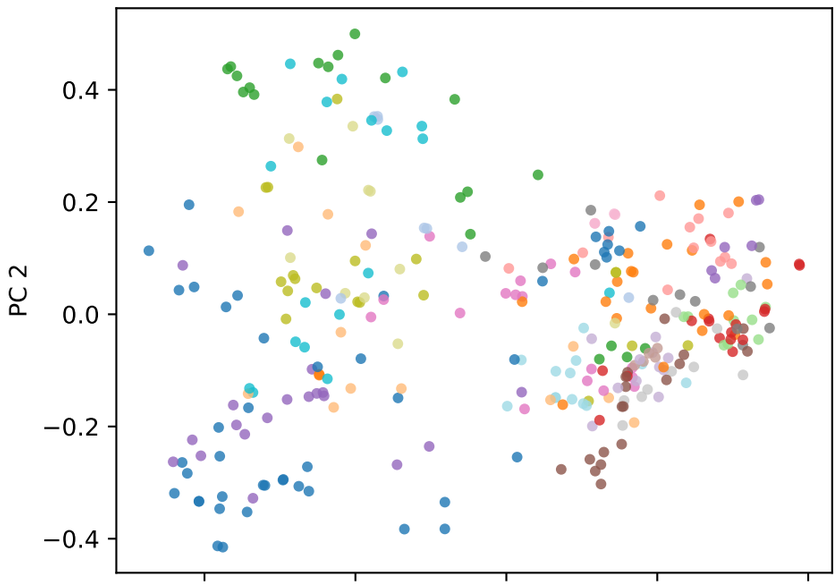


train_single · finetuned_unconditioned · K=31, seed=5; PCA explains 12.1%

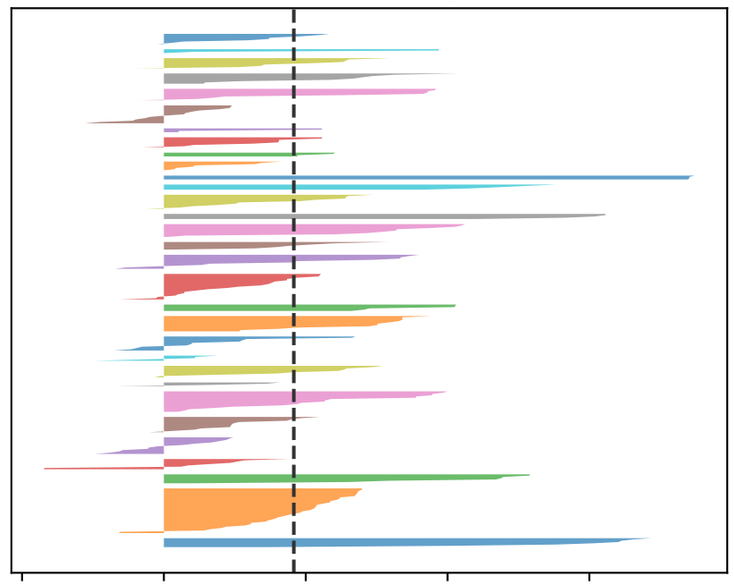
PCA: source color labels



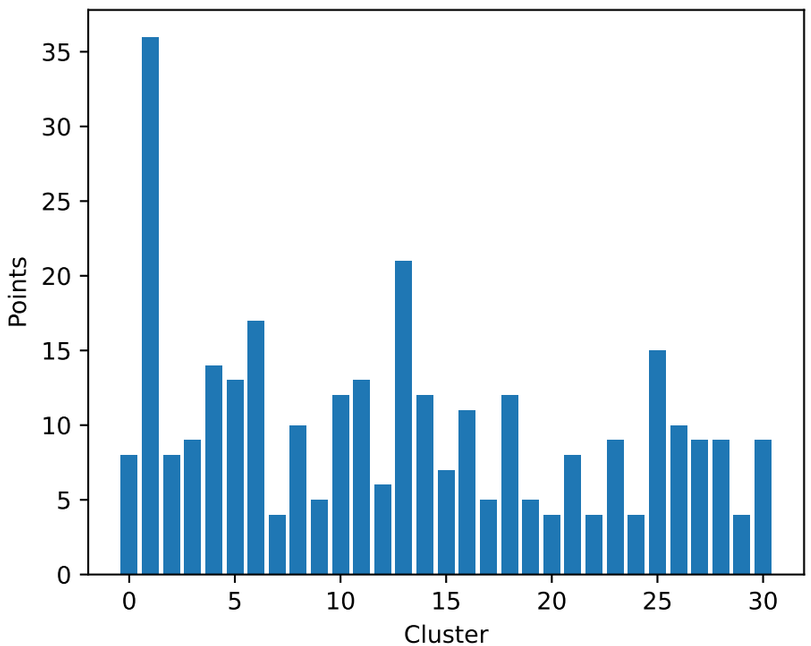
PCA: fitted cluster assignments



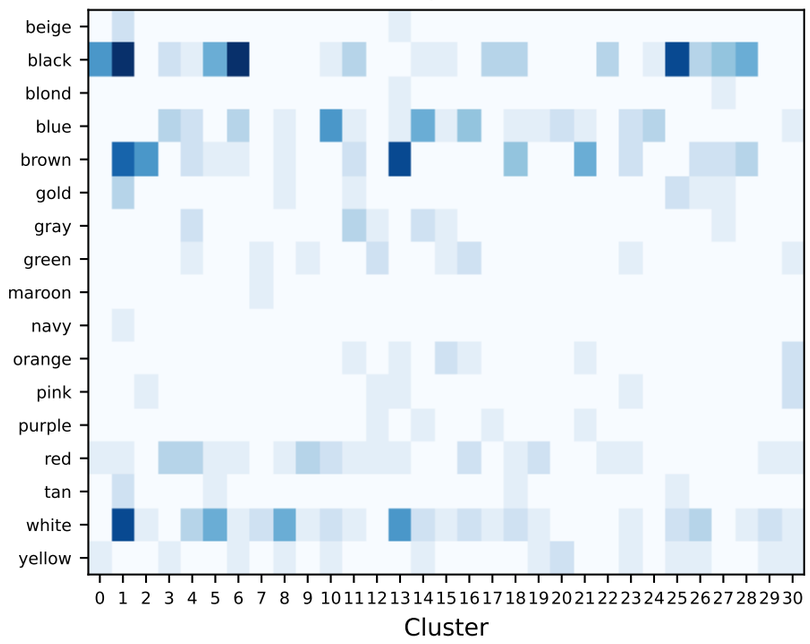
Cosine silhouette by cluster



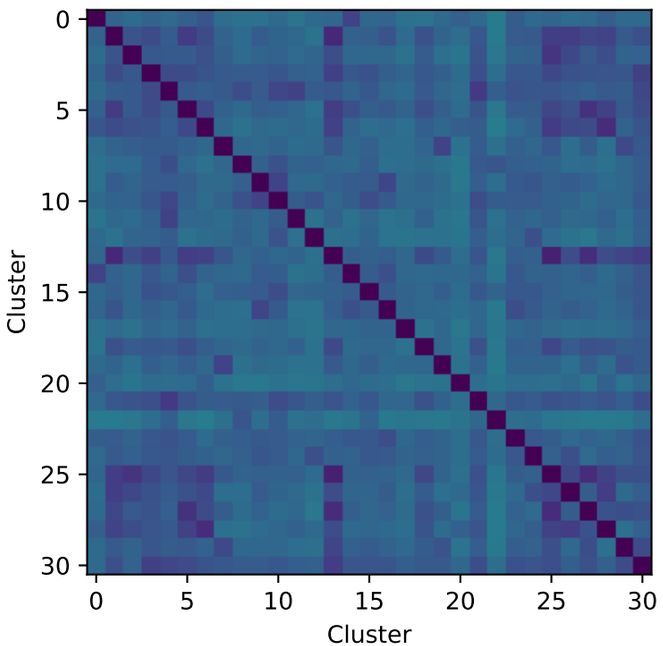
Cluster sizes

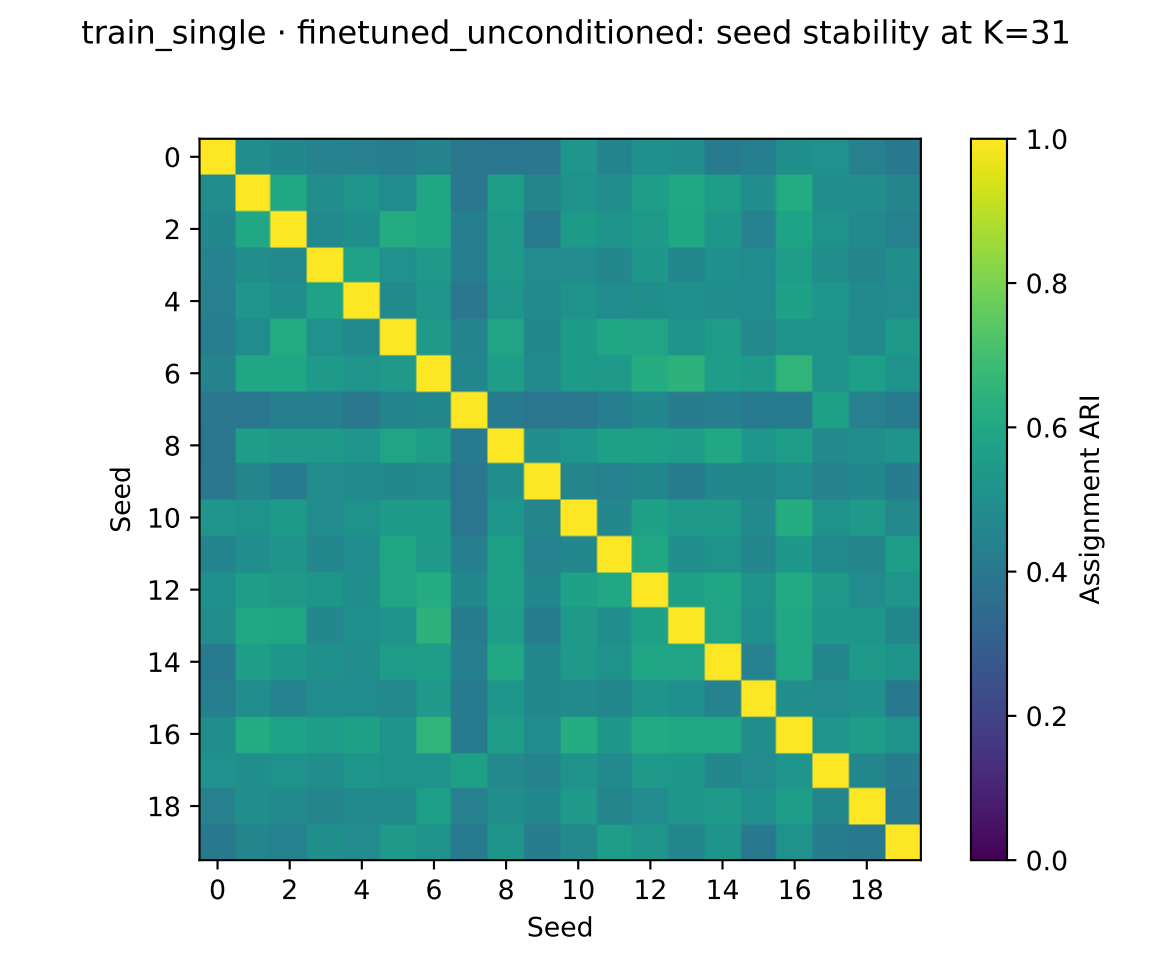


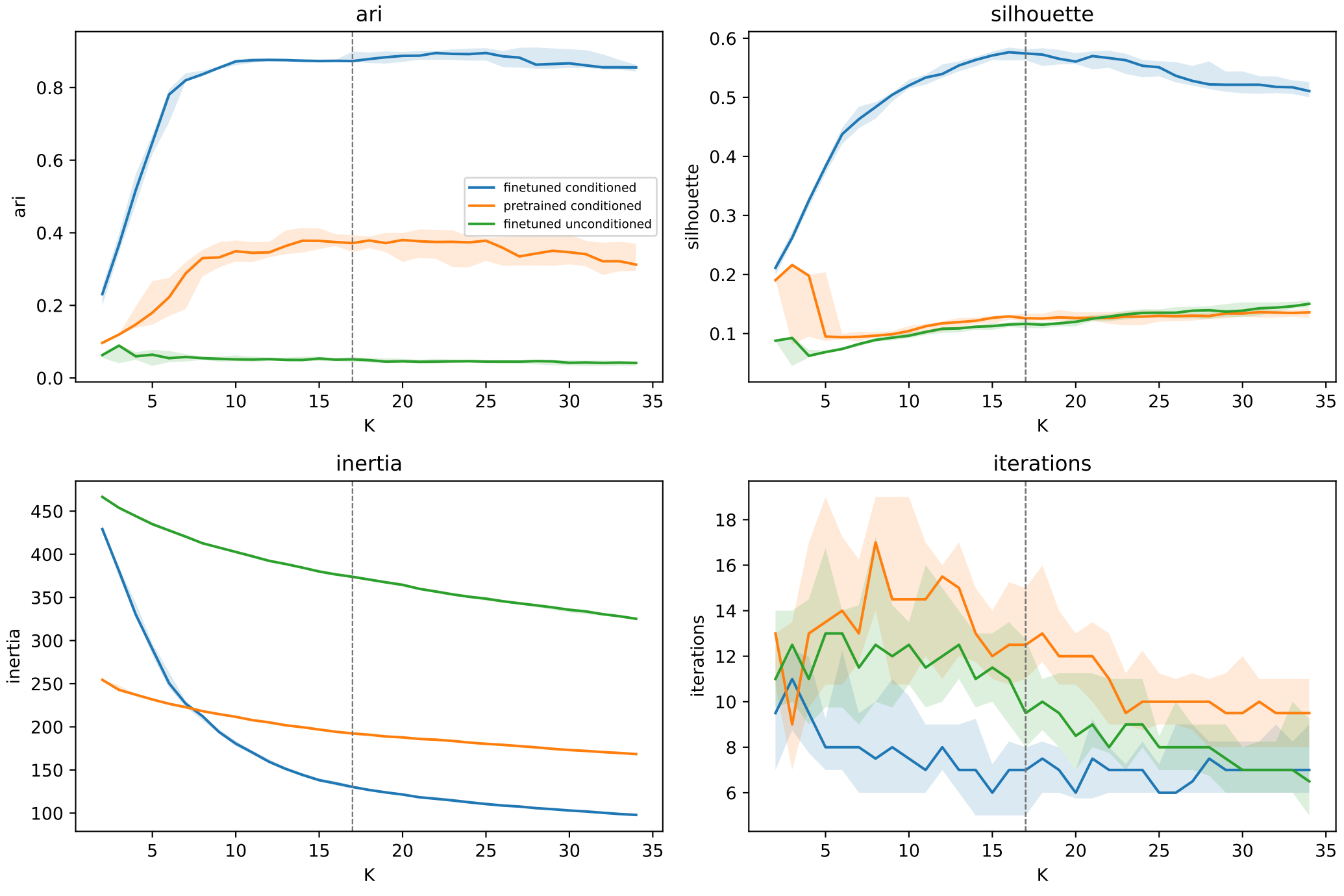
Source-color / cluster counts



Centroid cosine distances

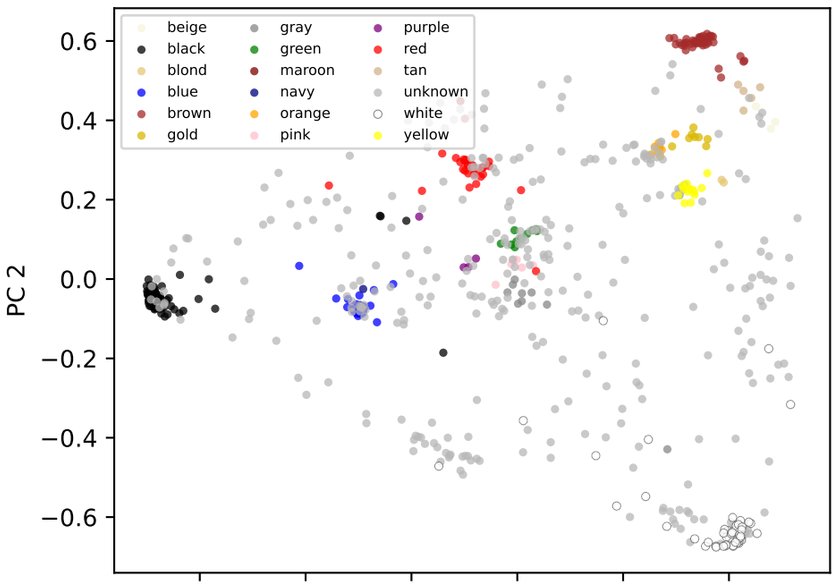



train_single · finetuned_unconditioned: seed stability at K=31
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

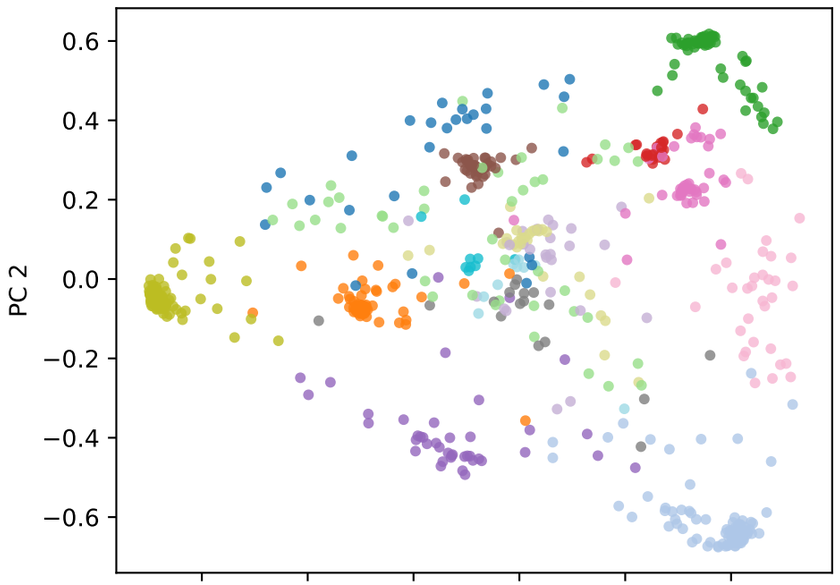

train_all: complete K sweep; bands show seed IQR
ari
silhouette
inertia
iterations
finetuned conditioned
pretrained conditioned
finetuned unconditioned
K

train_all · finetuned_conditioned · K=16, seed=19; PCA explains 32.2%

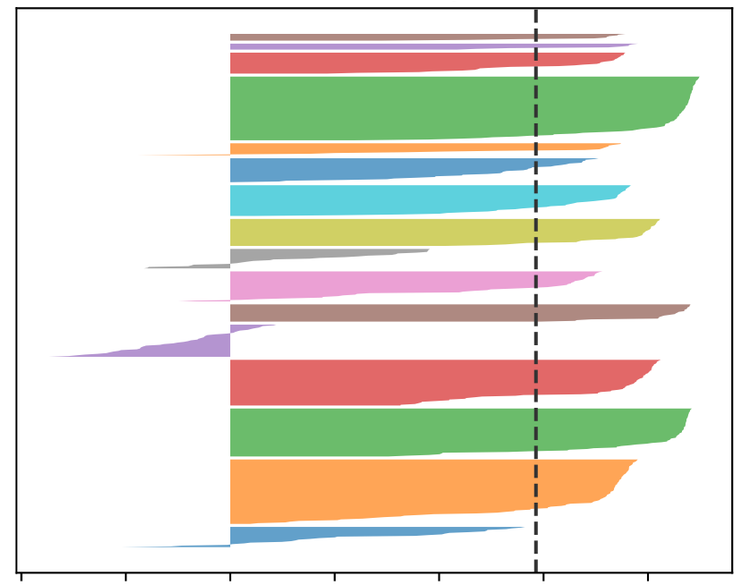
PCA: source color labels



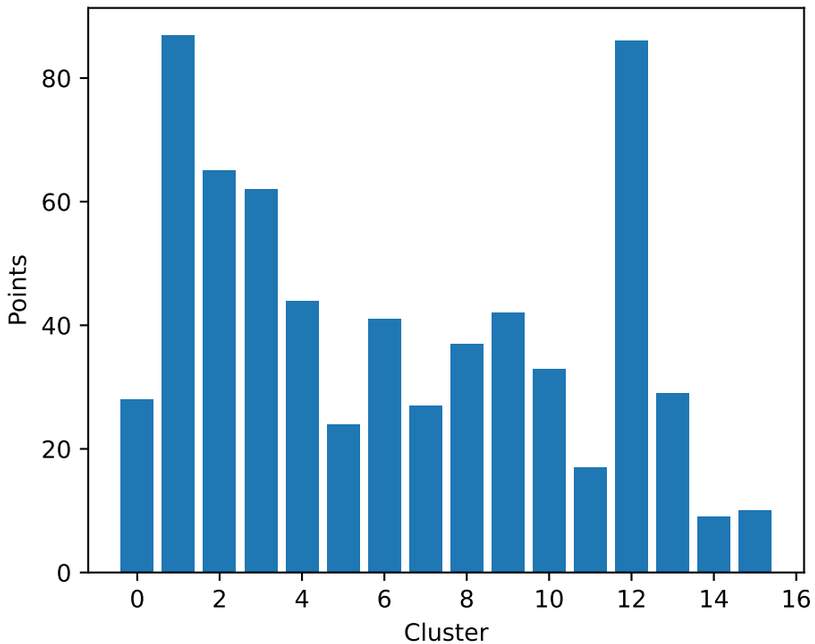
PCA: fitted cluster assignments



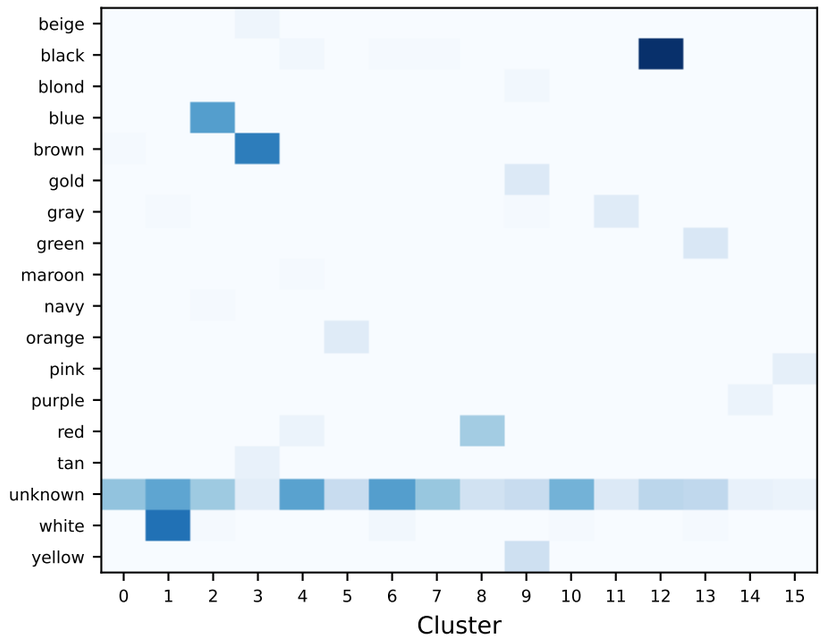
Cosine silhouette by cluster



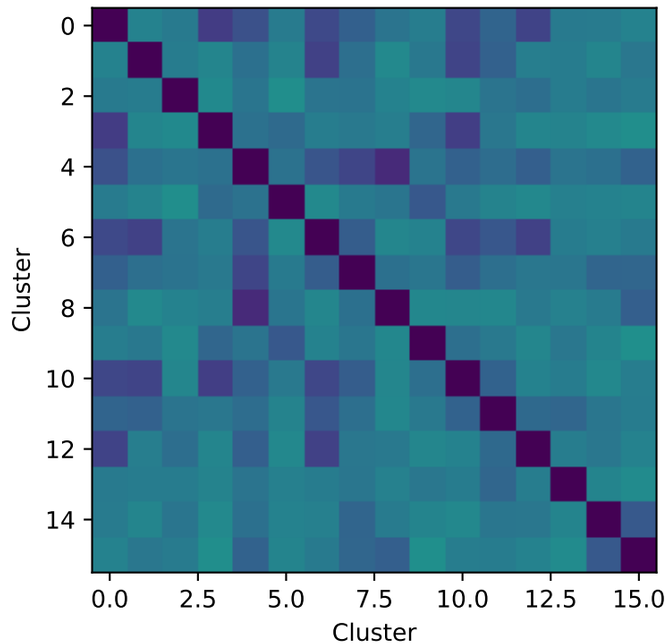
Cluster sizes



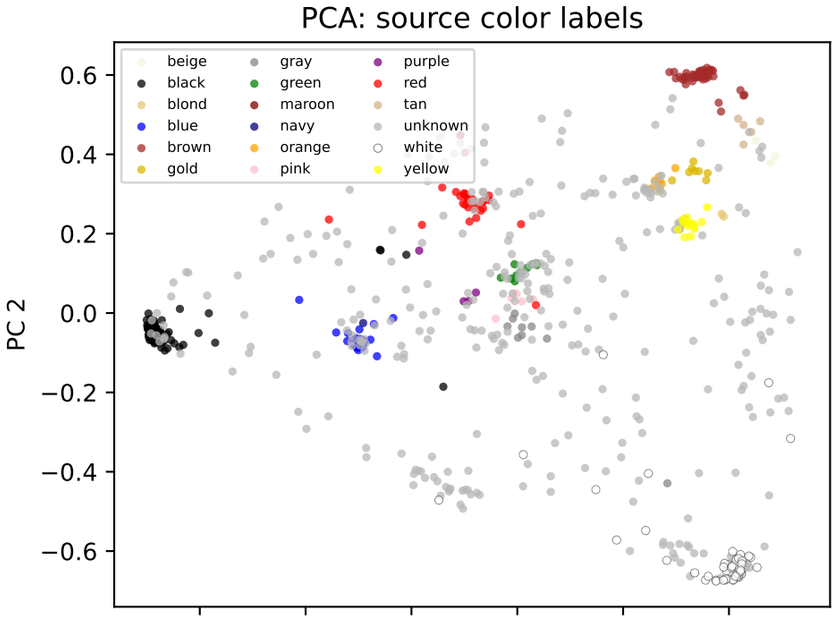
Source-color / cluster counts

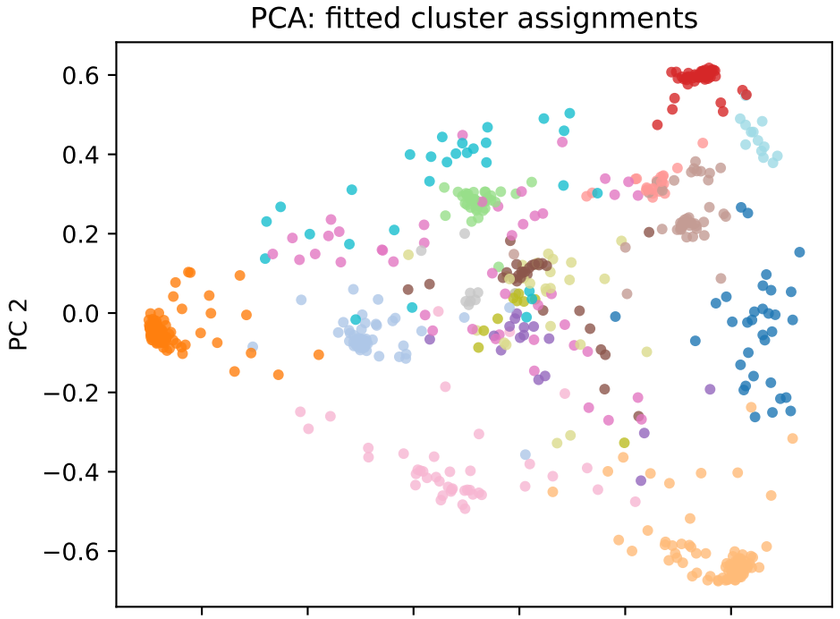


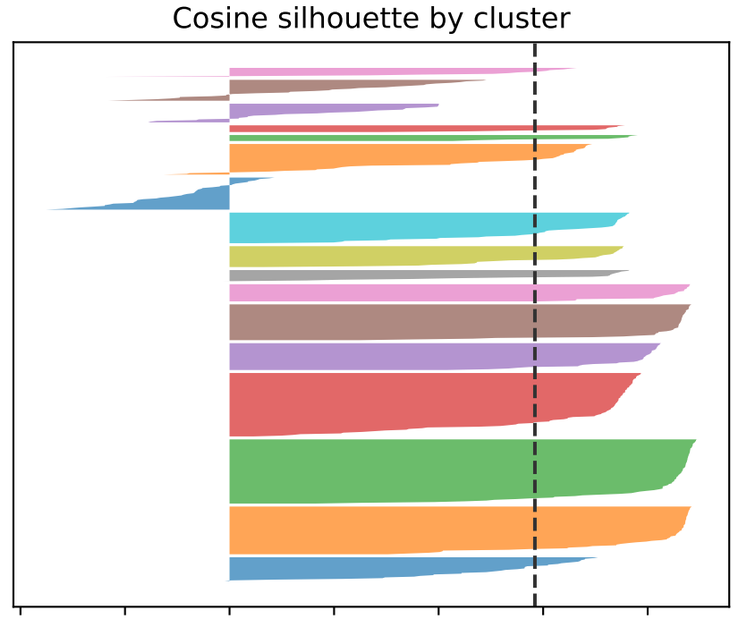
Centroid cosine distances

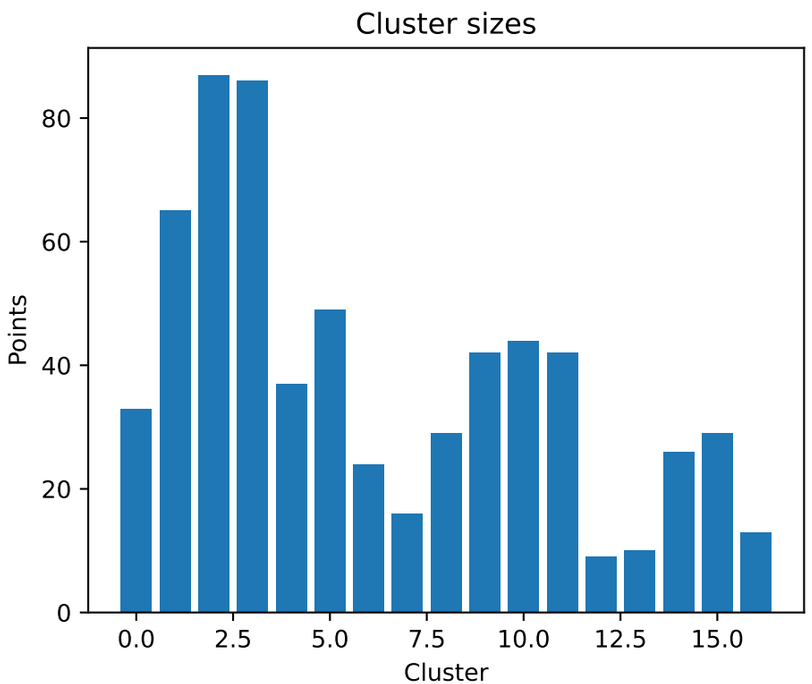


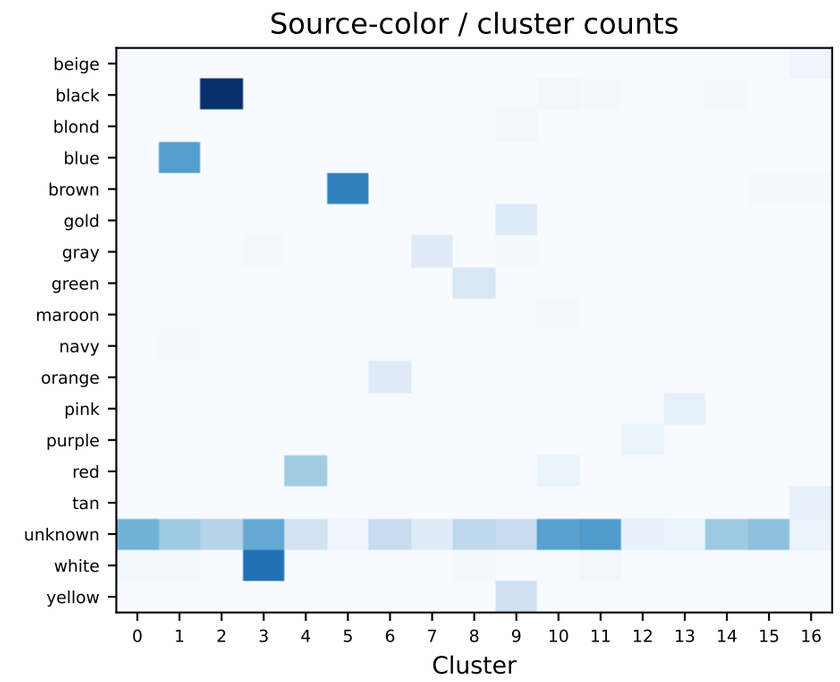
train_all · finetuned_conditioned · K=17, seed=1; PCA explains 32.2%

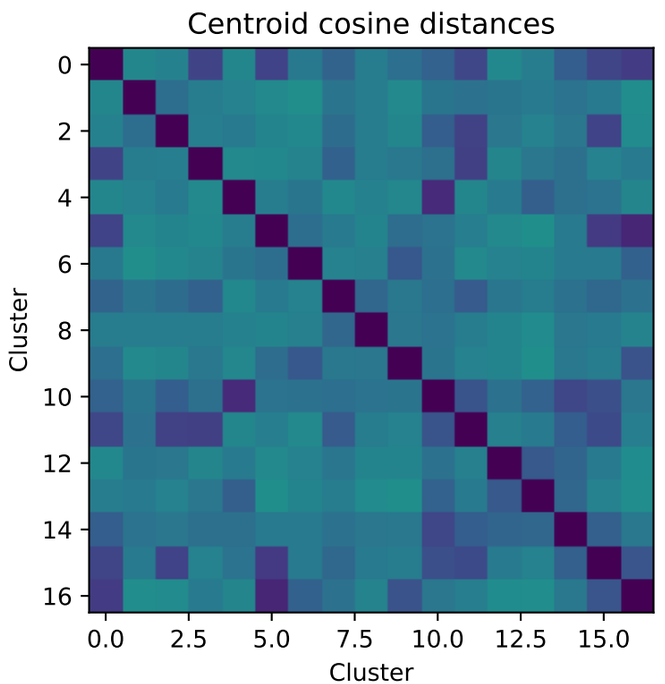


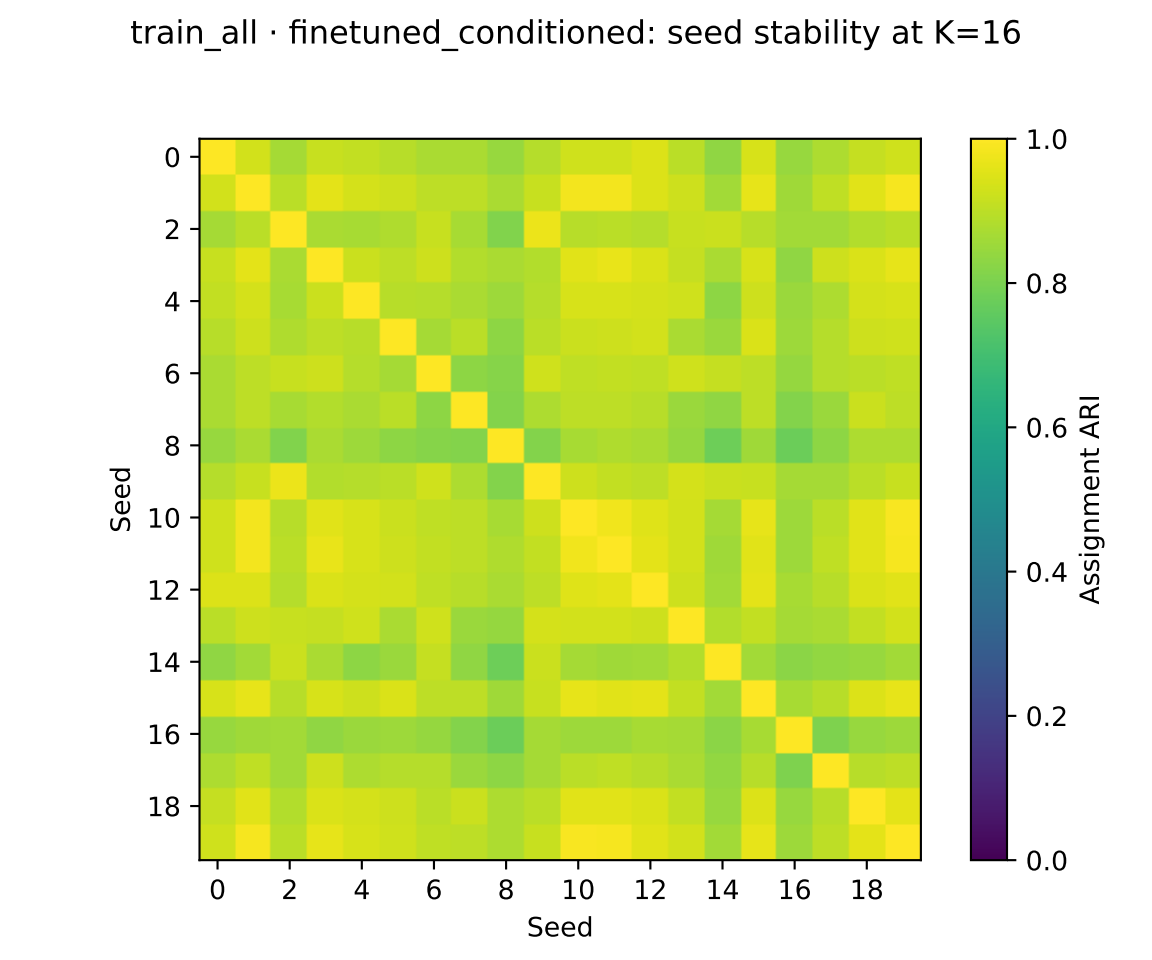






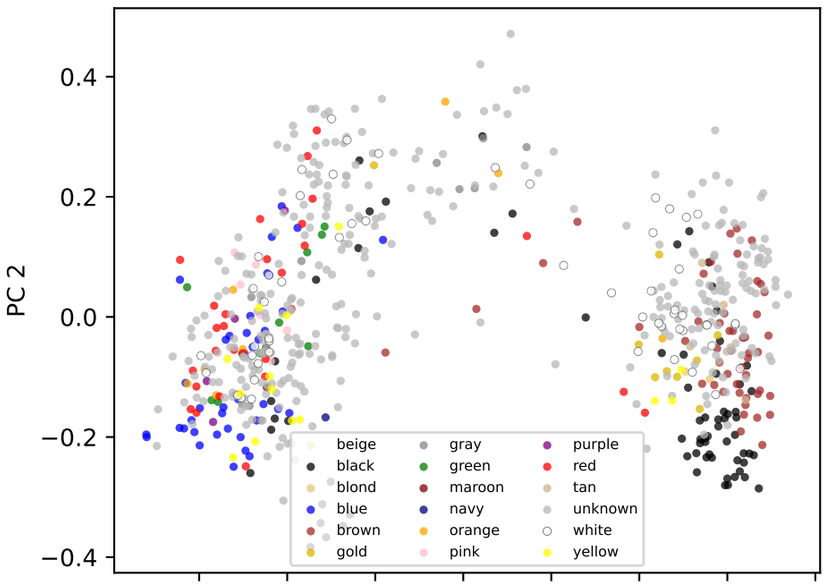




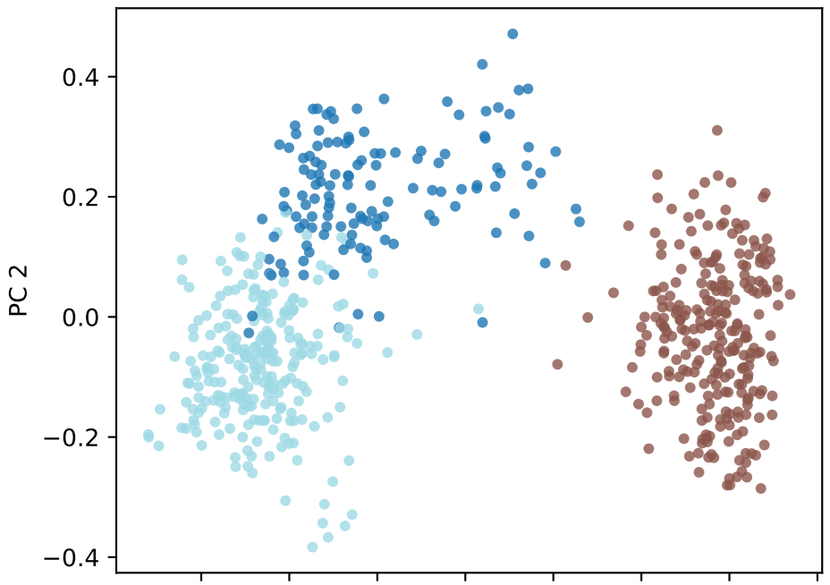

train_all · finetuned_conditioned: seed stability at K=16
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

train_all · pretrained_conditioned · K=3, seed=11; PCA explains 18.2%

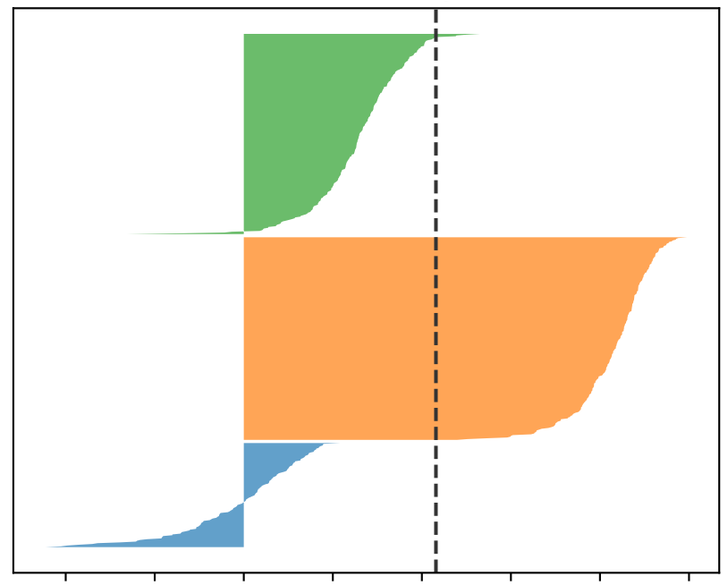
PCA: source color labels



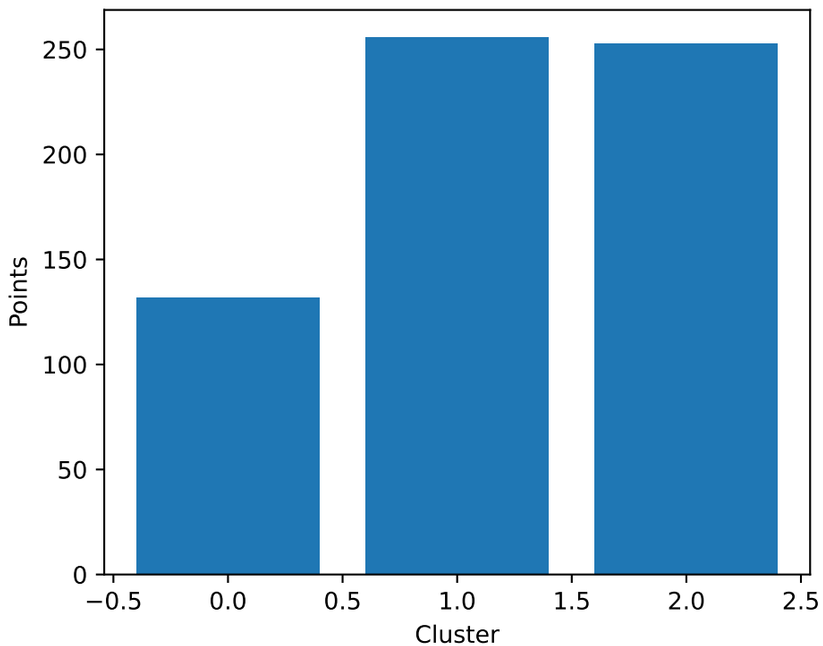
PCA: fitted cluster assignments



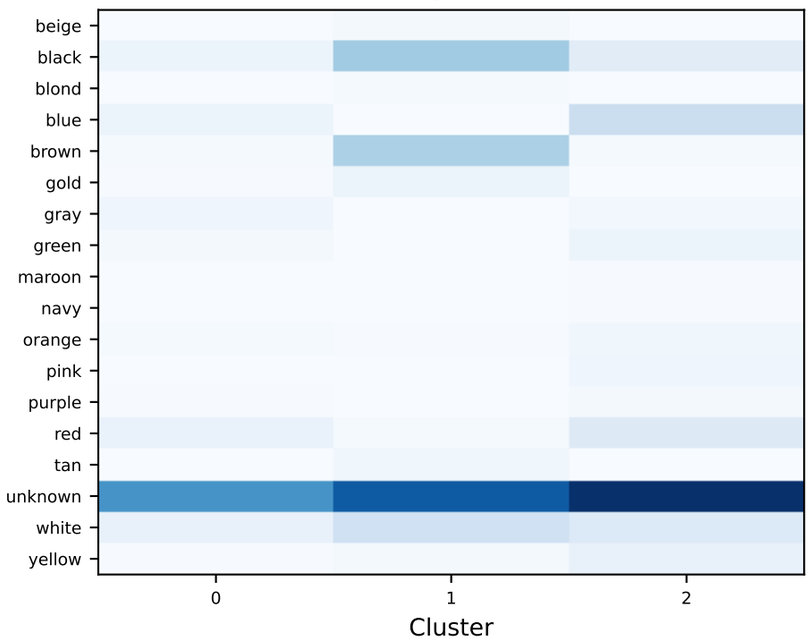
Cosine silhouette by cluster



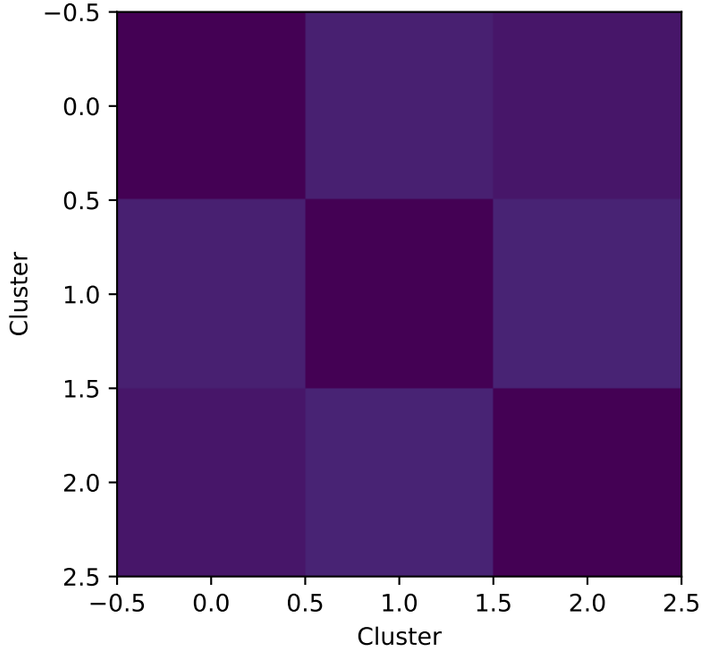
Cluster sizes



Source-color / cluster counts

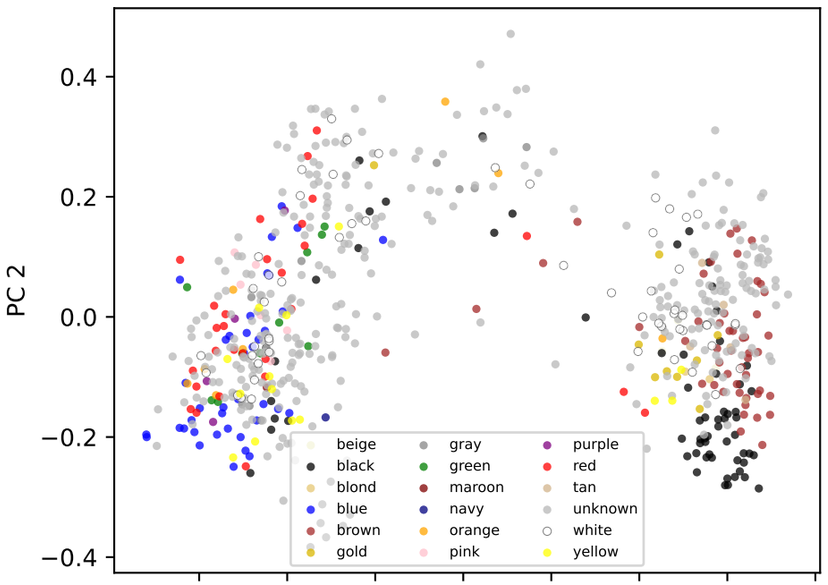


Centroid cosine distances

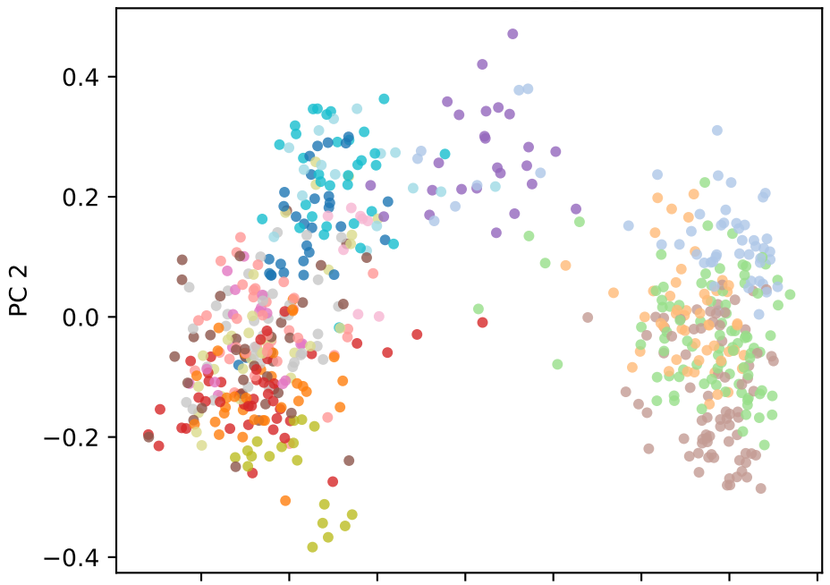


train_all · pretrained_conditioned · K=17, seed=2; PCA explains 18.2%

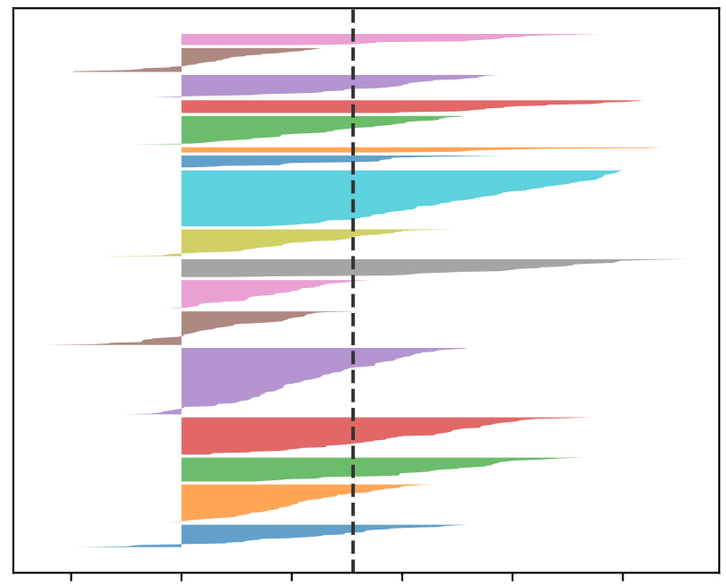
PCA: source color labels



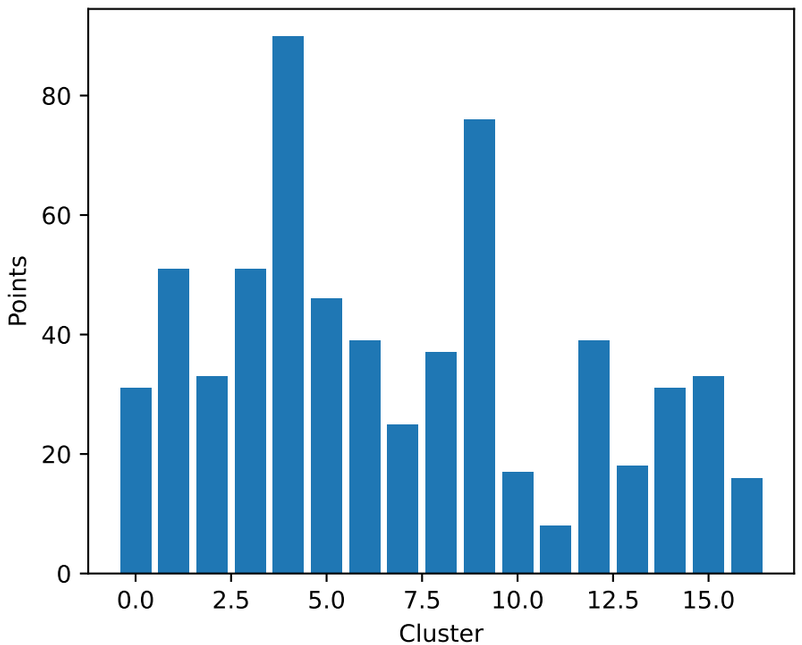
PCA: fitted cluster assignments



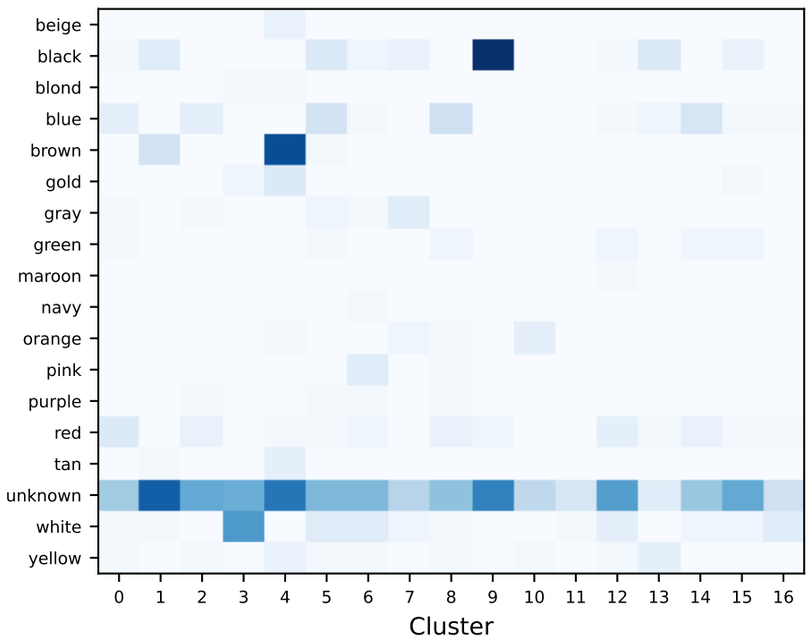
Cosine silhouette by cluster



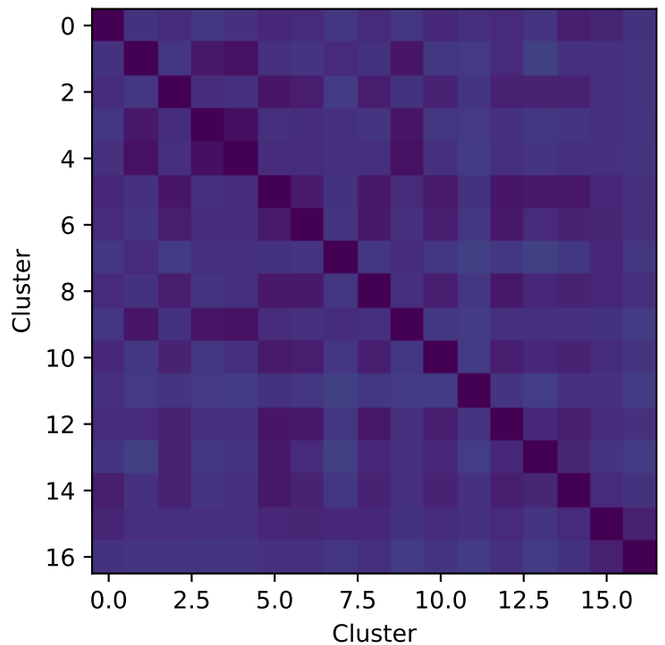
Cluster sizes

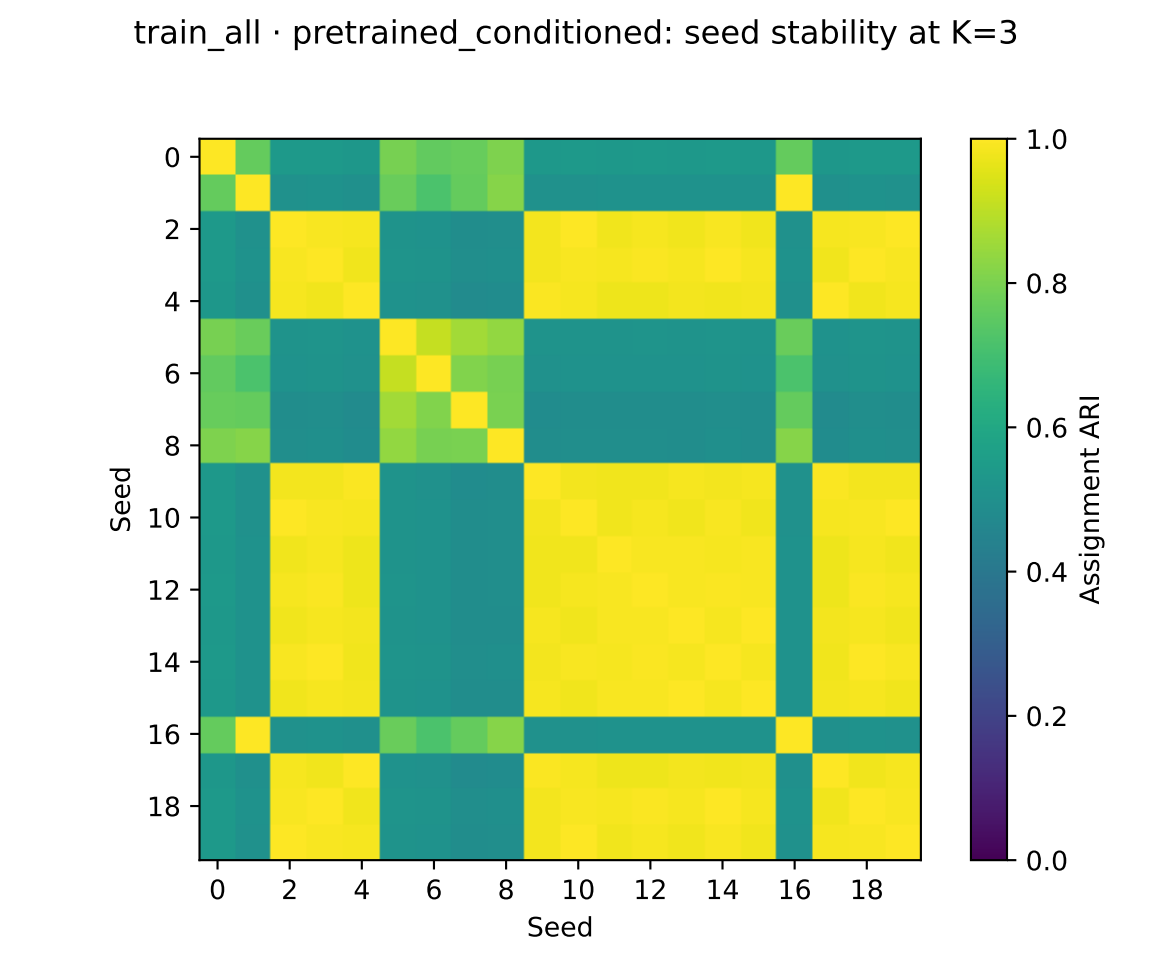


Source-color / cluster counts



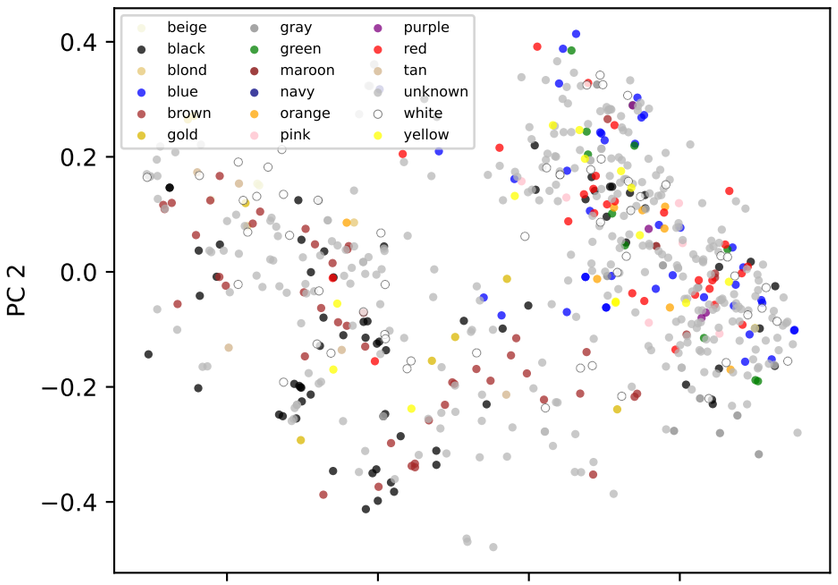
Centroid cosine distances



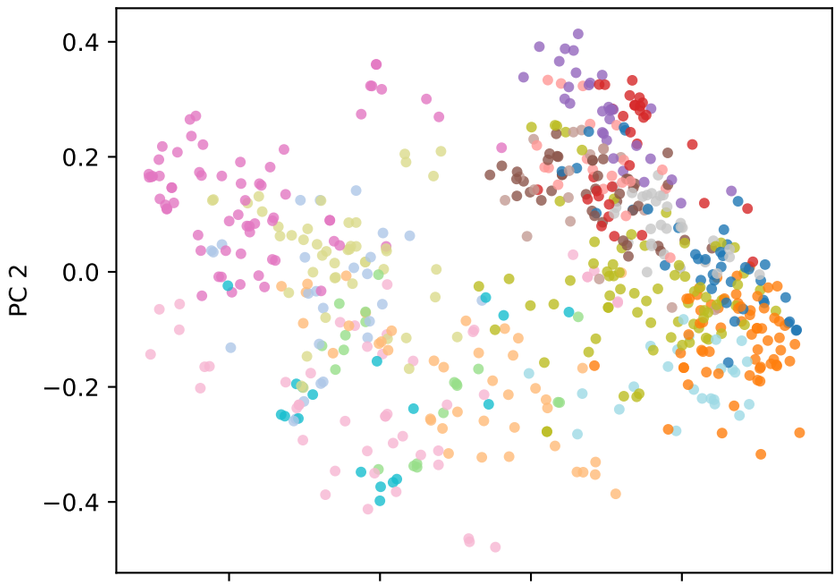

train_all · pretrained_conditioned: seed stability at K=3
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

train_all · finetuned_unconditioned · K=17, seed=10; PCA explains 10.9%

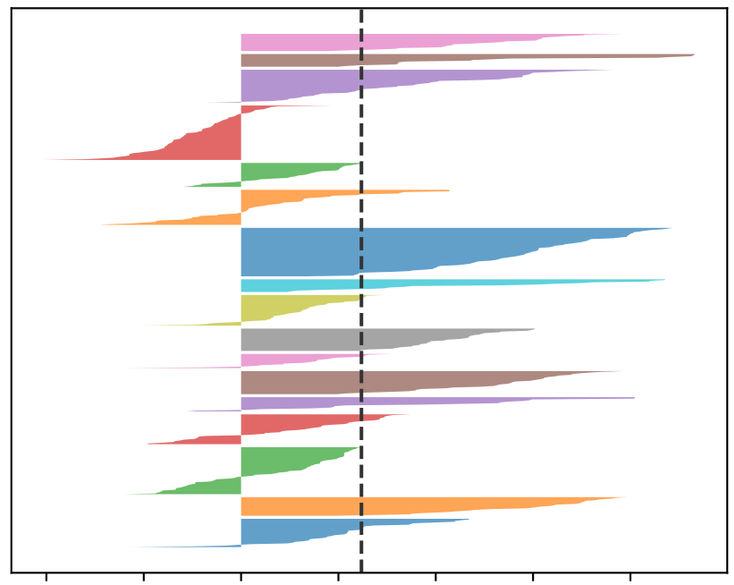
PCA: source color labels



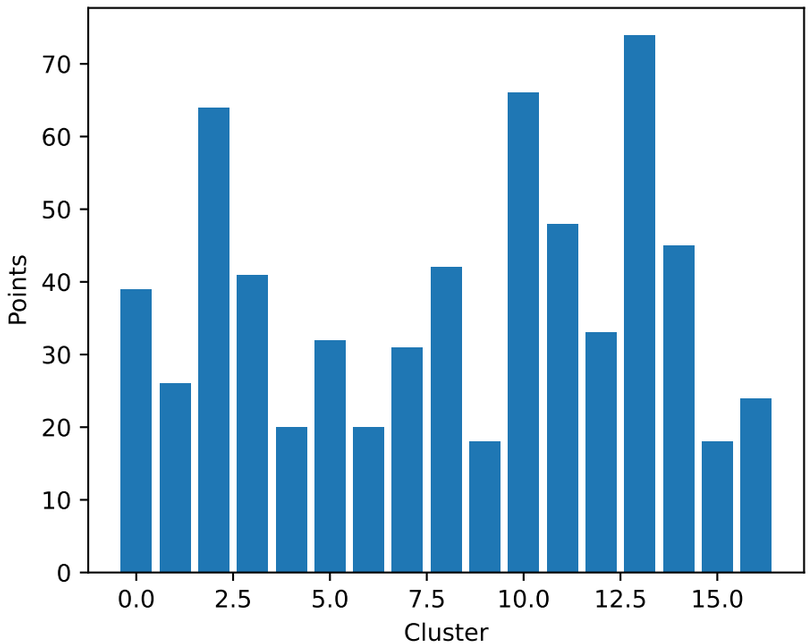
PCA: fitted cluster assignments



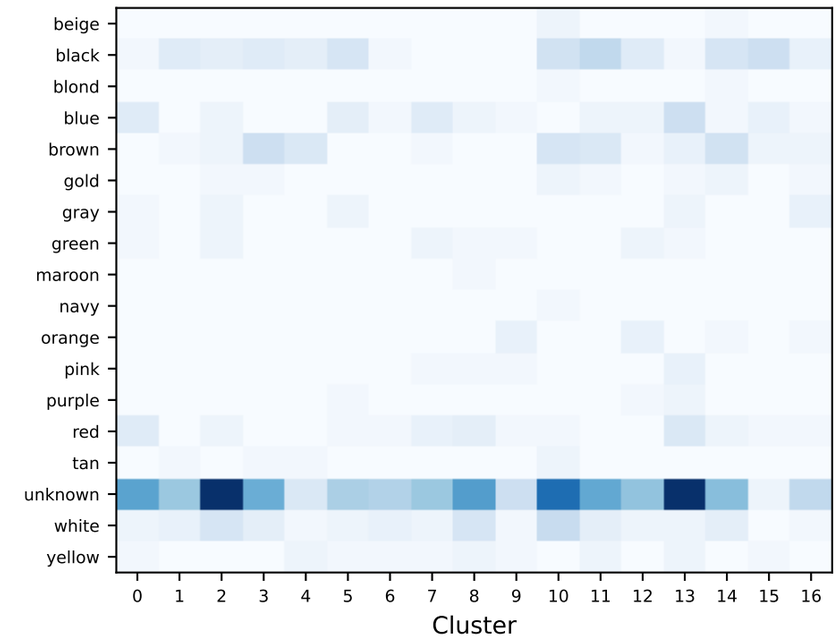
Cosine silhouette by cluster



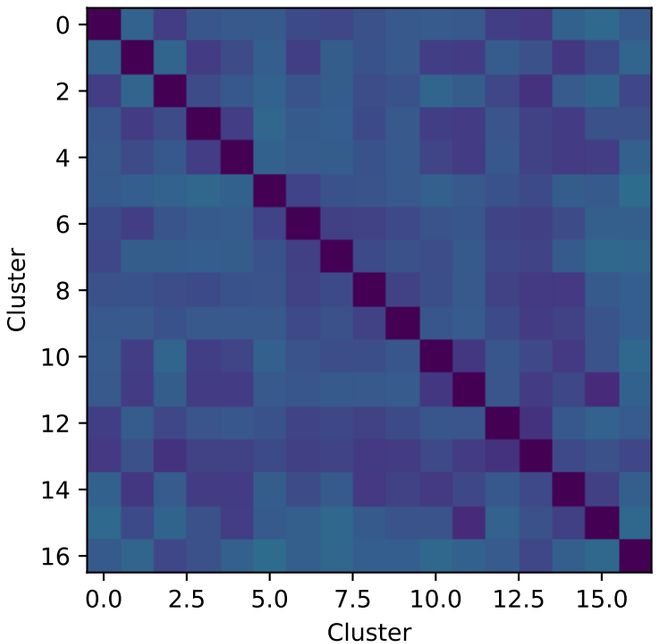
Cluster sizes



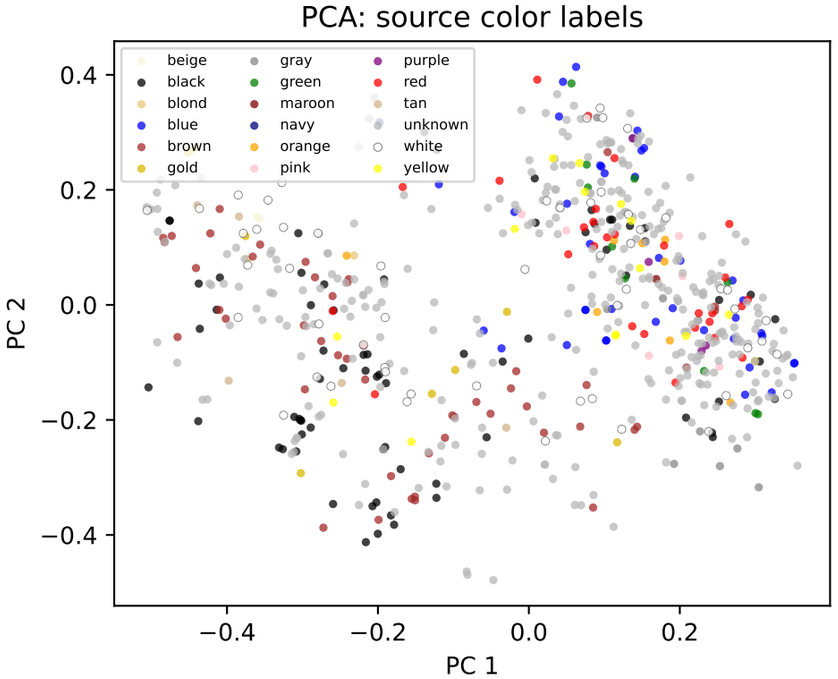
Source-color / cluster counts

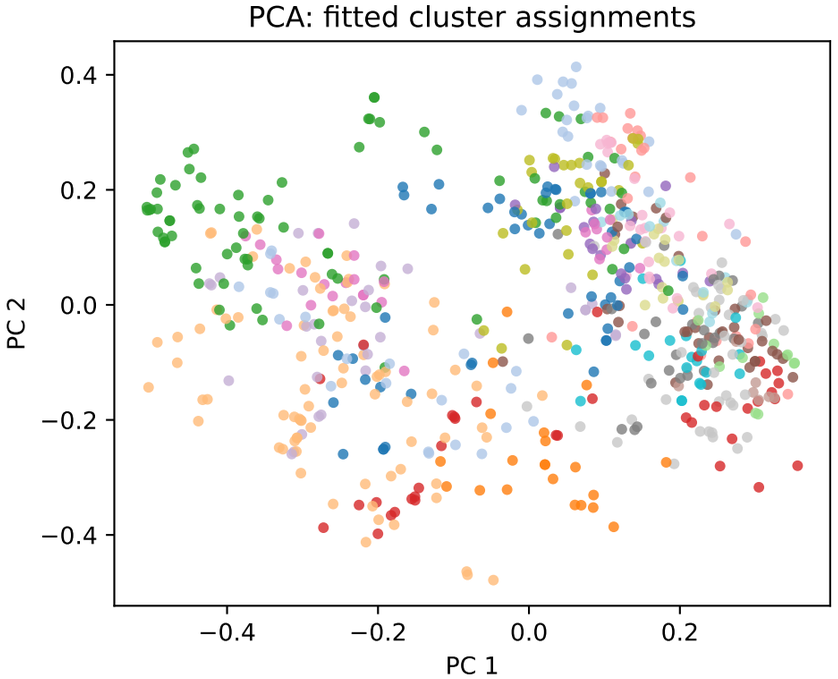


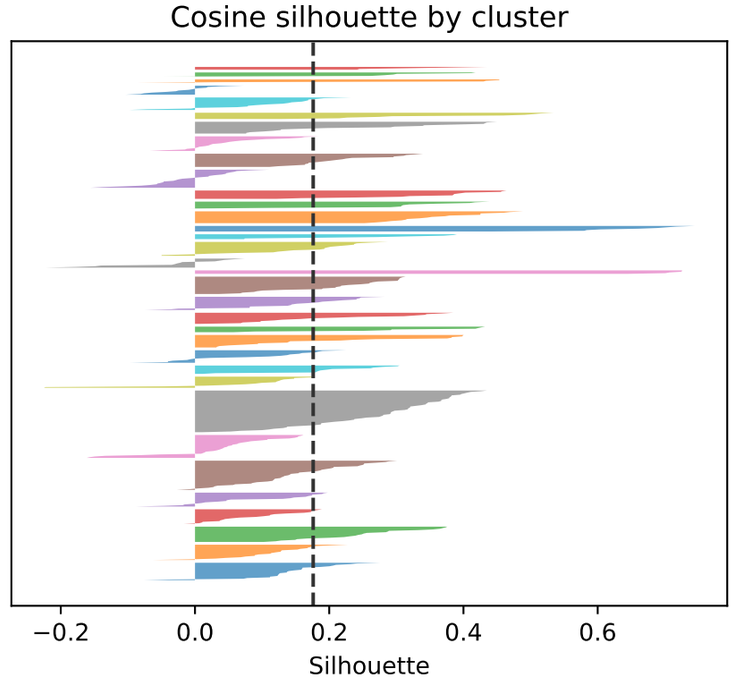
Centroid cosine distances

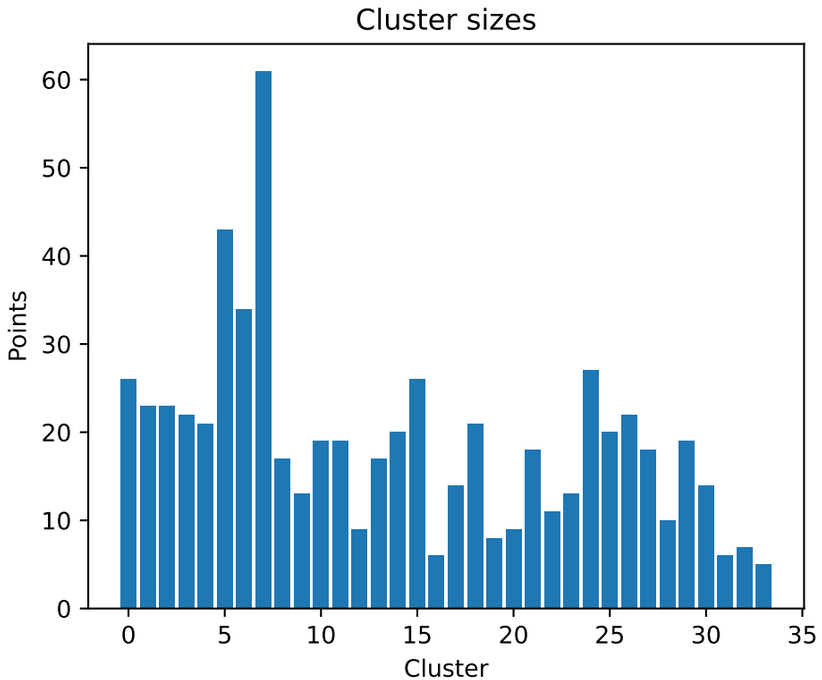


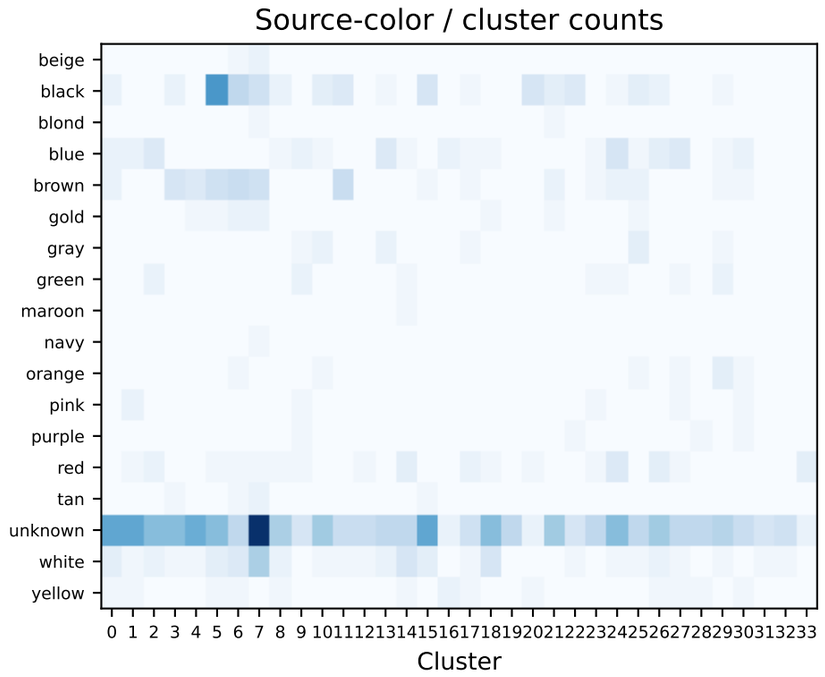
train_all · finetuned_unconditioned · K=34, seed=6; PCA explains 10.9%

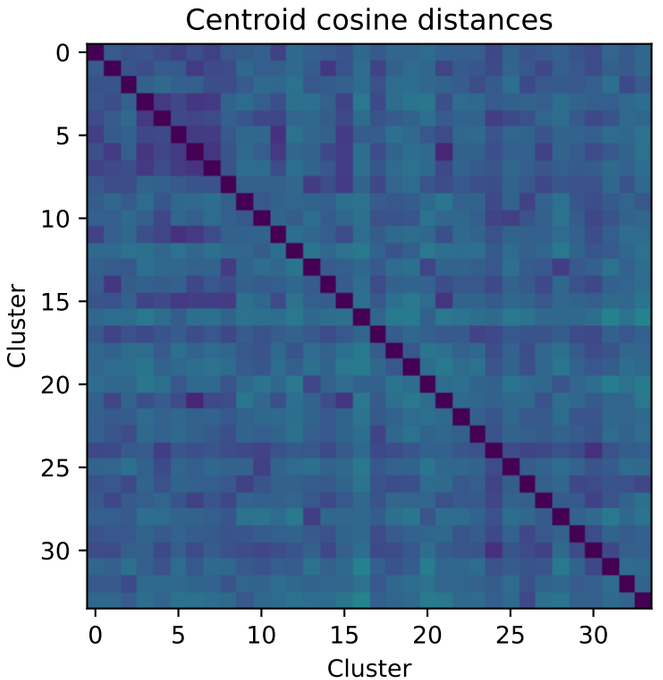


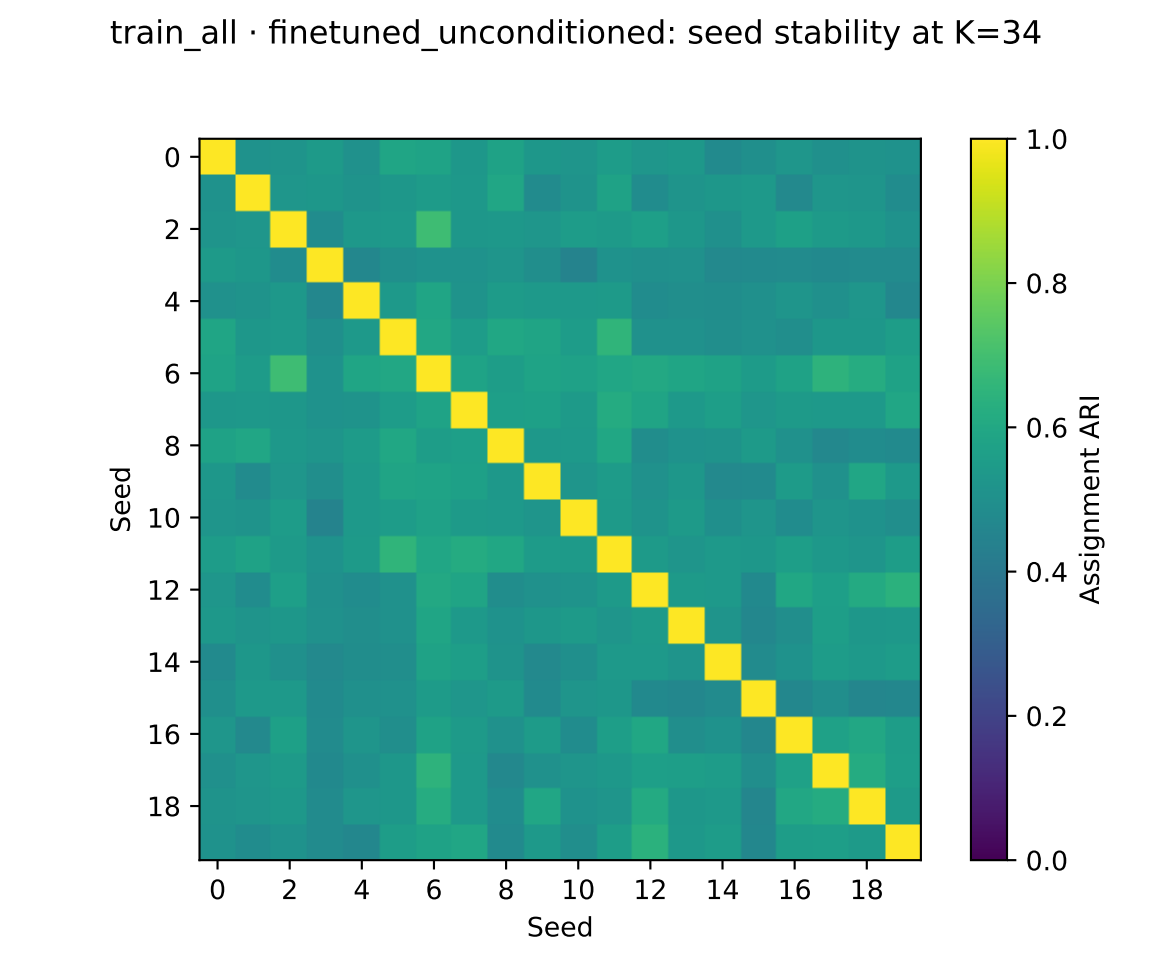


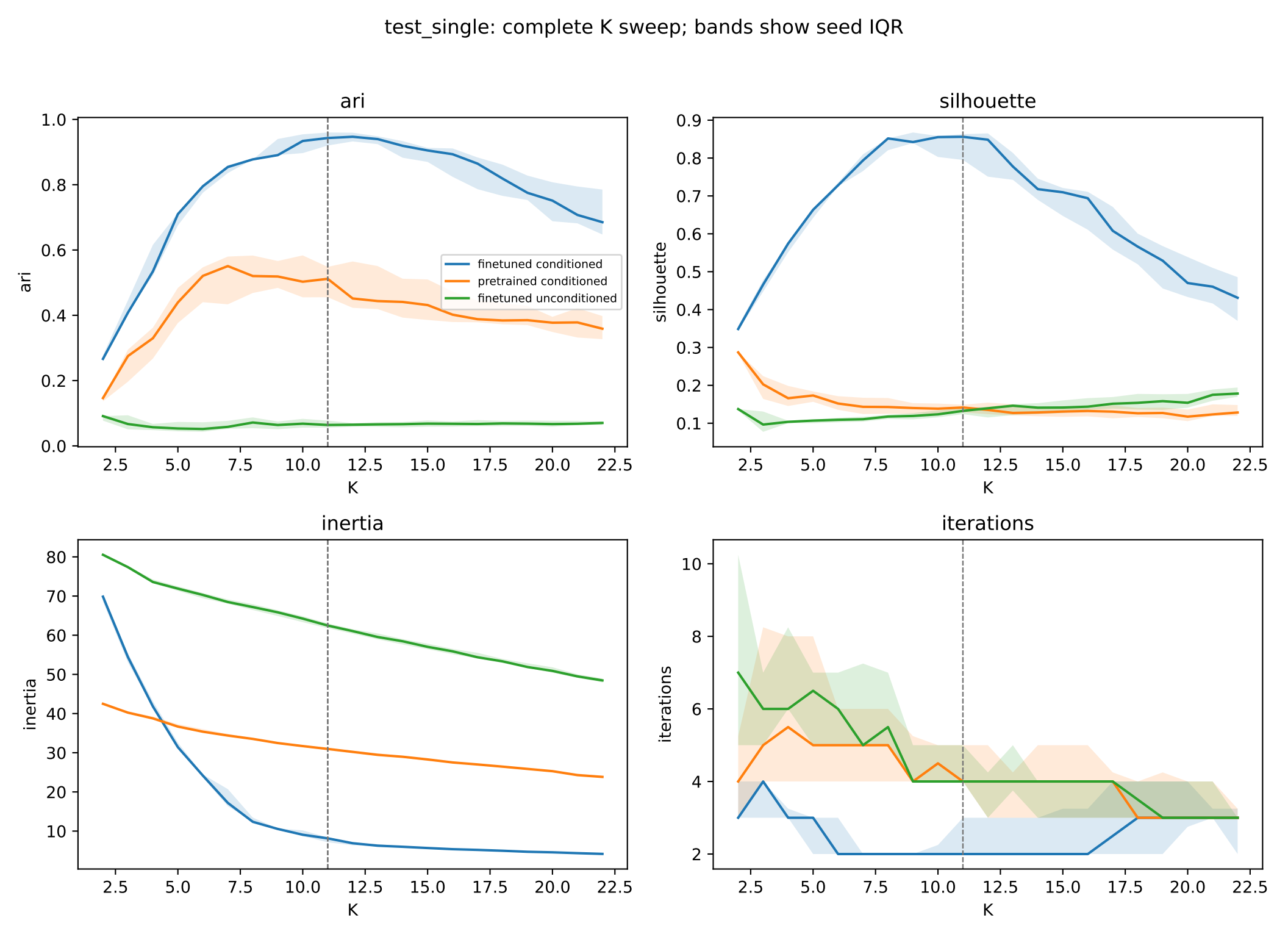




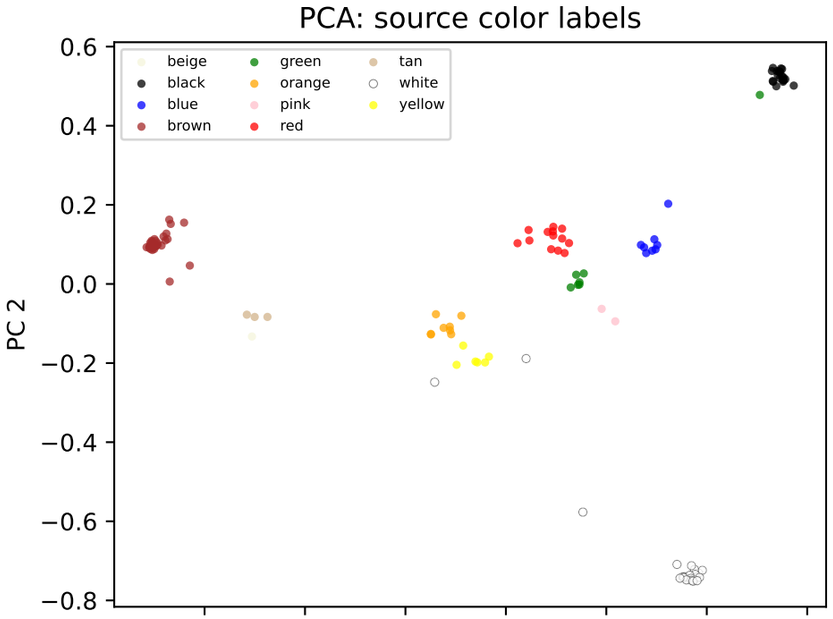


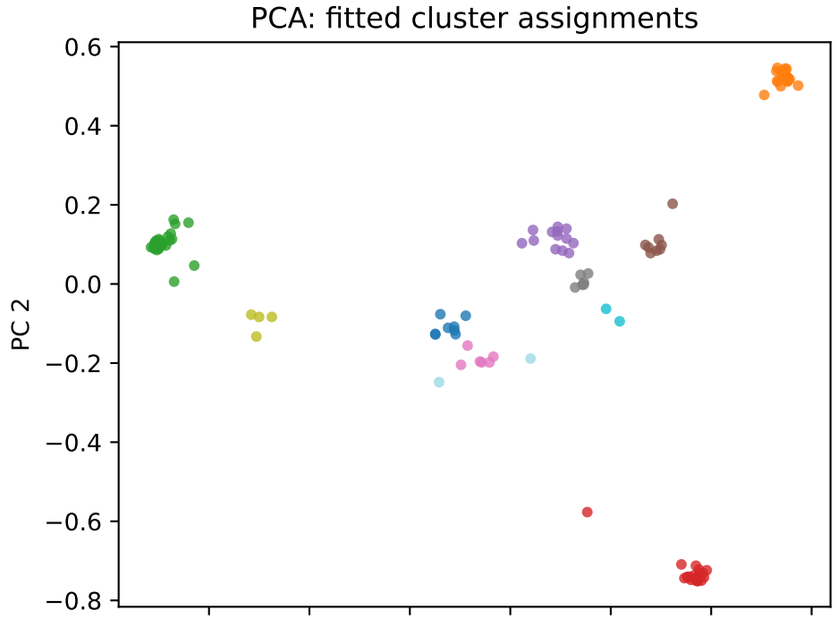


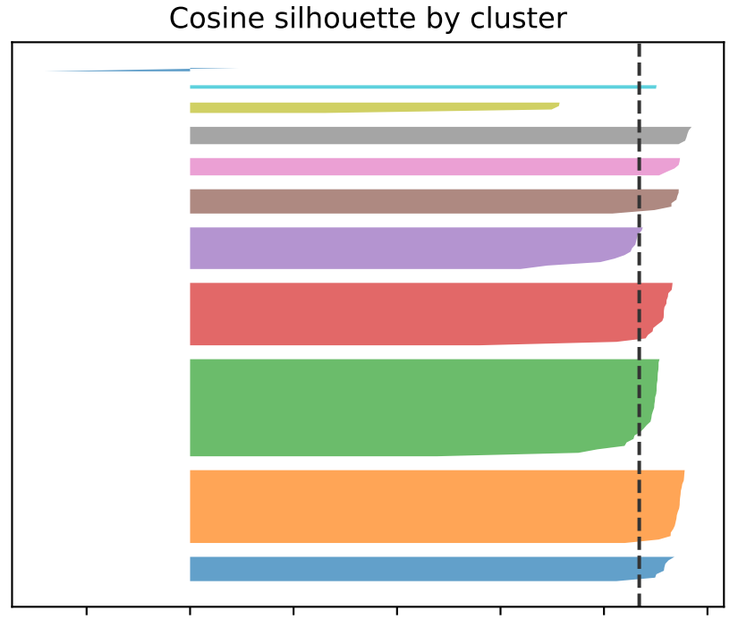

train_all · finetuned_unconditioned: seed stability at K=34
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

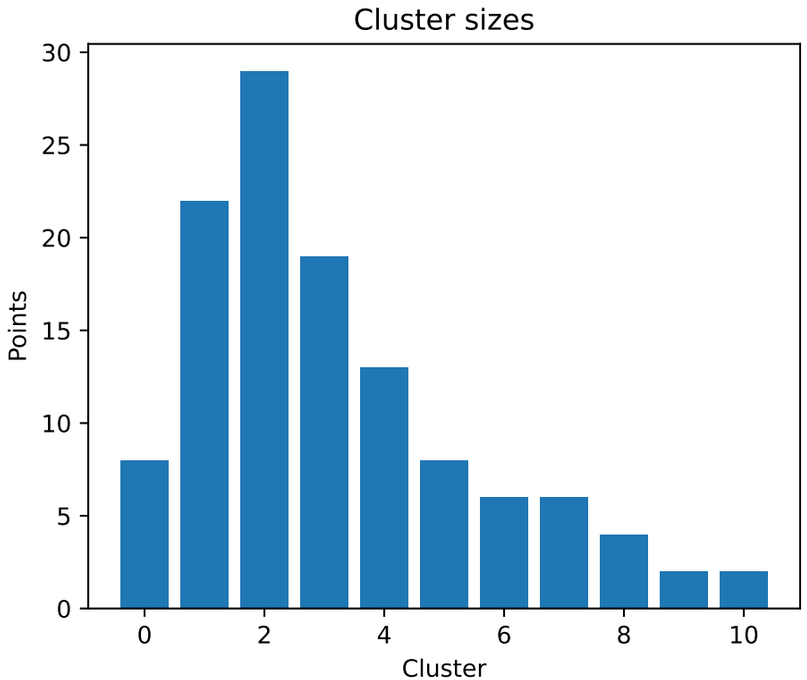

test_single: complete K sweep; bands show seed IQR
ari
silhouette
inertia
iterations
finetuned conditioned
pretrained conditioned
finetuned unconditioned
K

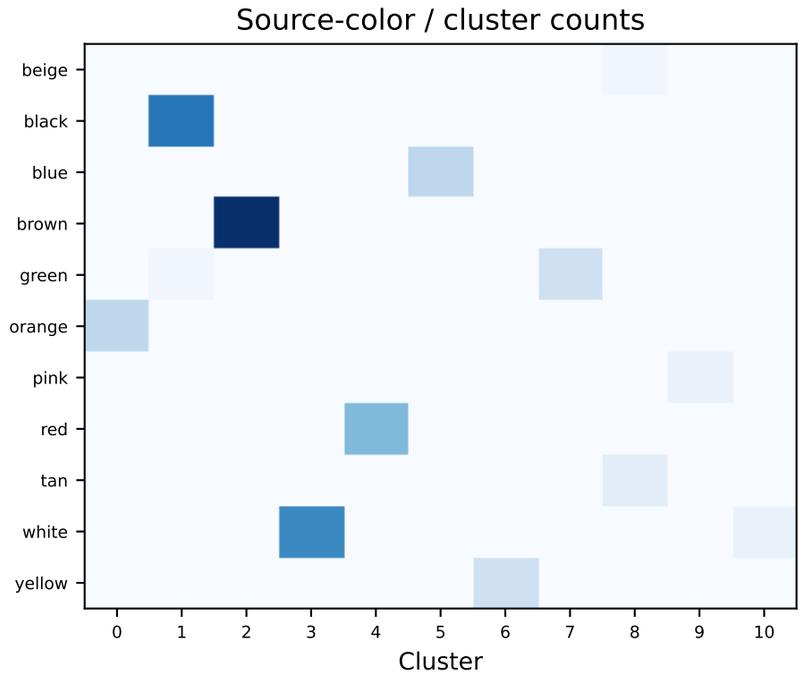
test_single · finetuned_conditioned · K=11, seed=0; PCA explains 47.1%

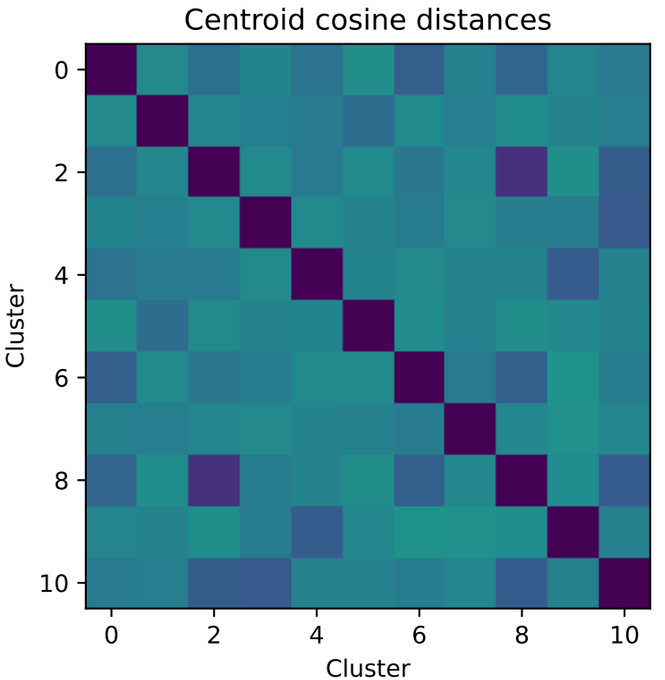


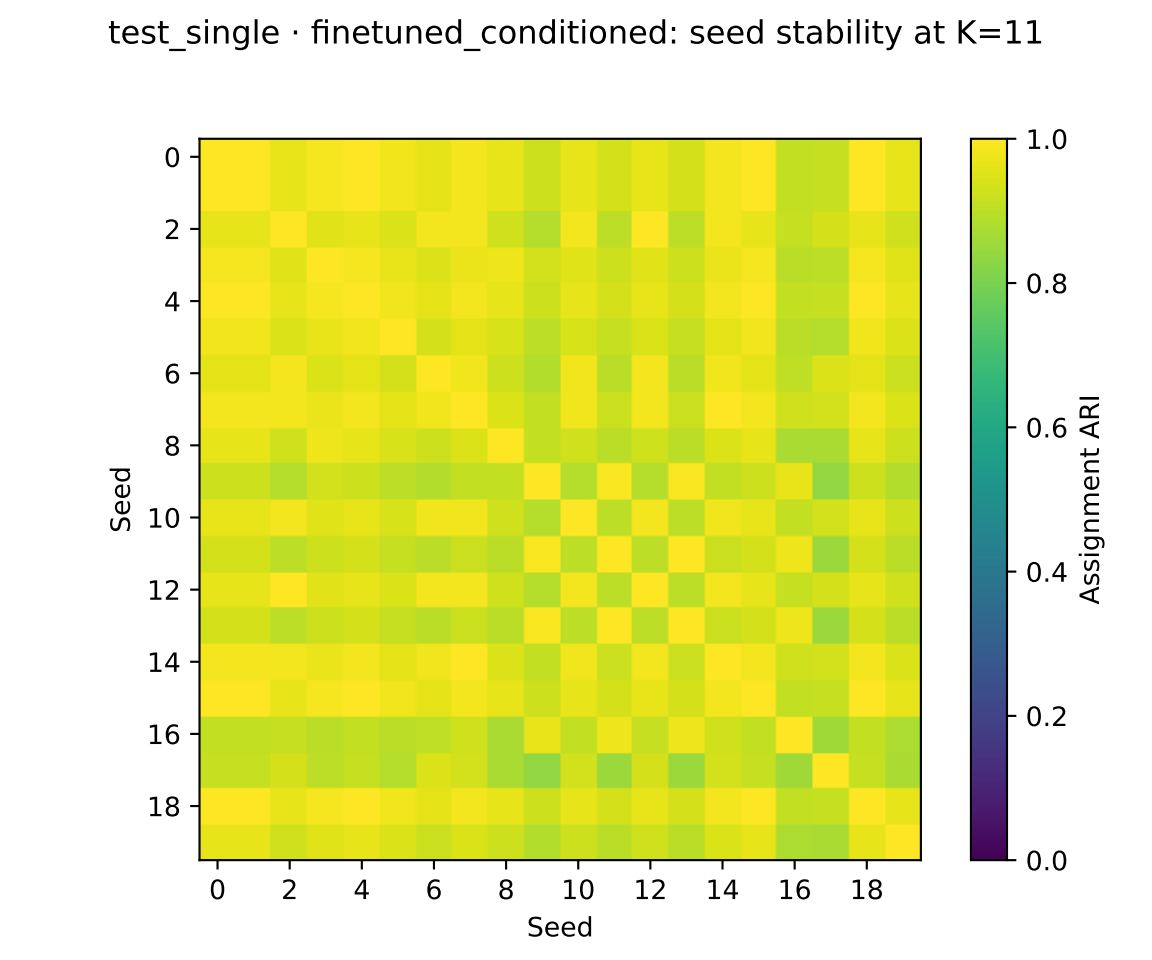




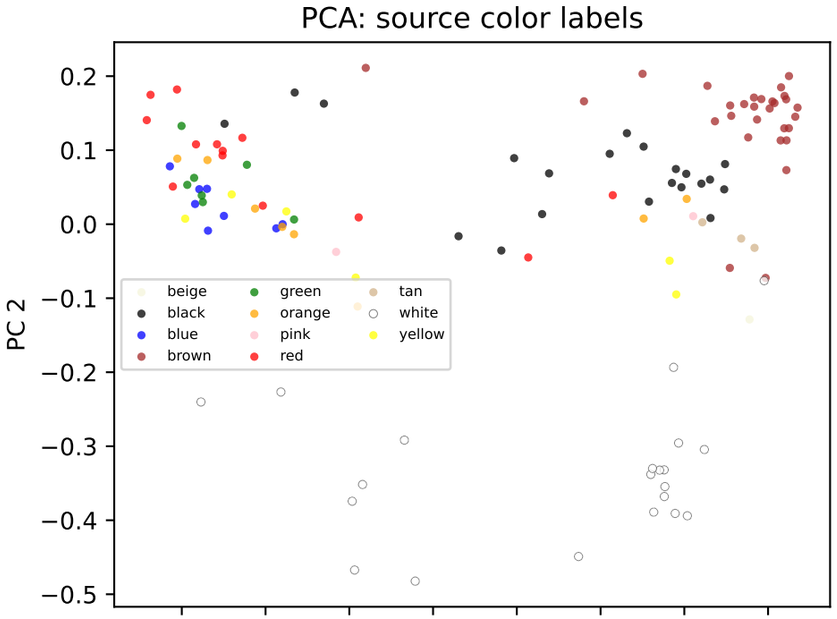


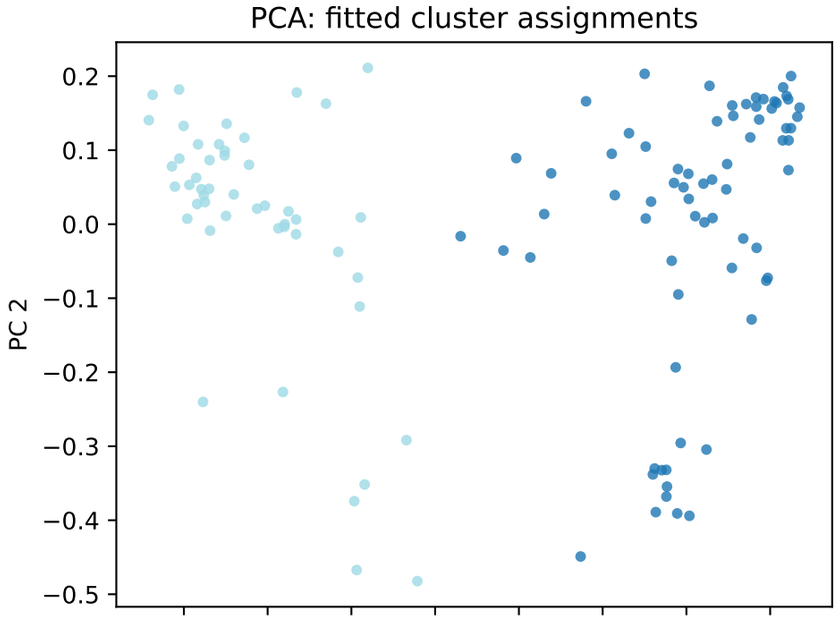


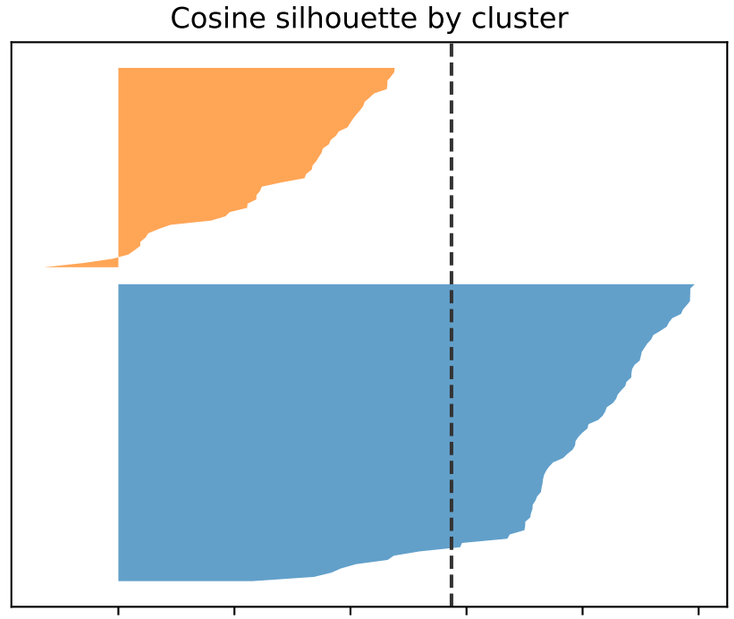


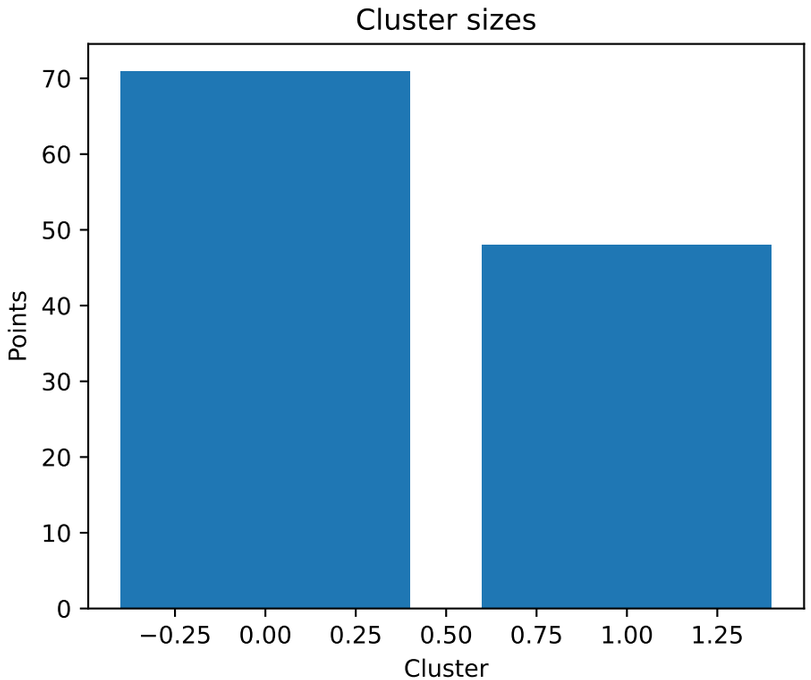

test_single · finetuned_conditioned: seed stability at K=11
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

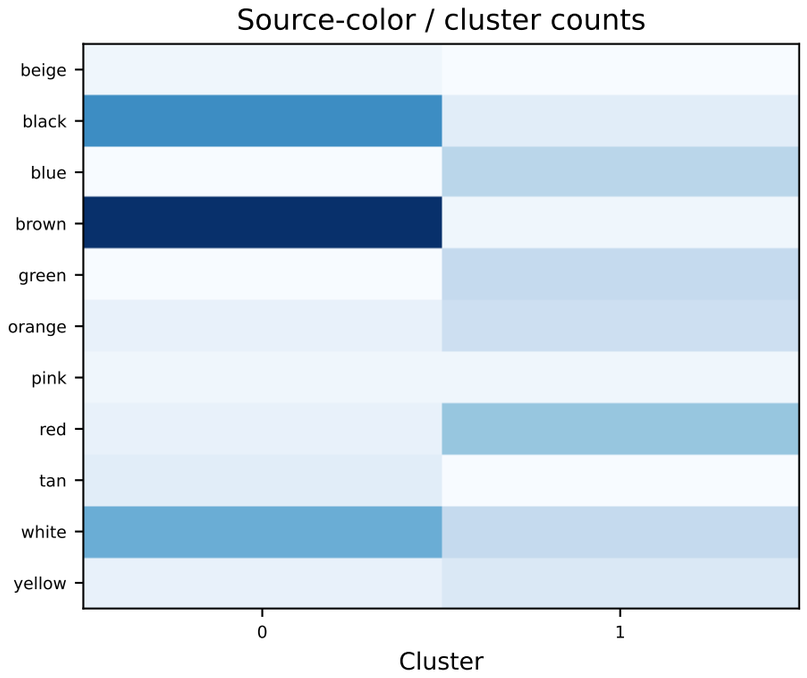
test_single · pretrained_conditioned · K=2, seed=0; PCA explains 24.7%

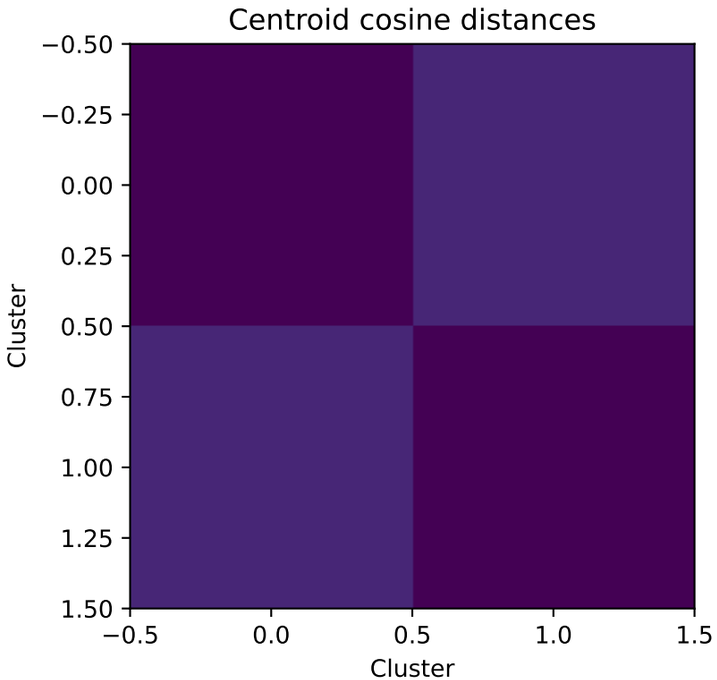




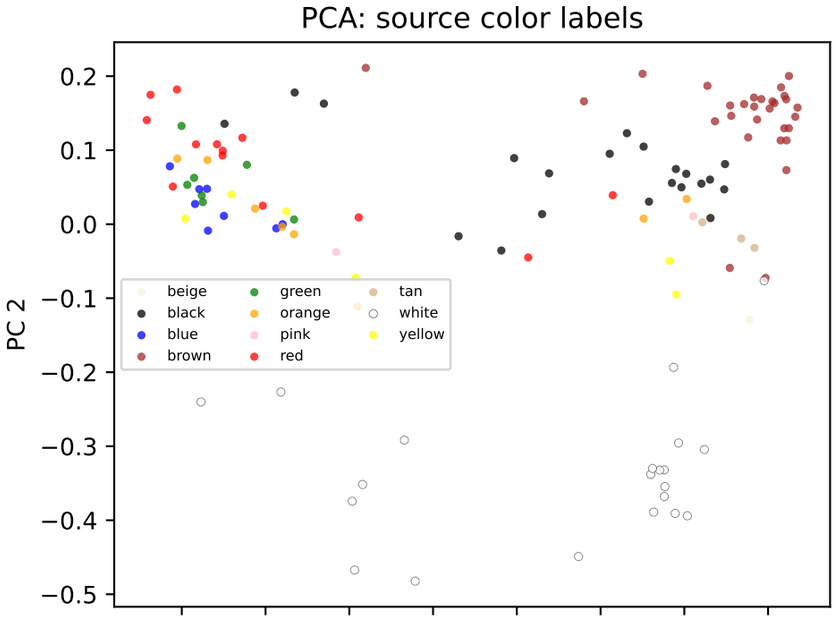


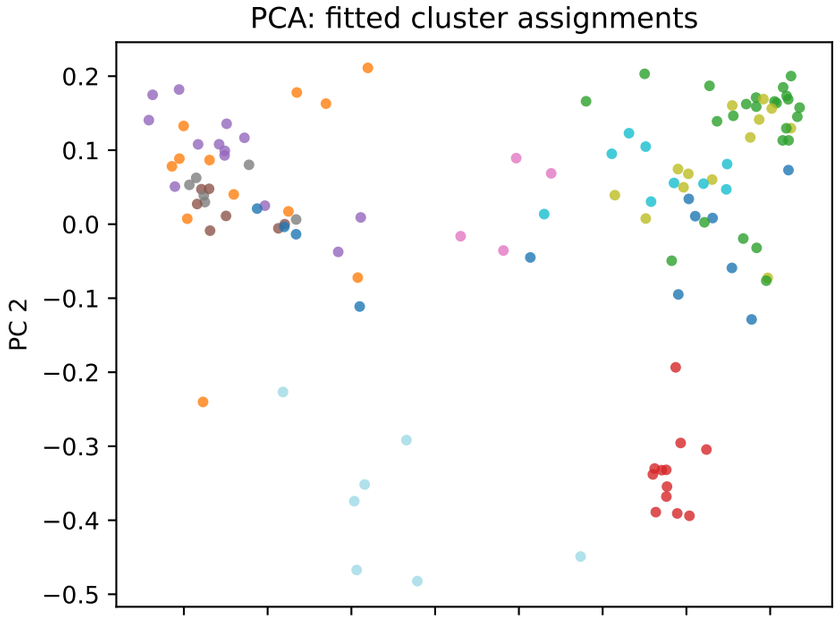


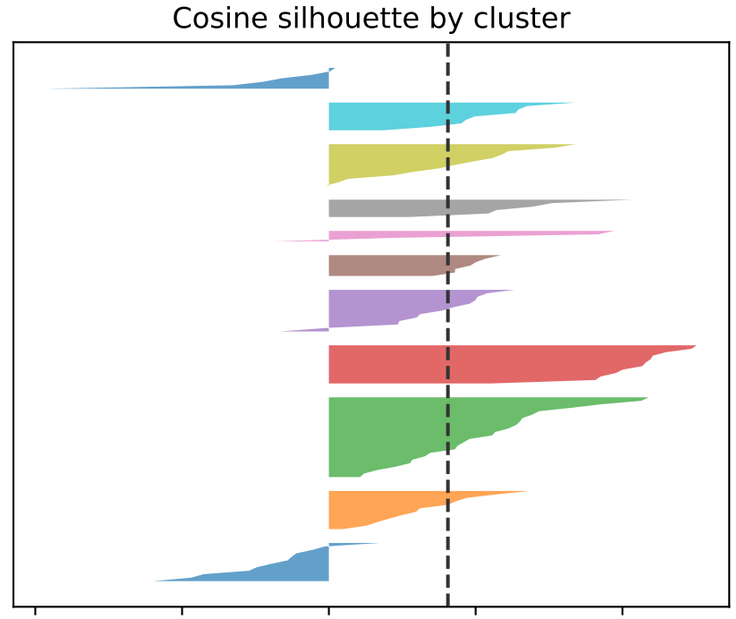


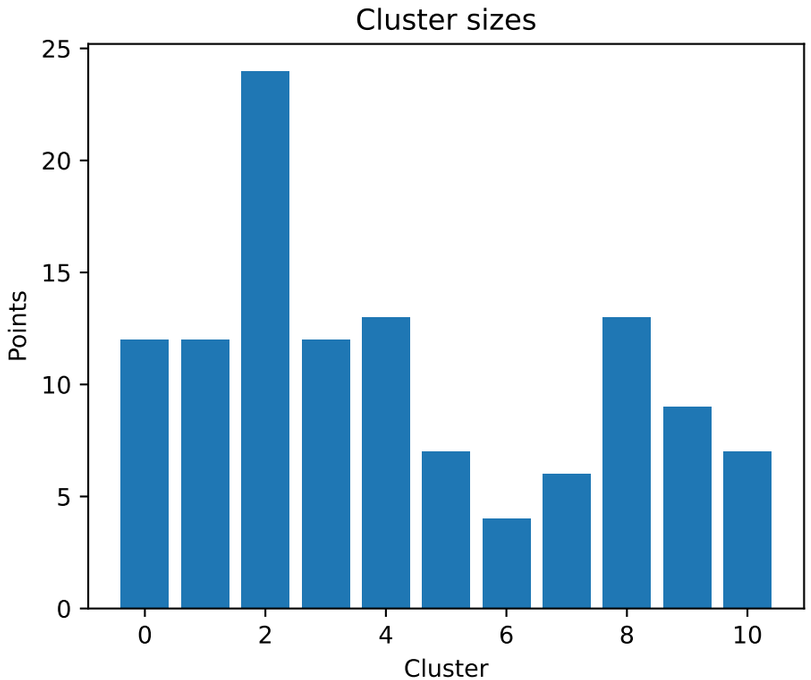


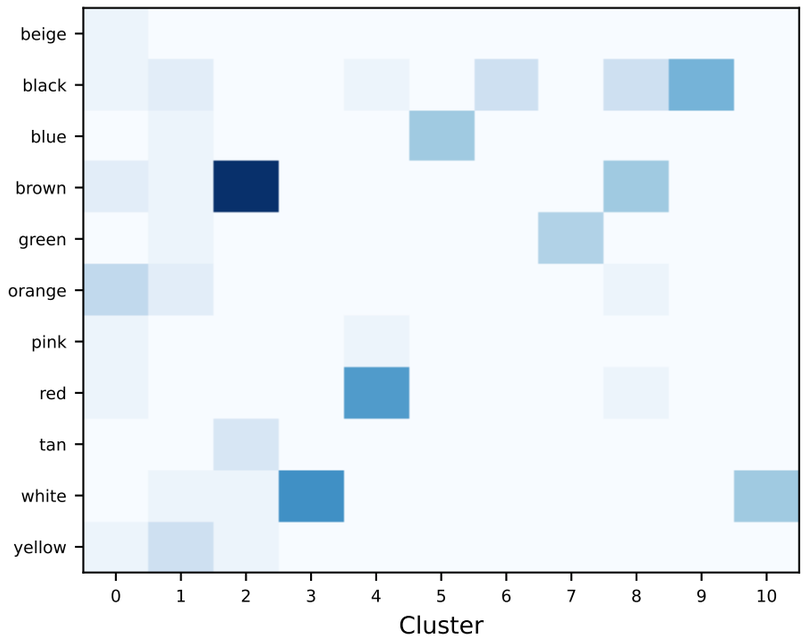
test_single · pretrained_conditioned · K=11, seed=0; PCA explains 24.7%

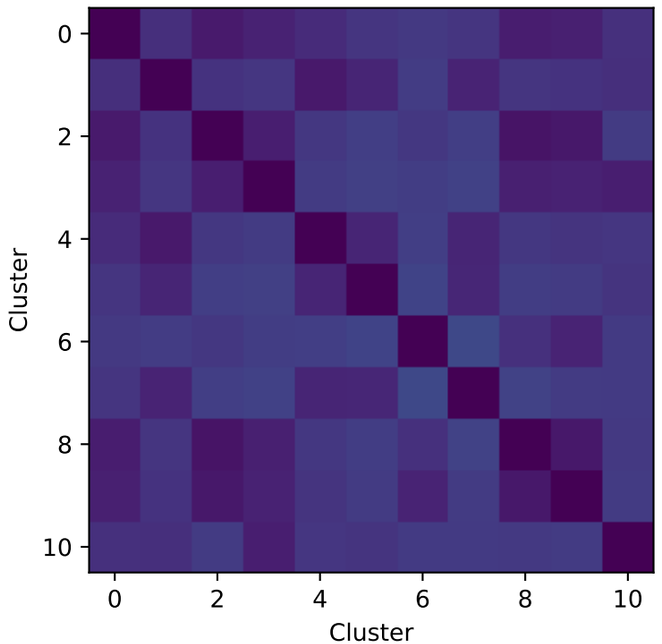


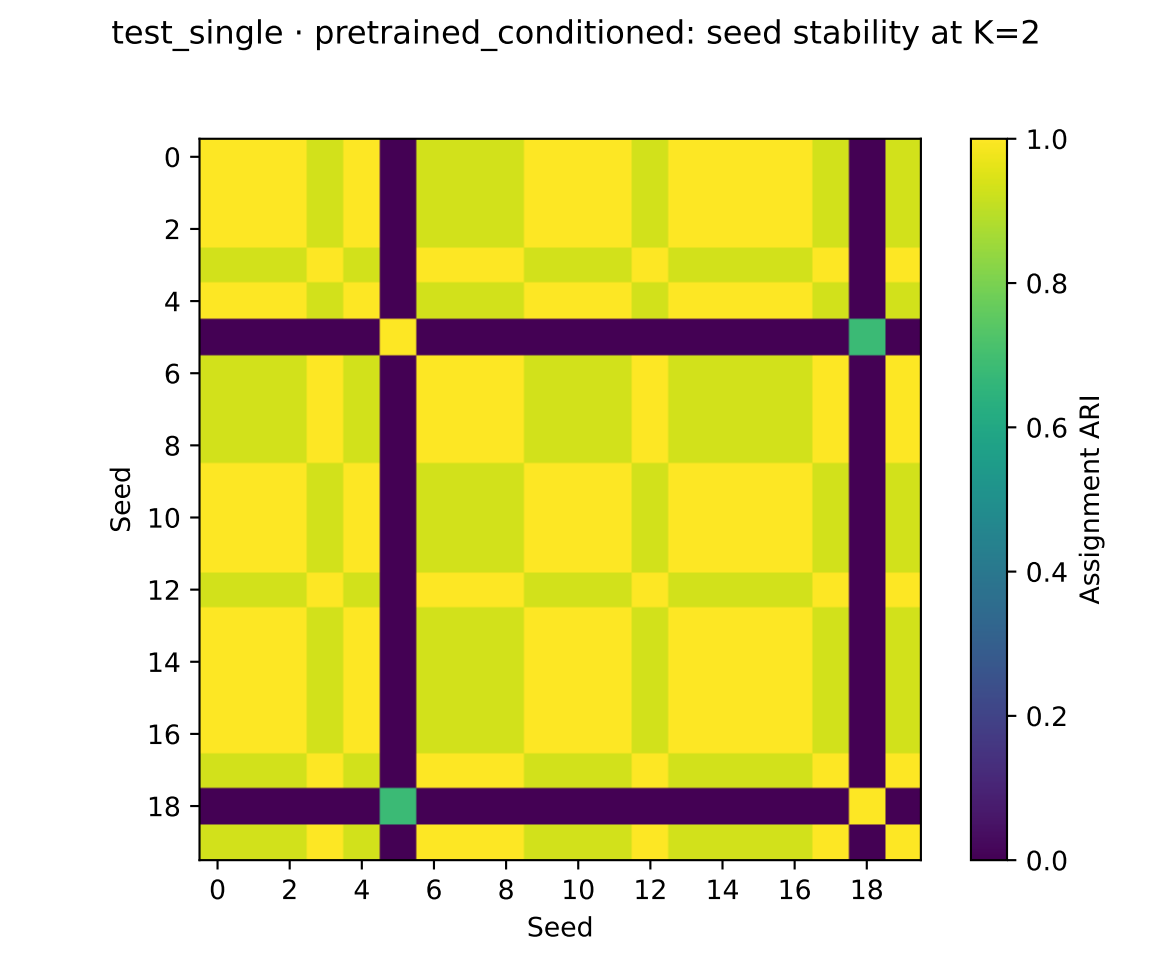






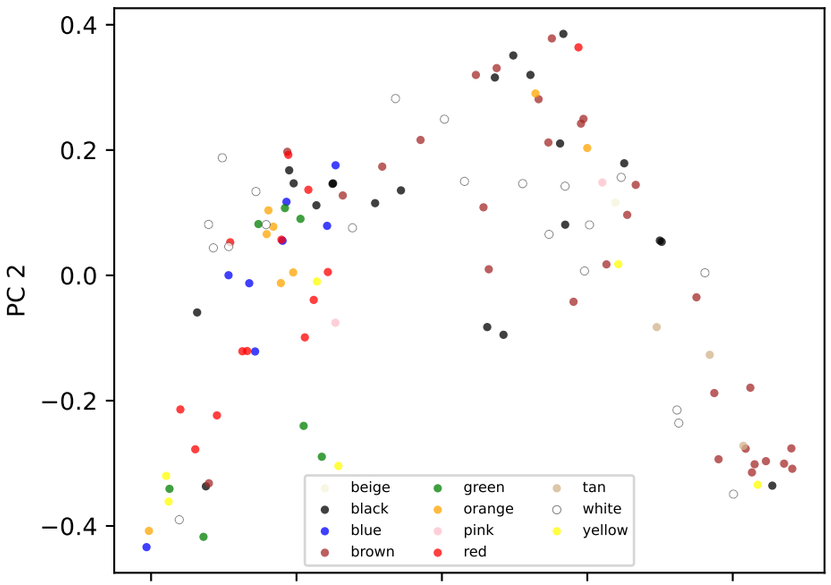




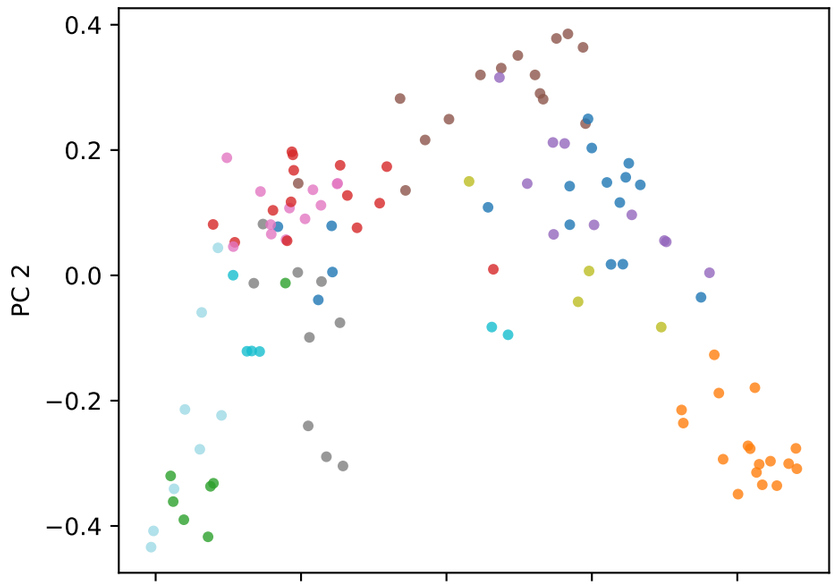

test_single · pretrained_conditioned: seed stability at K=2
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

test_single · finetuned_unconditioned · K=11, seed=18; PCA explains 15.5%

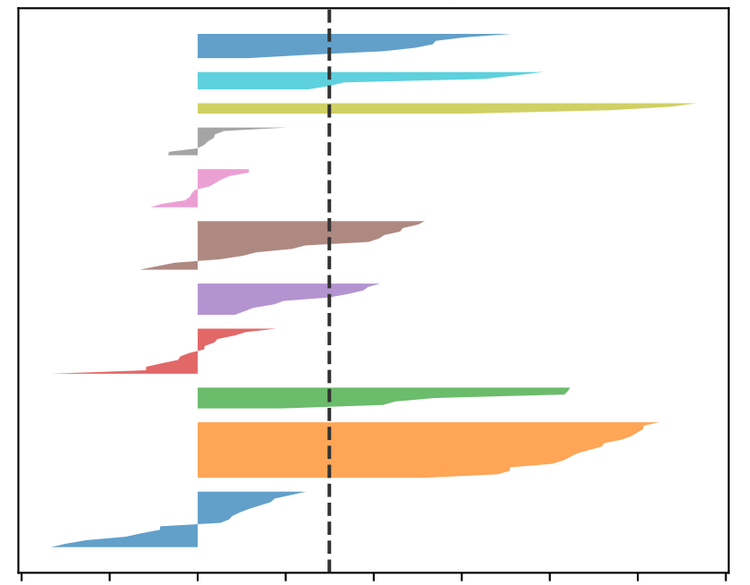
PCA: source color labels



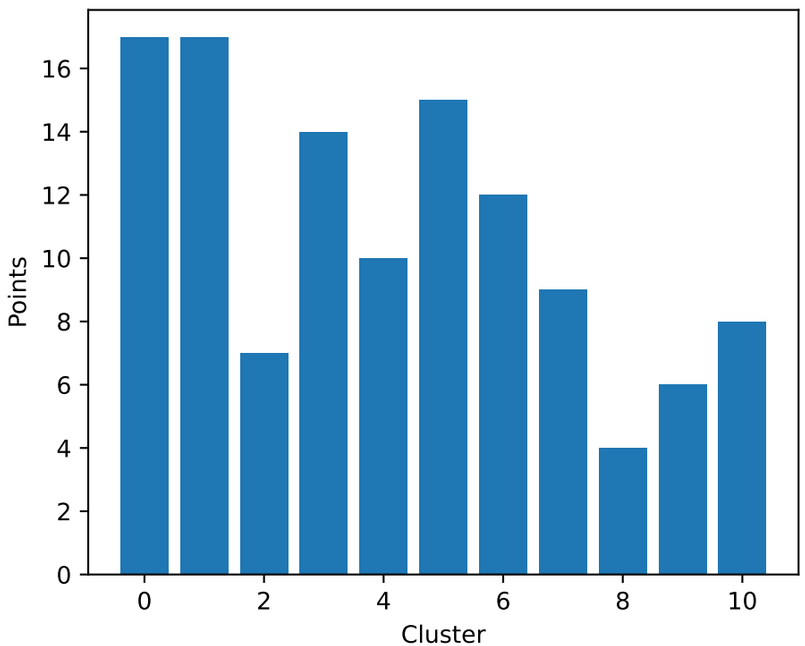
PCA: fitted cluster assignments



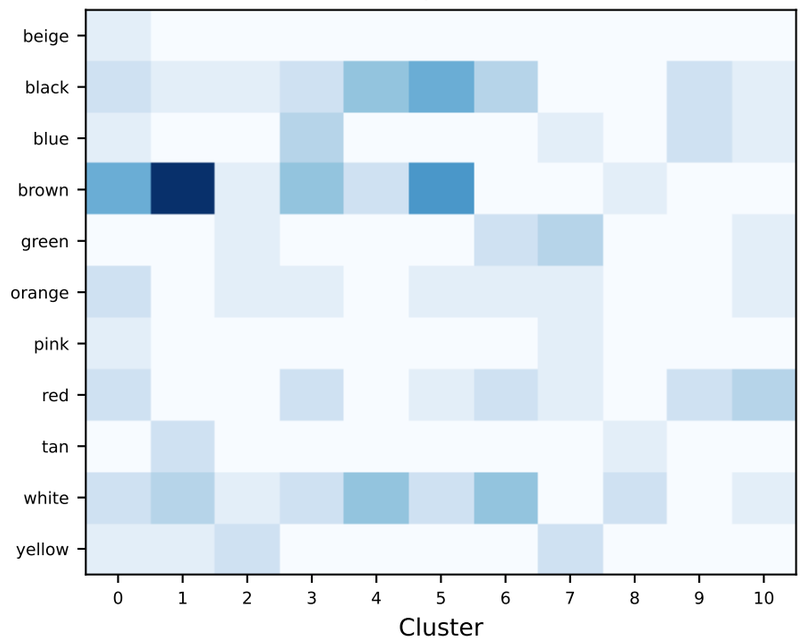
Cosine silhouette by cluster



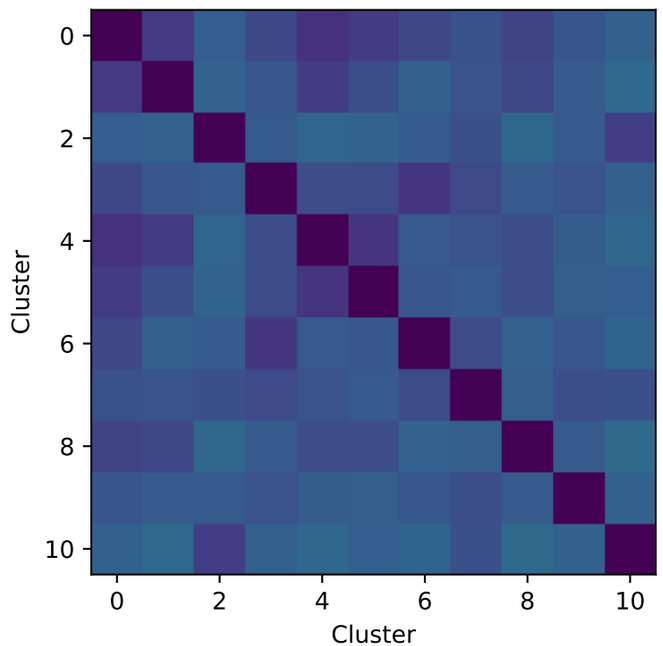
Cluster sizes



Source-color / cluster counts

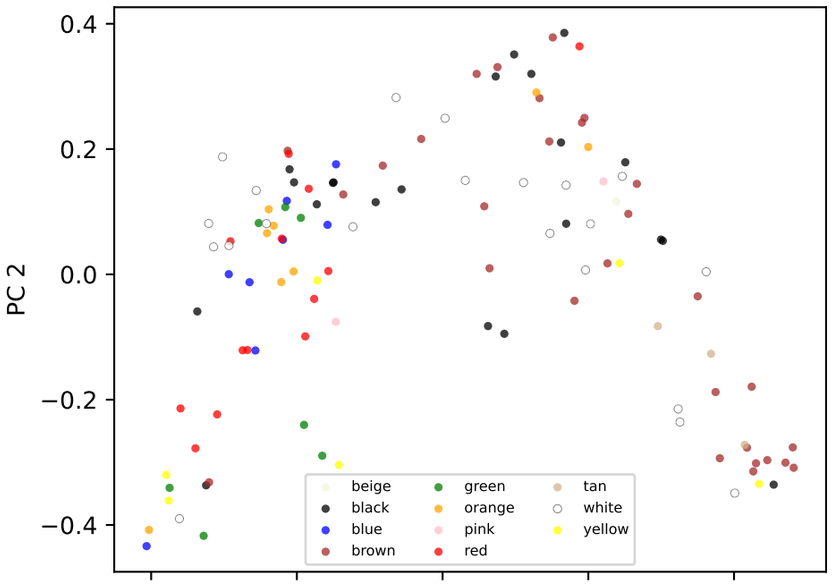


Centroid cosine distances

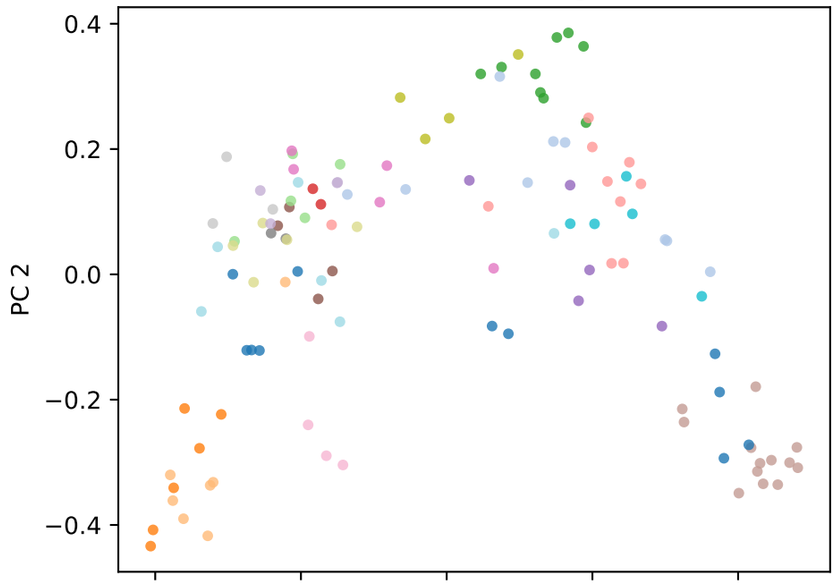


test_single · finetuned_unconditioned · K=22, seed=11; PCA explains 15.5%

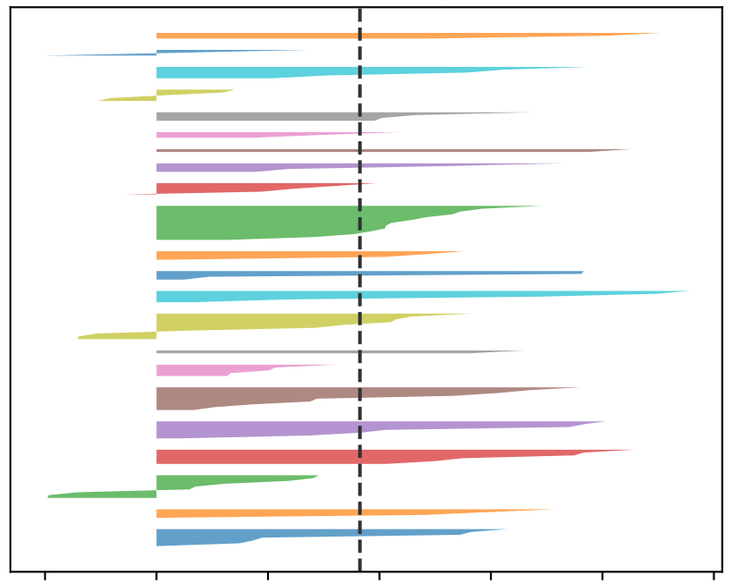
PCA: source color labels



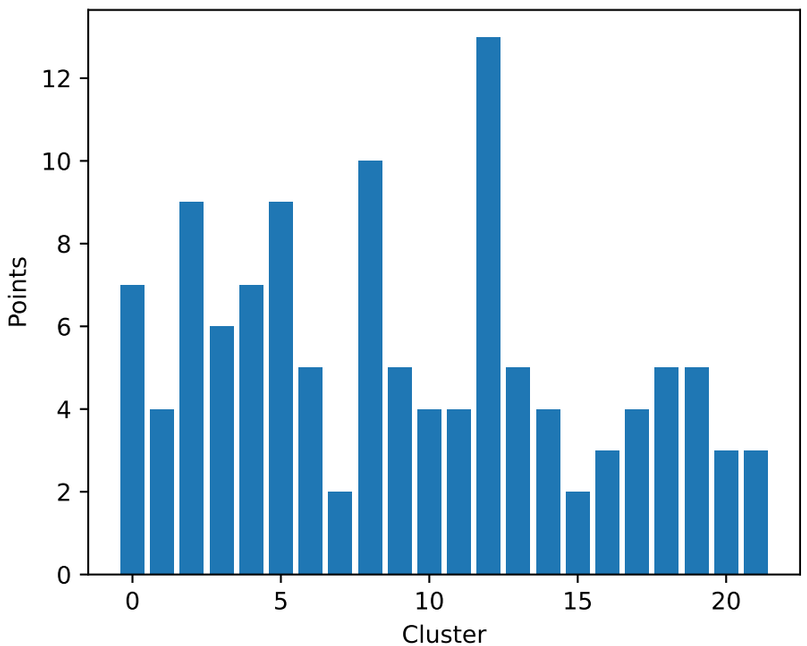
PCA: fitted cluster assignments



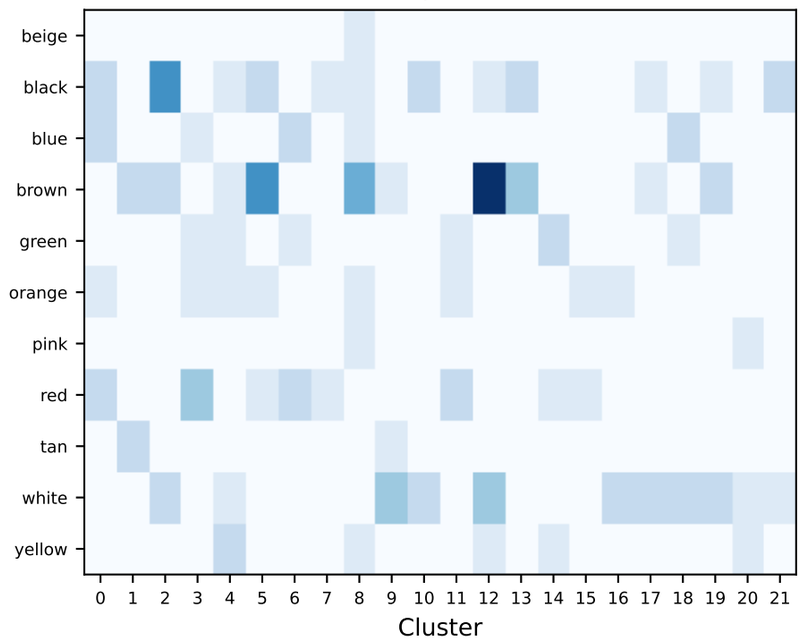
Cosine silhouette by cluster



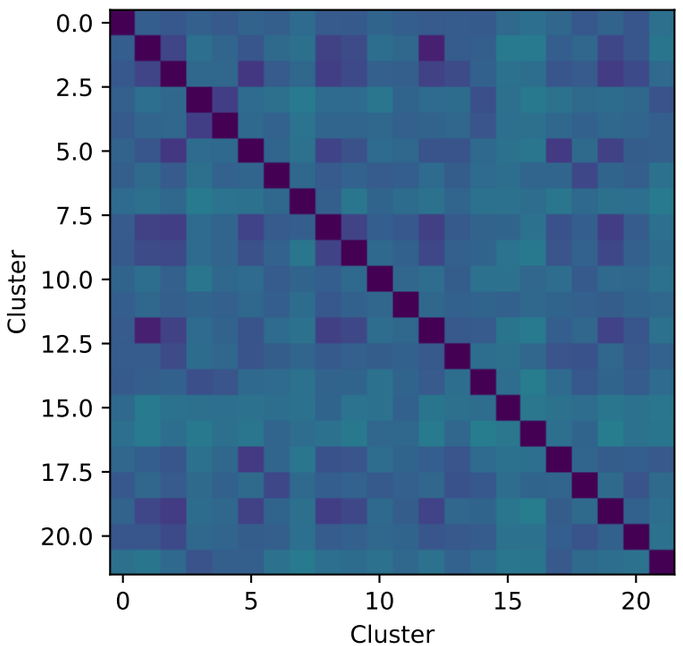
Cluster sizes

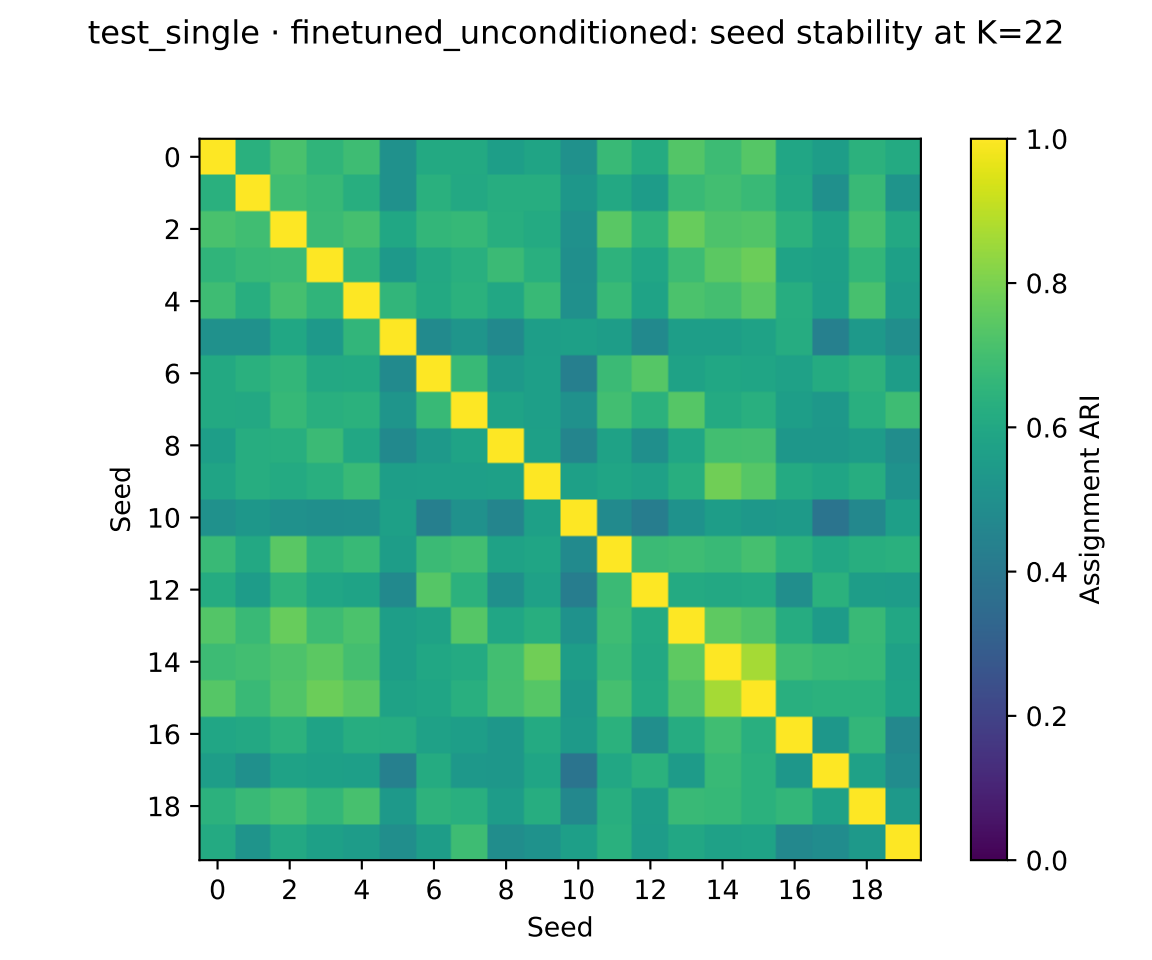


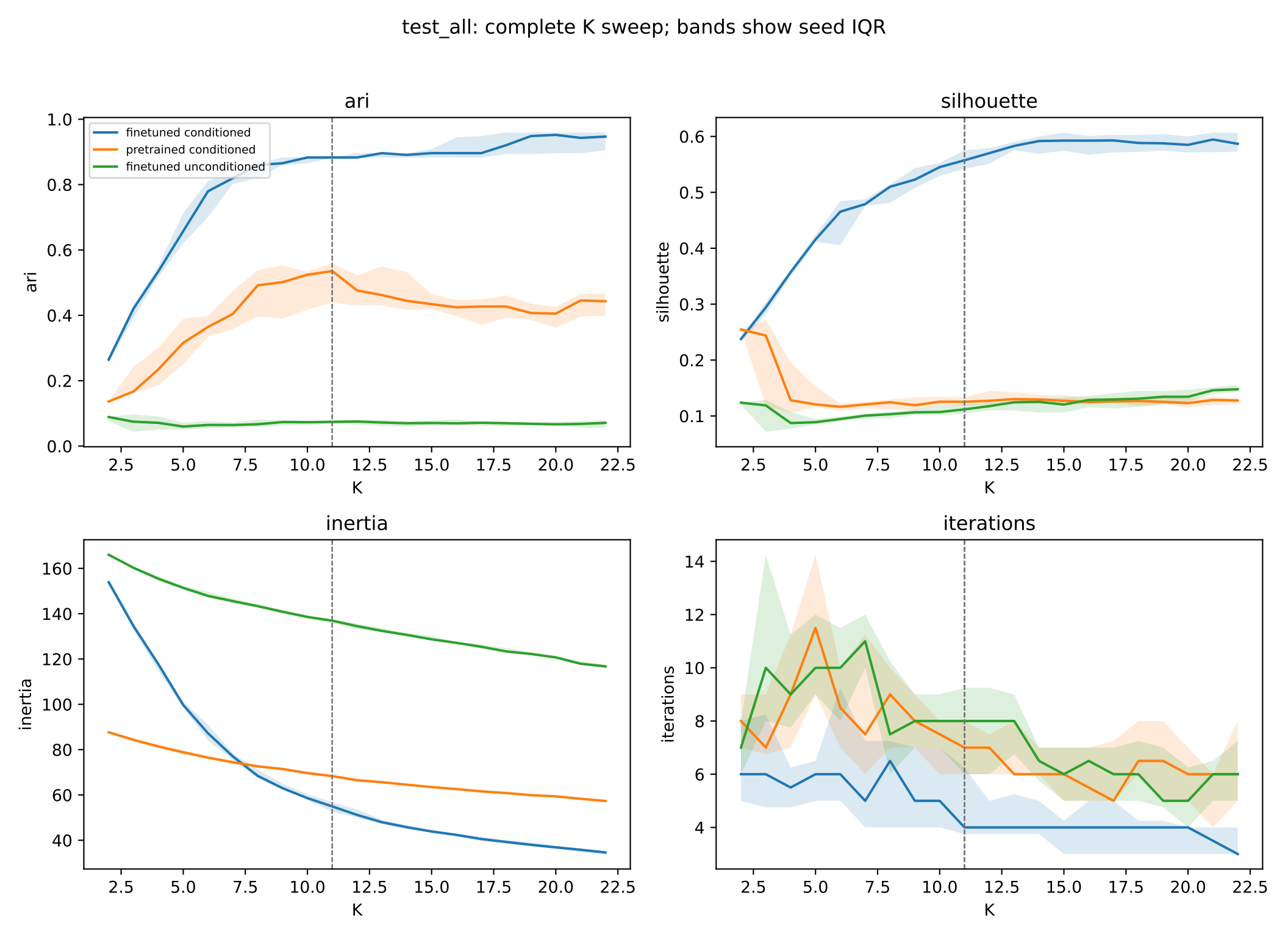
Source-color / cluster counts



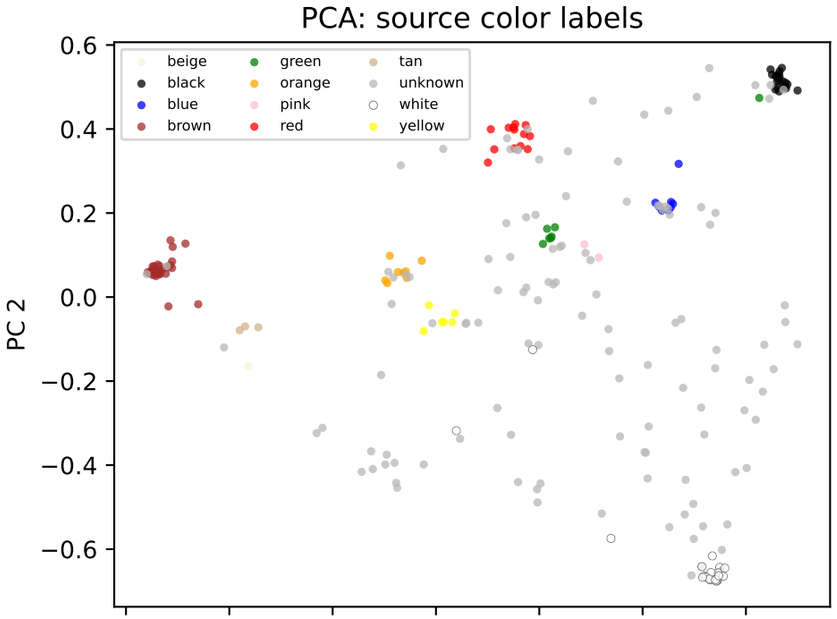
Centroid cosine distances

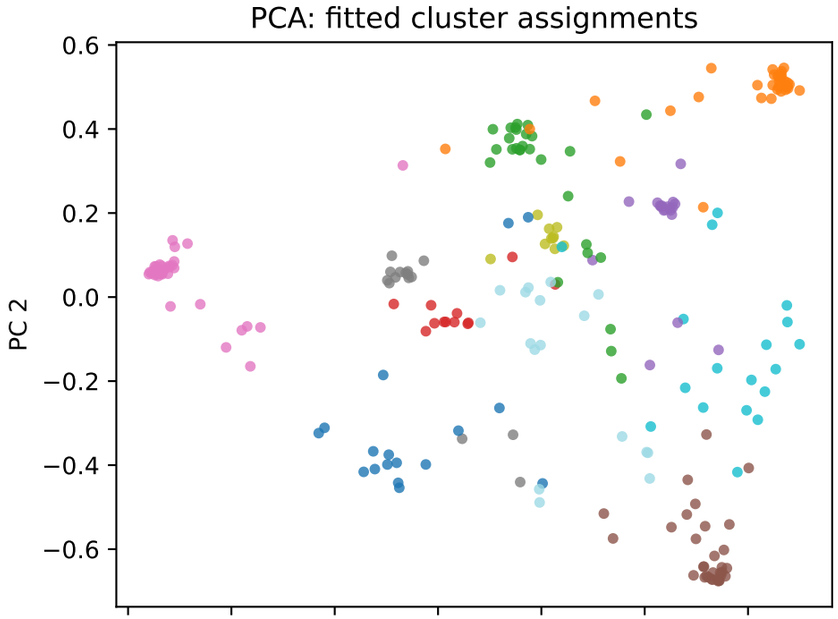


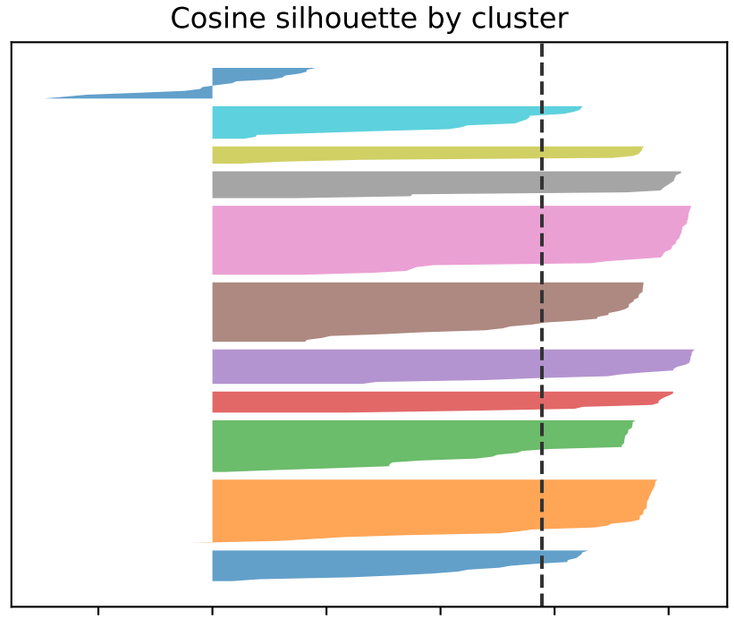

test_single · finetuned_unconditioned: seed stability at K=22
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

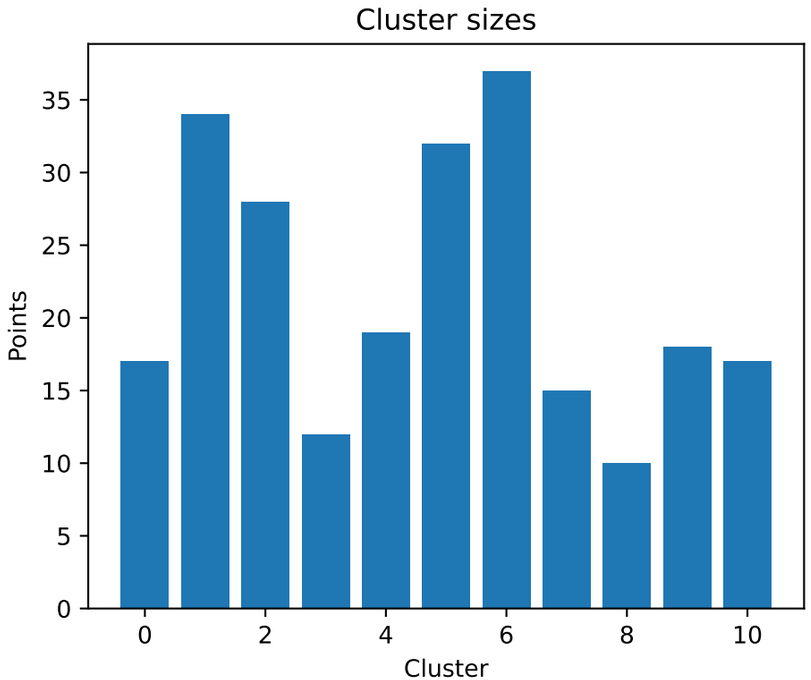

test_all: complete K sweep; bands show seed IQR
ari
silhouette
inertia
iterations
finetuned conditioned
pretrained conditioned
finetuned unconditioned
K

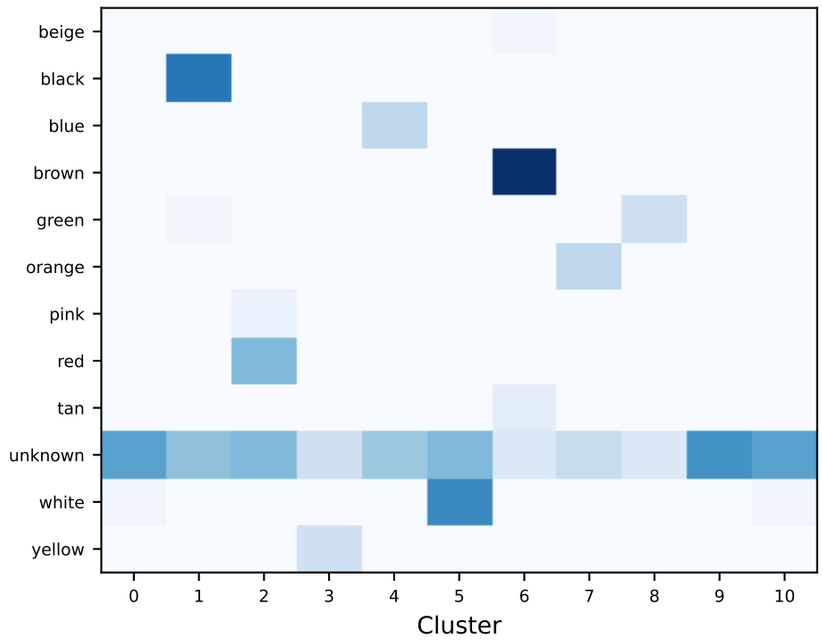
test_all · finetuned_conditioned · K=11, seed=4; PCA explains 35.2%

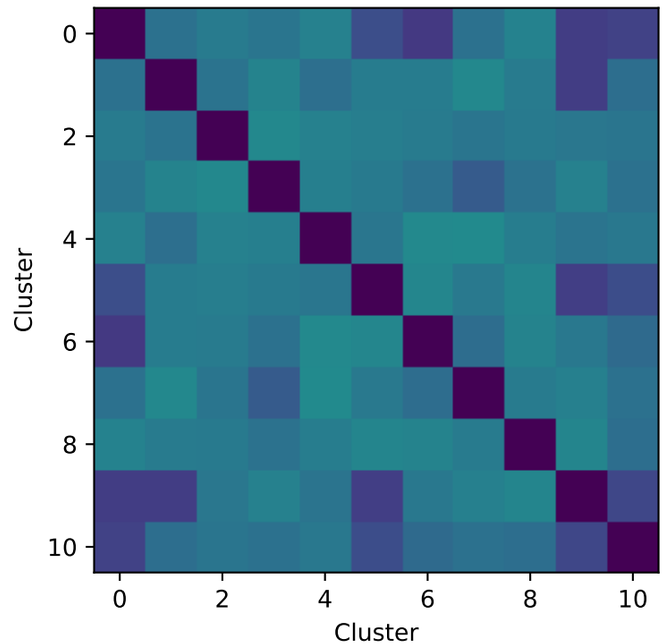




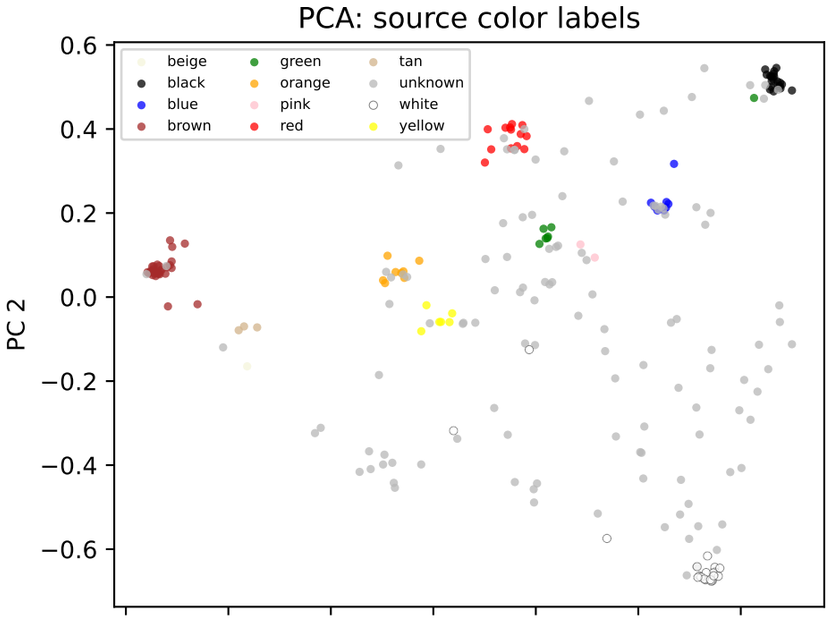


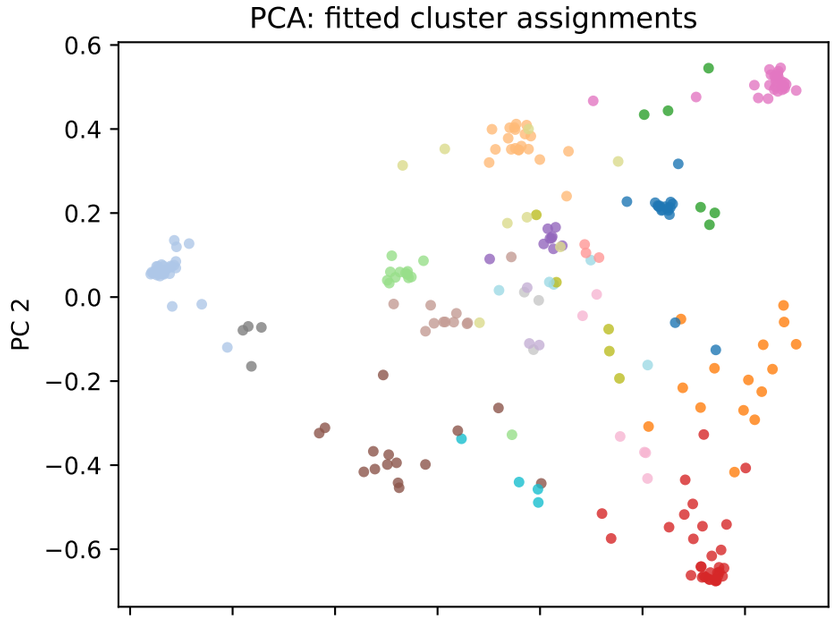


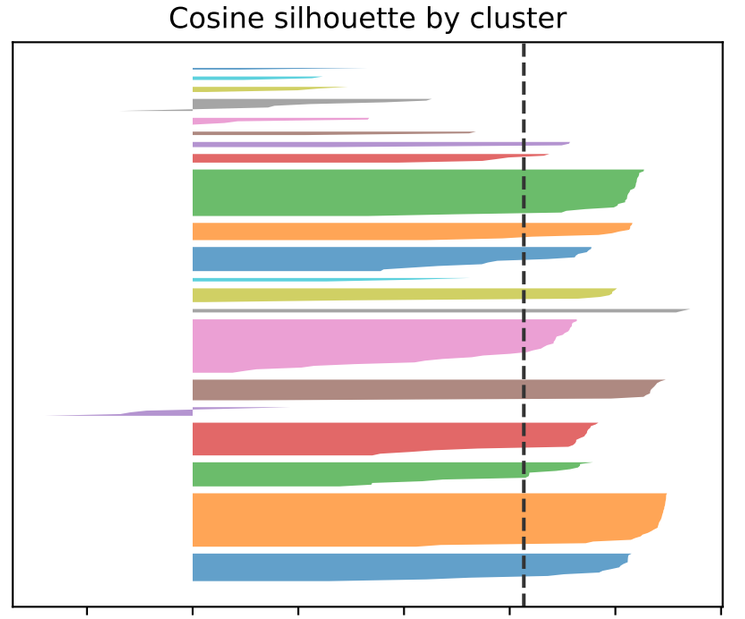




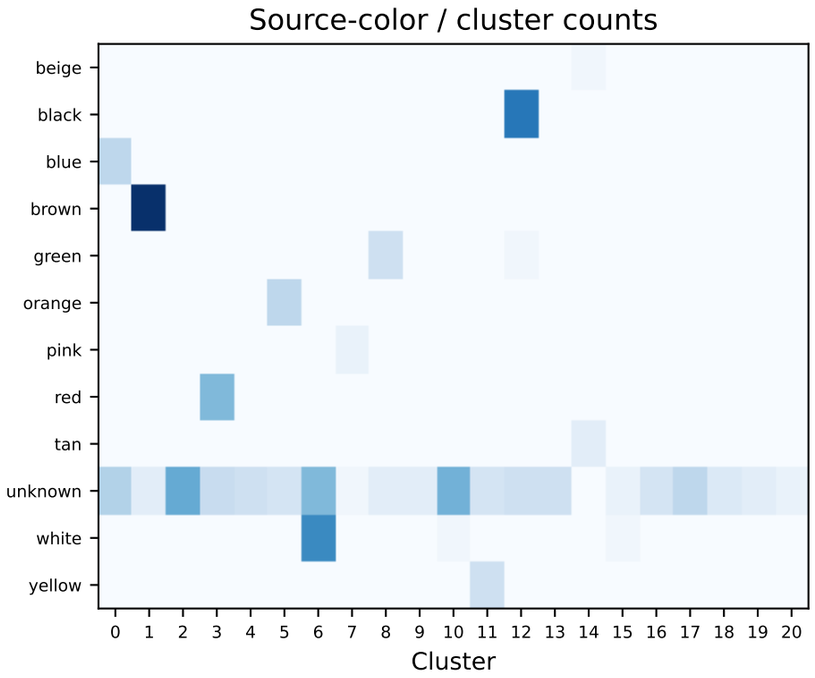
test_all · finetuned_conditioned · K=21, seed=2; PCA explains 35.2%

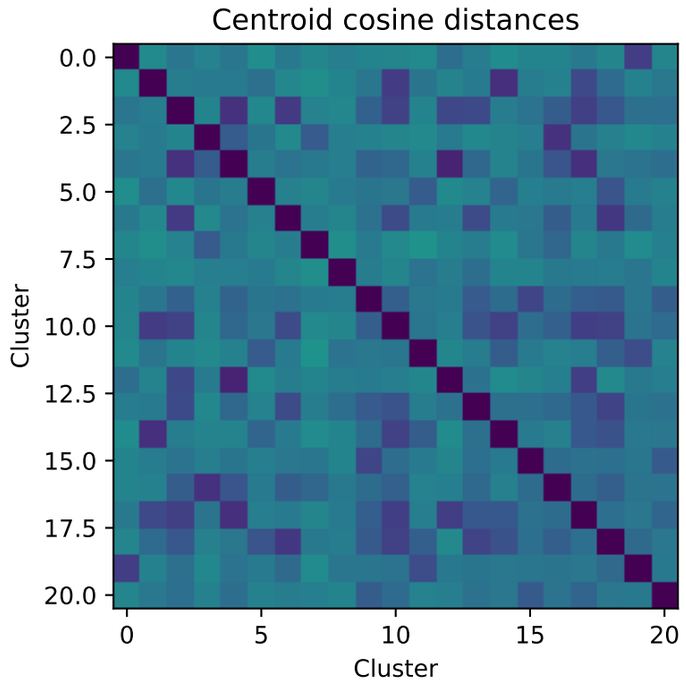


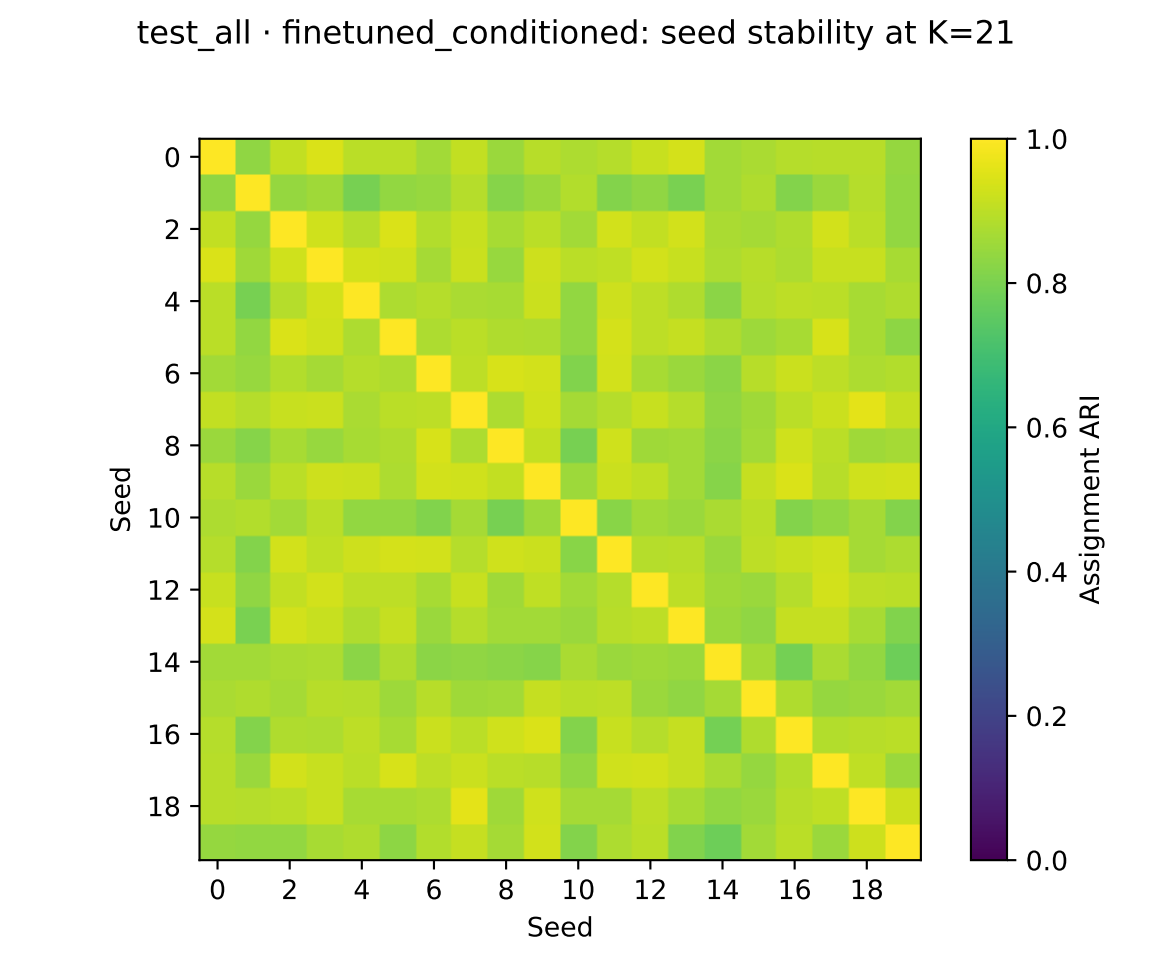




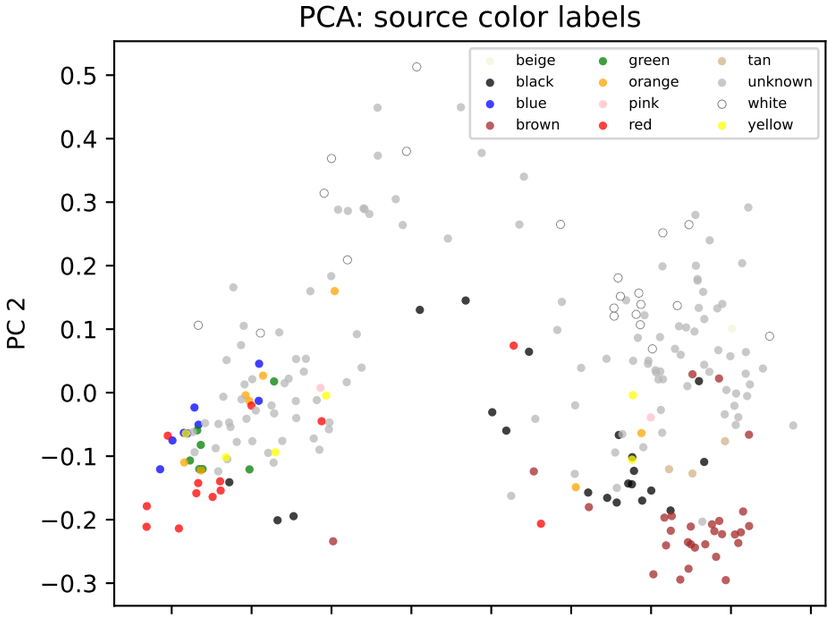


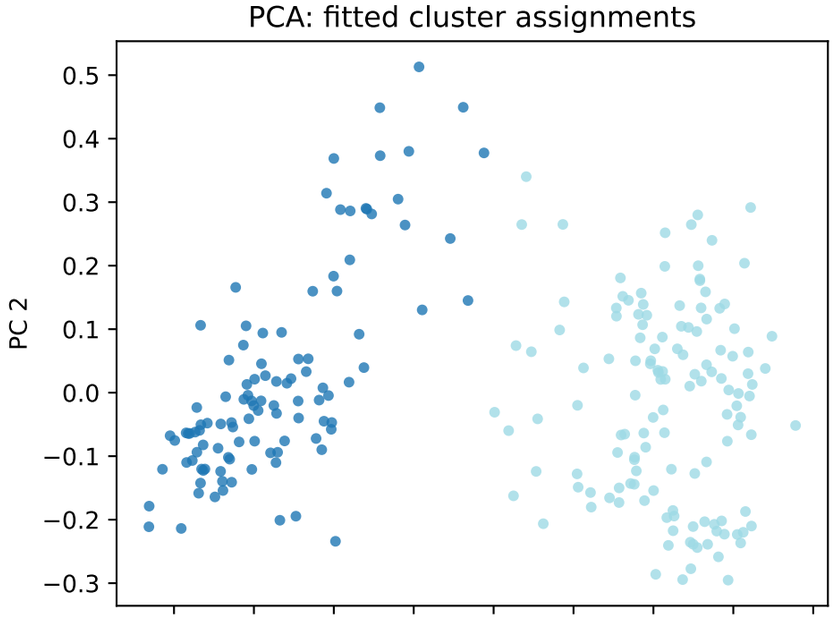


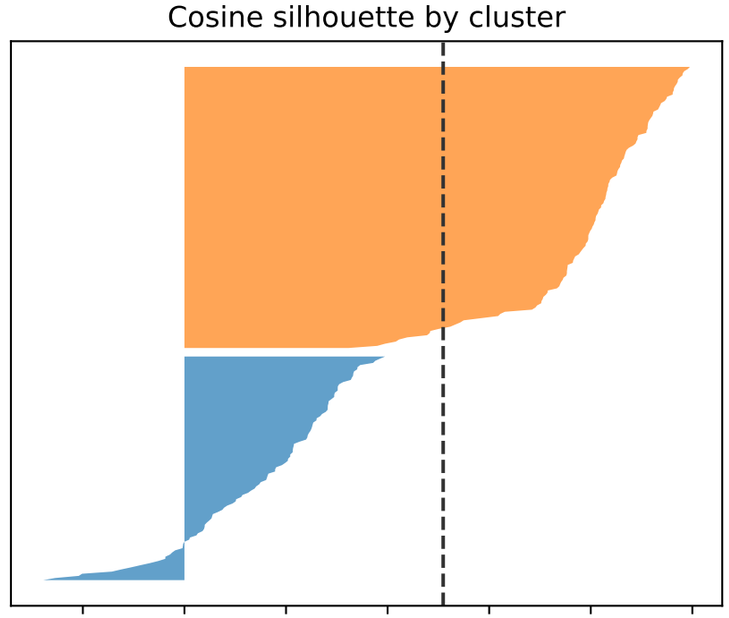


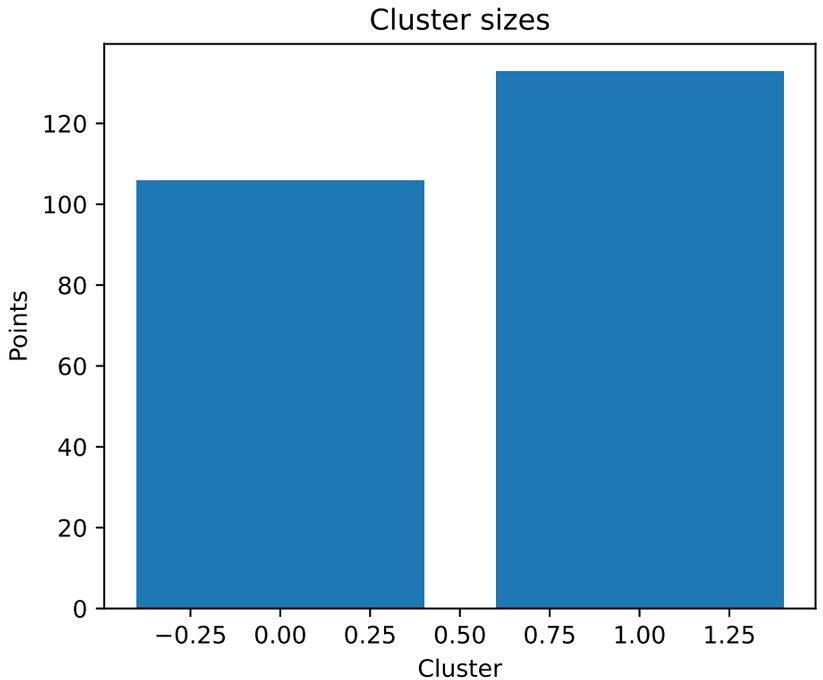

test_all · finetuned_conditioned: seed stability at K=21
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

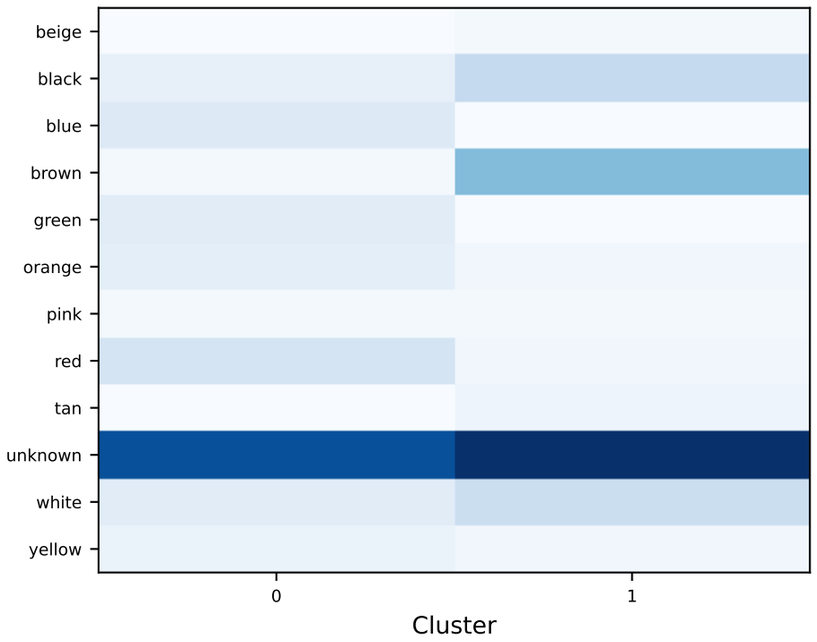
test_all · pretrained_conditioned · K=2, seed=0; PCA explains 21.4%

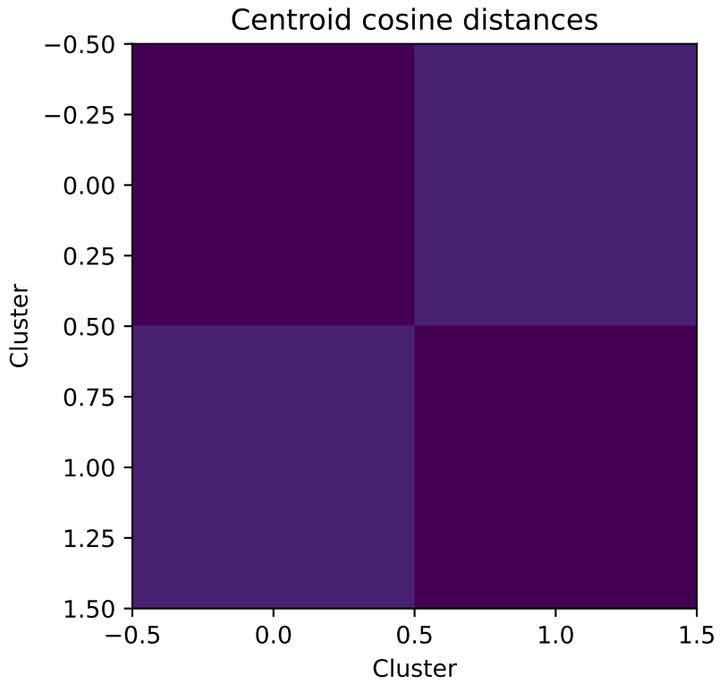




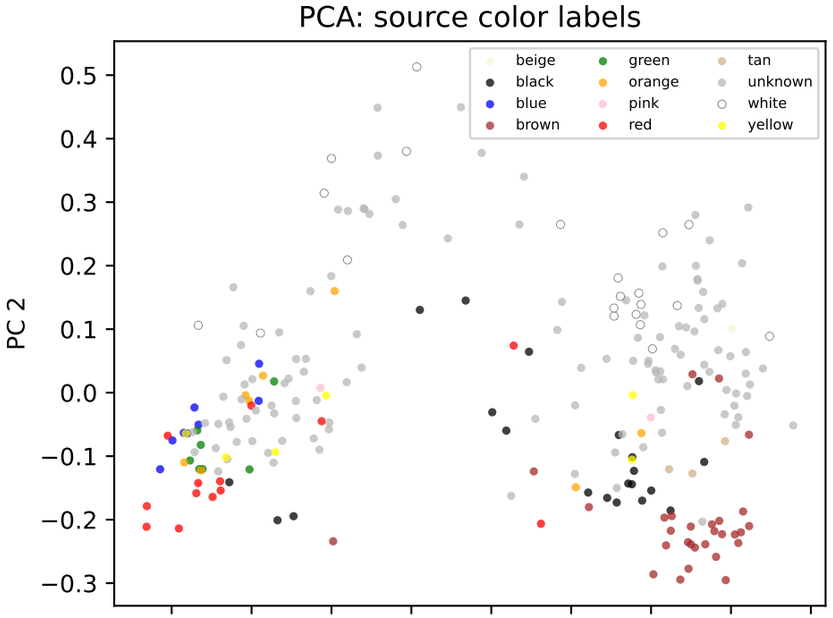


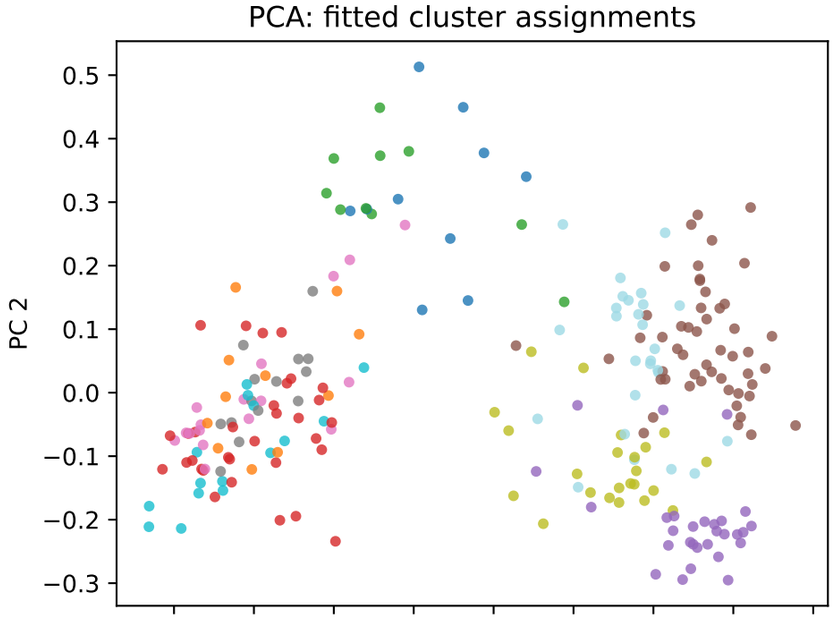


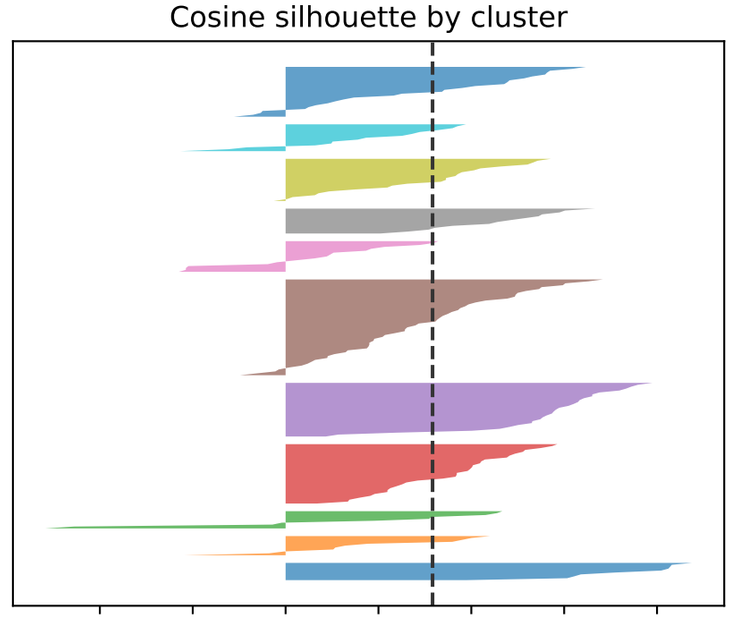


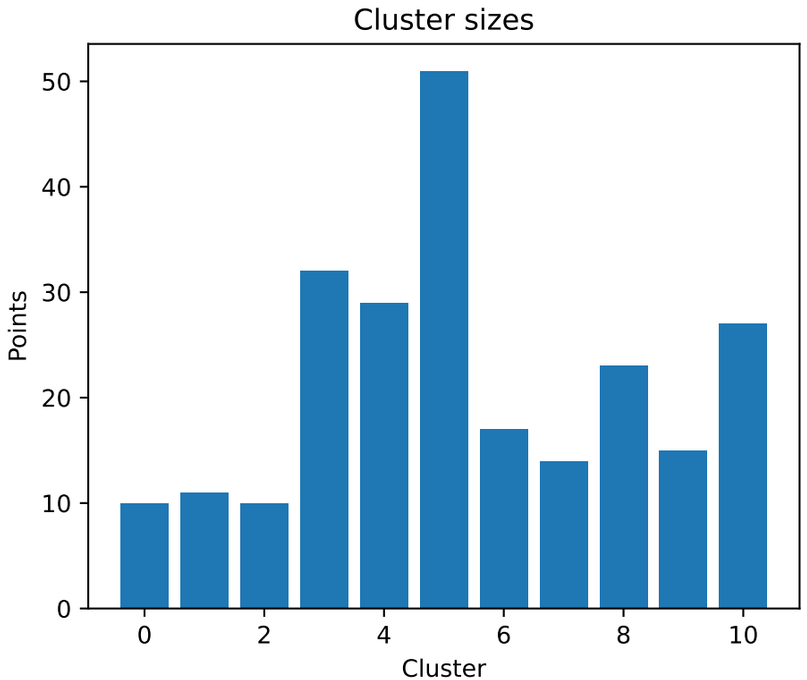


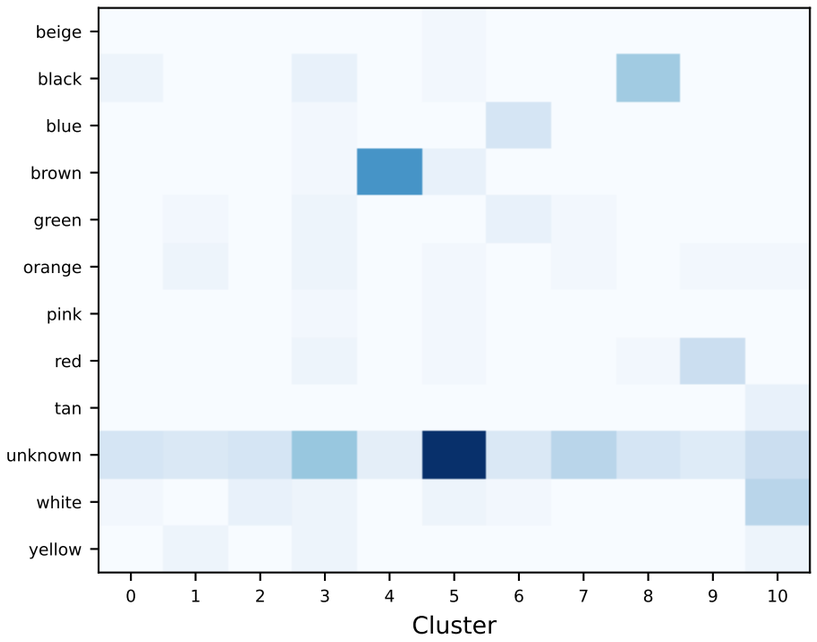
test_all · pretrained_conditioned · K=11, seed=7; PCA explains 21.4%

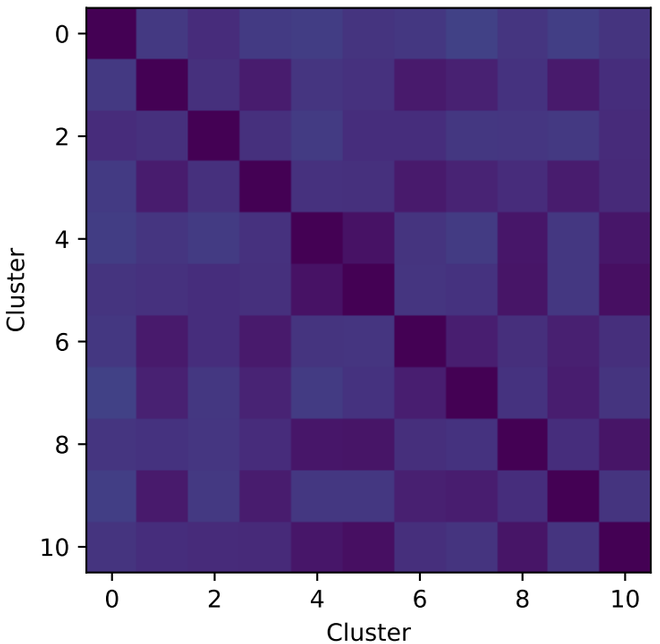


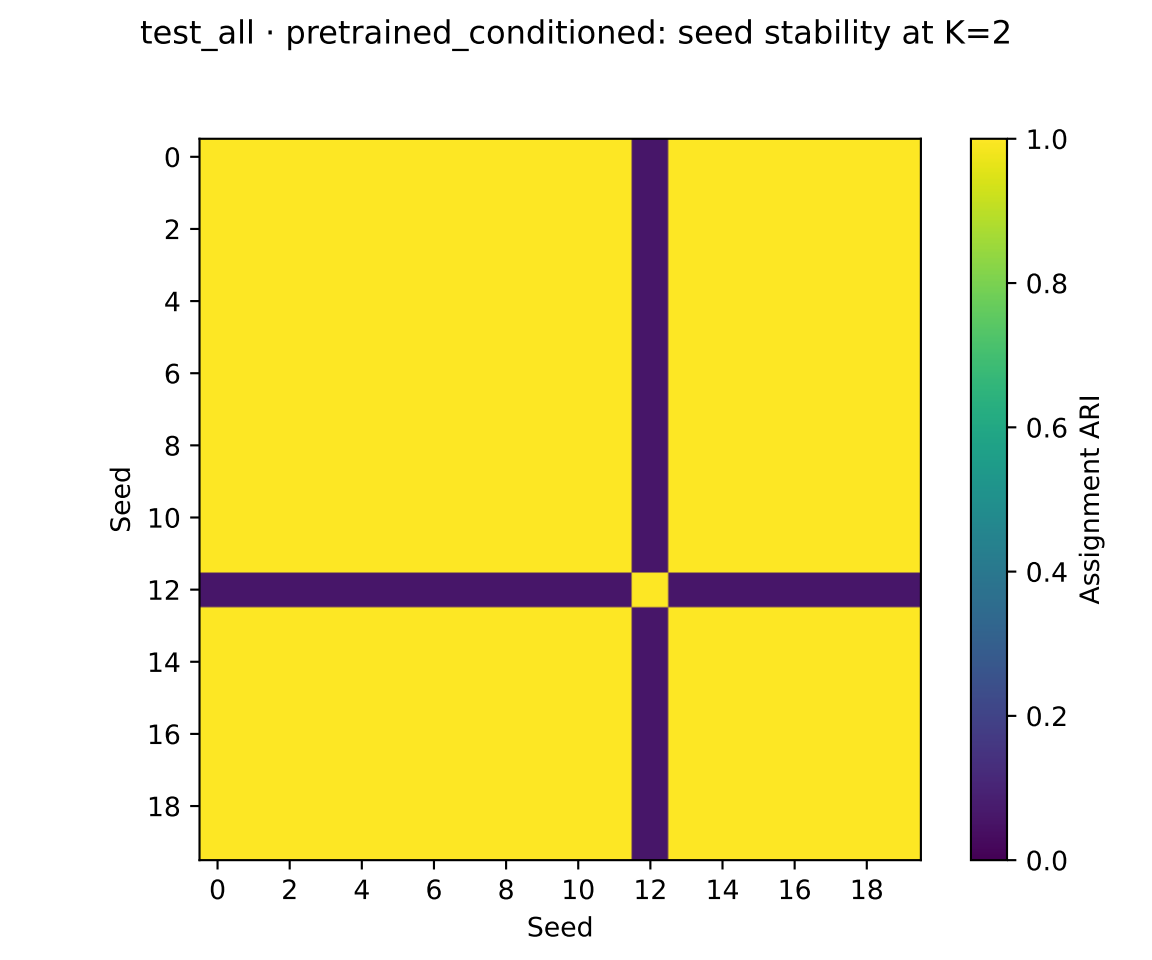




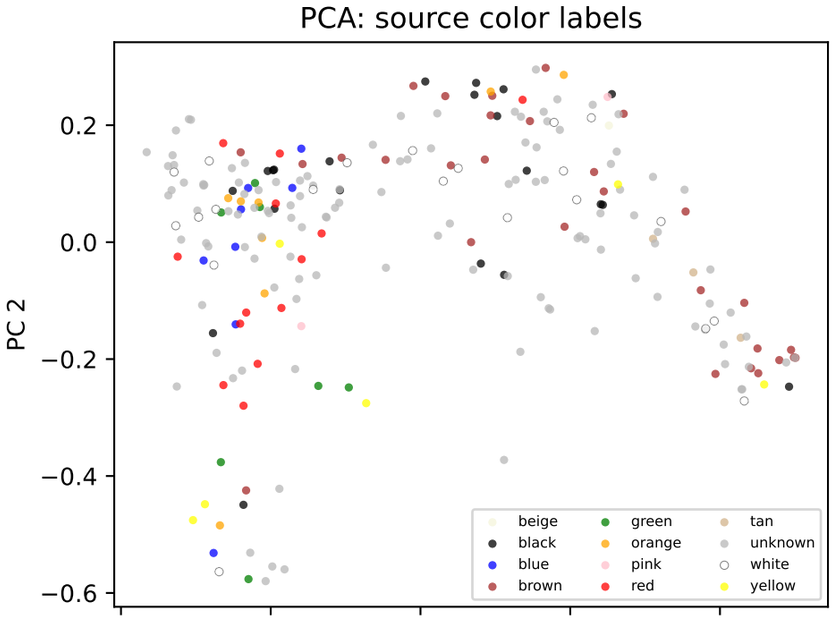


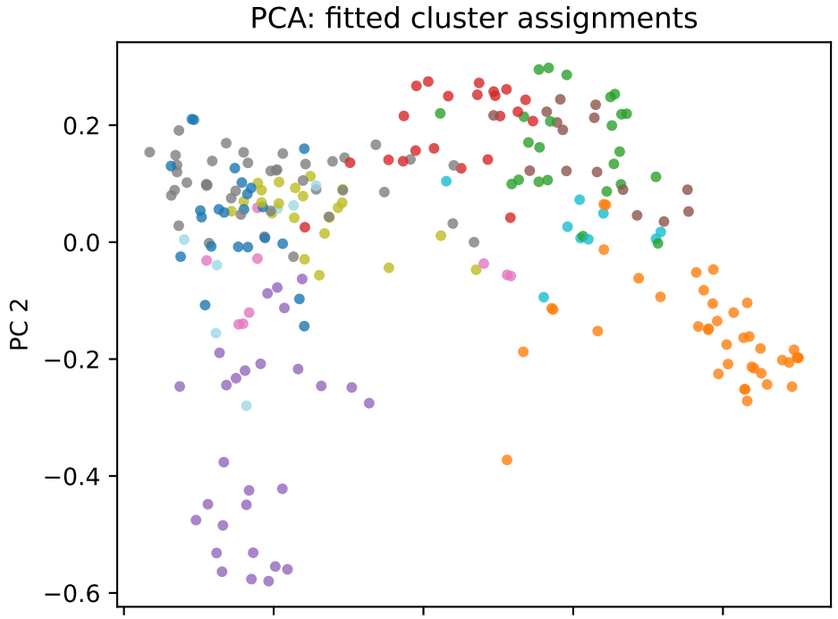


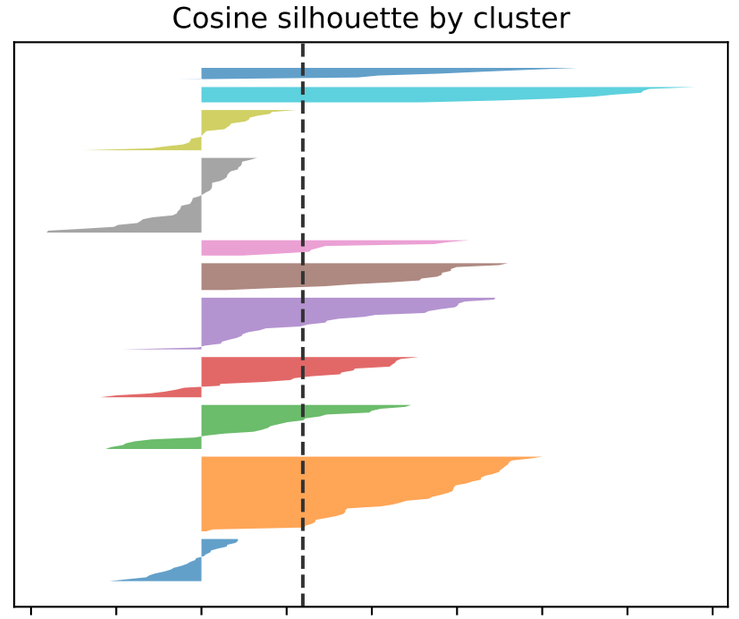


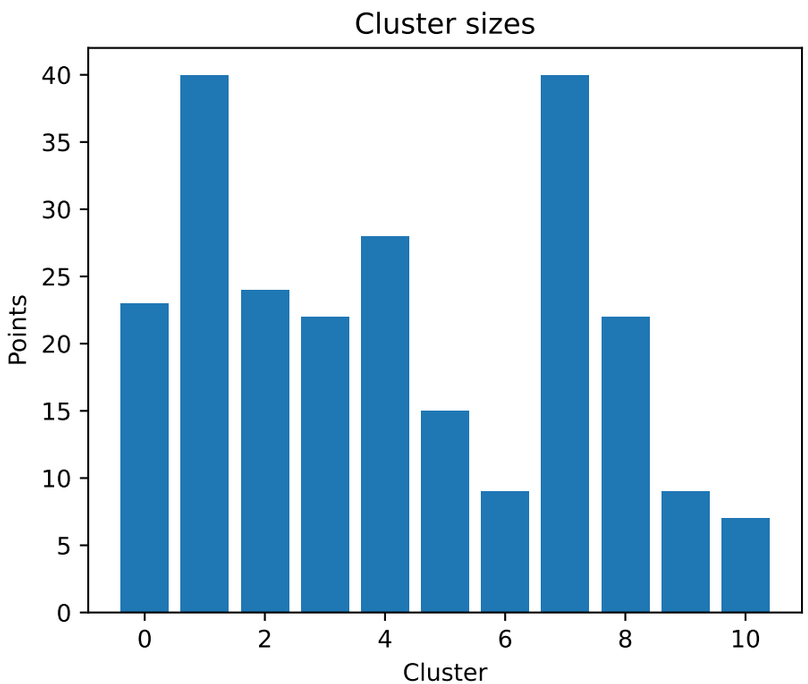

test_all · pretrained_conditioned: seed stability at K=2
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

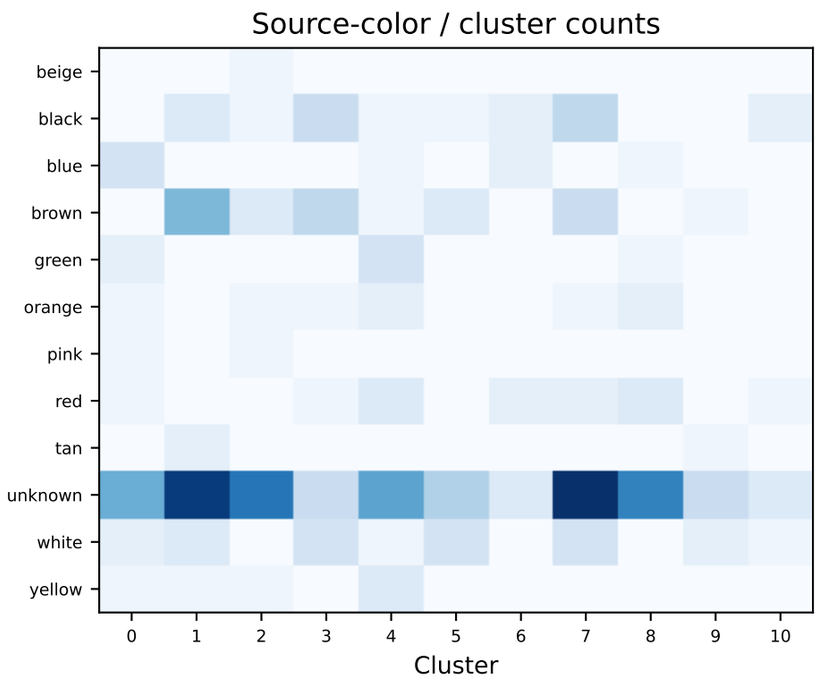
test_all · finetuned_unconditioned · K=11, seed=13; PCA explains 13.3%

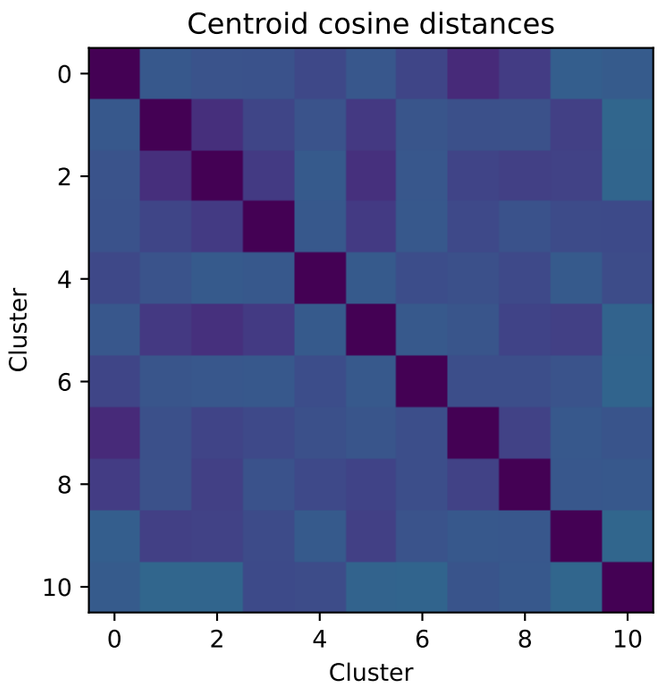






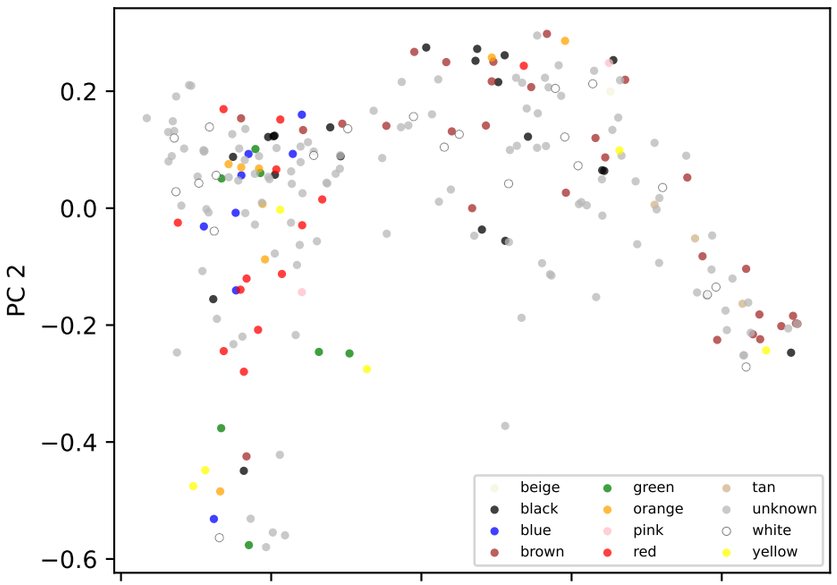




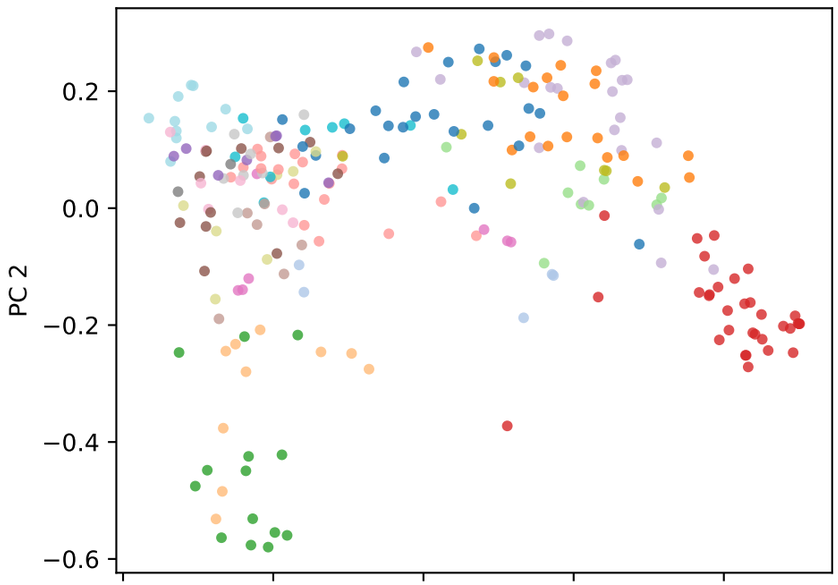


test_all · finetuned_unconditioned · K=22, seed=5; PCA explains 13.3%

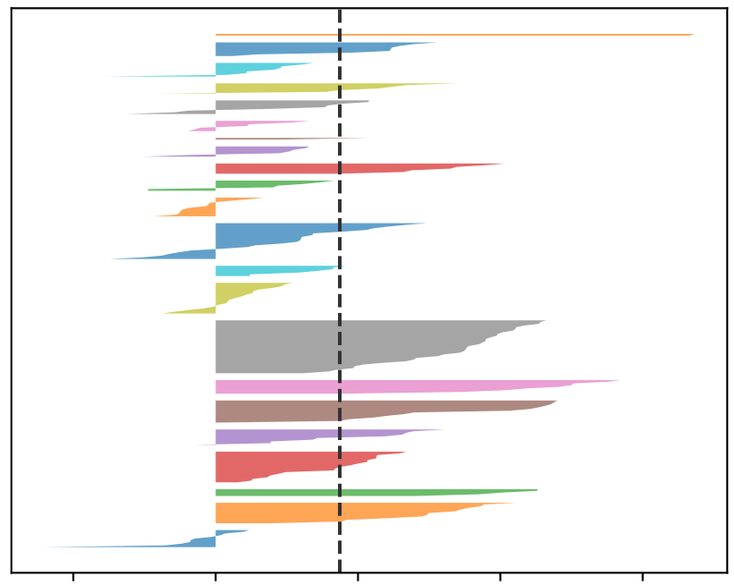
PCA: source color labels



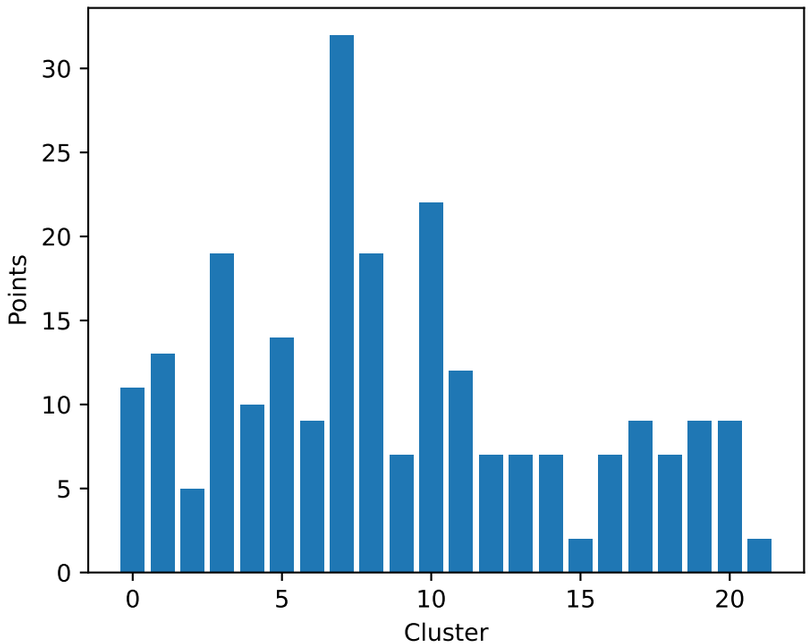
PCA: fitted cluster assignments



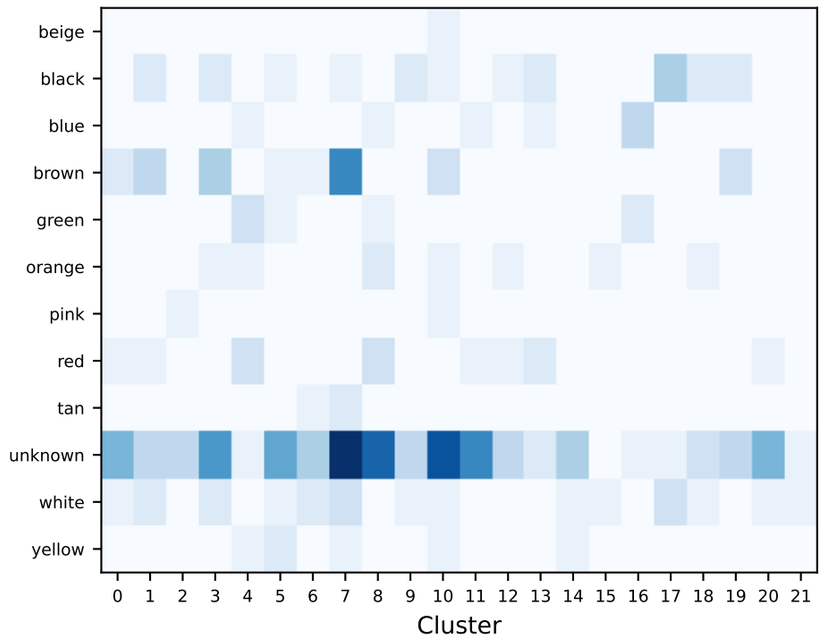
Cosine silhouette by cluster



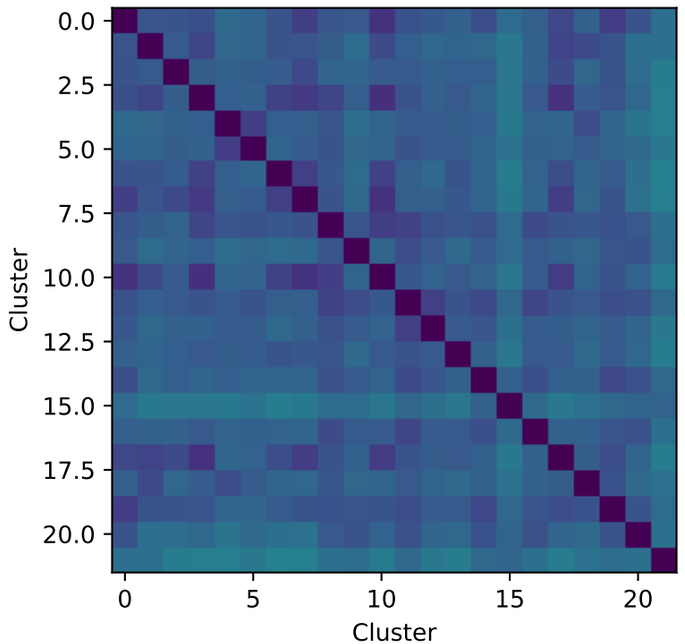
Cluster sizes

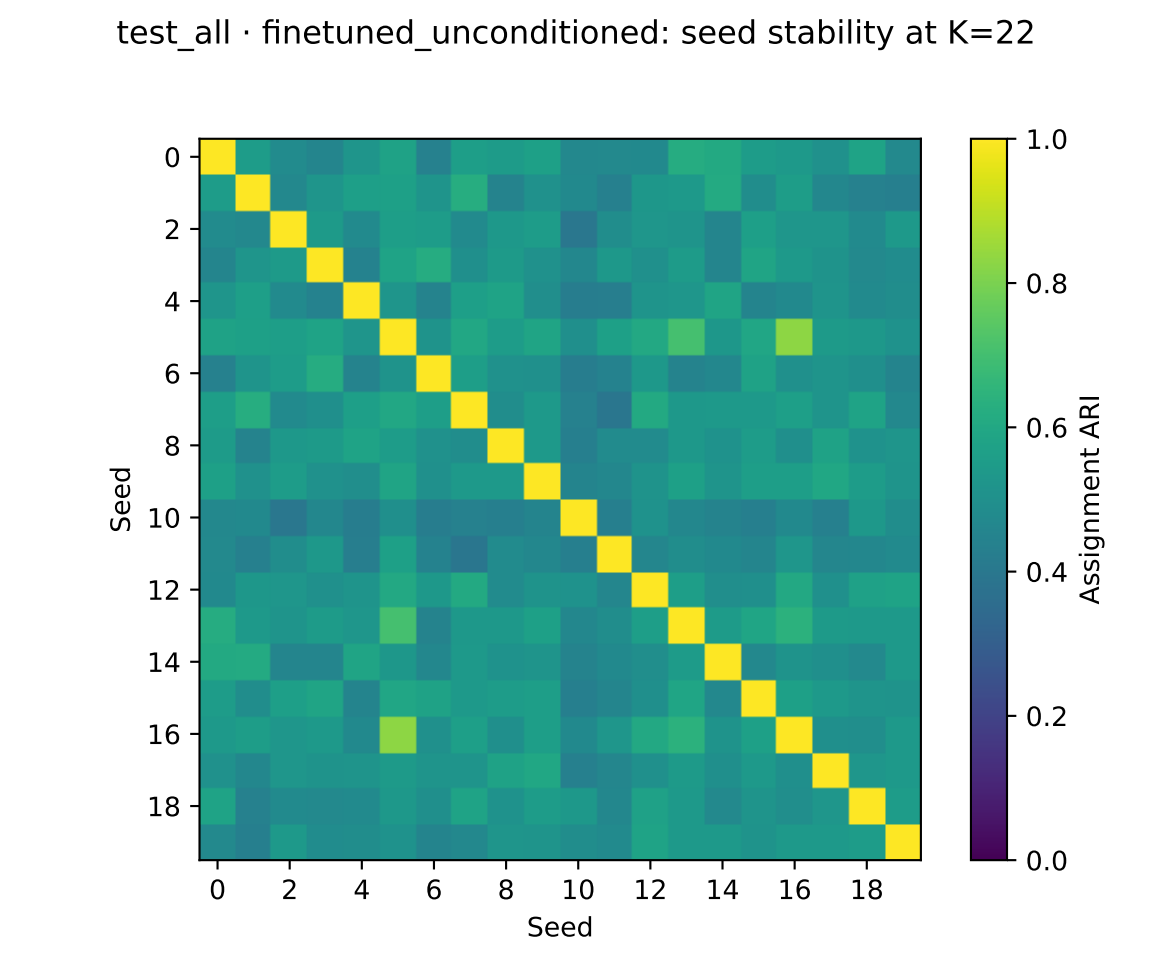


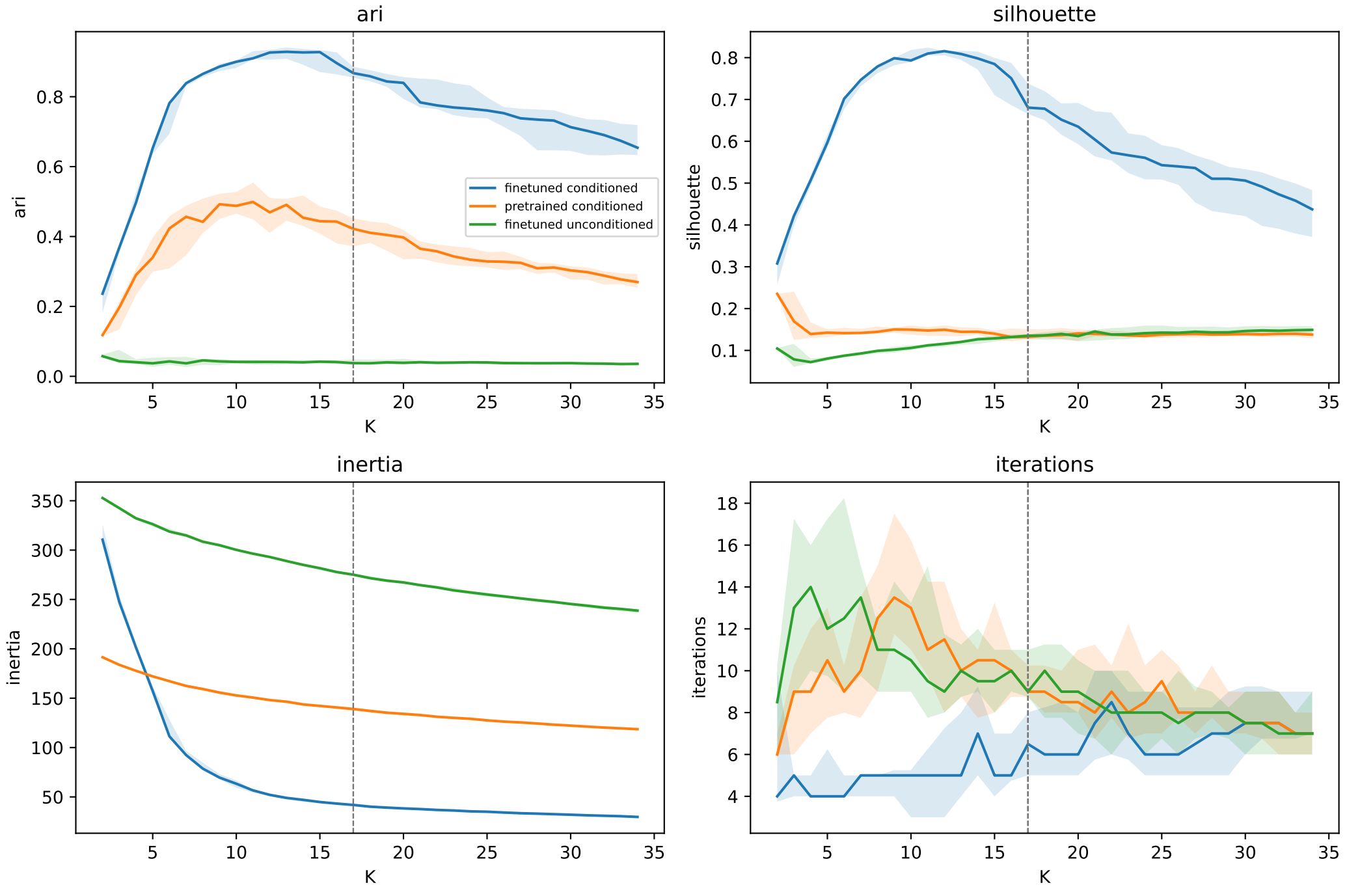
Source-color / cluster counts



Centroid cosine distances

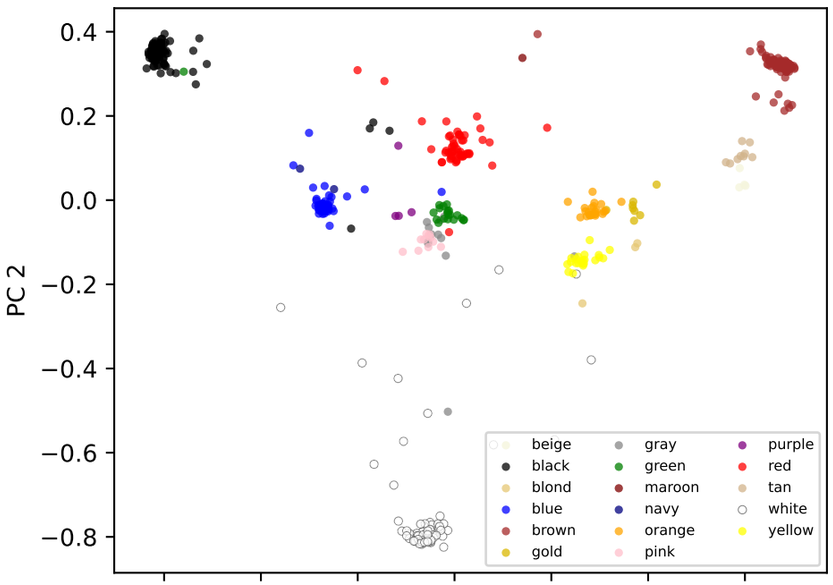



test_all · finetuned_unconditioned: seed stability at K=22
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

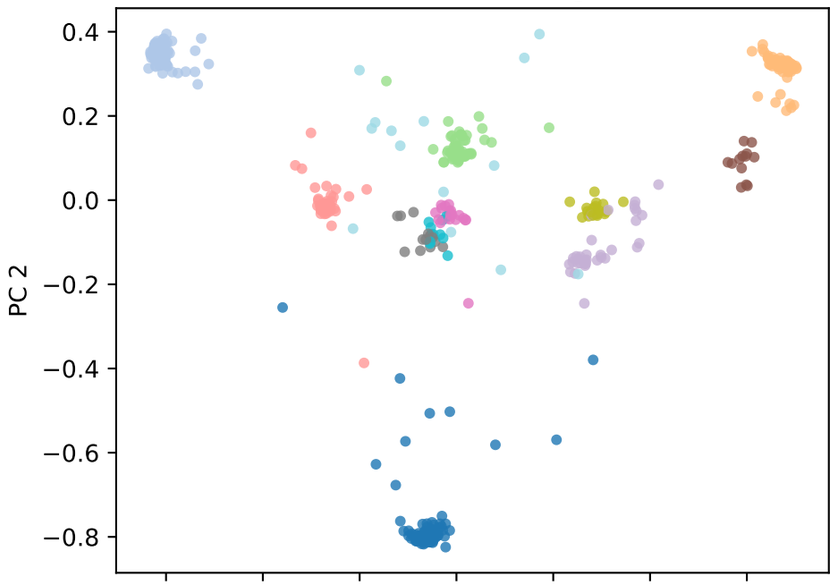

pooled_single: complete K sweep; bands show seed IQR
ari
silhouette
inertia
iterations
finetuned conditioned
pretrained conditioned
finetuned unconditioned
K

pooled_single · finetuned_conditioned · K=12, seed=16; PCA explains 42.1%

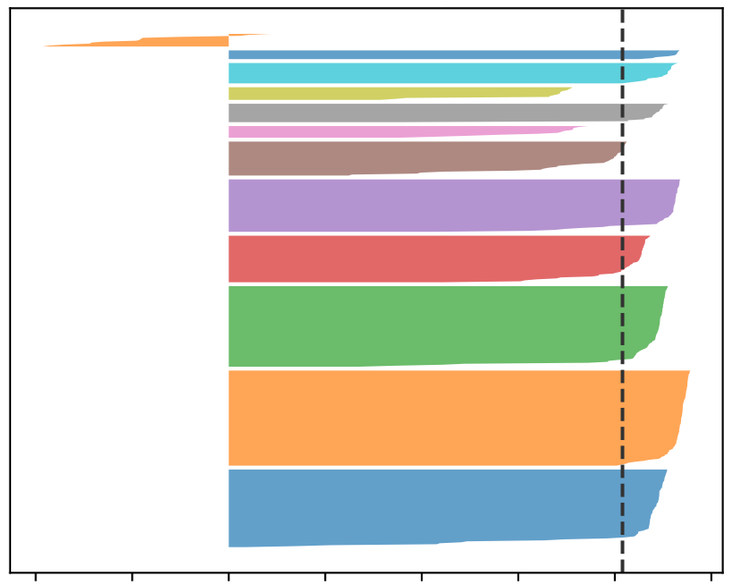
PCA: source color labels



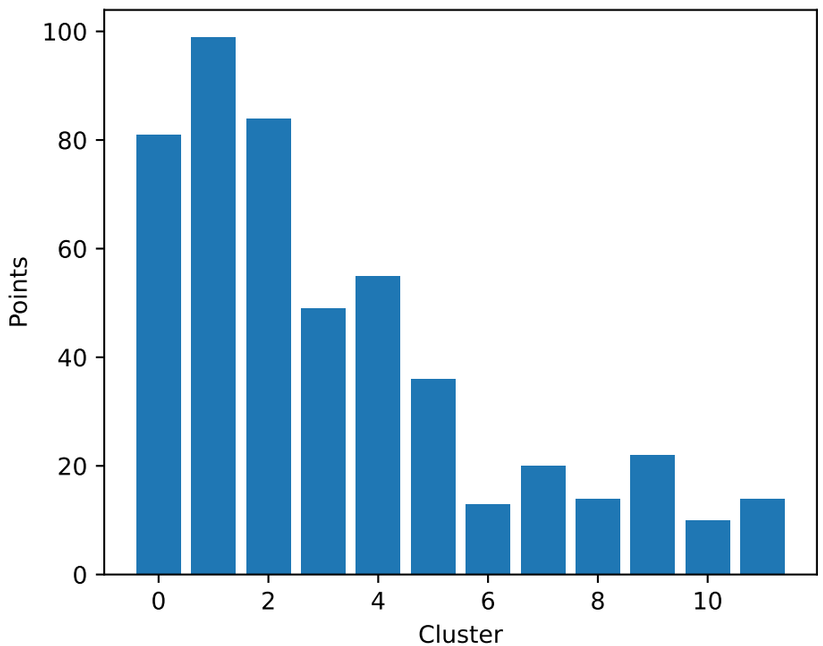
PCA: fitted cluster assignments



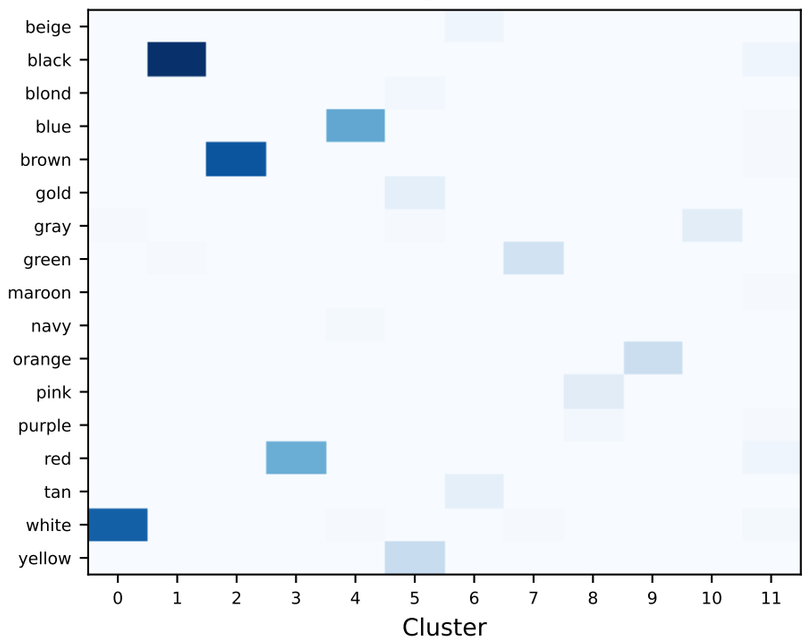
Cosine silhouette by cluster



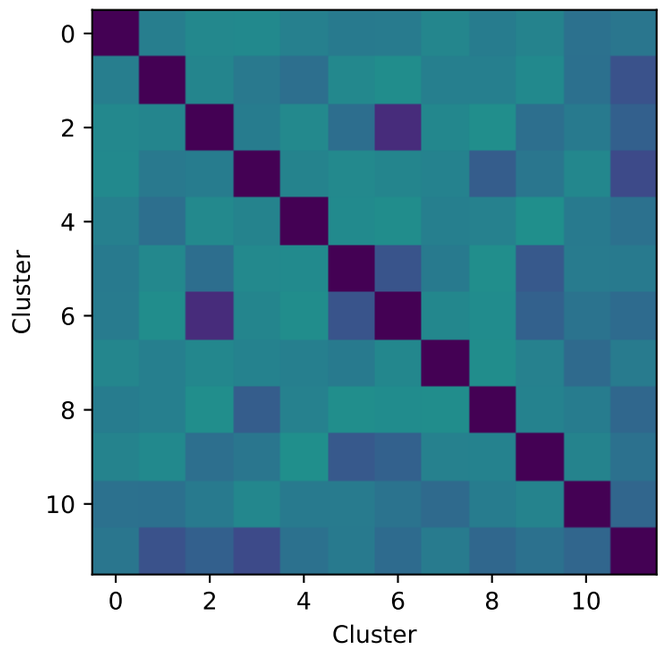
Cluster sizes



Source-color / cluster counts

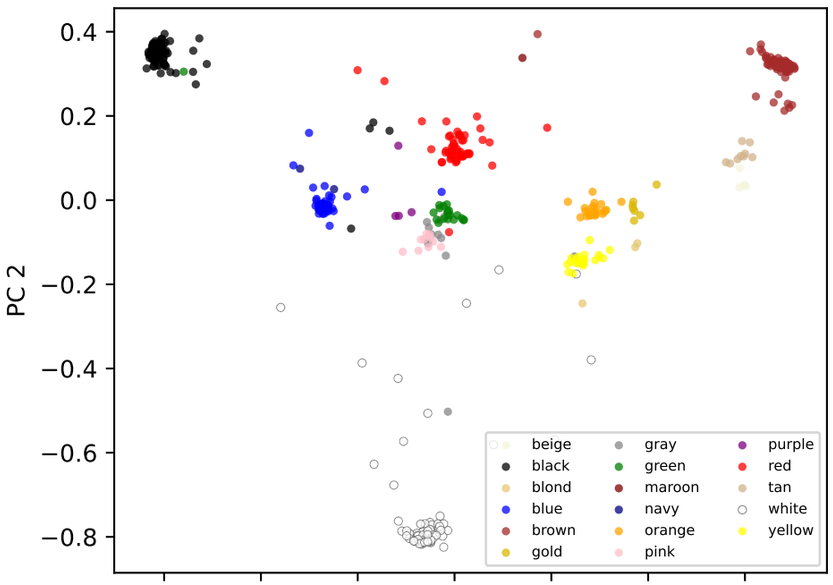


Centroid cosine distances

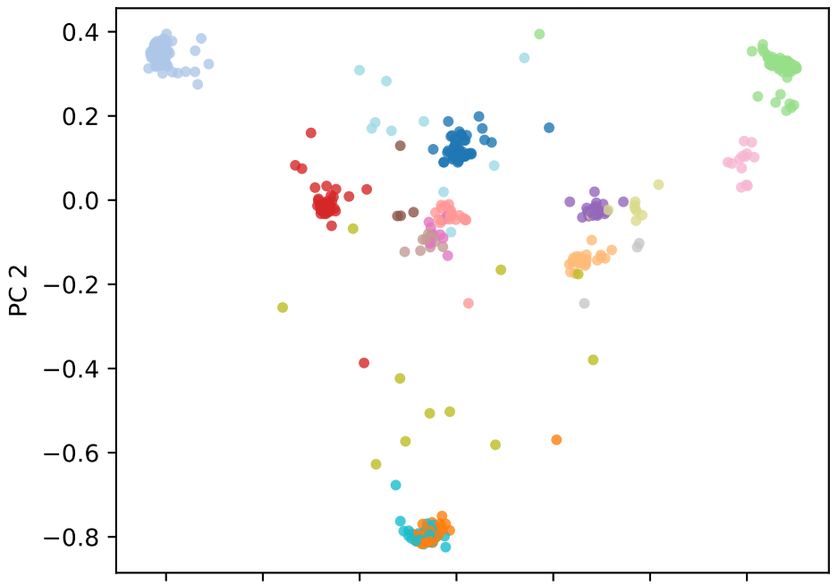


pooled_single · finetuned_conditioned · K=17, seed=9; PCA explains 42.1%

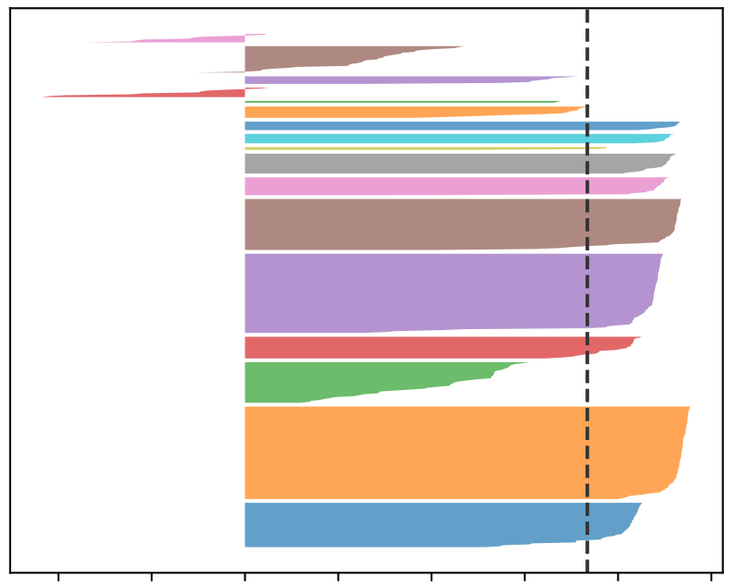
PCA: source color labels



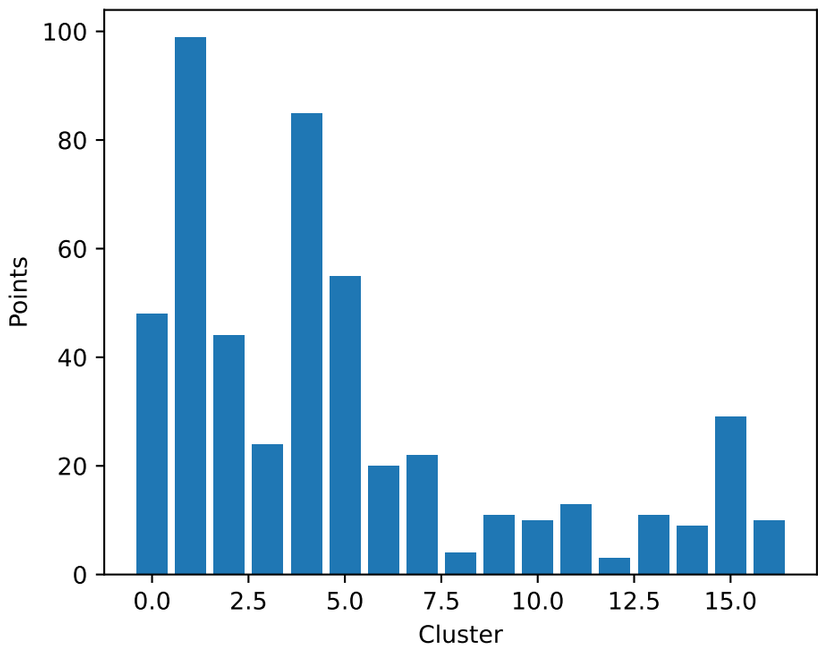
PCA: fitted cluster assignments



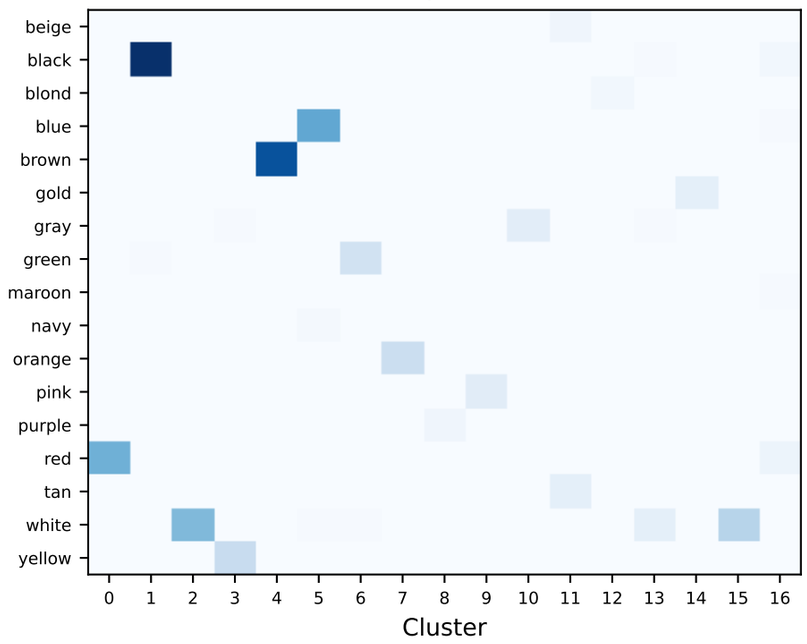
Cosine silhouette by cluster



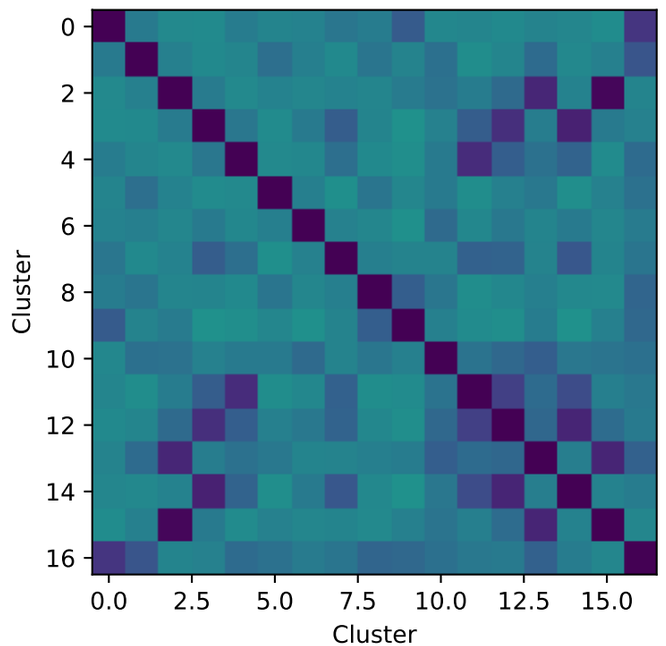
Cluster sizes

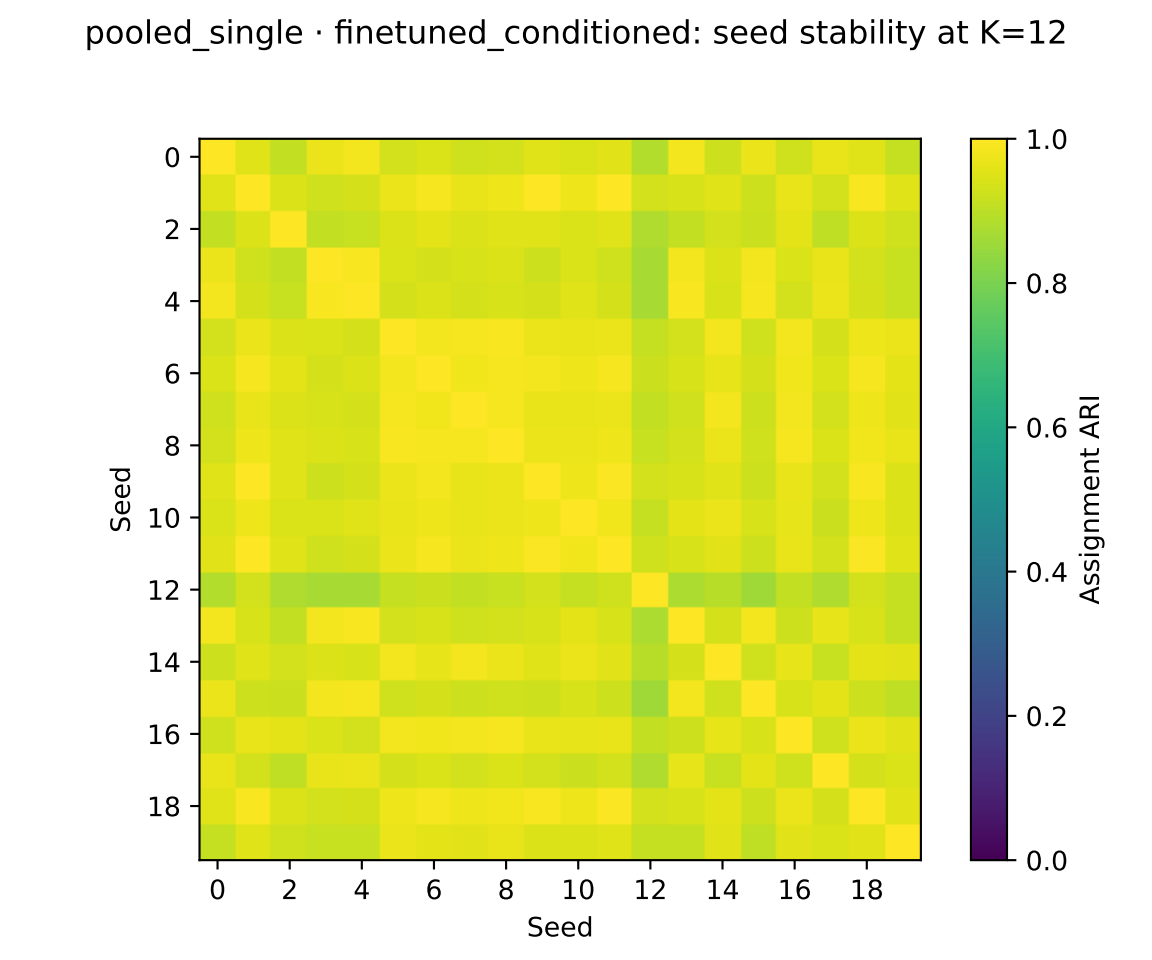


Source-color / cluster counts

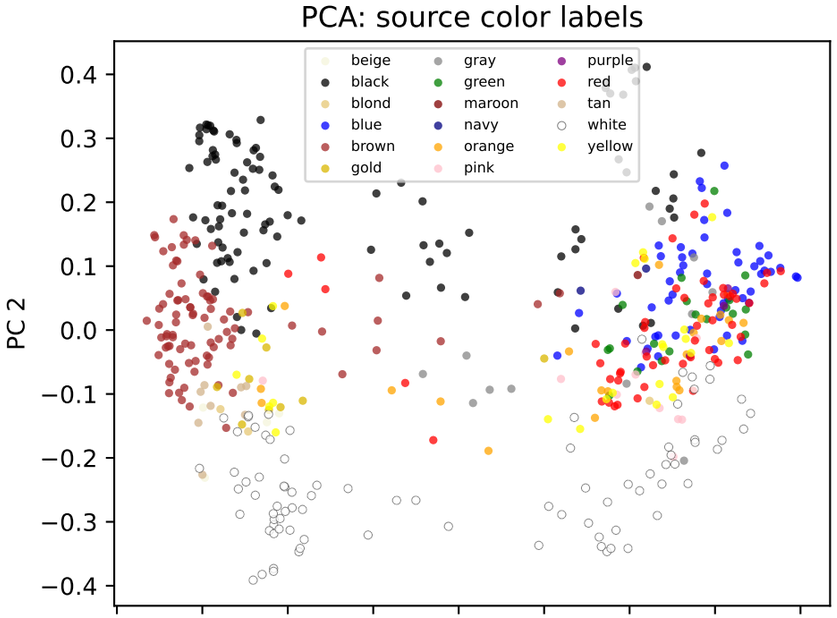


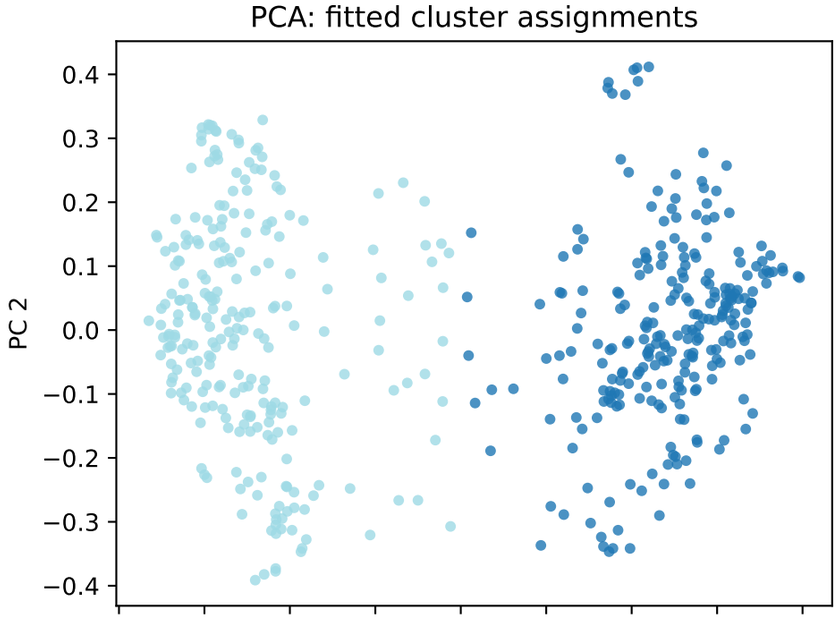
Centroid cosine distances

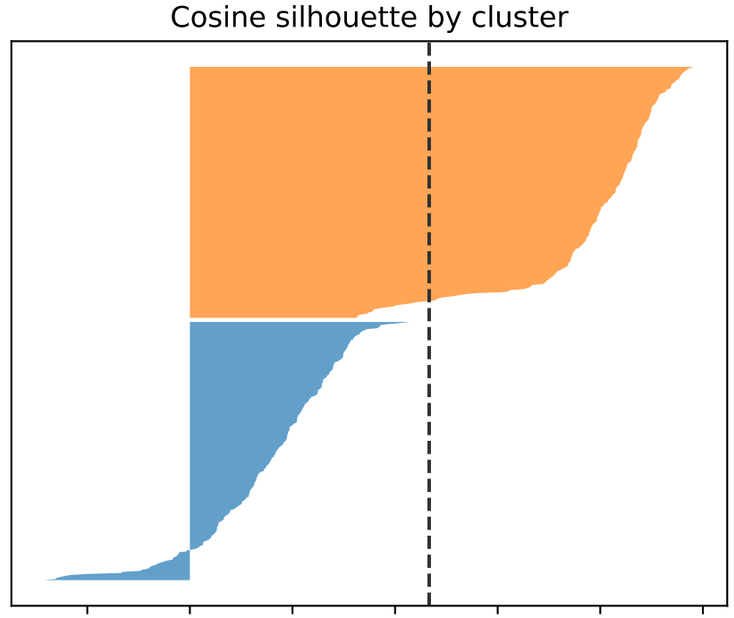


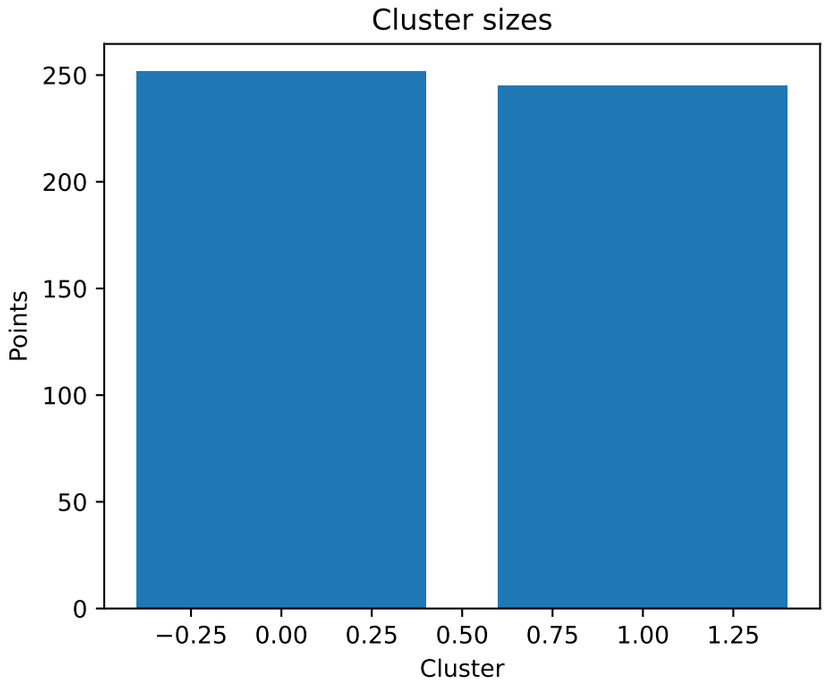

pooled_single · finetuned_conditioned: seed stability at K=12
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

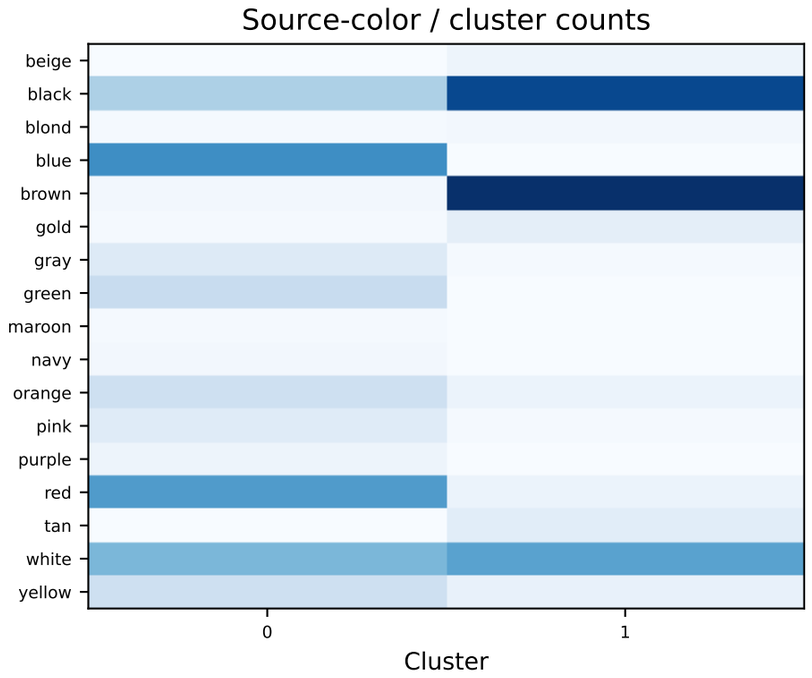
pooled_single · pretrained_conditioned · K=2, seed=5; PCA explains 20.6%

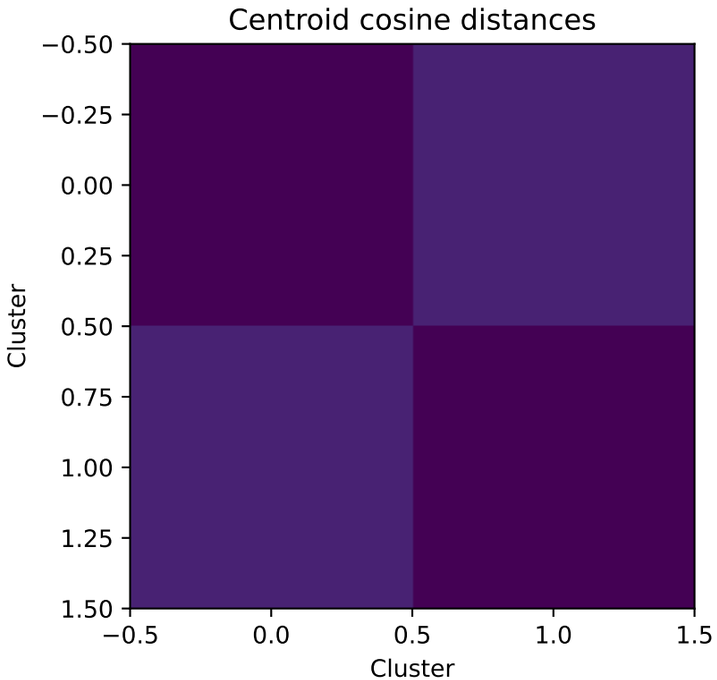






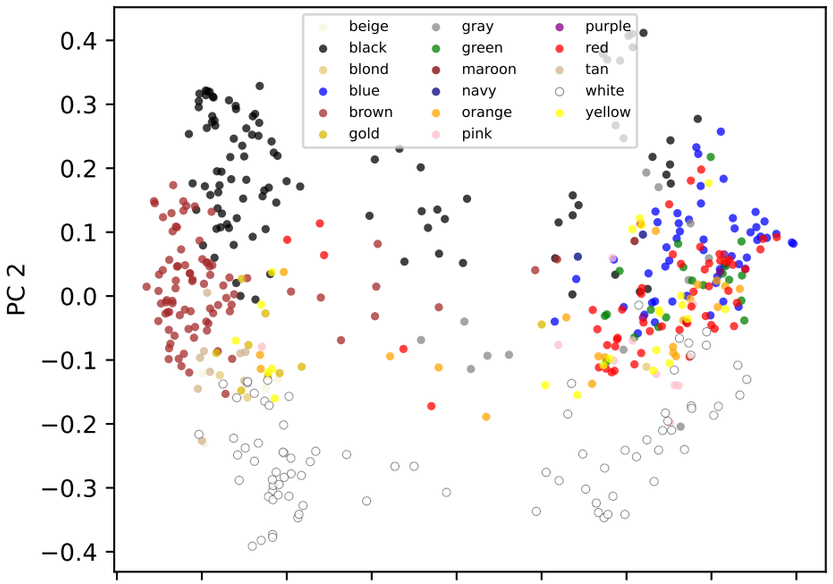




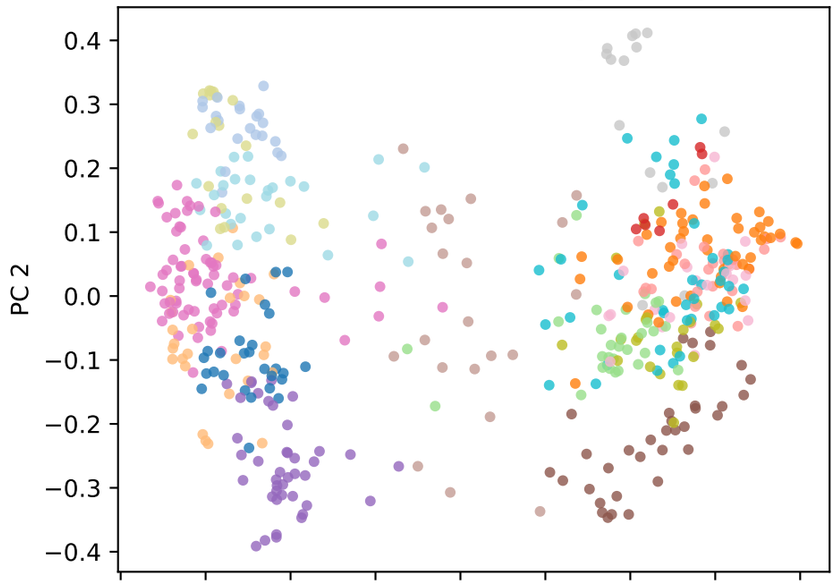


pooled_single · pretrained_conditioned · K=17, seed=14; PCA explains 20.6%

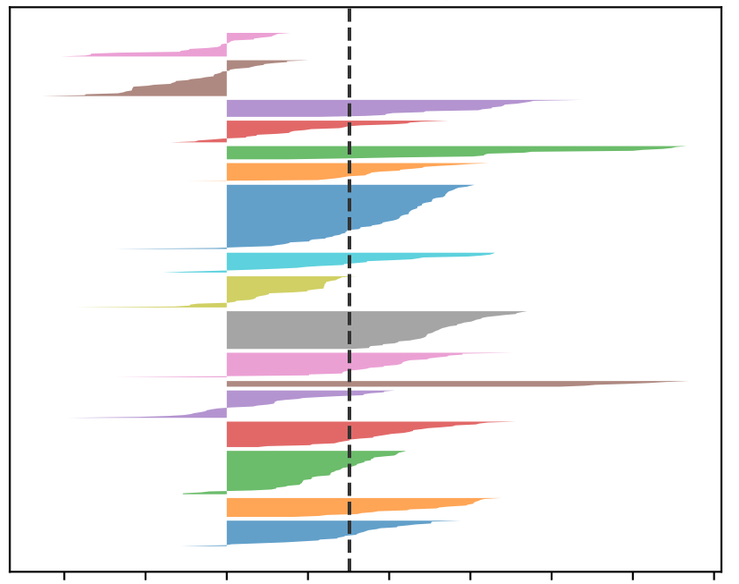
PCA: source color labels



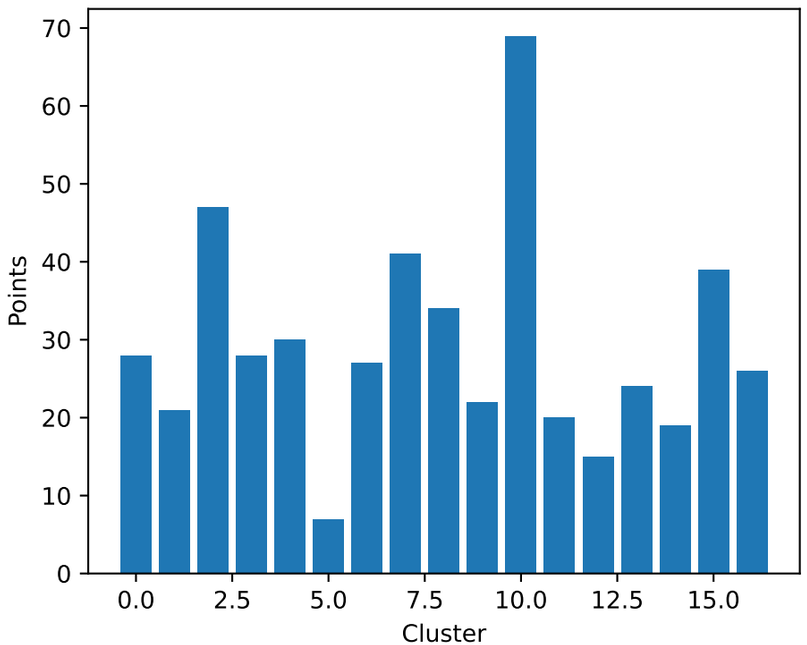
PCA: fitted cluster assignments



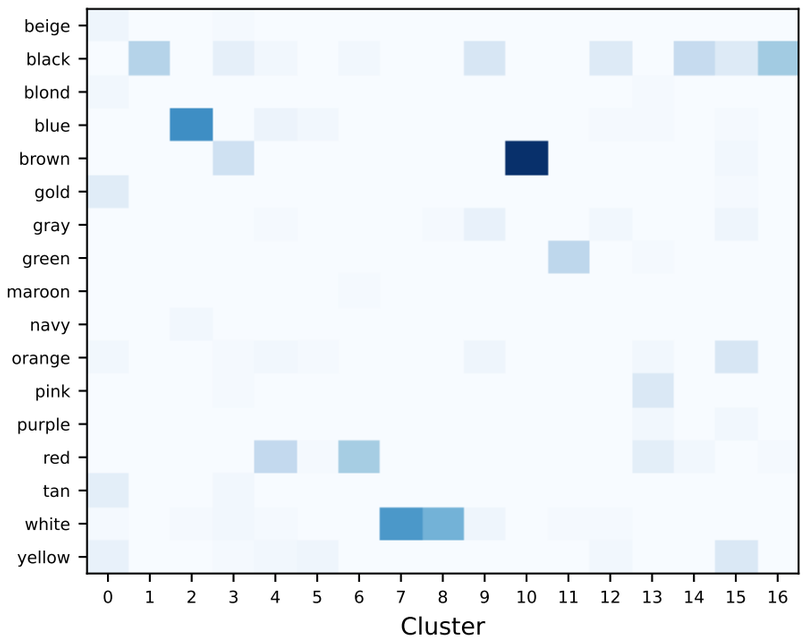
Cosine silhouette by cluster



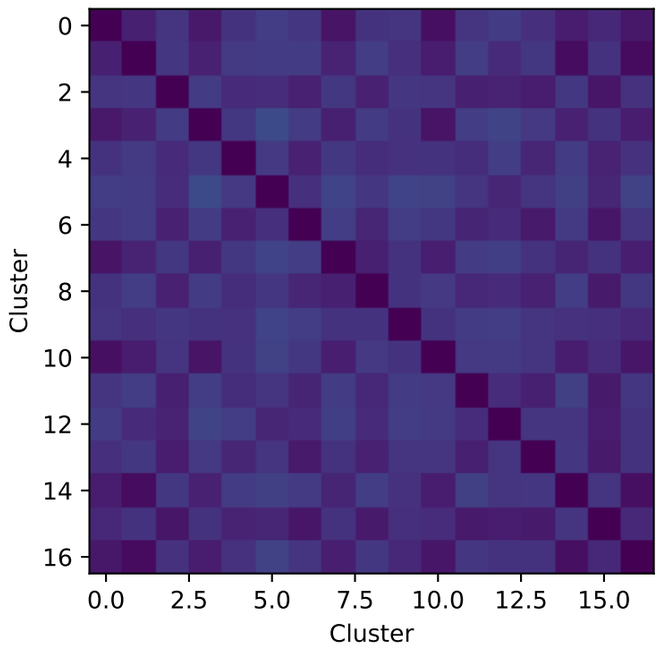
Cluster sizes

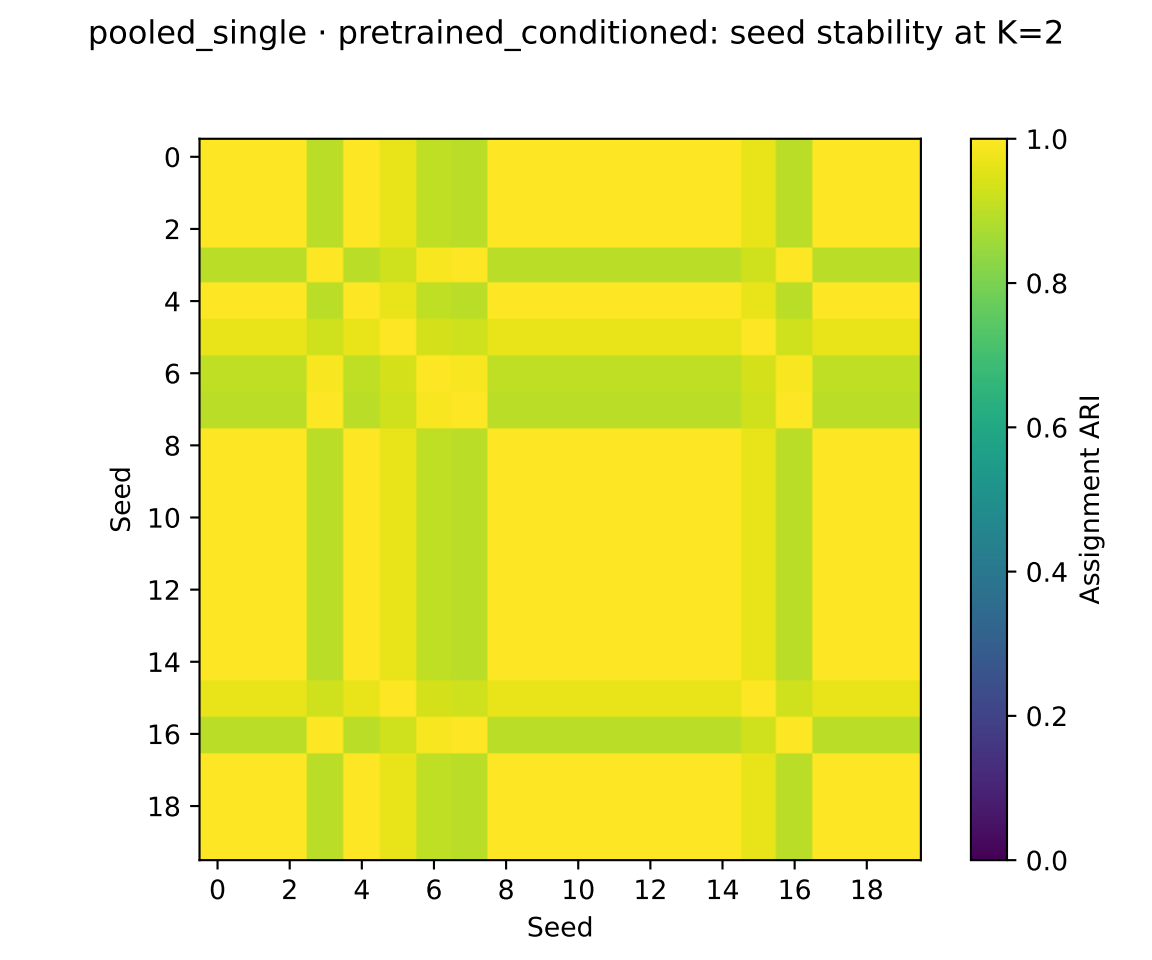


Source-color / cluster counts



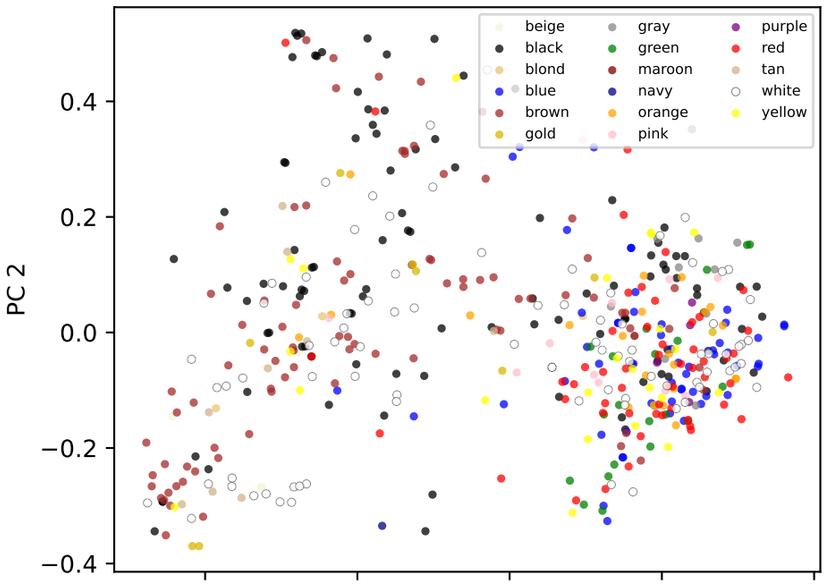
Centroid cosine distances



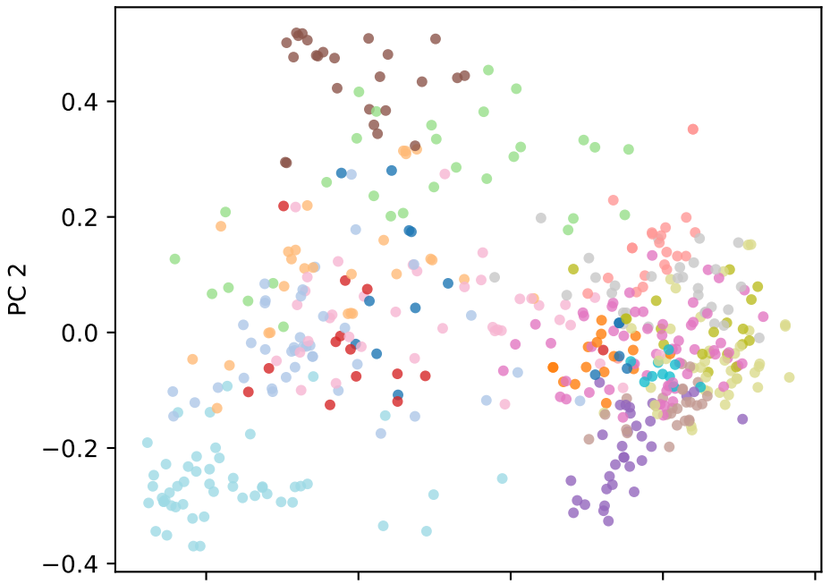

pooled_single · pretrained_conditioned: seed stability at K=2
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

pooled_single · finetuned_unconditioned · K=17, seed=4; PCA explains 11.9%

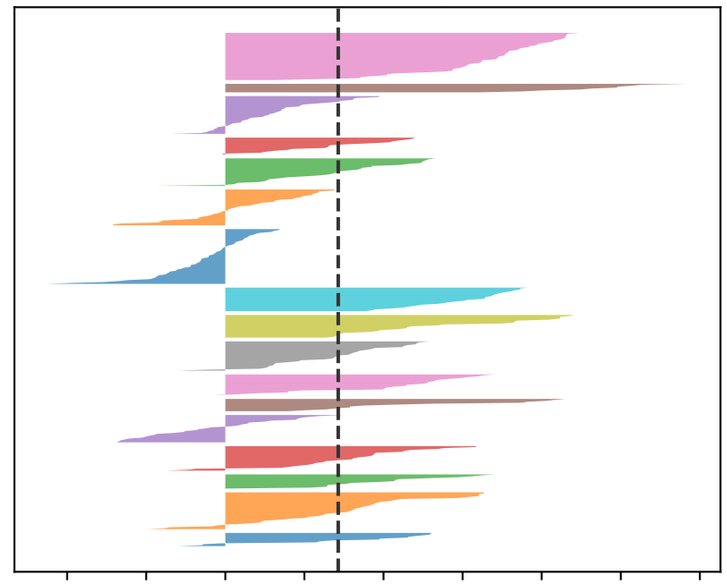
PCA: source color labels



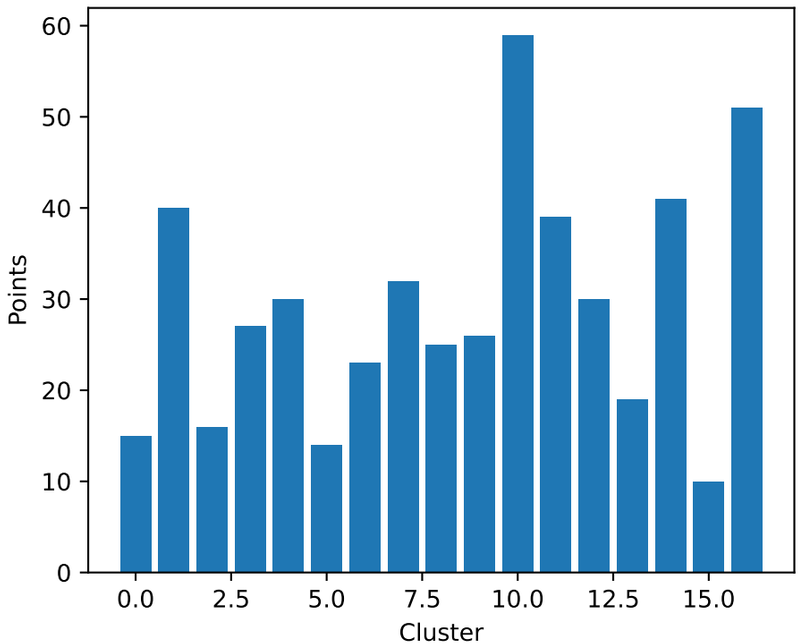
PCA: fitted cluster assignments



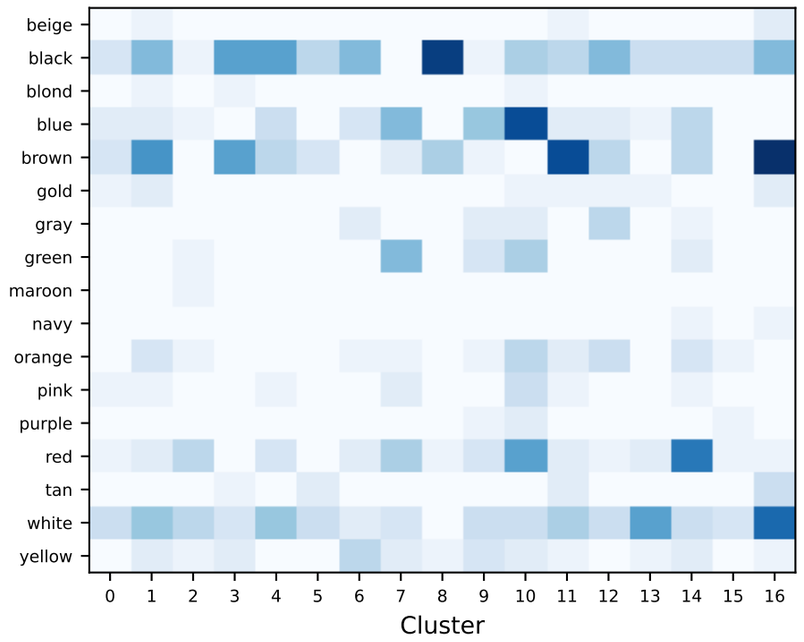
Cosine silhouette by cluster



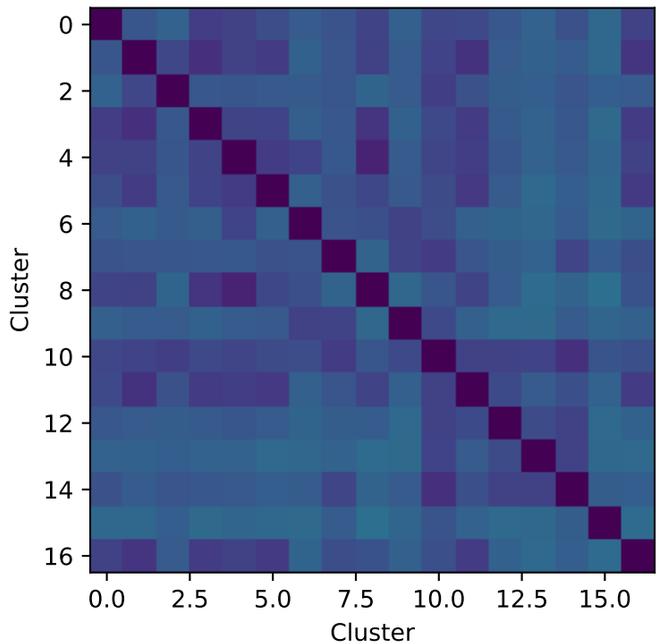
Cluster sizes



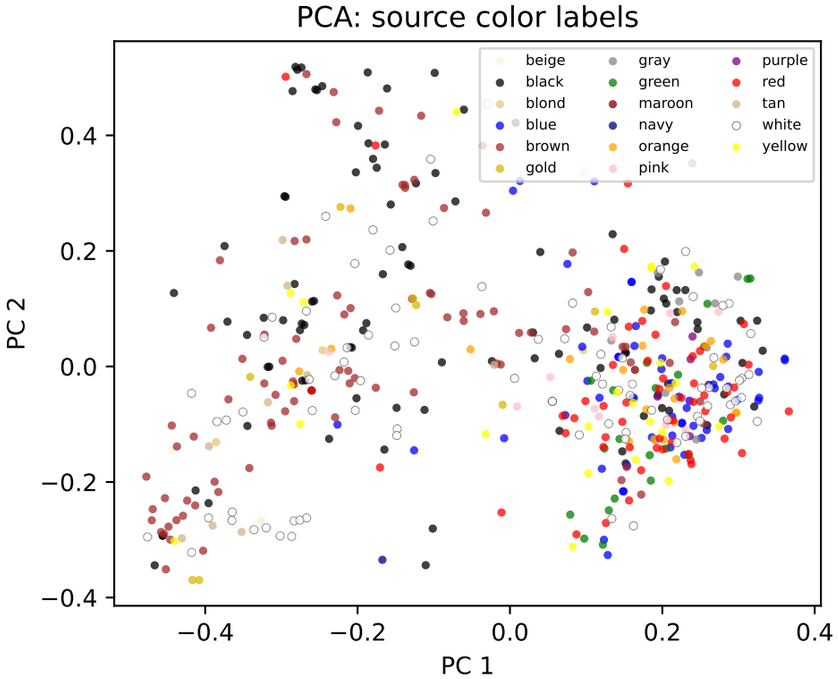
Source-color / cluster counts

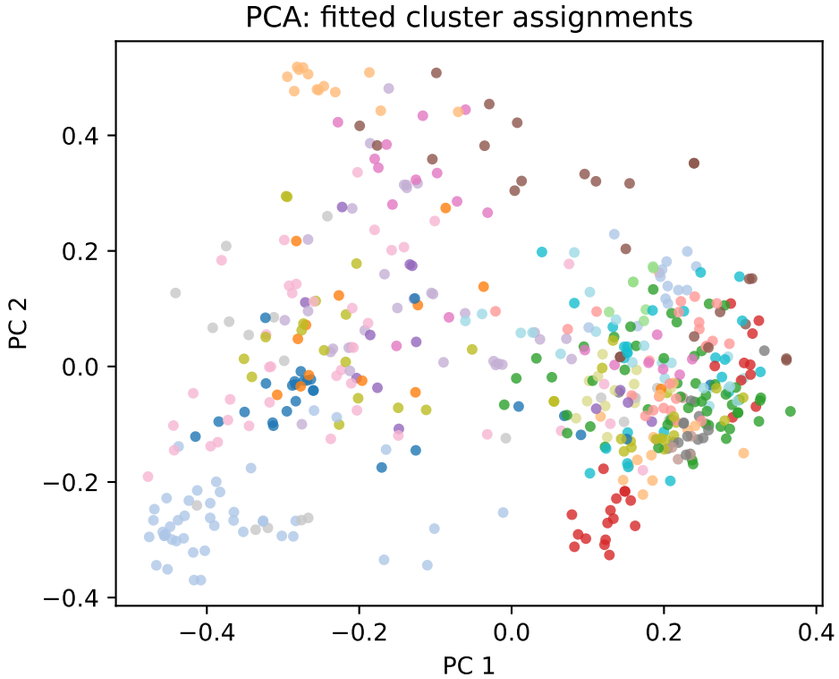


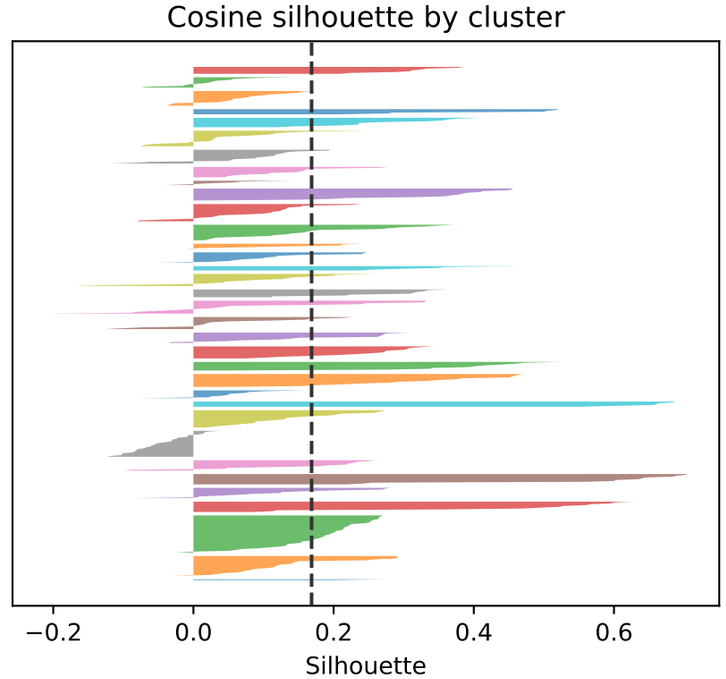
Centroid cosine distances

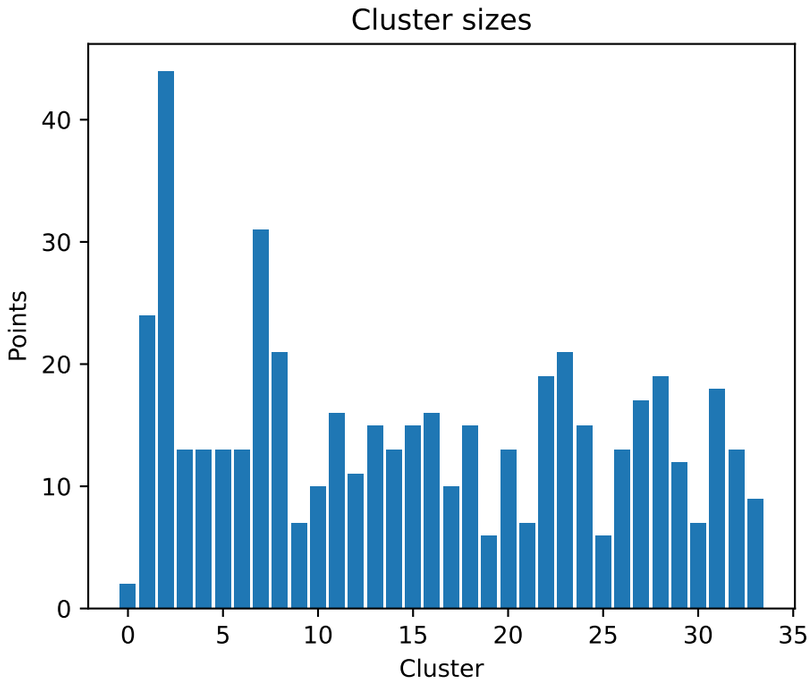


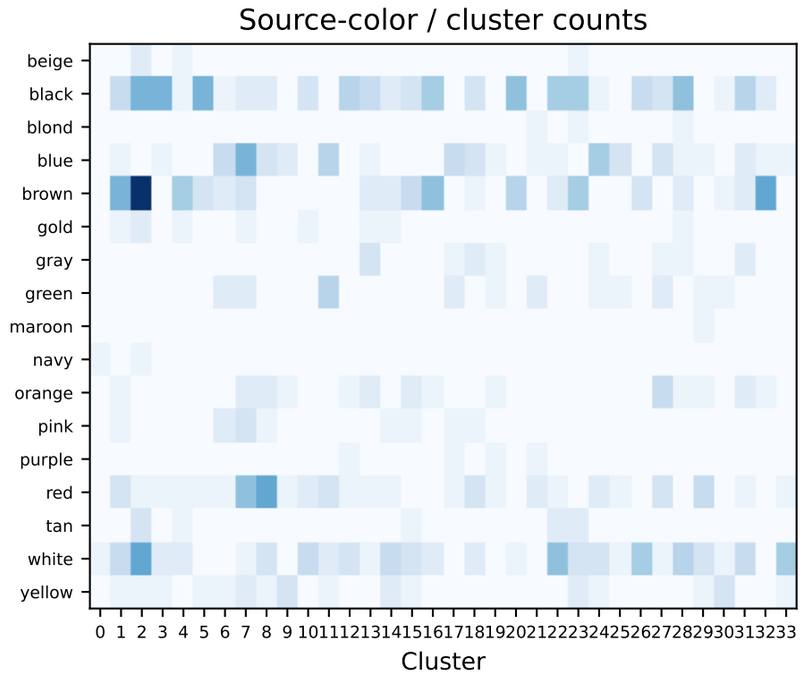
pooled_single · finetuned_unconditioned · K=34, seed=2; PCA explains 11.9%

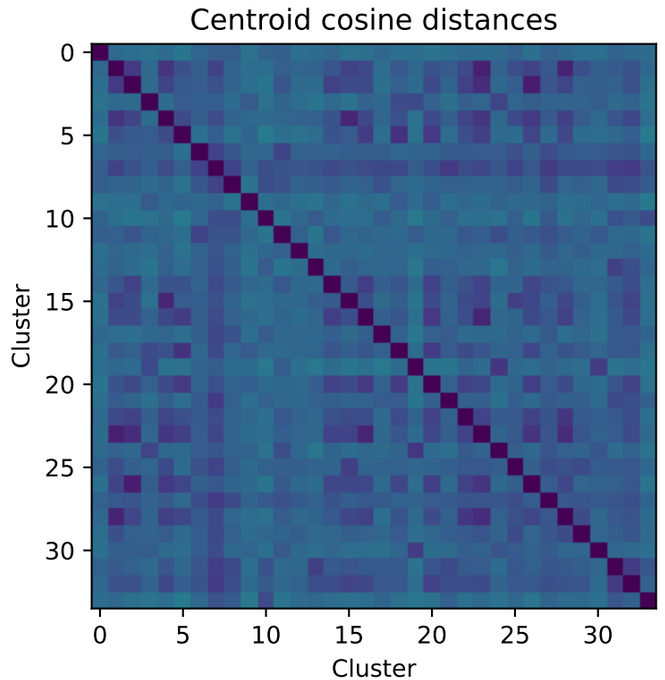


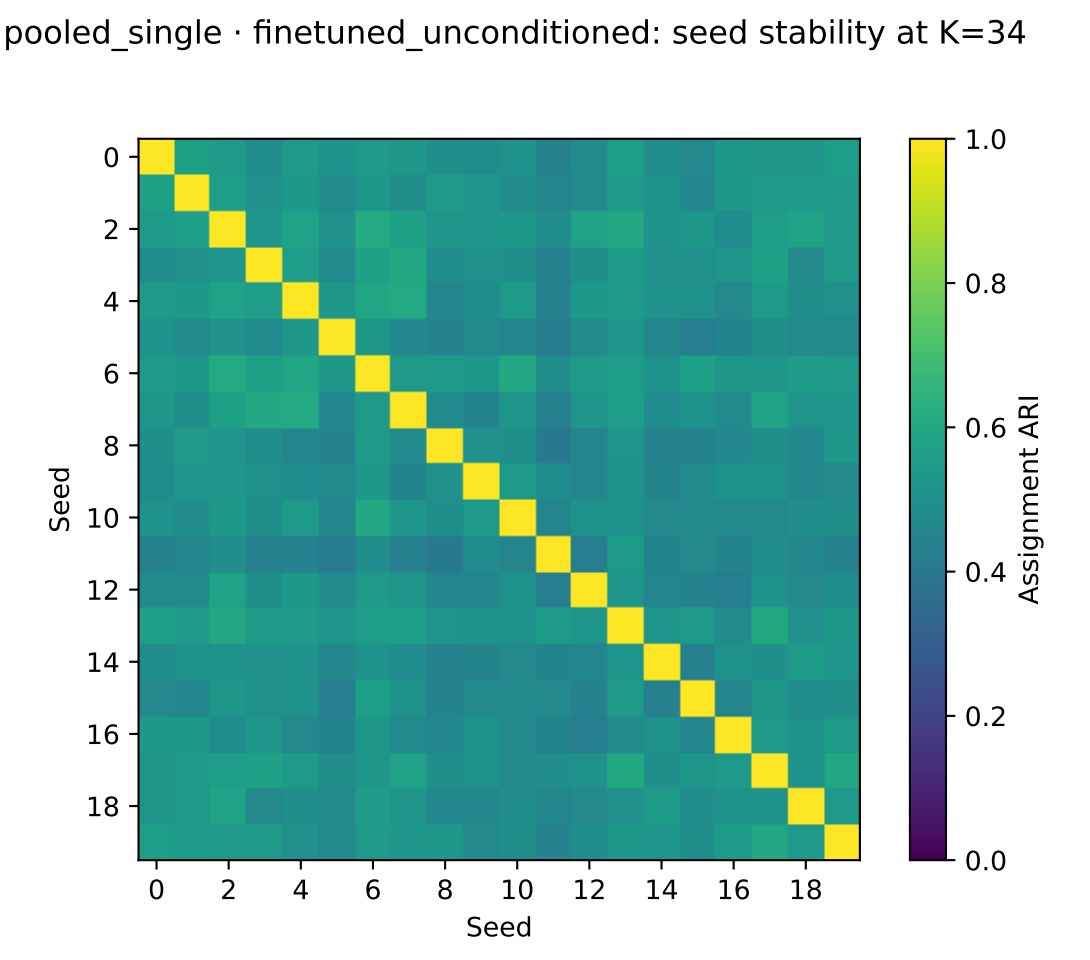


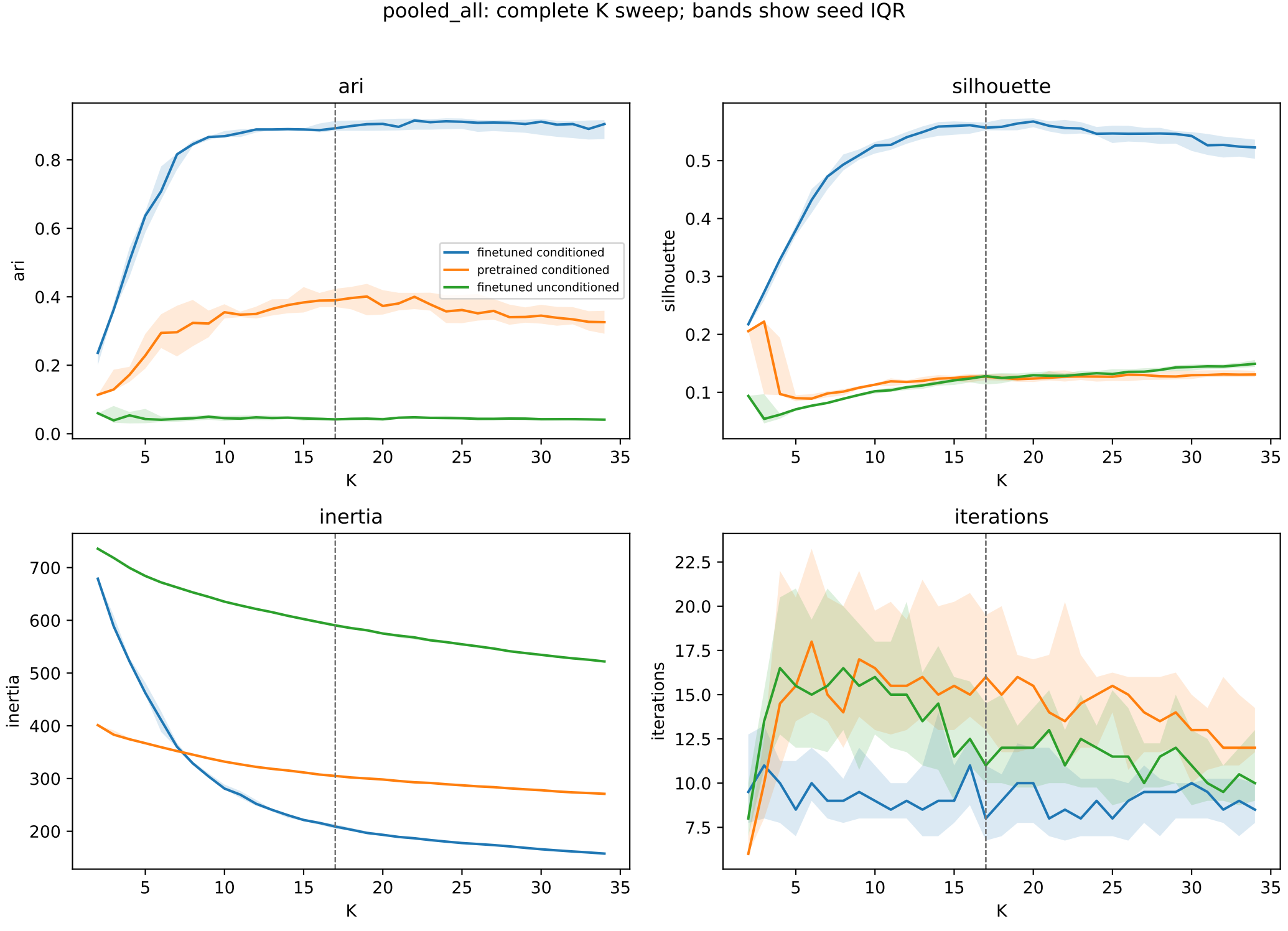






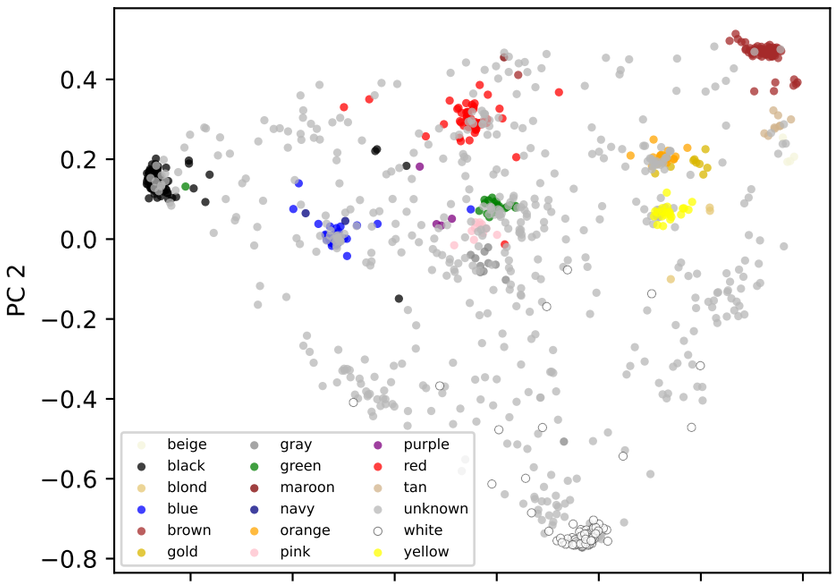



pooled_single · finetuned_unconditioned: seed stability at K=34
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

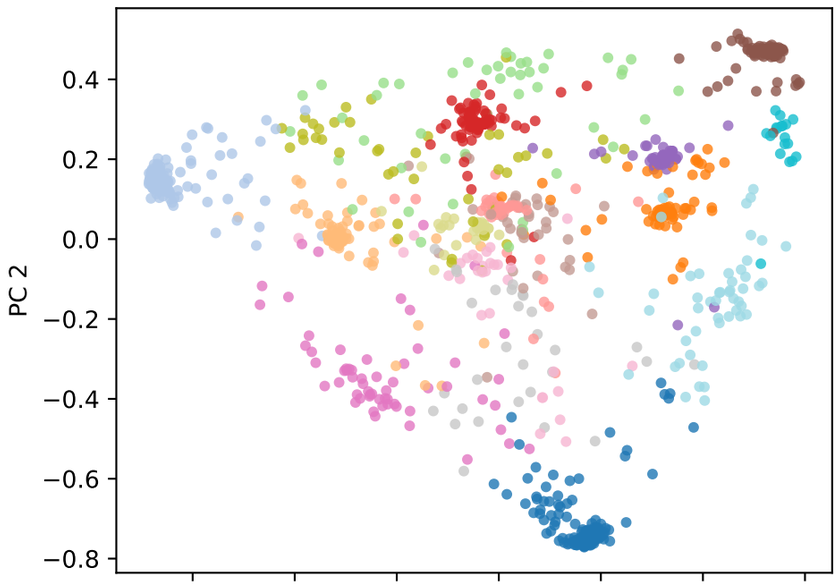

pooled_all: complete K sweep; bands show seed IQR
ari
silhouette
inertia
iterations
finetuned conditioned
pretrained conditioned
finetuned unconditioned
K

pooled_all · finetuned_conditioned · K=17, seed=4; PCA explains 32.1%

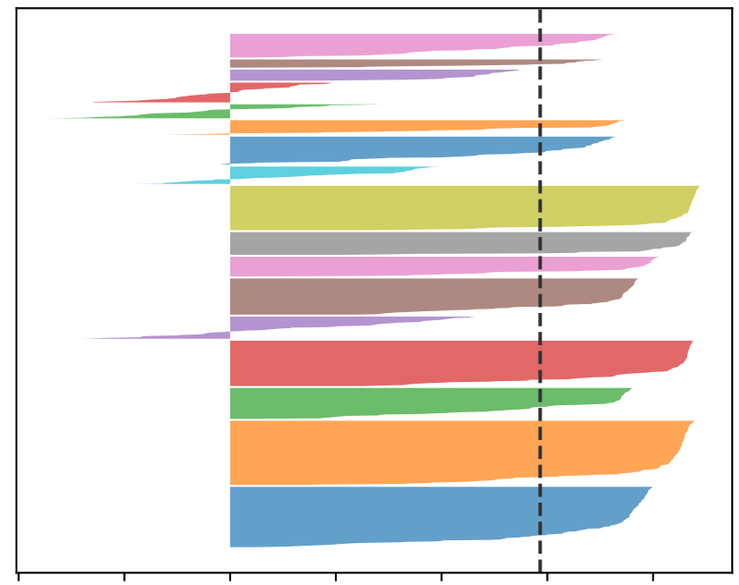
PCA: source color labels



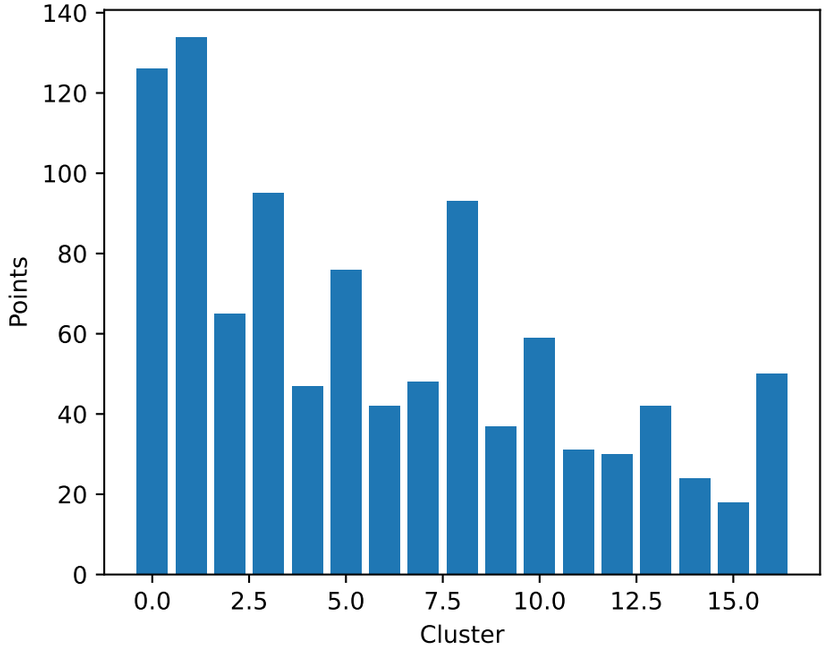
PCA: fitted cluster assignments



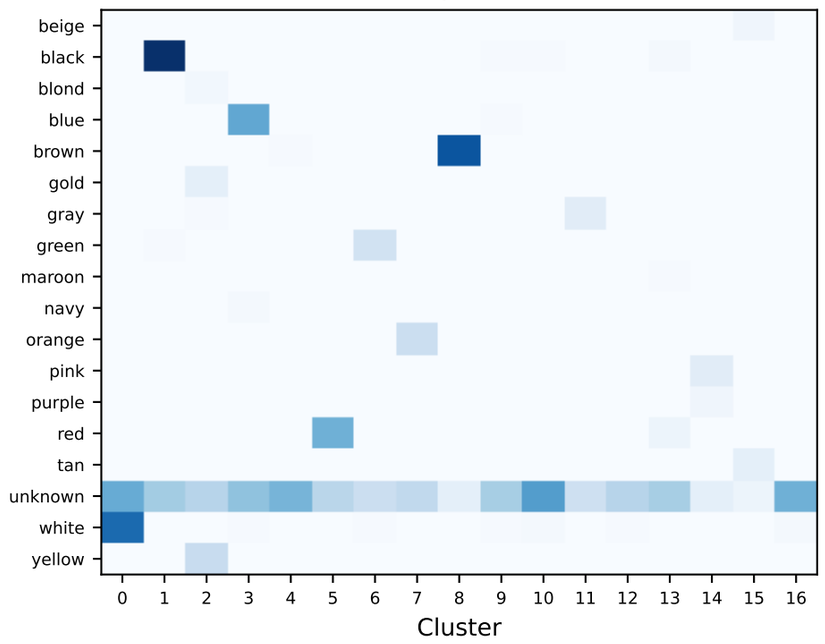
Cosine silhouette by cluster



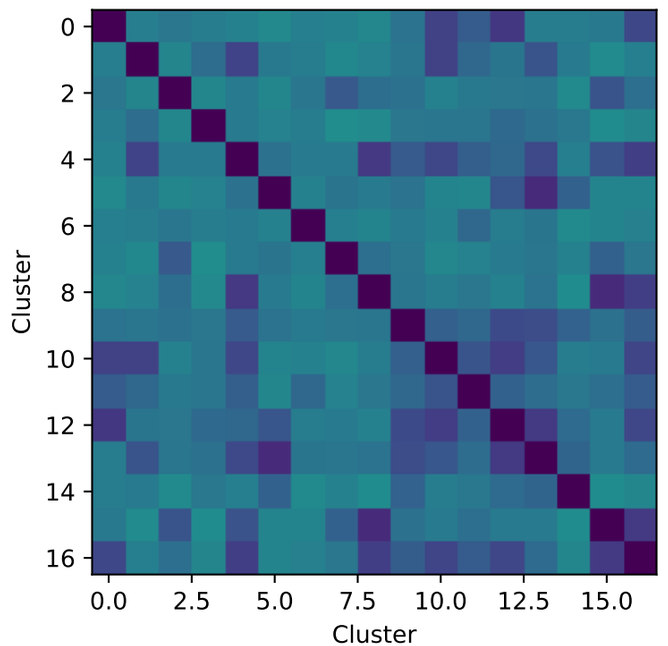
Cluster sizes



Source-color / cluster counts

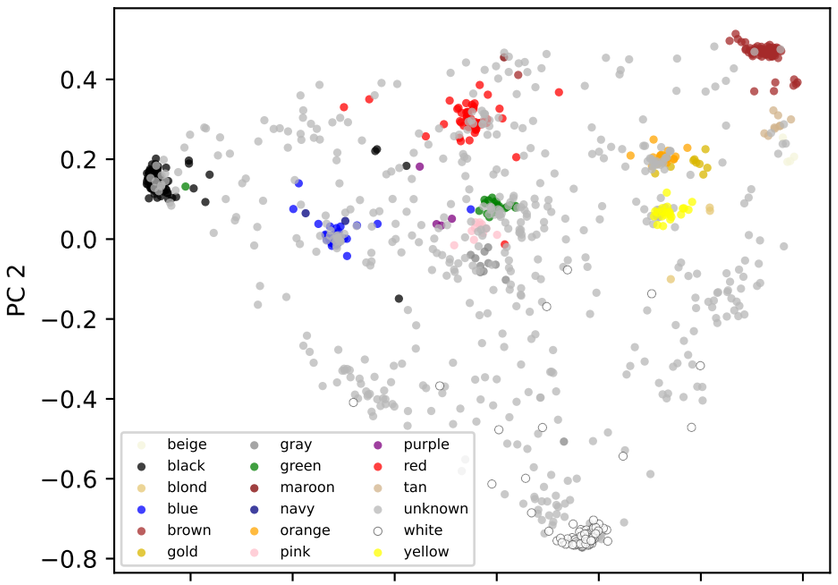


Centroid cosine distances

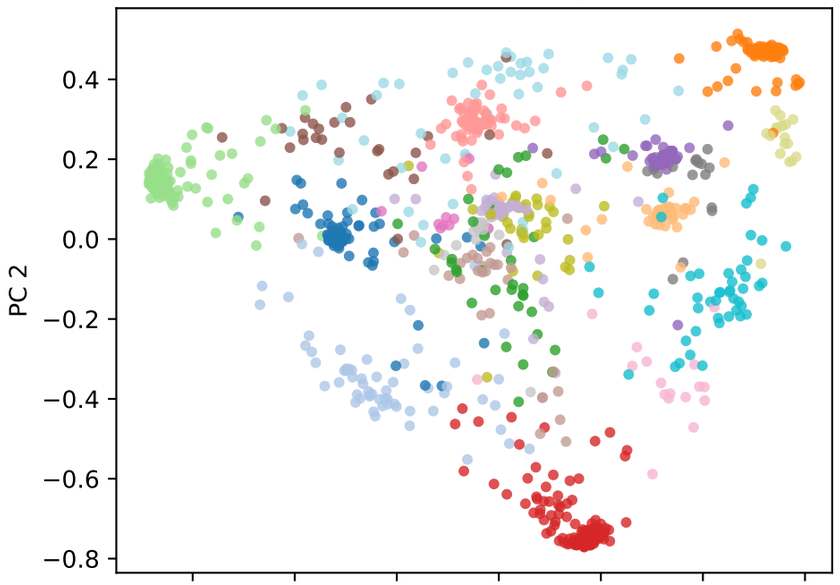


pooled_all · finetuned_conditioned · K=20, seed=0; PCA explains 32.1%

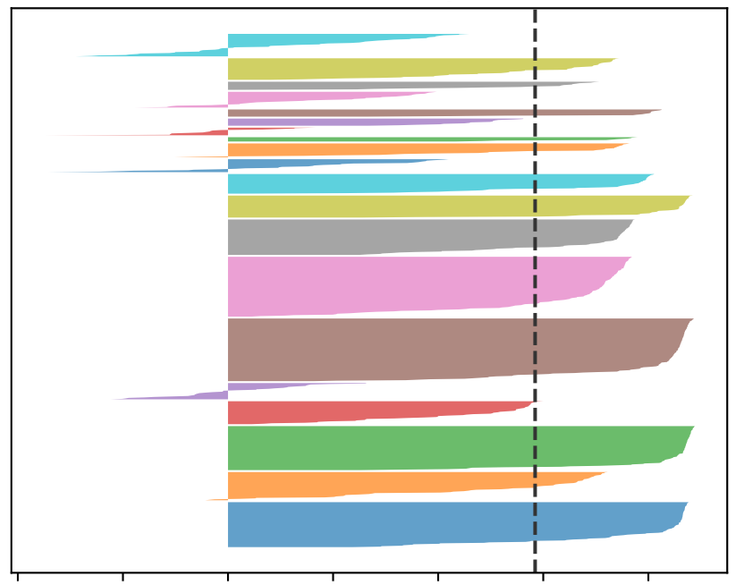
PCA: source color labels



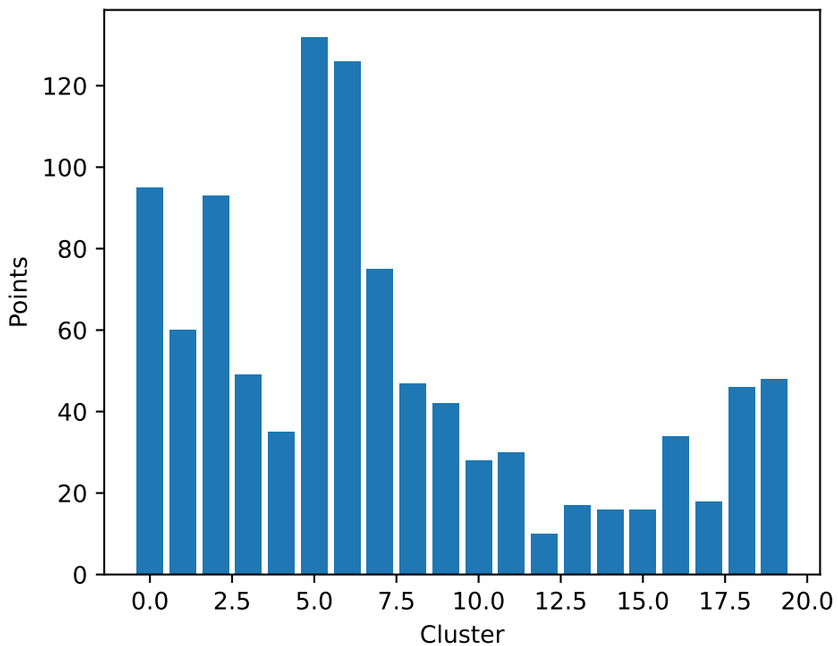
PCA: fitted cluster assignments



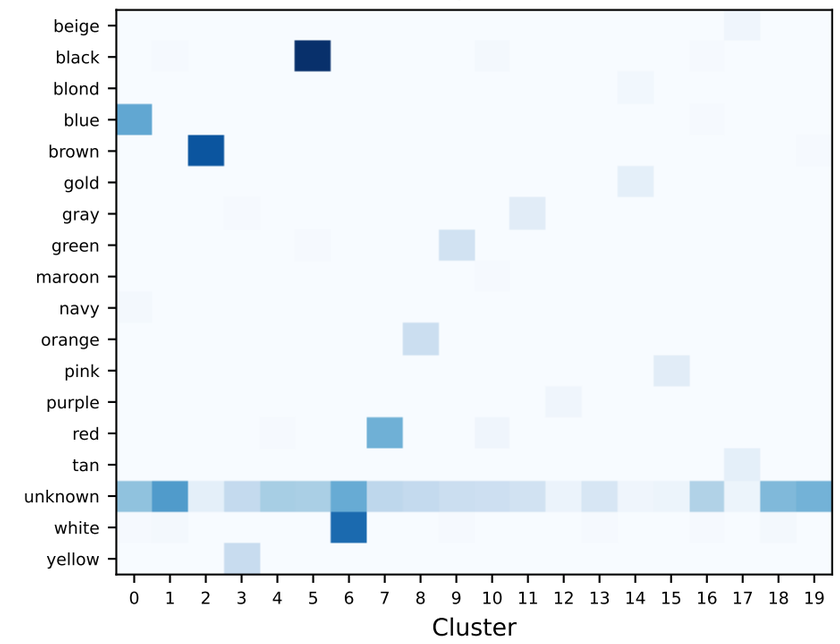
Cosine silhouette by cluster



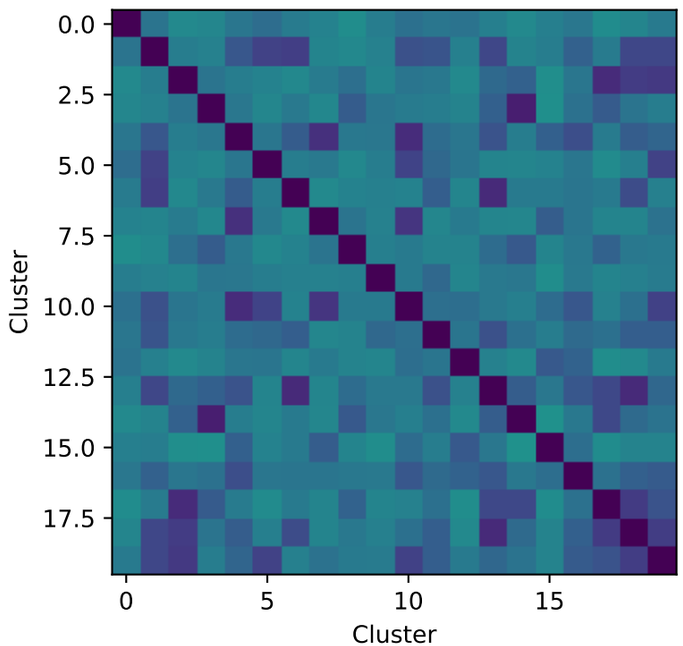
Cluster sizes

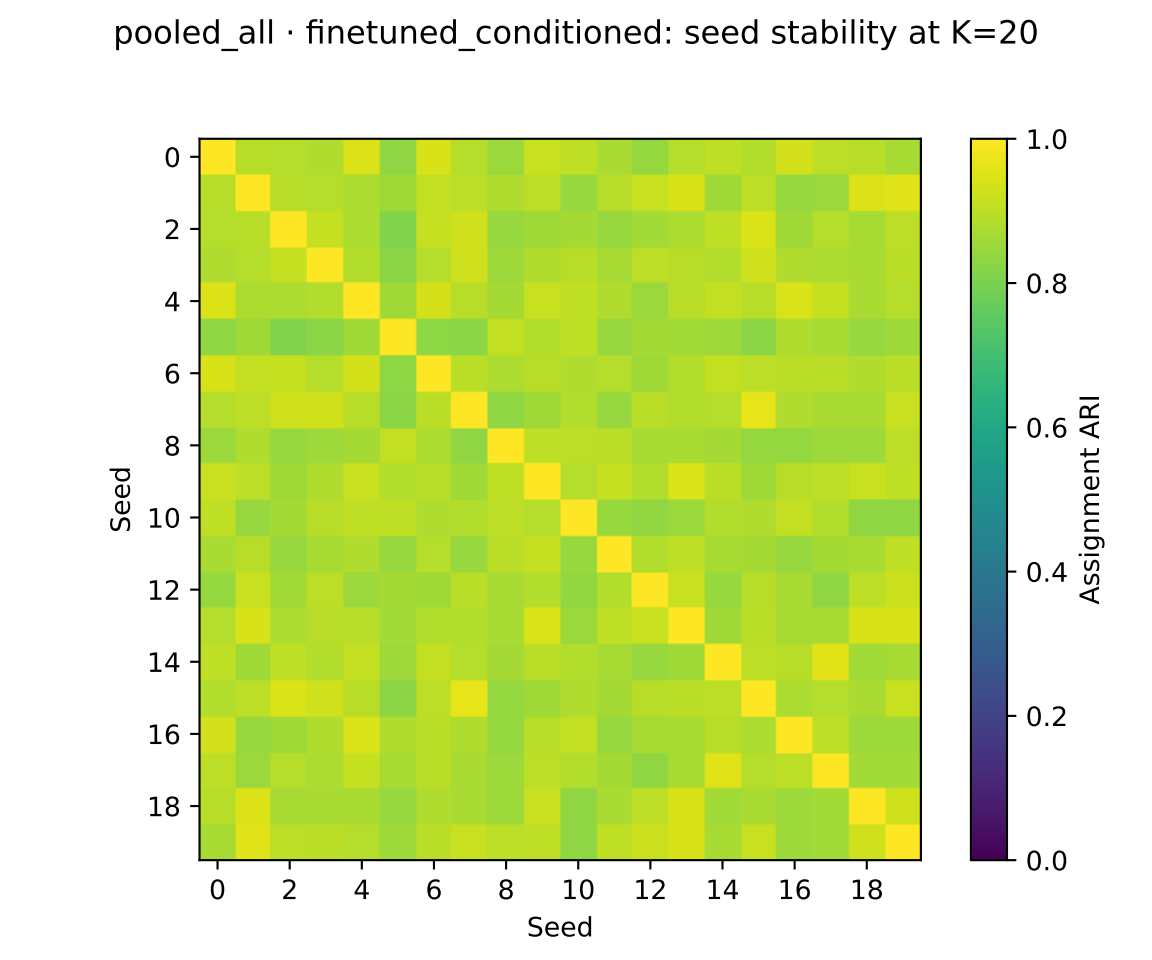


Source-color / cluster counts



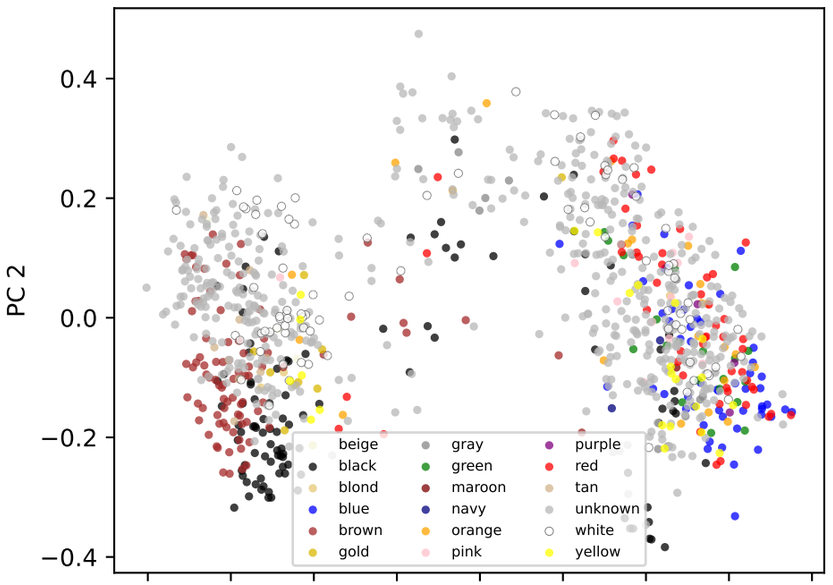
Centroid cosine distances



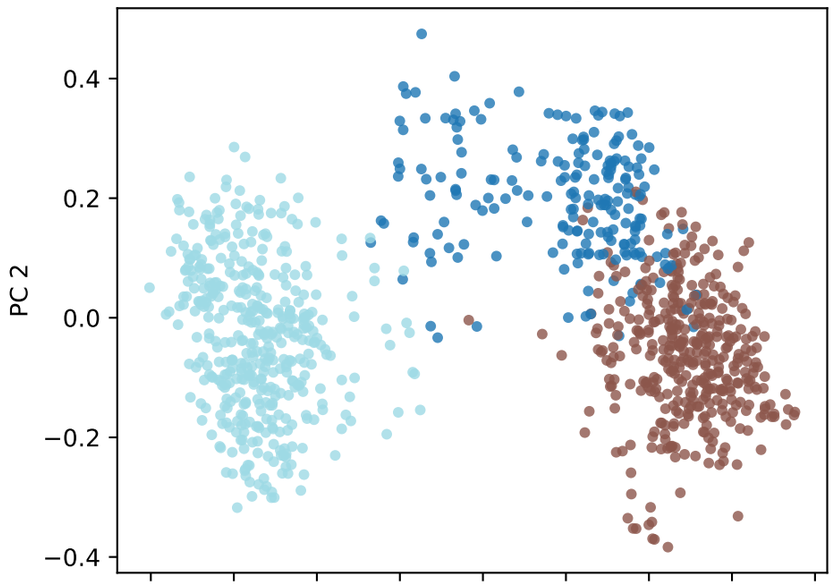

pooled_all · finetuned_conditioned: seed stability at K=20
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

pooled_all · pretrained_conditioned · K=3, seed=8; PCA explains 18.4%

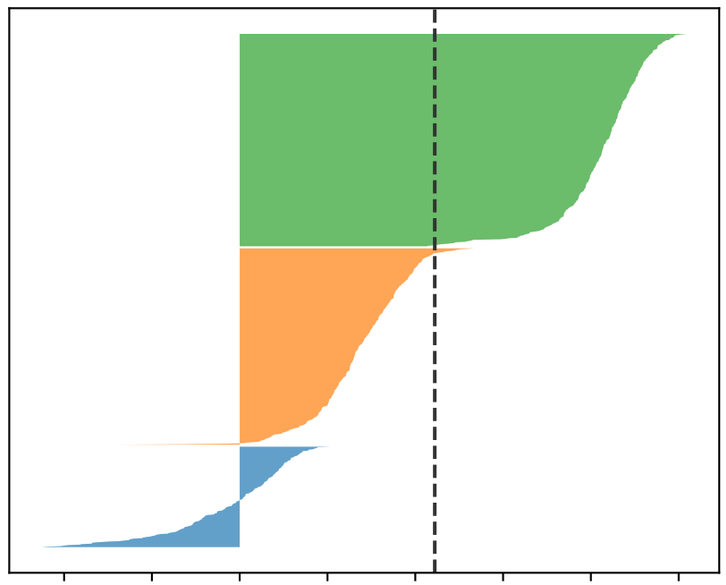
PCA: source color labels



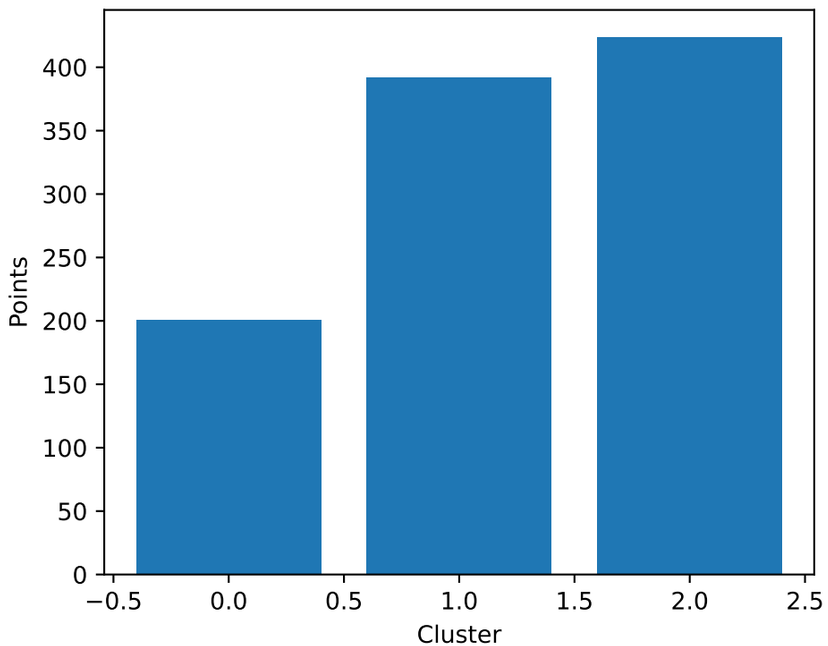
PCA: fitted cluster assignments



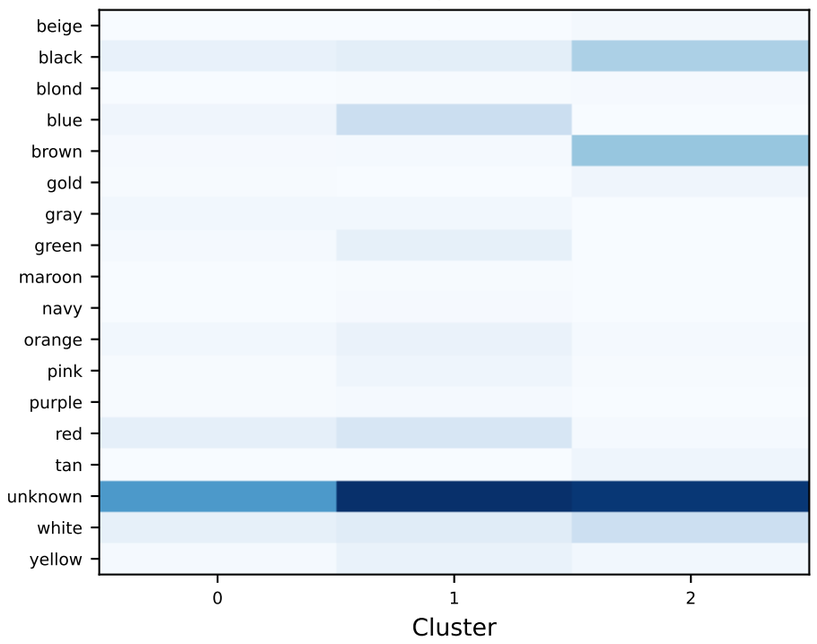
Cosine silhouette by cluster



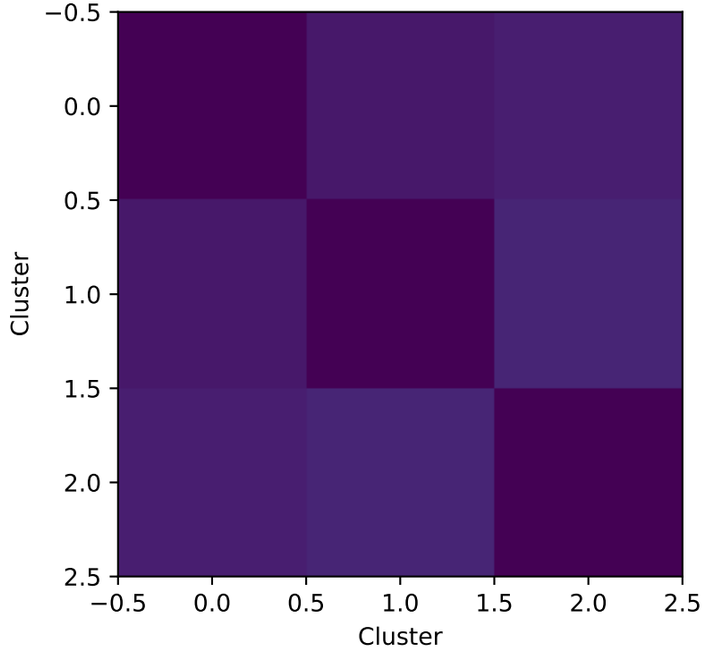
Cluster sizes



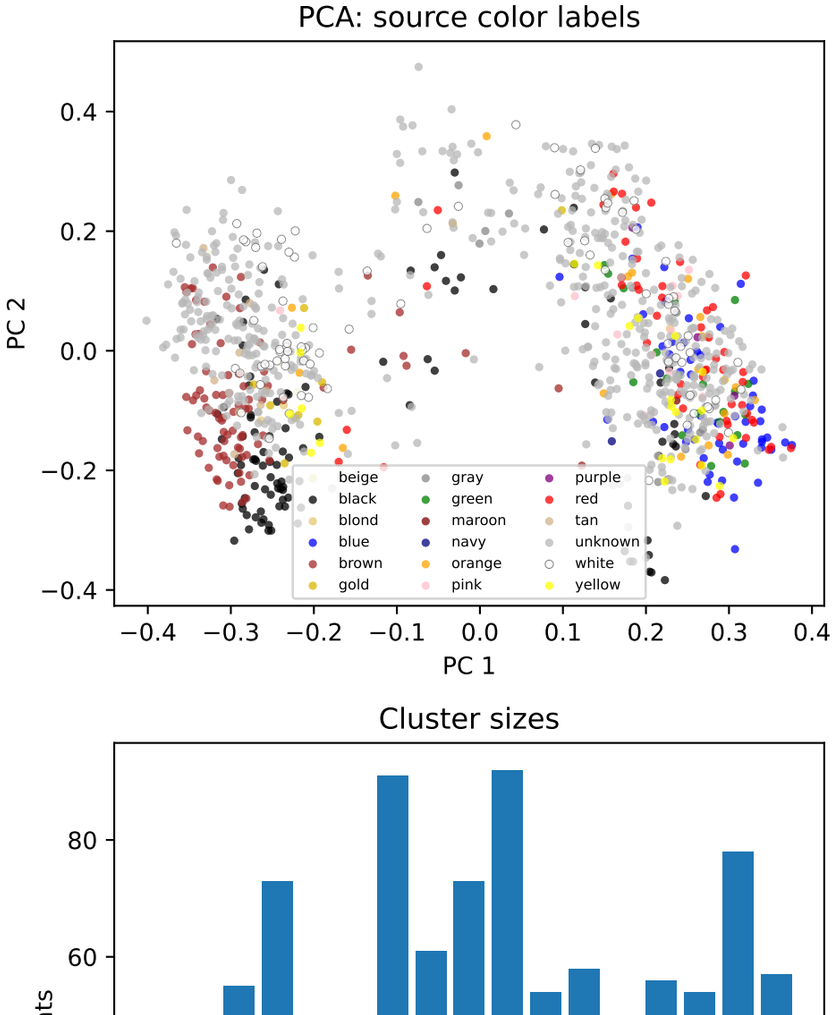
Source-color / cluster counts

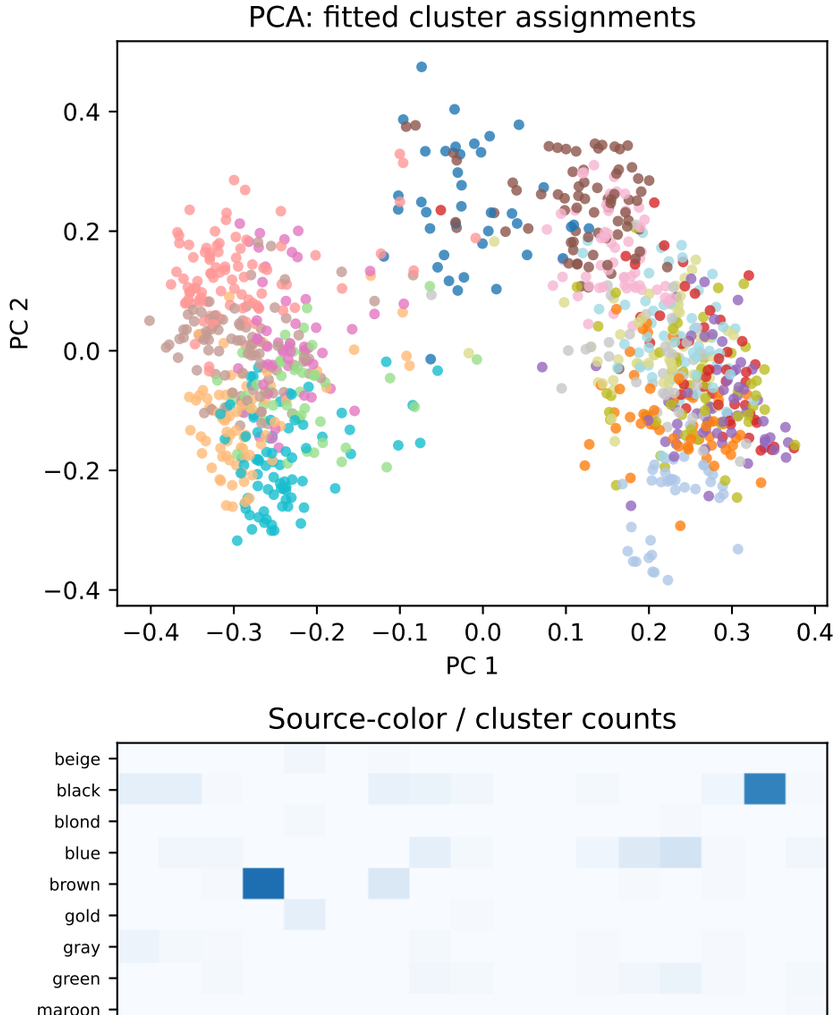


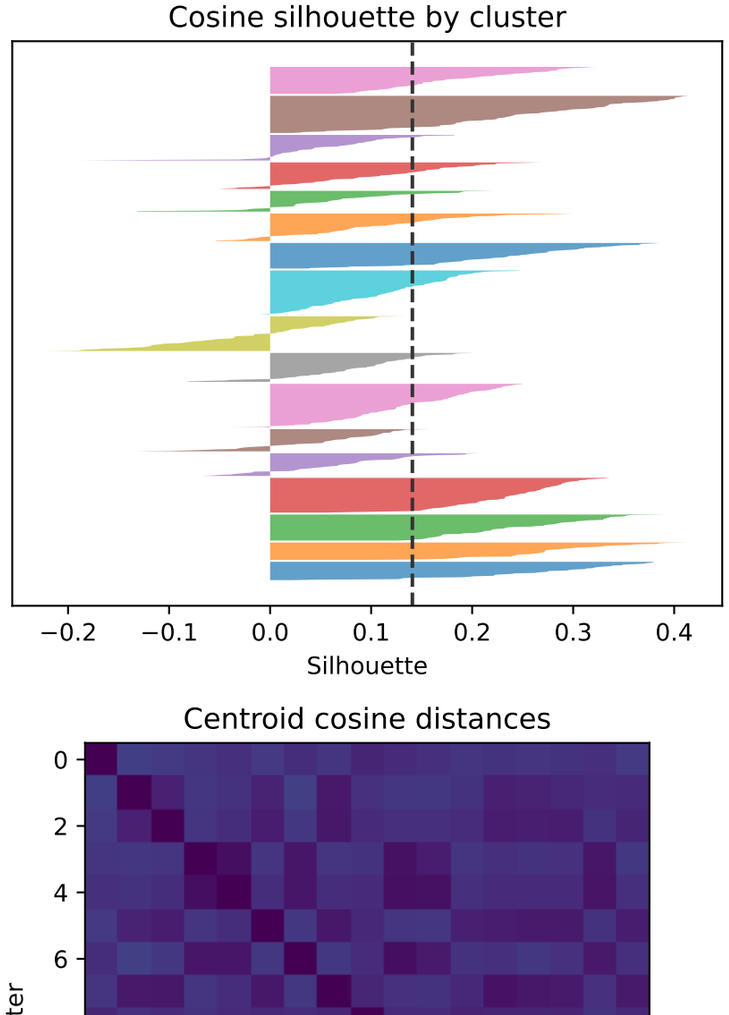
Centroid cosine distances

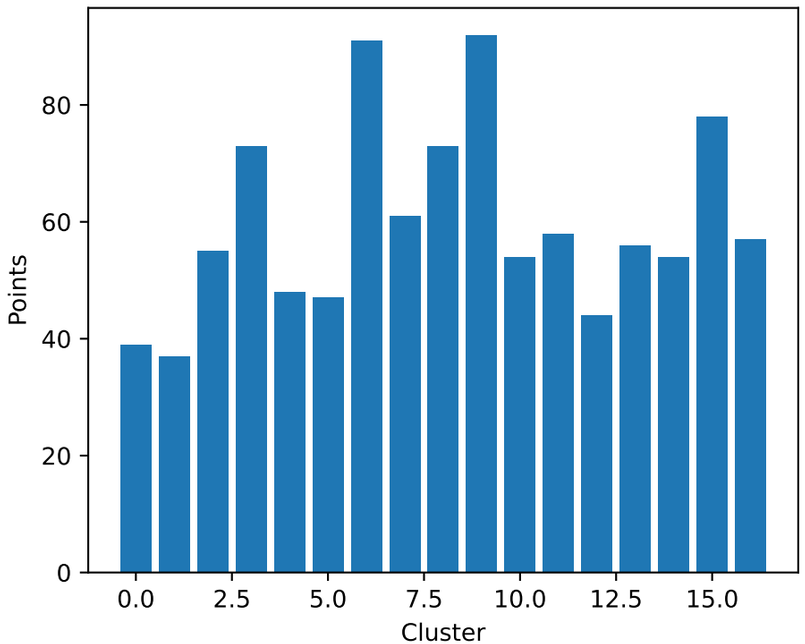


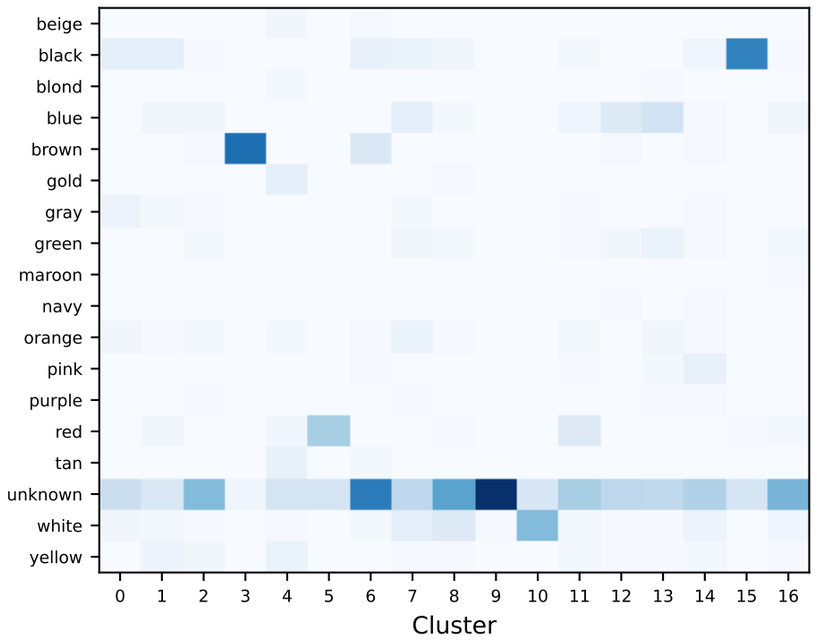
pooled_all · pretrained_conditioned · K=17, seed=9; PCA explains 18.4%

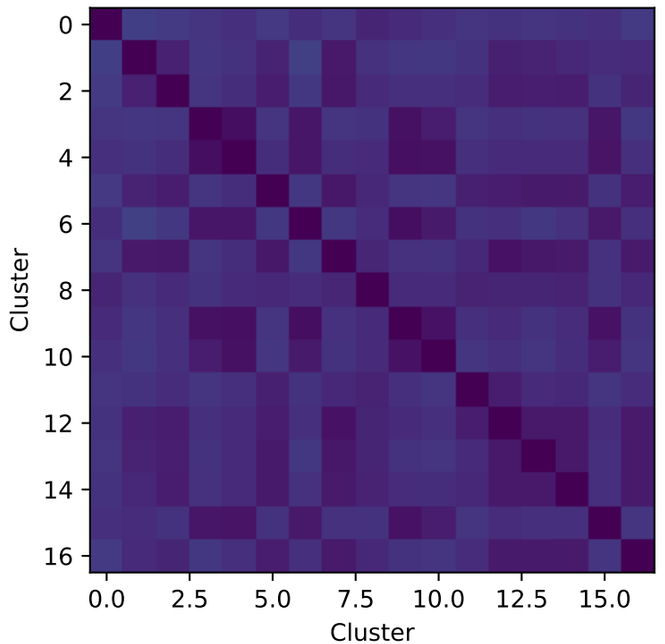


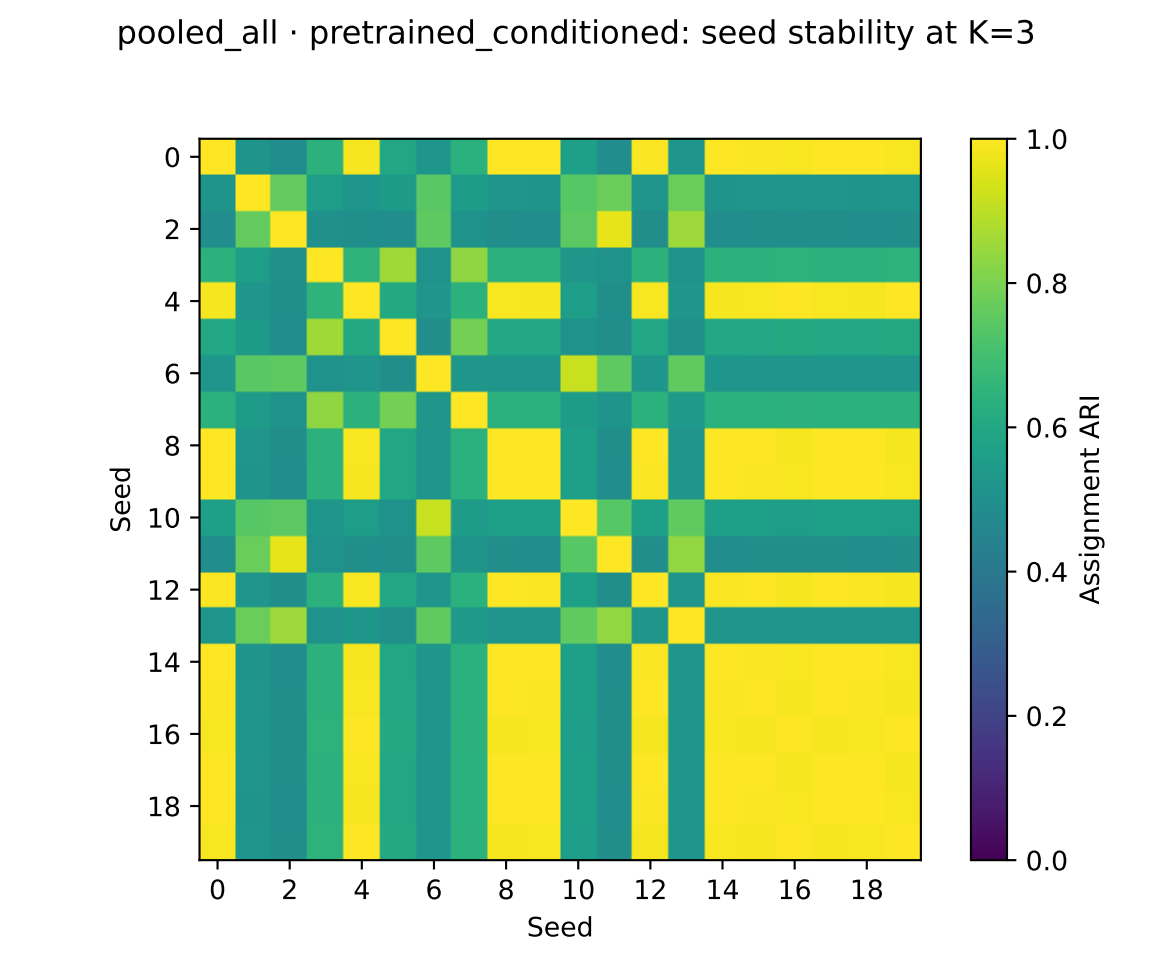






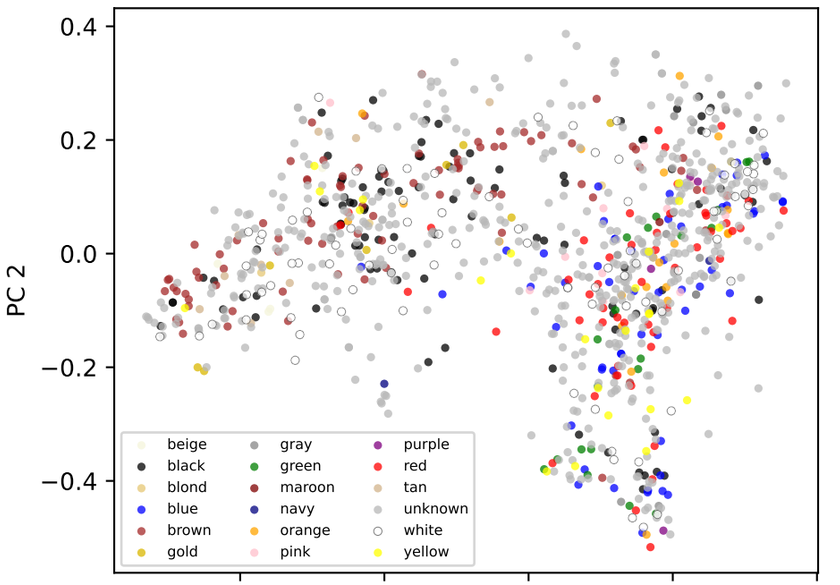




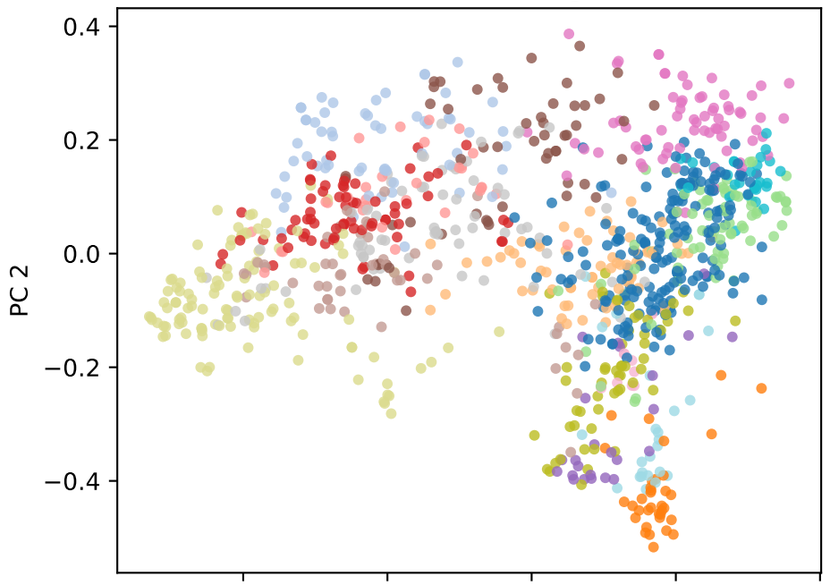

pooled_all · pretrained_conditioned: seed stability at K=3
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

pooled_all · finetuned_unconditioned · K=17, seed=19; PCA explains 11.0%

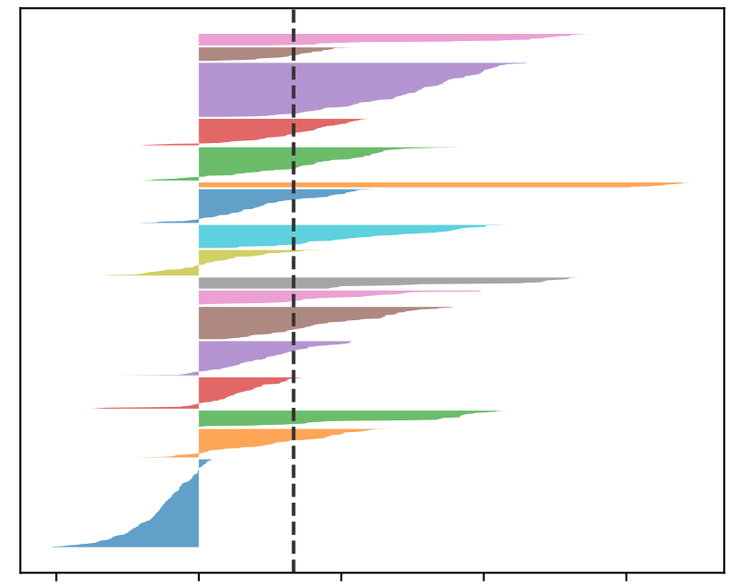
PCA: source color labels



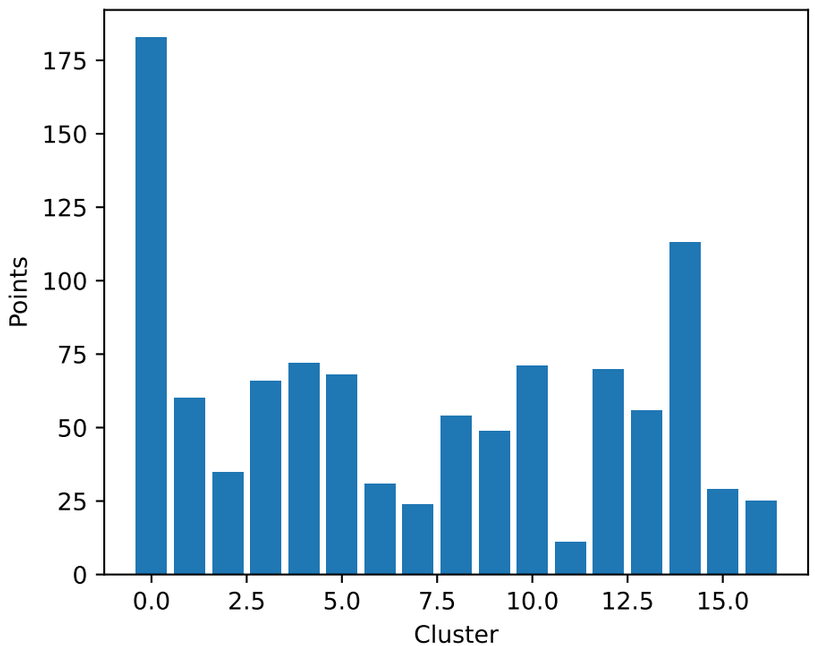
PCA: fitted cluster assignments



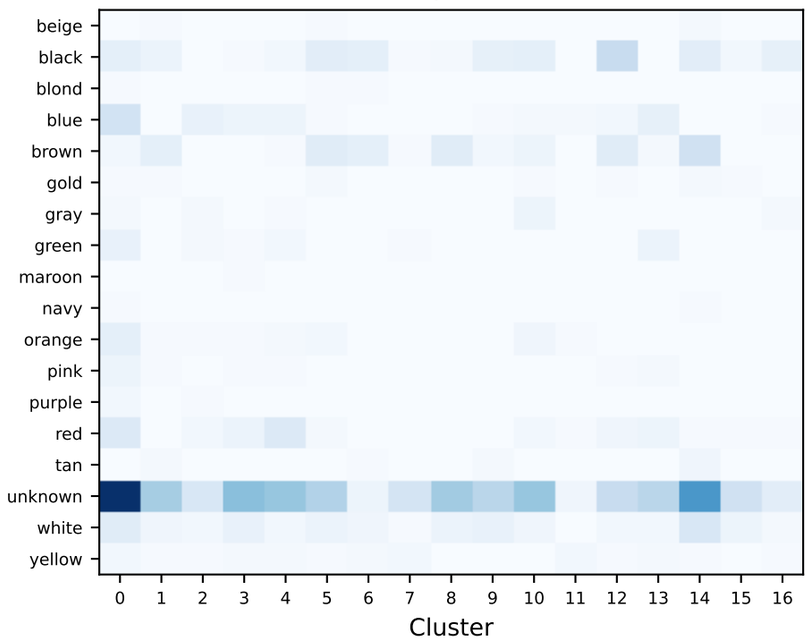
Cosine silhouette by cluster



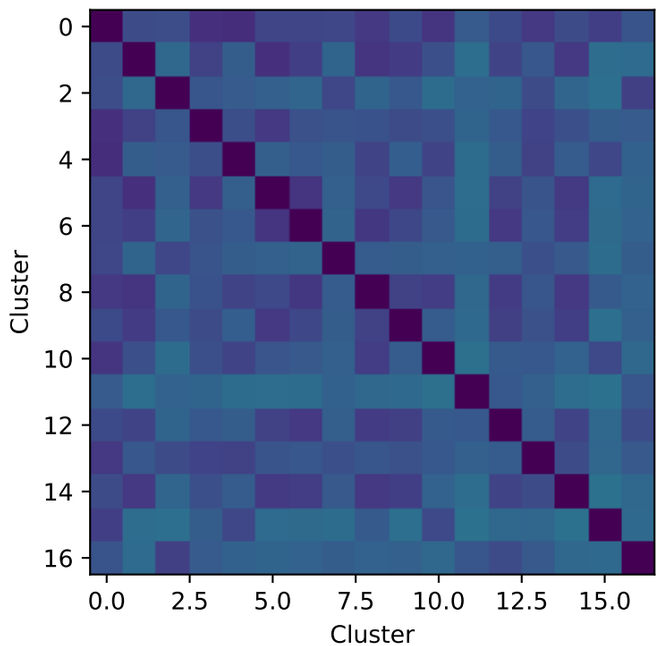
Cluster sizes



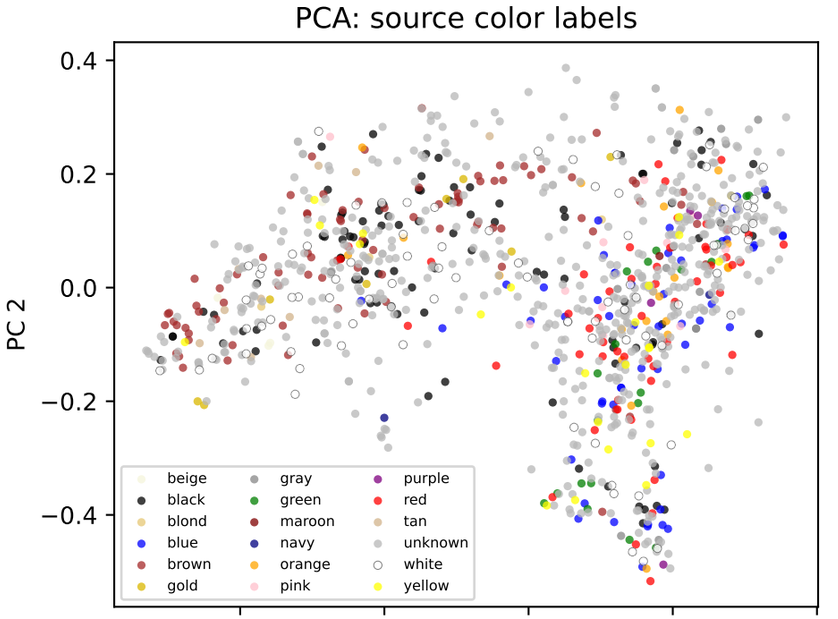
Source-color / cluster counts

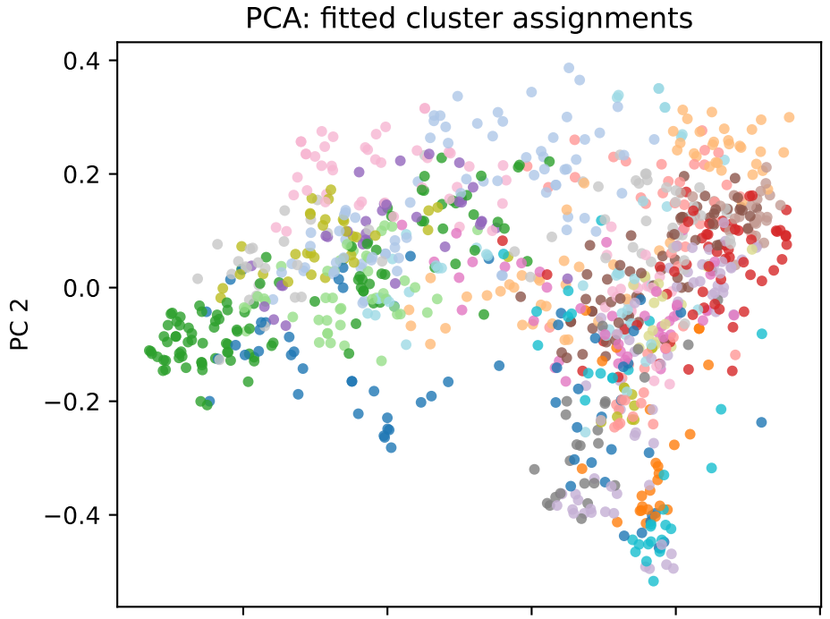


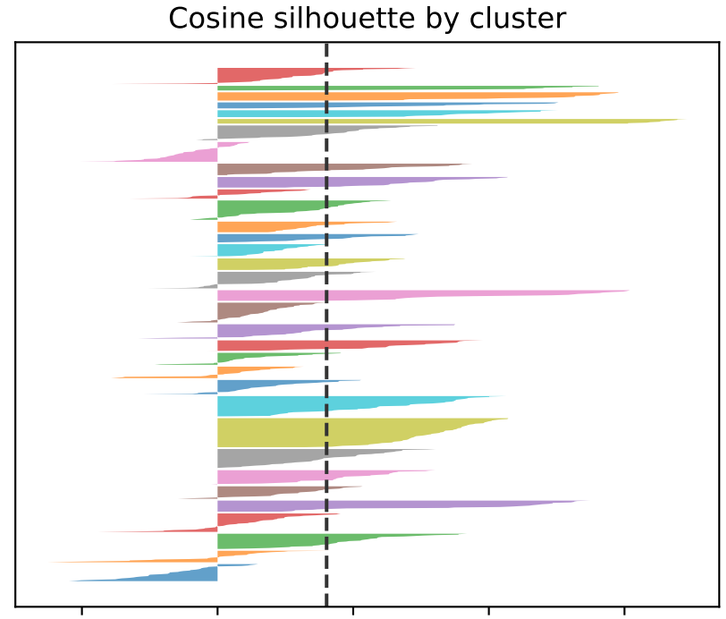
Centroid cosine distances

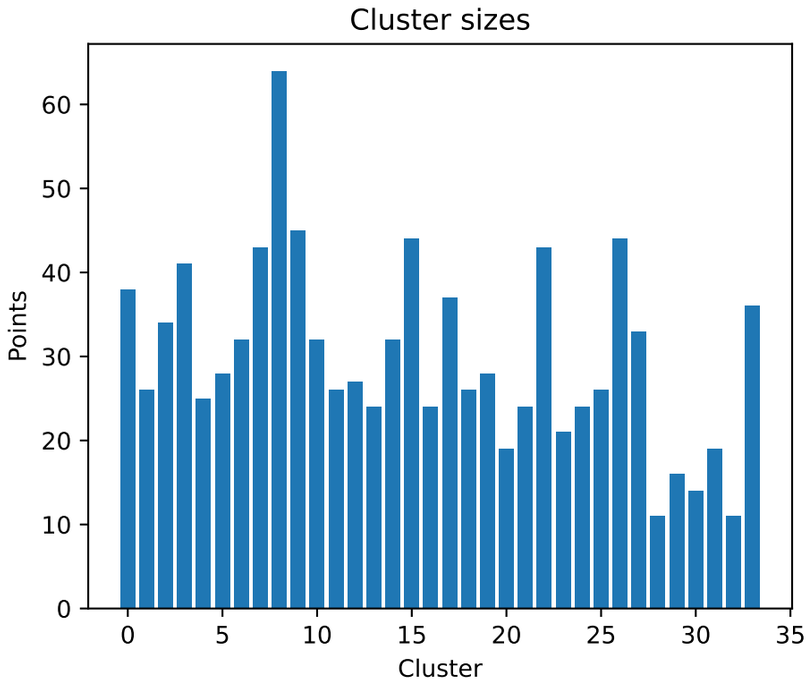


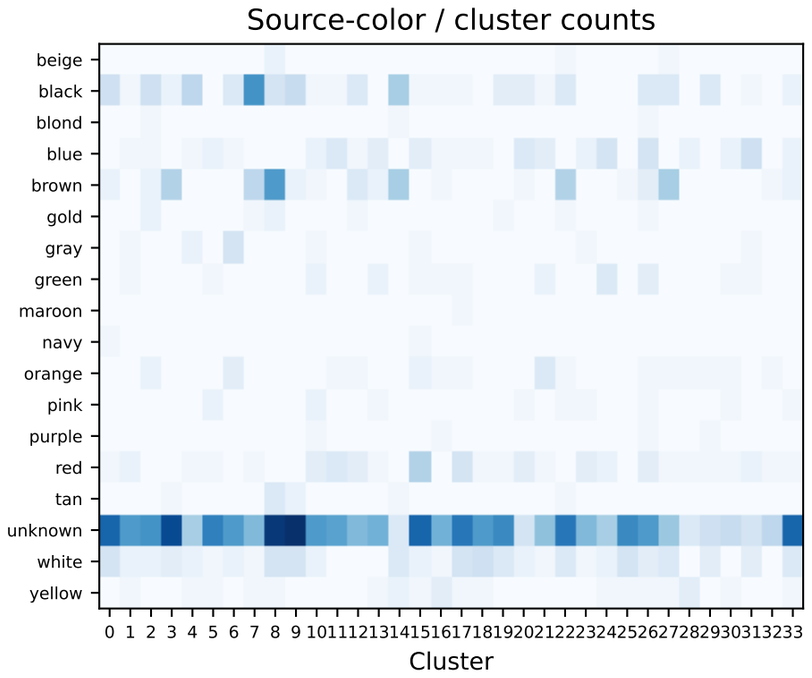
pooled_all · finetuned_unconditioned · K=34, seed=13; PCA explains 11.0%

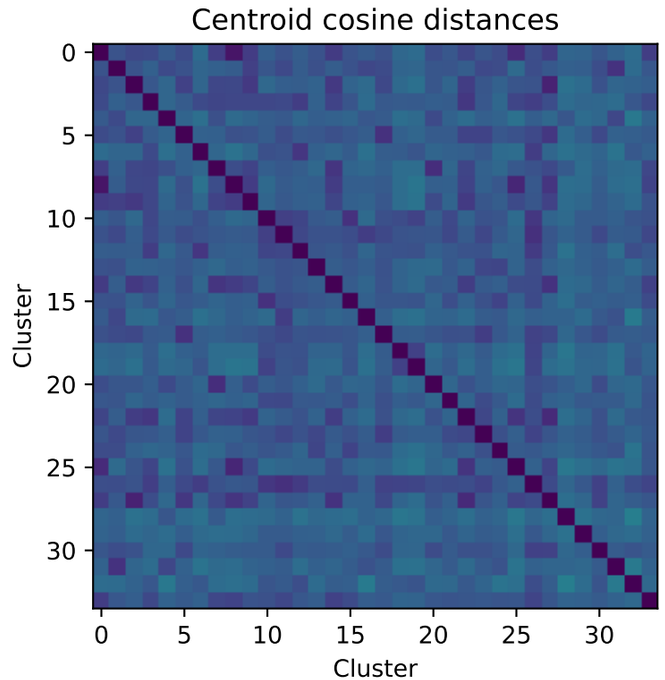


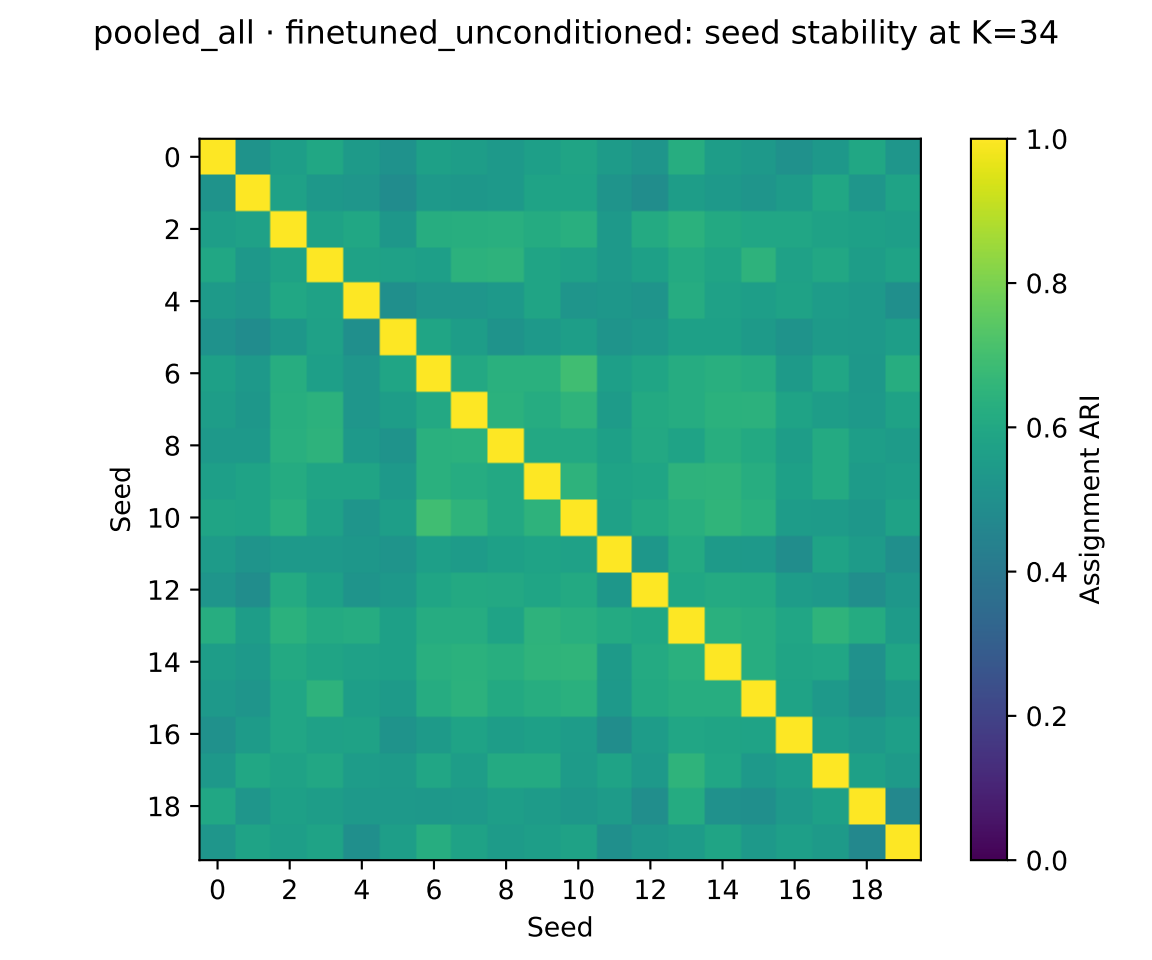


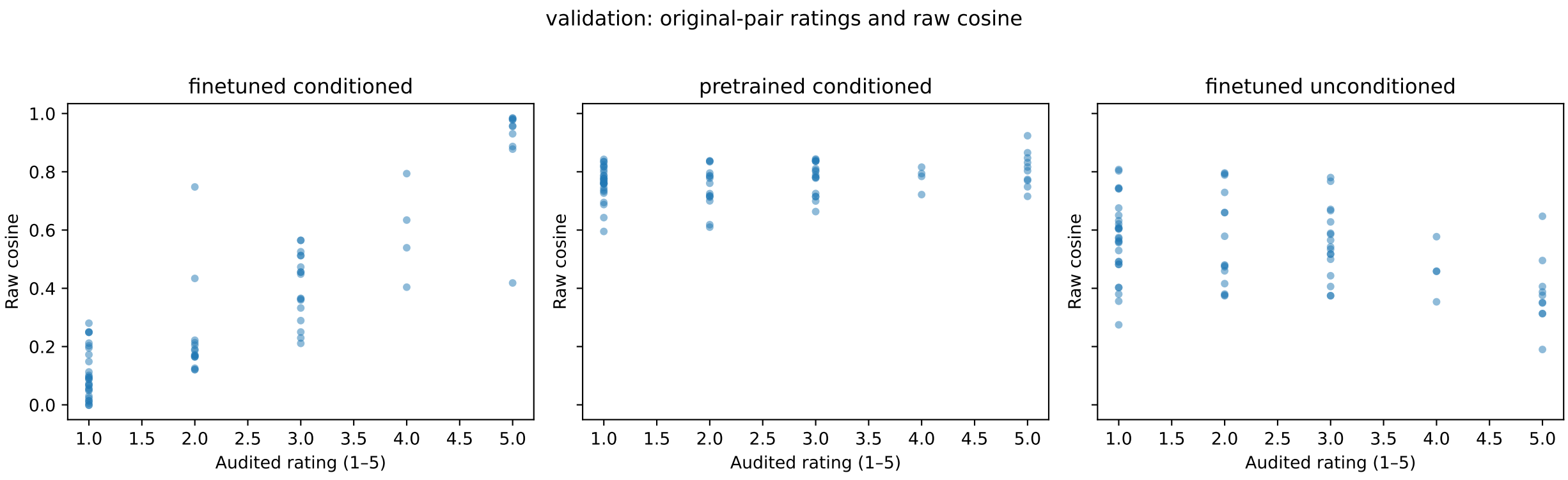


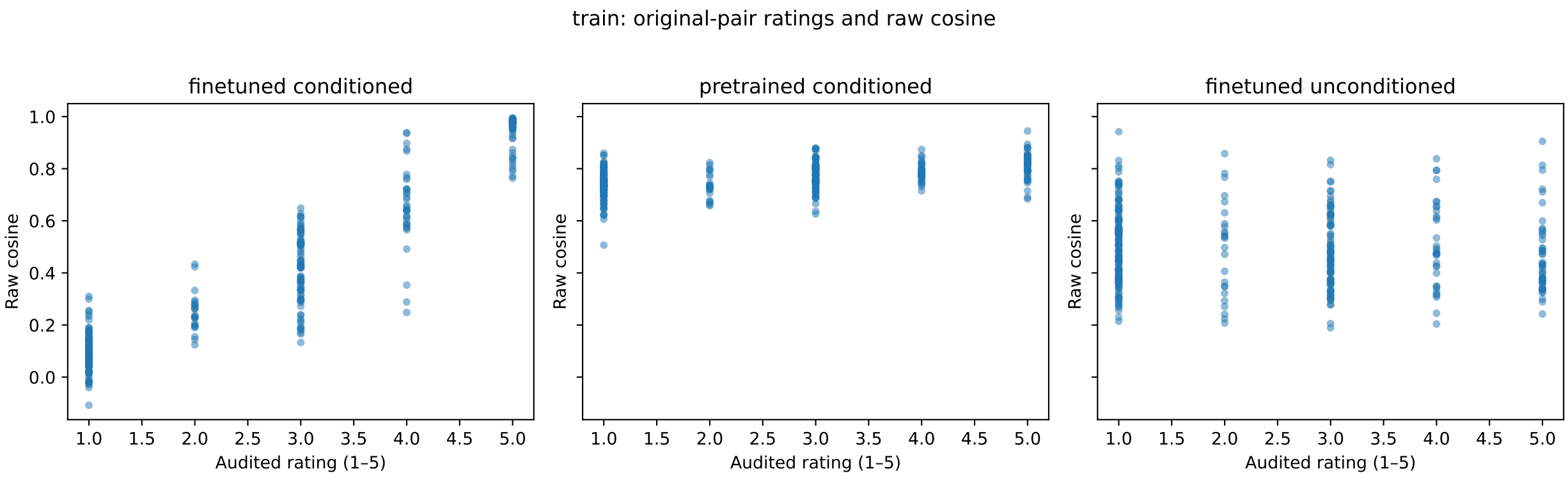


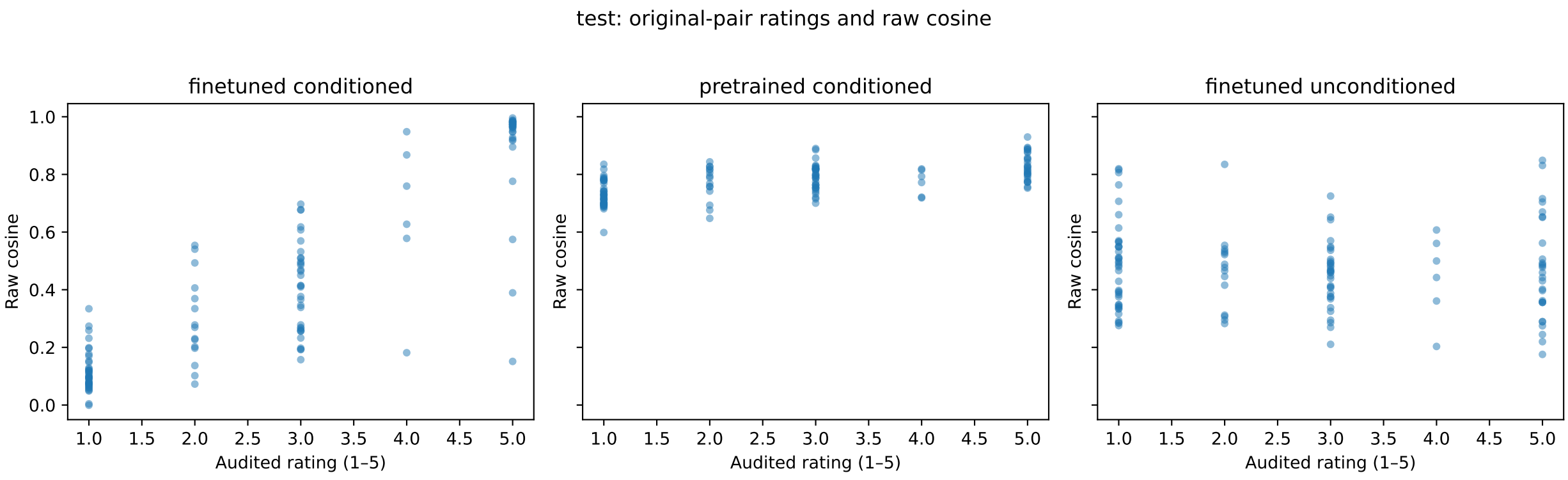


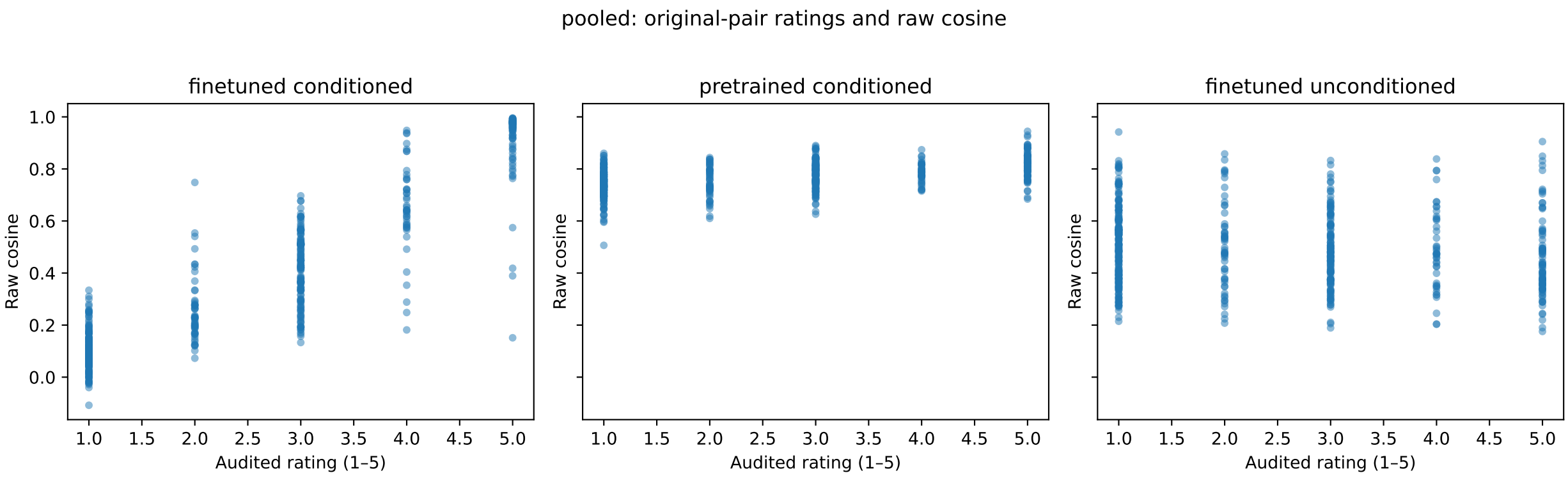


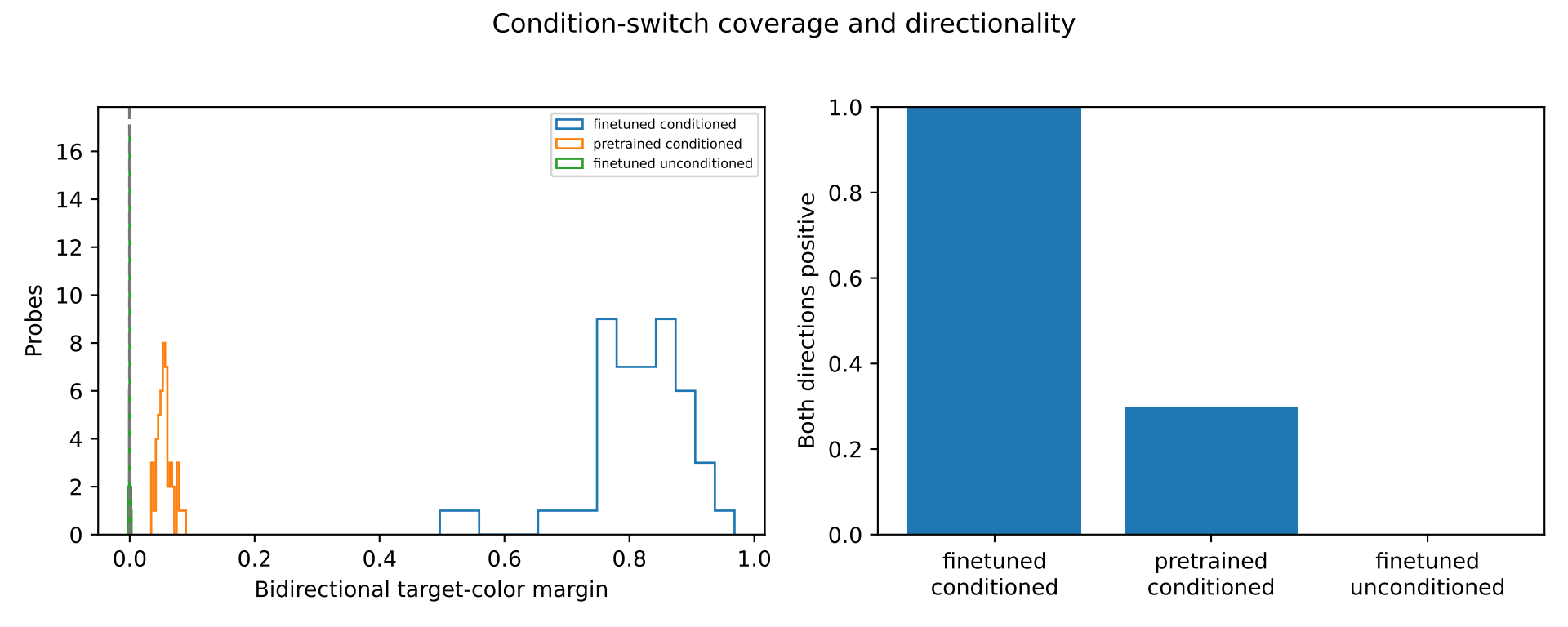

pooled_all · finetuned_unconditioned: seed stability at K=34
Seed
0
2
4
6
8
10
12
14
16
18
Seed
0
2
4
6
8
10
12
14
16
18
Assignment ARI
1.0
0.8
0.6
0.4
0.2
0.0

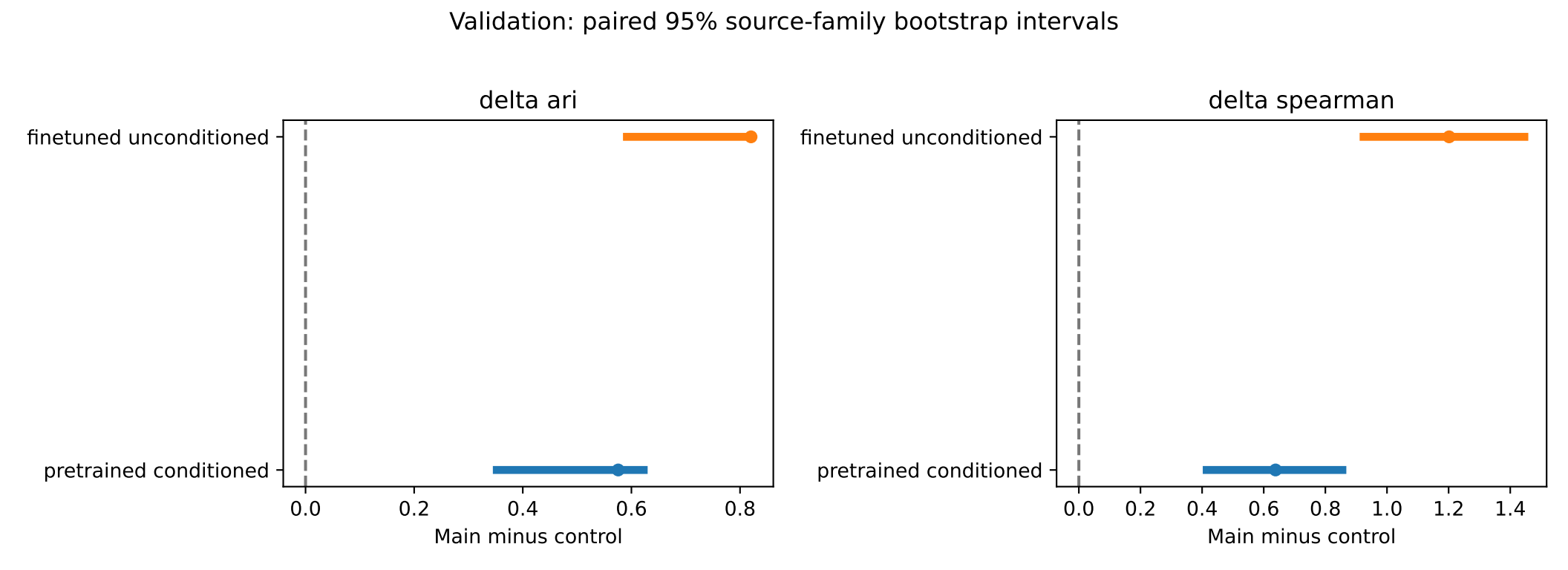

validation: original-pair ratings and raw cosine
finetuned conditioned
pretrained conditioned
finetuned unconditioned
Raw cosine
Audited rating (1–5)
0.0
0.2
0.4
0.6
0.8
1.0
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0


train: original-pair ratings and raw cosine
finetuned conditioned
pretrained conditioned
finetuned unconditioned
Raw cosine
Audited rating (1–5)
0.0
0.2
0.4
0.6
0.8
1.0
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0


test: original-pair ratings and raw cosine
finetuned conditioned
pretrained conditioned
finetuned unconditioned
Raw cosine
Audited rating (1–5)
0.0
0.2
0.4
0.6
0.8
1.0
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0


pooled: original-pair ratings and raw cosine
finetuned conditioned
pretrained conditioned
finetuned unconditioned
Raw cosine
Audited rating (1–5)
1.0
0.8
0.6
0.4
0.2
0.0
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0


Condition-switch coverage and directionality
finetuned conditioned
pretrained conditioned
finetuned unconditioned
Probes
Bidirectional target-color margin
Both directions positive
finetuned conditioned
pretrained conditioned
finetuned unconditioned


Validation: paired 95% source-family bootstrap intervals
delta ari
finetuned unconditioned
pretrained conditioned
0.0
0.2
0.4
0.6
0.8
Main minus control
delta spearman
finetuned unconditioned
pretrained conditioned
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Main minus control